Igor Potapov · Dmytro Chumachenko ·
Volodymyr Proshkin ·
Olexandr Konovalov · Vira Shendryk ·
Oleksandr Berezko ·
James McCafferty (Eds.)

Communications in Computer and Information Science 2647

Digitalisation and Digital Transformation

First Research Twinning Conference, RTC-Digital 2023
Liverpool, UK, March 27–30, 2023
Proceedings

Springer

Communications in Computer and Information Science 2647

Series Editors

Gang Li, *School of Information Technology, Deakin University, Burwood, VIC, Australia*
Joaquim Filipe, *Polytechnic Institute of Setúbal, Setúbal, Portugal*
Zhiwei Xu, *Chinese Academy of Sciences, Beijing, China*

Rationale
The CCIS series is devoted to the publication of proceedings of computer science conferences. Its aim is to efficiently disseminate original research results in informatics in printed and electronic form. While the focus is on publication of peer-reviewed full papers presenting mature work, inclusion of reviewed short papers reporting on work in progress is welcome, too. Besides globally relevant meetings with internationally representative program committees guaranteeing a strict peer-reviewing and paper selection process, conferences run by societies or of high regional or national relevance are also considered for publication.

Topics
The topical scope of CCIS spans the entire spectrum of informatics ranging from foundational topics in the theory of computing to information and communications science and technology and a broad variety of interdisciplinary application fields.

Information for Volume Editors and Authors
Publication in CCIS is free of charge. No royalties are paid, however, we offer registered conference participants temporary free access to the online version of the conference proceedings on SpringerLink (http://link.springer.com) by means of an http referrer from the conference website and/or a number of complimentary printed copies, as specified in the official acceptance email of the event.

CCIS proceedings can be published in time for distribution at conferences or as post-proceedings, and delivered in the form of printed books and/or electronically as USBs and/or e-content licenses for accessing proceedings at SpringerLink. Furthermore, CCIS proceedings are included in the CCIS electronic book series hosted in the SpringerLink digital library at http://link.springer.com/bookseries/7899. Conferences publishing in CCIS are allowed to use our online conference service (Meteor) for managing the whole proceedings lifecycle (from submission and reviewing to preparing for publication) free of charge.

Publication process
The language of publication is exclusively English. Authors publishing in CCIS have to sign the Springer CCIS copyright transfer form, however, they are free to use their material published in CCIS for substantially changed, more elaborate subsequent publications elsewhere. For the preparation of the camera-ready papers/files, authors have to strictly adhere to the Springer CCIS Authors' Instructions and are strongly encouraged to use the CCIS LaTeX style files or templates.

Abstracting/Indexing
CCIS is abstracted/indexed in DBLP, Google Scholar, EI-Compendex, Mathematical Reviews, SCImago, Scopus. CCIS volumes are also submitted for the inclusion in ISI Proceedings.

How to start
To start the evaluation of your proposal for inclusion in the CCIS series, please send an e-mail to ccis@springer.com

Igor Potapov · Dmytro Chumachenko ·
Volodymyr Proshkin · Vira Shendryk ·
Oleksandr Berezko · James McCafferty ·
Olexandr Konovalov
Editors

Digitalisation and Digital Transformation

First Research Twinning Conference, RTC-Digital 2023
Liverpool, UK, March 27–30, 2023
Proceedings

Springer

Editors
Igor Potapov
University of Liverpool
Liverpool, UK

Volodymyr Proshkin
Borys Grinchenko Kyiv Metropolitan University
Kyiv, Ukraine

Oleksandr Berezko
Lviv Polytechnic National University
Lviv, Ukraine

Olexandr Konovalov
School of Computer Science
University of St Andrews
St Andrews, UK

Dmytro Chumachenko
National Aerospace University – Kharkiv Aviation Institute
Kharkiv, Ukraine

Vira Shendryk
Sumy State University
Sumy, Ukraine

James McCafferty
Wellcome Sanger Institute
Hinxton, UK

ISSN 1865-0929 ISSN 1865-0937 (electronic)
Communications in Computer and Information Science
ISBN 978-3-032-04730-4 ISBN 978-3-032-04731-1 (eBook)
https://doi.org/10.1007/978-3-032-04731-1

© The Editor(s) (if applicable) and The Author(s), under exclusive license to Springer Nature Switzerland AG 2026

This work is subject to copyright. All rights are solely and exclusively licensed by the Publisher, whether the whole or part of the material is concerned, specifically the rights of translation, reprinting, reuse of illustrations, recitation, broadcasting, reproduction on microfilms or in any other physical way, and transmission or information storage and retrieval, electronic adaptation, computer software, or by similar or dissimilar methodology now known or hereafter developed.
The use of general descriptive names, registered names, trademarks, service marks, etc. in this publication does not imply, even in the absence of a specific statement, that such names are exempt from the relevant protective laws and regulations and therefore free for general use.
The publisher, the authors and the editors are safe to assume that the advice and information in this book are believed to be true and accurate at the date of publication. Neither the publisher nor the authors or the editors give a warranty, expressed or implied, with respect to the material contained herein or for any errors or omissions that may have been made. The publisher remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

This Springer imprint is published by the registered company Springer Nature Switzerland AG
The registered company address is: Gewerbestrasse 11, 6330 Cham, Switzerland

If disposing of this product, please recycle the paper.

Preface

The Research Twinning Conference on Digitalisation and Digital Transformation promoted research in Computer Science, Digital Technologies, Mathematics, and Computational Engineering that can impact Digitalisation and Digital Transformation and address challenges in modern society and technologies. In addition to the main objective, this conference was also a cornerstone of a broader initiative to integrate Ukrainian academics and research institutions into the global research network, focusing on developing Ukraine's digital infrastructure, economy, and IT sector. In this respect, the event helped to strengthen the collaboration between the UK and Ukraine in the expansive digitalisation and digital transformation fields.

Aligned with the OECD Policy Response "Digitalisation for recovery in Ukraine," our conference sought to harness the potential of digital technologies as a driving force for recovery and sustainable growth. This event brought together top-tier researchers from the UK, Ukraine, and international communities to present and discuss the most pressing challenges and opportunities in digitalisation and digital transformation.

Our comprehensive program included invited talks from leading global experts who introduced the latest research directions, addressed long-term challenges, and provided insights into the landscape of research centres in their fields. Ukrainian academics from various universities and the National Academy of Sciences were also invited to highlight the active research areas and capabilities of Ukrainian research centres and groups.

This conference featured highlight presentations from British and Ukrainian participants, offering a platform for academics, early-career researchers, and PhD students to engage in quick, impactful pitches that opened doors to new research collaborations. Furthermore, our poster sessions facilitated rich, interactive discussions and network-building among participants.

The Research Twinning Conference on Digitalisation and Digital Transformation 2023 served as the starting point for what we expect will be a long and fruitful collaboration. We hope this and subsequent conferences will foster scientific dialogue and play a crucial role in Ukraine's technological and economic advancement.

All accepted papers were single-blindly peer-reviewed by three reviewers and evaluated on the basis of technical quality, relevance, originality, significance, and clarity. The Program Committee and the Editorial Team of this volume were composed of internationally recognised experts who reviewed submissions and selected papers for acceptance and inclusion in the proceedings. 101 papers were submitted, 51 were accepted for oral presentations, and 22 were accepted for poster presentations. Among them 29 papers were selected for publication in this Communications in Computer and Information Science (CCIS) post-proceedings volume. All accepted papers were presented during the conference.

We express our profound appreciation to all the speakers, participants, and organisers who contributed their time and expertise to make this conference successful. We also thank the many supporting institutions that recognised the importance of international

cooperation in fostering technological advancement and recovery during challenging times.

Looking ahead, we are excited about the future of this initiative and the positive changes it will bring to the global and Ukrainian digital landscapes. We eagerly anticipate the continued growth of this conference and the lasting impact of the collaborations it has initiated.

July 2025

Igor Potapov
Dmytro Chumachenko
Olexandr Konovalov
Volodymyr Proshkin
Vira Shendryk
Oleksandr Berezko
James McCafferty

Organization

Program Committee Chairs

Igor Potapov	University of Liverpool, UK
Olexandr Konovalov	University of St Andrews, UK

Program Committee

Vitalina Babenko	V. N. Karazin Kharkiv National University, Ukraine
Oleksandr Letychevskyi	Institute of Cybernetics, NAS of Ukraine, Ukraine
Inna Skarga-Bandurova	Oxford Brookes University, UK
Nataliia Kotenko	Kyiv National University of Trade and Economics, Ukraine
Oleg Pikhurko	University of Warwick, UK
James McCafferty	Wellcome Sanger Institute, UK
Gennadi Malaschonok	National University of Kyiv-Mohyla Academy, Ukraine
Dmytro Chumachenko	National Aerospace University "Kharkiv Aviation Institute", Ukraine
Alaa Alzoubi	University of Derby, UK
Alexei Lisitsa	University of Liverpool, United Kingdom
Shuai Wang	Vrije University Amsterdam, Netherlands
Viktoria Kondratenko	University of Edinburgh, UK
Kateryna Osadcha	Bogdan Khmelnitsky Melitopol State Pedagogical University, Ukraine
Kapil Dev	Liverpool Hope University, UK
Tetiana Zhylenko	Sumy State University, Ukraine
Vira Shendryk	Sumy State University, Ukraine
Oleksandr Makarenko	National Technical University of Ukraine "Igor Sikorsky Kyiv Polytechnic Institute", Ukraine
Leszek Gasieniec	University of Liverpool, UK
Raj Mani Shukla	Anglia Ruskin University, UK
Miao Yu	University of Lincoln, UK
Jordan Bird	Nottingham Trent University, UK
Volodymyr Proshkin	Borys Grinchenko Kyiv University, Ukraine
Kostas Margellos	University of Oxford, UK

James H. Davenport	University of Bath, UK
Iryna Vasylieva	University of Klagenfurt, Austria
Andrij Vasylenko	University of Liverpool, UK
Giuseppe Claudio Guarnera	University of York, UK
Ruslan Davidchack	University of Leicester, UK
Ali Sadiq	Nottingham Trent University, UK
Asma Adnane	Loughborough University, UK
Lei Shi	Newcastle University, UK
Oleksii Ignatenko	Ukrainian Catholic University, Ukraine
Oleksandr Berezko	Lviv Polytechnic National University, Ukraine

Contents

Data Science and Machine Learning Techniques

Application of Machine Learning for Wheat Yield Prediction 3
Petro Hrytsiuk, Tetiana Babych, Oksana Kardash, and Maksym Havryliuk

Research on the Use of the Convolutional Neural Network for Speech Impairment Detection 12
Olha Pronina, Olena Piatykop, Tetiana Levytska, and Irina Fedosova

Learning Perceptual Properties of 3D Shapes: A Brief Synopsis 20
Kapil Dev

Application Surrogate Models to Study Stress State of Dispersion Strengthened Composite 26
Ruslan Babudzhan, Mariia Shapovalova, and Oleksii Vodka

Deep Learning for Space Applications 39
Serhii Khoroshylov and Mykhailo Redka

A Framework for Decision-Support in Landmine Surveying: Spatio-Temporal Analysis of Land Cover Change Using Sentinel-2 47
Alaa AlZoubi and Nasir Ibrahim

Pipeline for Automatic Segmentation of Epithelial Cells in Buccal Swabs Slides with Custom Cellpose-Based Model 54
Dmytro Horyslavets, Oleksandr Skorobohatov, Yaroslav Ryndyk, Pavlo Areshkov, Olena Romantsova, Charlotte Oliver, Anna Paterson, Wei Cope, Emily Joslin, Emily O'Dea, Jon Teague, Peter Clapham, Moritz Przybilla, Mykhailo Tukalo, James McCafferty, Inigo Martincorena, and Alina Frolova

Improving Short Utterance Dialogue Act Classification Using a Transformer Ensemble 60
Harry Maltby, Julie Wall, Tracy Gooodluck Constance, Mansour Moniri, Cornelius Glackin, Marvin Rajwadi, and Nigel Cannings

Mathematical Foundations for Digital Technologies

The Exploration of Elastic SH Waves and Their Second Harmonics in Anisotropic Waveguide in the Form of a Layer on a Halfspace 69
Nadia Zhogoleva

Development of Computational Algorithm of Investigation Self-heating Processes in Elastomers 76
Yuliia Viazovychenko and Oleksiy Larin

Application of the Discrete Models of the Topologies on a Finite Set to the Enumeration and Calculation of All Labeled T_0-Topologies 84
Anna Skryabina and Polina Stegantseva

Crystal Structure Prediction, A Challenging and Fundamental Problem 90
Duncan Adamson

Parallel Methods for Composing Difference Approximations in Simulation Dynamic Processes 97
Olga Dmytriyeva and Vira Huskova

Classification of Transformations of Equivalent Kernels of Some Determinantal Point Processes 104
Harry Sapranidis Mantelos

On the Computational Complexity of the Interval Dependence Problem in Credal Networks 113
Marco de Angelis, Hector Estrada Lugo, Scott Ferson, and Edoardo Patelli

Homogenization as a Method for Modeling Processes in Photonic Crystals 119
Gennadiy V. Sandrakov

New Cellular Automata. Models, Algorithms, Applications and Anticipation Accounting 127
Aleksander Makarenko

The Maximum Cover for Passive Localisation 135
Igor Potapov, Jason F. Ralph, and Theofilos Triommatis

Mathematical Formalisms for Relational (Table) Databases 146
Iryna Lysenko

Digital Technologies and Applications

Problems of Implementation of the Centre for Parallel Simulation Technologies 155
Serhii Kovalov, Oleksandr Miroshkin, Michael Resch, and Volodymyr Svjatnyj

Application of Digital Twins for High-Tech Enterprises and Infrastructure Objects in the Context of Digital Transformation 161
Oleksandr Prokhorov, Oleg Fedorovich, and Evgeniy Druzhinin

Development and Implementation of Digital Transformation Tools in the Domain of Apparel Design 169
Oksana Zakharkevich, Tetiana Zhylenko, Julia Koshevko, and Galina Shvets

Mathematical Learning Experiment Using Digital Technologies 175
Mariia Astafieva, Dmytro Bodnenko, Oleksandra Lokaziuk, Oksana Lytvyn, and Volodymyr Proshkin

Mathpar Project 182
Gennadi Malaschonok

Two-Step Singular Value Decomposition on a Single GPU with CUDA 190
Serhii Sukharskyi

New Evolutionary Methods for Solving Optimization Problems in the Digital Transformation Processes 197
Vitaliy Snytyuk and Nataliia Tmienova

Algebraic Approach to Molecular Modelling and Its Application in Studies of the Effect of Radiation Therapy on Cancer Treatment 205
Yuliia Tarasich

Computer Design and 3D Printing of New-Generation Multifractals 212
Ivan Grabar and Yuri Kubrak

Application of an Algebraic Approach to the Formalization of Complex Biochemical Processes at the Level of Quantum Interactions 221
Hanna Soloshenko

Author Index 229

Data Science and Machine Learning Techniques

Application of Machine Learning for Wheat Yield Prediction

Petro Hrytsiuk(✉), Tetiana Babych, Oksana Kardash, and Maksym Havryliuk

The National University of Water and Environmental Engineering, Soborna Str., 11, Rivne 33028, Ukraine
p.m.hrytsiuk@nuwm.edu.ua

Abstract. The progressive informatization of society allows us to receive an increasing amount of agriculture yield data and climate data. This allows to use of available data for more accurate forecasting of grain yields, grain prices, analysis of crop losses, etc. Climatic factors play a decisive role in grain yield fluctuations. Using climate databases makes it possible to build yield prognostic models, which allow in advance (in 3 months) to estimate the future yield. Pre-harvest yield forecasting can assist grain producers in making the necessary arrangements for the storage and marketing of the crop.

In this work, a study of the influence of climatic factors on wheat yield fluctuations was carried out using machine learning techniques. We divided the detrended yield values into two groups, labeled as "low yield" and "high yield". Five machine-learning models were trained on the available data and were used as classifiers. The random forest model and support vector machine are the most effective classifiers and provide 85% classification accuracy.

Keywords: Wheat yield · climatic factors · machine learning · classification · cross-validation

1 Introduction

Grain production is one of Ukraine's most important agricultural production branches. Average annual grain production in Ukraine for 2019–2021 reached 75 million tons. In 2021, a record crop of 84 million tons of grain was harvested in Ukraine. The average annual export during this time was 50 million tons.

However, the gross harvest of grain and grain yield in Ukraine experience significant fluctuations. This is due to the influence of changing climatic factors, which have undergone significant changes over the past 30 years. The warming of the climate led to a change in the assortment of cultivated grain crops and the geography of their location [2]. In recent years, against climate change, the share of wheat in the total grain harvest has decreased from 50% to 40%, and the share of corn has increased from 15% to 42% [1]. In the chernozem zone of Ukraine and the Polissia zone, there is a significant increase in the production of heat-loving crops, such as corn, soybeans, and sunflowers. The area of grain crops in the western regions of Ukraine has significantly expanded.

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2026
I. Potapov et al. (Eds.): RTC-Digital 2023, CCIS 2647, pp. 3–11, 2026.
https://doi.org/10.1007/978-3-032-04731-1_1

The basis for making business decisions in grain production is the forecasting of crop yields [3]. Predicting crop yields is a complex task, given the random nature of many influencing factors. At the current stage, state-of-the-art mathematical modeling methods, with a prominent place given to machine learning techniques and artificial neural network technologies, are employed in constructing predictive models. In this research, there was used such a powerful machine-learning tool as a classification method.

Based on the statistical analysis results, the wheat yield in each region of Ukraine was separated into a trend component and the detrended yield. Detrended yield is calculated as the difference between the actual yield and the yield trend. The trend's emergence is attributed to financial factors (investments in grain production) and technological factors (innovations). Climatic factors are the primary cause of fluctuations in detrended yield. Investigating the influence of climatic factors on detrended yield fluctuations in the western region of Ukraine is the main focus of this article.

2 Literature Review

The yield of cereal crops is largely influenced by weather and climate. The impact of climate change on winter wheat yield in three ecological zones of Ukraine was examined in an analytical review conducted as part of the German-Ukrainian agricultural policy dialogue project [4]. The authors argue that future climate changes could exacerbate the negative impact on grain production in the Ukraine steppe zone and contribute to this region's desertification.

The progress in science and technology has accumulated extensive statistical data from diverse agricultural sectors. This has significantly enhanced the potential for using computer technologies to analyze and model the impact of climate on crop yields. The advent of big data technology has led to the emergence of powerful new analytical tools, including machine learning techniques, which have demonstrated their effectiveness in fields such as medicine, finance, and biology. In recent years, many publications have discussed the application of machine learning methods in forecasting agricultural crop yields.

The paper [5] aims to identify the best yield prediction model to help farmers decide which crop to grow based on climate conditions and nutrients present in the soil. In an analysis of yield prediction by three different supervised machine learning models, the authors concluded that the best accuracy was achieved with the Random Forest Classifier in Entropy and Gini Criterion.

Early and reliable seasonal yield forecasts are critical at the state level and for farmers. In the study [6], the authors developed an improved wheat yield forecasting system based on multiple linear regression and machine learning. The paper [7] proposed predictive models for state-level wheat yield variations in Australia. Using machine learning techniques, the models combine large-scale climate factors with more localized environmental predictions. The results demonstrate reliable forecasting of wheat yield fluctuations at the state level three months before harvest. A demonstration of the more effective use of machine learning and regression methods compared to process-based models can also be seen in [8]. The authors analyzed the expediency of using various methods for forecasting long-term trends, the average level, and extreme corn yield values in the USA example.

The research authors [9] concentrated on the problem that is an obstacle to using machine learning in seasonal forecasting applications. This is a limited sample size of observational data for training models. To avoid this problem, the expediency of training various machine learning approaches on a large ensemble of climate models is considered. A new approach to seasonal precipitation forecasting is shown using the example of the western part of the USA. Several separate machine learning approaches were trained on simulations of large climate models, and then their predictions were combined into an ensemble to predict large-scale patterns of precipitation anomalies further.

The work [10] established that the dependence of crop yield on weather conditions is non-linear. Machine learning methods allow modeling nonlinear dependencies much more effectively than linear statistical methods. The significant impact of extreme weather indicators on the yield of agricultural crops has been confirmed, which justifies the need to include them in the model as predictors. The artificial neural network model for yield dynamics modeling was also used in the works [11, 12].

3 Methodology and Results

3.1 Data Collection

In terms of significance, wheat is central to Ukraine's grain crops. Ukraine's average annual wheat production from 2019 to 2021 was 26.5 million tons. Wheat accounted for 38% of the total grain exports during this period.

From the perspective of natural-climatic characteristics influencing the cultivation of grain crops, the territory of Ukraine can be conditionally divided into three major zones: the steppe zone (Kirovohrad, Dnipropetrovsk, Zaporizhzhia, Kherson, Mykolaiv, and Odesa subregions), the forest-steppe chernozem zone (Chernihiv, Sumy, Poltava, Kharkiv, Kyiv, Cherkasy, and Vinnytsia subregions), and the western region (Volyn, Rivne, Zhytomyr, Lviv, Ternopil, and Khmelnytskyi subregions). Transcarpathian, Ivano-Frankivsk, and Chernivtsi regions exhibit distinct climatic characteristics that set them apart from the previously mentioned clusters, and they do not fall within any of the aforementioned groups.

This study used statistical climate data [13] and wheat yield data [1] from 2000 to 2021 for the Volyn, Rivne, Lviv, Ternopil, Khmelnytskyi, and Zhytomyr subregions, which are located in the western region of Ukraine. To assess the climate's impact on wheat yield, we used average decadal temperature values for April, May, and June (factors t1–t9) and monthly precipitation amounts (R10, R20, R30) during this period. Decadal temperature values allow for a more precise consideration of the air temperature's influence at various stages of plant growth. Monthly precipitation totals were used, taking into account the long-term aftereffects of precipitation on plant development. To meet the data requirements of machine learning techniques, we consolidated the data from the six aforementioned subregions into a single dataset. This was made possible by similar climatic and soil characteristics in these regions.

3.2 Analysis of Wheat Yield Dynamics

An analysis of wheat yield dynamics in the regions of Ukraine over the past 22 years shows that the yield is increasing. This resulted from significant investments in the grain industry in recent years. As a result, the seed base improved, the agrotechnical culture improved, and the logistics network of grain production developed. However, the tendency to increase grain yield is accompanied by significant fluctuations in yield, the cause of which is mostly the impact of weather and climate factors. The wheat yield dynamic in the Khmelnytskyi region can serve as an illustration (Fig. 1). The largest negative yield deviations from the trend were observed in 2006, 2010, and 2020. The weather conditions of these years (cold May, a significant amount of precipitation in May and June) were unfavorable for wheat vegetation.

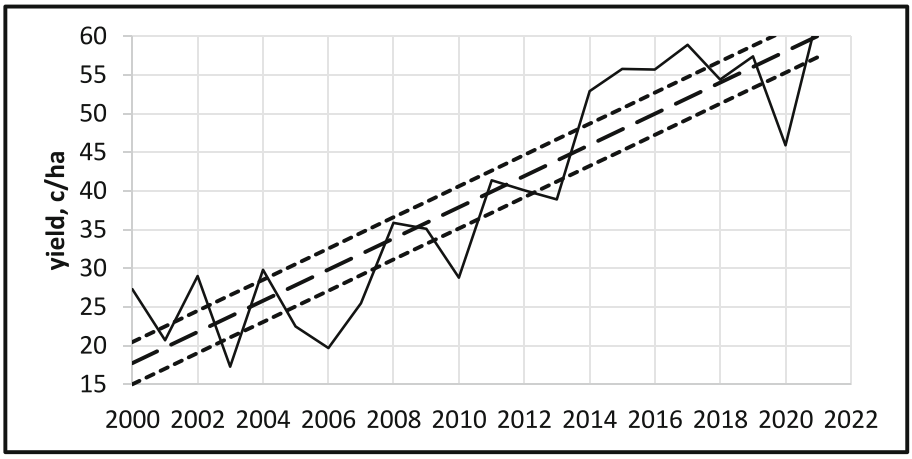


Fig. 1. Wheat yield dynamics in the Khmelnytskyi region. The dashed line is a linear trend. Dotted lines are high and low yield boundaries. Author's calculations according to [1].

To model yield dynamics, a linear trend model, we used

$$tr_t = a_0 + ta_1, \tag{1}$$

Here a_0, a_1 - the trend coefficients, determined by statistical data using the least squares method. Climatic impact can be detected by studying fluctuations in detrended yield *eps*

$$eps_t = y_t - tr_t, \tag{2}$$

To identify the distribution law of detrended yield, we used a combined sample of detrended yield for six regions of the western region of Ukraine (132 values). We tested the fit of the detrended yield distribution to three distribution laws: the normal distribution, the Laplace distribution, and the Cauchy distribution. The closest to the actual distribution of detrended yields is the normal distribution. However, the reliability level of the conclusion regarding the correspondence of the detrended yields to the normal distribution is slightly lower than 95%.

3.3 Binarization of Detrended Yield

When making investment decisions and planning the logistics of the agrarian business, it is not necessary to have an accurate yield forecast. Usually, in this case, it is enough to estimate the future yield in terms of "high yield" – "low yield." At the same time, "high yield" means such a value of yield that significantly exceeds the average yield level; "low yield" means a yield value significantly lower than the average yield value. This approach enables the use of classification methods in yield forecasting.

In this research, our main task is to forecast low wheat yield values. To the "low yield" group, we include those yield values with a probability of $p < 0.33$, located on the integral curve of the normal distribution of detrended yields. Yield values for which condition $p \geq 0.33$ is fulfilled will be assigned to the "high yield" group. To implement a classification approach to yield prediction, we introduce a binary variable eps1 that takes only two values: 1 ("low yield") and 0 ("high yield")

$$eps1 = \begin{cases} 1, if F(eps) < 0.33; \\ 0, if F(eps) \geq 0.33. \end{cases}, \tag{3}$$

3.4 Binarization of Detrended Yield

The most important impacting factors that determine grain yield fluctuation are climatic factors. In this work, we used the average decadal temperature values for April, May, and June and the monthly precipitation amounts across six regions in the western zone of Ukraine. Correlation analysis showed that cold May and June, moderate rainfall in May and June contributed to the successful vegetation of wheat in this region. Factors that play a decisive role in the fluctuation of wheat yields in different regions of Ukraine are listed in Table 1. The ' + ' sign indicates a positive impact of the factor, while the '-' sign denotes a negative influence.

Table 1. Influence of climatic factors on wheat yield.

	t1	*t2*	*t3*	*t4*	*t5*	*t6*	*t7*	*t8*	*t9*	*R10*	*R20*	*R30*
Steppe region					–		–			+		
Forest-steppe			+	–			–					
Western region					–							–

We will build a model according to which the wheat yield is formed under the impact of 12 climatic factors: t1, t2, t3, t4, t5, t6, t7, t8, t9, R10, R20, R30. We believe that climatic factors affect not the average yield but the yield deviation from the trend value (detrended yield). Thus, we will consider the detrended yield epst as a response to the action of these factors according to (2).

The linear multiple regression equation has the following form:

$$\begin{aligned} eps = {} & \beta_0 + \beta_1 t_1 + \beta_2 t_2 + \beta_3 t_3 + \beta_4 t_4 + \beta_5 t_5 + \beta_6 t_6 + \beta_7 t_7 \\ & + \beta_8 t_8 + \beta_9 t_9 + \beta_{10} R_{10} + \beta_{20} R_{20} + \beta_{30} R_{30} + \varepsilon, , \end{aligned} \tag{4}$$

Here *β0, β1, β2, β3, β4, β5, β6, β7, β8, β9, β10, β20, β30* are model parameters, *t1, t2, t3, t4, t5, t6, t7, t8, t9, R10, R20, R30* are model factors, *eps* is response, ε is model residual. One of the obstacles to using machine learning in seasonal forecasting tasks is the limited size of observational data samples. To solve this problem, we implemented model training on aggregated sample ensembles that combine data from neighboring subregions with similar weather, climatic conditions, and soil types [9]. As described above, we formed these samples by grouping Ukrainian subregions into agro-climatic zones. The corresponding data samples will range from 132 to 154 instances, each containing detrended yield (eps) and 12 climatic factors. We used the R programming environment with the RStudio extension for data processing (Table 3).

Table 2. Values of estimates of the linear regression model LM2.

	Estimate	Std. Error	t-value	Pr(>|t|)	Signif. Codes
(Intercept)	9.802	3.246	3.020	0.0032	**
t3	0.265	0.161	1.642	0.1038	
t5	–0.603	0.209	–2.882	0.0048	**
R30	–0.045	0.010	–4.363	3.1e-0.5	***

Table 3. Quality criteria of machine learning techniques.

	Accuracy	Sensitivity	Specificity	AUC
LDA	0.718	0.417	0.852	0.634
SVML	0.718	0.500	0.815	0.657
SVMR	0.846	0.583	0.963	0.773
RF	0.846	0.500	1.000	0.750
GLM	0.718	0.333	0.889	0.611

Guided by the principles of statistical modeling [14], we will divide all the data into two parts: the training sample (99 rows, randomly selected - 75% of the data) and the control sample (33 rows that are not included in the training sample –25% of the data). The linear regression model built on all factors is generally adequate, but most of the factors of this model are insignificant. After eliminating insignificant factors, we obtained the following model (Table 2). The linear regression model is adequate (F-statistic = 8.00 on 3 and 101 DF; p = 7.75e-05). The model shows that factors t3, t5, and R30 impact wheat yield fluctuations most.

3.5 Methods of Classification and Their Assessment

Recently, classification methods have been widely used for data-driven decision-making. We binarized the detrended yield for categorical wheat yield forecasting according to (3).

We applied several binary classification methods to the obtained data: logistic regression, linear discriminant analysis, decision tree model, random forest model, and support vector machine. Confusion matrix is used to evaluate the performance of the classifiers we built [15]. Four cases are distinguished in the matrix:

TP (True Positives) – the model correctly detected a low yield value; FP (False Positives) – the model wrongly recognized a high yield as a low yield; FN (False Negatives) – the model wrongly recognized a low yield as a high yield; TN (True Negatives) – the model correctly identified a case of high yield.

Usually used, the following performance indicators of the binary classifier can be determined:

- *Sensitivity* $SE = TP/(TP + FN)$, determines the share of correctly identified low-yield cases among all low-yield cases;
- *Specificity* $SP = TN/(TN + FP)$, determines the share of correctly identified high-yield cases among all high-yield cases;
- *Accuracy* $AC = (TP + TN)/(TP + FP + FN + TN)$, determines the overall probability of the test giving correct results.

Method of cross-validation. Even when a large data set and random sampling have been applied to constructing the training sample, the resulting model may be statistically unreliable. After all, another set of samples can lead to another model significantly different from the first. This shortcoming can be eliminated by the cross-validation method. Let's briefly outline its essence [16].

The original data set is divided into k groups ("folds") of approximately the same size. One of the folds is defined as the test sample. All other folds form the training sample. This process is repeated k times using a different fold as the test sample. We then calculate the total test MSE as the average of k test MSEs. We used the cross-validation method to implement all yield classification methods.

Comparison of the classification models effectiveness. We used five methods to build binary classification models: linear discriminant analysis (LDA), support vector machine with linear kernel function (SVML), support vector machine with radial kernel function (SVMR), random forest method (RF), logistic regression method (GLM). We compared them using the method described in [17]. The SVMR and RF methods showed the best accuracy among the tested machine-learning techniques. The SVMR method also demonstrated the highest sensitivity, which is especially important for predicting low yields.

A universal method for comparing the classifiers' accuracy is ROC analysis [18]. The area under the ROC curve AUC is a criterion for evaluating the classifier. According to the AUC criterion, the best classifiers are the support vector machine (radial kernel function) and the random forest method.

4 Conclusions

When making investment decisions and planning material and technical support for agrarian business, it is enough to estimate the future yield in the "high yield" - "low yield" categories. We could compare the accuracy of different machine learning classification algorithms by performing yield binarization.

The random forest model and support vector machine showed the best detrended wheat yield classification accuracy and provided 85% prediction accuracy. This classification accuracy is very good for the complex natural processes we are considering.

The methodology we proposed is quite simple in terms of implementation, and grain producers can use it for short-term yield forecasting. This technique can be extended to studying climate influence on agricultural crops in various regions and countries. In our recent article [19], we investigated the influence of climatic factors on wheat yield fluctuations in the chernozem zone of Ukraine. This study aimed to investigate the influence of climate on grain production processes in the western region of Ukraine. Hence, this work represents the concluding segment of a research cycle dedicated to studying the influence of climate on wheat yield fluctuations across various regions of Ukraine.

Our research contributes to solving the problem of ensuring the sustainability of grain production in Ukraine. The obtained results can be used to support Ukraine's economic development and solve the world's food problem.

References

1. State Statistics Service of Ukraine (2022). http://www.ukrstat.gov.ua
2. Adamenko, T.: Climate change and agriculture in Ukraine: what farmers should know. German-Ukrainian agropolitical dialogue, Zapovit, Kyiv (2019)
3. Hrytsiuk, P., Babych, T., Mandziuk, O.: Region sown areas portfolio optimization taking into account crop production economic risk. Glob. J. Environ. Sci. Manage. **5**(SI), 140–150 (2019)
4. Muller, D., Jungandreas, A., Koch, F., Shirhorn, F.: The impact of climate change on wheat production in Ukraine. Report on agricultural policy (APD) (2016)
5. Kalimuthu, M., Vaishnavi, P., Kishore, M.: Crop prediction using machine learning. In: Proceedings of the Third International Conference on Smart Systems and Inventive Technology, ICSSIT '2020, pp. 926–932, IEEE (2020). https://doi.org/10.1109/ICSSIT48917.2020.9214190
6. Feng, P., et al.: Dynamic wheat yield forecasts are improved by a hybrid approach using a biophysical model and machine learning technique. Agric. Forest Meteorol. 285–286 (2020). https://doi.org/10.1016/j.agrformet.2020.107922
7. Wang, B., Feng, P., Waters, C., Cleverly, J., Liu, D.L., Yu, Q.: Quantifying the impacts of pre-occurred ENSO signals on wheat yield variation using machine learning in Australia. Agric. Forest Meteorol. **291** (2020). https://doi.org/10.1016/j.agrformet.2020.108043
8. Leng, G., Hall, J.W.: Predicting spatial and temporal variability in crop yields: an inter-comparison of machine learning, regression and process-based models. Environ. Res. Lett. **15**, 4 (2020). https://doi.org/10.1088/1748-9326/ab7b24
9. Gibson, P.B., Chapman, W.E., Altinok, A., Monache, L.D., DeFlorio, M.J., Waliser, D.E.: Training machine learning models on climate model output yields skillful interpretable seasonal precipitation forecasts. Commun. Earth Environ. **2** (2021). https://doi.org/10.1038/s43247-021-00225-4
10. Konduri, V.S., Vandal, T.J., Ganguly, S., Ganguly, A.R.: Data science for weather impacts on crop yield. Front. Sustain. Food Syst. **4** (2020). https://doi.org/10.3389/fsufs.2020.00052
11. Crane-Droesch, A.: Machine learning methods for crop yield prediction and climate change impact assessment in agriculture. Environ. Res. Lett. **13**(11), 1–12 (2018). https://doi.org/10.1088/1748-9326/aae159

12. van Klompenburg, T., Kassahun, A., Catal, C.: Crop yield prediction using machine learning: a systematic literature review. Comput. Electr. Agric. **177** (2020). https://doi.org/10.1016/j.compag.2020.105709
13. Meteorological data archive. [Electronic resource] – Access mode: https://meteopost.com/weather/archive/
14. Gareth, J., Witten, D., Hastie, T., Tibshirani, R.: An Introduction to Statistical Learning, Springer, New York, NY (2013). https://doi.org/10.1007/978-1-4614-7138-7
15. Zumel, N., Mount, J.: Practical Data Science with R. Manning Publications Co., New York, NY (2020)
16. Berrar, D.: Cross-Validation. In: Ranganathan, S., Gribskov, M., Nakai, K., Schönbach, C., (eds.), The Encyclopedia of Bioinformatics and Computational Biology. Academic Press, Oxford, pp. 542–545 (2019). https://doi.org/10.1016/B978-0-12-809633-8.20349-X
17. Kuhn, M., Johnson, K.: Applied Predictive Modeling, Springer, New York, NY (2013). https://doi.org/10.1007/978-1-4614-6849-3
18. Fawcett, T.: An Introduction to ROC Analysis. Pattern Recogn. Lett. **27**(8), 861–874 (2006). https://doi.org/10.1016/j.patrec.2005.10.010
19. Hrytsiuk, P., Babych, T., Baranovsky, S., Havryliuk, M. Assessing of climate impact on wheat yield using machine learning techniques. In: CEUR Workshop Proceedings, vol. 3513, pp. 314–329 (2023)

Research on the Use of the Convolutional Neural Network for Speech Impairment Detection

Olha Pronina, Olena Piatykop(✉), Tetiana Levytska, and Irina Fedosova

State Higher Educational Institution «Pryazovskyi State Technical University»,
19 Dmytro Yavornytskyi Av, Dnipro, Ukraine
piatykop_o_ye@pstu.edu

Abstract. The development of human speech begins in early childhood. It is very important to identify a child's speech disorders in a timely manner. Because speech impediments can impact a child's learning and social interactions. Classification of speech defects at an early age will allow correction and treatment to be carried out in the most effective way. Analysis of the literature showed that CNN models can successfully detect speech defects. Therefore, this paper explores the architecture of a convolutional neural network to solve the problem of speech defect detection. To use a CNN, audio signals of a child's speech are first converted into spectrograms (images representing the frequency and time characteristics of the audio signal). The convolved neural network layers are then used to automatically extract essential features from the spectrogram. The article describes the architecture of the CNN for identifying four types of speech defects (dyslexia, stuttering, dysphonia, or dyslalia), which made it possible to obtain recognition of defects with recognition results of 77–79%.

Keywords: Deep Learning · CNN · speech defects · model of a convolutional neural network

1 Introduction

To successfully exist in society, a person needs the most important skill - the ability to communicate with others. When communicating, a person conveys some information to the interlocutor using gestures and facial expressions. But the bulk of it comes from speech. Normal development of children certainly includes the acquisition of coherent speech. It is not an innate skill for people. The baby must master it independently. The basics of using speech and its development occur in early childhood. That is, from one year to 5–6 years, the basis is laid on which the child will subsequently build his entire life in society. Speech activity in children is directly related to the formation of the psyche and affects the intellectual, sensory, emotional, and volitional spheres of life. An important step is the timely diagnosis of speech disorders in a child. Speech disorders in childhood are many disorders that differ in the mechanism of formation, structure, and means of overcoming them.

Currently, there is no unified classification of speech disorders. However, all violations are divided into their own classes, and the following defects are more common:

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2026
I. Potapov et al. (Eds.): RTC-Digital 2023, CCIS 2647, pp. 12–19, 2026.
https://doi.org/10.1007/978-3-032-04731-1_2

dyslexia, stuttering, dysphonia, or dyslalia. The task of identifying speech defects can be classified as a classification task. To solve these problems, information technologies based on Artificial Intelligence and Machine Learning can be successfully applied. Intelligent data processing capabilities will allow you to recognize and determine the type of speech defect. Speech defect classification models can then be used to create specialized software. Such software will allow for initial diagnostics.

2 Literature Review

The rapid spread of deep learning methods has also affected the field of knowledge processing. The problem of using convolutional neural networks for speech processing is relevant and many authors are studying this problem. The rapid proliferation of deep learning methods has also affected the field of knowledge processing. Thus, the article [1] provides an overview of modern deep learning methods for processing audio signals: deep learning models are considered, including convolutional neural networks and neural networks more specific to audio models.

The authors in their article [2] consider the problem of identifying and classifying different forms of stuttering. The authors propose a model that is based on acoustic characteristics. This model allows the classification of several types of stuttering without the speech recognition process. The model proposed by the authors uses a deep residual network and bidirectional layers of long short-term memory. The same authors describe in the article [3] their successes in training a deep neural network on the publicly available UCLASS dataset.

The article [4] provides an up-to-date, comprehensive review of various researchers' acoustic characteristics and classification methods to identify and recognize stuttering disorders. The authors believe that real-time stuttering detection is not yet available, and more powerful models and datasets are needed to detect stuttering on multilingual platforms.

The article's authors describe the relevance of using convolutional neural networks for speech recognition [5]. Their article is devoted to solving the problem of automatic detection of pathological voice disorders. Spectrograms of normal and pathological speech recordings are used as input data for the convolutional neural network.

Their work [6] compares convolutional and recurrent neural networks to identify voice pathologies. The Python language was used using the TensorFlow package to conduct the research. As a result, the authors obtained an accuracy of the convolutional neural network, which was 87.11%.

The work [7] is also devoted to the task of identifying voice pathology. The authors describe successful experience in using convolutional neural networks for speech processing. The authors optimize a convolutional neural network using the artificial bee colony (ABC) algorithm.

Thus, today, convolutional neural networks and deep learning have proven themselves quite well for solving problems of voice processing and identifying its various deviations. Many authors are already improving and optimizing models for their tasks [8]. Therefore, further research in this area is urgent. Therefore, this paper explores the architecture of a convolutional neural network to solve the problem of speech defect

detection. Classification of speech defects at an early age will allow correction and treatment to be carried out in the most effective way.

3 Research of Convolutional Neural Network for Speech Impairment Classification

3.1 Stages of Classification of Speech Defects

Convolutional neural networks are widely used in image processing, but they can also be adapted to work with serial data such as audio signals, making them valuable tools in language processing. To use a CNN for detecting speech defects, you need to convert audio signals into spectrograms (images representing an audio signal's frequency and time characteristics). The convolved neural network layers can automatically extract essential features from the spectrogram. Therefore, convolutional neural networks (CNNs) can detect speech defects in children. To successfully use a CNN, you need good data labeling and training with a sufficient number of different examples. This will allow the network to establish an image of patterns characteristic of different types of speech defects.

The process of identifying speech defects includes several stages. In the first stage, the data with which the system will work is prepared. For the study, publicly available video materials were selected for all types of disorders: dyslexia, stuttering, dysphonia, or dyslalia. This data can be obtained from the global network: a video of speech therapists talking or working with children with speech disorders. All videos were filmed with speech disorders of children who have not yet started working with a speech therapist. Each video is approximately ten seconds long. The input data for the model is audio signals selected and processed from those found in the public domain, with labels of the type of speech defect established by the speech therapist.

In the second stage, audio signals are processed and divided into phrases. Splitting the speech audio file uses Python libraries, namely pydup, and Librosa. This allows you to break the replicas into words and build a spectrogram for further analysis. The signals from these video materials are processed and cut into phrases and words. These audio signals are then converted into a spectrogram. A spectrogram is a visual representation of the spectrum at a signal's frequencies and how it changes over time. An example is shown in Fig. 1.

A sound spectrogram is used to phonetically identify spoken words and analyze them. It is displayed as a heat map, an image whose intensity is displayed by changing color or brightness. Frequencies are shown on the vertical axis, and time is shown on the horizontal axis. Moreover, the color intensity increases with increasing density. After this, the spectrogram data was fed to the neural network as a training sample.

The next stage is creating and training a convolutional neural network model. After this, the model is trained and trained. With high-quality preparation and confirmation of performance, its further use occurs.

Thus, the general algorithm for data processing and the process of working with a model for identifying speech defects consists of the following stages:

- Receiving sound audio data from the child

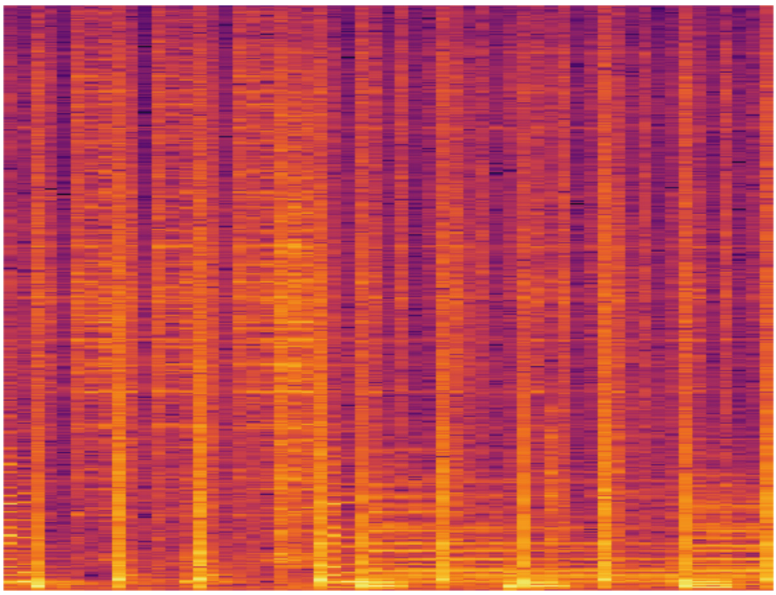


Fig. 1. Example of a spectrogram.

- Audio signal processing
- Break into phrases
- Conversion to spectrogram
- Creating a model
- Model training on defect samples
- Using the model

3.2 CNN Structure

Based on the features of the task and analysis of the literature, a convolutional neural network was chosen as a neural network. Since it has a number of advantages for solving the problem of speech recognition, when using a convolutional neural network, you can significantly reduce the number of training parameters and get a high-quality classification. Another reason why a convolutional neural network was chosen is because it provides partial resistance to scale changes, displacements, rotations, angle changes, and other distortions. Based on the advantages, the structure of a convolutional neural network was developed, which is used to train the model to solve the problem of determining the type of speech impairment. The zero layer of a convolutional neural network is the layer with the input data. The first layer of a convolutional neural network is the Convolution (Conv2D) layer. It performs the convolution process. Its primary input parameters are the number of filters, the size of the convolution window, and an array with the value of the input data size. The value for the number of filters was set to 32, and the size of the convolution window was set to 3 by 3. The value for the size of the input was set to 64 by 64.

The second layer reduces the data dimension and gets essential information using the AveragePooling2D() function. AveragePooling2D(), in turn, acquires the size of the pooling window; the value for it was set to 2 by 2. The third layer is the activated layer. The relu activation function was chosen as the primary activation function. Its main advantage is that it avoids and fixes the disappearing gradient problem and is less computationally intensive. The fourth layer is the Conv2D convolution layer. It had the following values: 64 for filters, 3 by 3 convolution window size, and offset type - same. The image size remains unchanged if you select the "same" offset.

The fifth layer consists of the AveragePooling2D() function, which reduces the data. This layer's inputs are: The pool window size is two. The sixth layer is the activated layer. The relu activation function is chosen for this activation layer. This function returns 0 if it accepts an opposing argument; if it accepts a positive argument, it returns it. The seventh layer is convolutional. It has the same input values and parameters as the fourth layer. The eighth layer serves to reduce the dimension of the data. It consists of the function AveragePooling2D and takes values such as the size of the decrease window, which has been set to 2 by 2. The ninth layer is the activation layer; like the sixth layer, the primary activation function is relu.

The tenth layer is the pull-out layer. The Flatten function performs a tensor flattening operation, changing the tensor's shape to equal the number of elements contained in the tensor, regardless of the batch size. The value was left at the default. The eleventh layer is the model adjustment layer. Using the Dropout() function, we indicate that the layer is regulating. The parameter to be passed to Dropout indicates how likely the neurons will be disabled. It was set to 0.5, meaning half of the neuron will be turned off. The twelfth layer is a fully connected Dense layer. The primary function of this layer is to receive information from all nodes of the previous layer. The input value is the dimension of the output space; for this layer, the dimension has been set to 64.

The thirteenth layer is the active layer; like the ninth layer, it is the activation layer with relu activation function. This function was chosen because there are no special requirements for the output value of the neuron within the framework of the task. The fourteenth layer is the learning adjustment layer. Like the eleventh layer, it takes only one parameter, and it is equal to 0.5, which means that half of the neurons will not be involved in training. The fifteenth layer is fully connected and receives information from all other layers. The dimension of this method depends on the number of classes in which the model is trained. The number of training classes was chosen to be 4, so the dimension value was set to 4. The last layer is the activation layer. The softmax activation function was installed for it. Softmax turns a vector of real numbers into a vector of probabilities for the selected classes. The deviation turns them into values between 0 and 1 to be interpreted as probabilities. However, if one of the inputs is significantly tiny or negative, the softmax function turns it into a low probability.

Conversely, if the input is significant, softmax turns it into a significant probability. In this case, the values will always remain between 0 and 1, suitable for the problem statement for the selected subject area.

Thus, the Softmax function transforms the feature vector obtained in the neural network for each image, a vector consisting of numbers, which in turn are in the range from 0 to 1, and the sum of all elements of each such vector is equal to one. Each element

of this vector can be estimated as the probability that the image in question belongs to the corresponding element of the class.

A study was conducted on a convolutional neural network. The study explores the model with different numbers of hidden convolutional layers. Table 1 presents the test results. The "loss" column is the value of the loss function, the lower this indicator, the better. The auc metric is used to evaluate the model's weight tuning, i.e. How accurate is the model's prediction compared to the ground truth. The "Val_loss" metric is the value of the loss function, but unlike "loss", it is performed at the end of the epoch and on the test sample. The "Val_acc" metric, like "val_loss", performs its own operations, namely validating the configured weights on the test sample. The "L" column value indicates the number of hidden convolutional layers in the neural network. Column "E" indicates the number of epochs specified during training.

Three hidden convolutional layers turned out to be optimal; if you take lower, the model has a low validation score acc; if you take more than three layers, the model begins to overtrain.

Table 1. The network parameters research.

№	loss	auc	val_loss	val_auc	L	E
1	4.6534	0.1032	4.2643	0.1544	1	50
2	4.1234	0.1432	3.8043	0.1943	1	55
3	3.7665	0.1898	3.4766	0.2367	1	60
4	3.2854	0.2243	3.1032	0.2665	1	65
5	2.8554	0.2598	2.6553	0.3332	1	70
6	2.4544	0.2943	2.502	0.3698	1	75
7	2.6522	0.2153	2.2522	0.3754	2	50
8	1.8440	0.3553	1.5440	0.4923	2	55
9	1.2522	0.4553	0.9586	0.5676	2	60
10	0.9522	0.5	0.7808	0.6953	2	65
11	0.6926	0.6176	0.6926	0.7498	2	70
12	0.5382	0.6912	0.5382	0.8123	2	75
13	1.8775	0.4865	1.5434	0.5143	3	50
14	1.6543	0.5676	1.2343	0.5832	3	55
15	1.2546	0.6354	0.9543	0.6643	3	60
16	0.9543	0.7323	0.6436	0.7643	3	65
17	0.6764	0.7654	0.3454	0.7943	3	70
18	0.4231	0.7954	0.1723	0.8565	3	75
19	0.2343	0.6423	0.1023	0.7253	4	50
20	0.1032	0.7823	0.0854	0.8543	4	55
21	0.0634	0.8543	0.0432	0.9631	4	60
22	0.0154	0.983	0.0090	1	4	65
23	0.0075	1	0.0090	1	4	70
24	0.0034	1	0.0090	1	4	75

After training the model, a data set that had not previously been used in training was taken, and the model was tested. A total of 32 experiments were conducted 8 experiments for each type of speech disorder. The results for determining dyslalia were 78.78%, dyslexia −78.74%, stuttering –77.55%, and dysphonia –76.25%.

4 Conclusions

Over the past few decades, much research has been conducted on using machine learning for pattern recognition based on convolutional neural networks (CNN). Convolutional neural networks are widely used in image processing, but they can also be adapted to work with serial data such as audio signals, making them valuable tools in language processing.

Literature analysis has shown that deep learning-based approaches show successful results in speech recognition, and therefore, they have attracted a lot of research and studies. Various voice and speech abnormalities are shared among the world's population. Many researchers have conducted research on the identification and classification of voice disorders based on machine learning.

This is especially true for identifying speech defects in children. Early intervention can significantly increase the chances of successful correction. This can help professionals such as speech therapists and doctors work effectively with children.

In this work, the architecture of a convolutional neural network was investigated to solve the problem of detecting speech defects: dyslexia, stuttering, dysphonia, or dyslalia. To use the CNN, audio signals with the child's speech were first converted into spectrograms. Next, the network was trained based on a data set with labels for each class of defects. An optimal architecture, including three hidden convolutional layers, was selected during the research process.

This network can identify four types of speech defects (dyslexia, stuttering, dysphonia, or dyslalia) with 77–79% recognition results.

The study results can further be used to develop unique software for detecting speech defects. The development and use of such software will enable the speech therapist to conduct high-quality diagnostics. This program will also be helpful for parents. This will make it possible to conduct an initial diagnosis at home and contact a specialist in a timely manner.

References

1. Purwins, H., Li, B., Virtanen, T., Schlüter, J., Chang, S.J., Sainath, T.: Deep learning for audio signal processing. IEEE J. Select. Top. Sign. Process. **13**(2), 206–219 (2019). https://doi.org/10.1109/JSTSP.2019.2908700
2. Kourkounakis, T., Hajavi, A., Etemad, A.: Detecting multiple speech disfluencies using a deep residual network with bidirectional long short-term memory. In.: ICASSP 2020 - 2020 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), pp. 6089–6093 (2020). https://doi.org/10.1109/ICASSP40776.2020.9053893
3. Kourkounakis, T., Hajavi, A., Etemad, A.: Fluentnet: End-to-end detection of speech disfluency with deep learning (2020). arXiv:2009.11394

4. Sheikh, S.A., Sahidullah, M., Hirsch, F., Ouni, S.: Machine learning for stuttering identification: review, challenges and future directions. Neurocomputing **514**, 385–402 (2022). https://doi.org/10.1016/j.neucom.2022.10.015.DOI:10.1016/j.neucom.2022.10.015
5. Wu, H., Soraghan, J., Lowit, A., Di-Caterina, G.: A deep learning method for pathological voice detection using convolutional deep belief networks. In: Proceedings Interspeech 2018, pp. 446–450 (2018). https://doi.org/10.21437/Interspeech.2018-1351
6. Abid Syed, S., Rashid, M., Hussain, S., Zahid, H.: Comparative analysis of CNN and RNN for voice pathology detection. BioMed Res. Int. 6635964, **8** (2021). https://doi.org/10.1155/2021/6635964
7. Jegan, R., Jayagowri, R.: Voice pathology detection using optimized convolutional neural networks and explainable artificial intelligence-based analysis. Comput. Methods Biomech. Biomed. Engin. (2023). https://doi.org/10.1080/10255842.2023.2270102
8. Peng, X., Xu, H.X., Liu, J., et al.: Voice disorder classification using convolutional neural network based on deep transfer learning. Sci. Rep. **13**, 7264 (2023). https://doi.org/10.1038/s41598-023-34461-9
9. Dawodi, M., Baktash, J.A., Wada, T., Alam, N., Joya, M.Z.: Dari speech classification using deep convolutional neural network. In: 2020 IEEE International IOT, Electronics and Mechatronics Conference (IEMTRONICS), pp. 1–4 (2020)

Learning Perceptual Properties of 3D Shapes: A Brief Synopsis

Kapil Dev(✉)

Liverpool Hope University, Liverpool, UK
devk@hope.ac.uk
https://www.linkedin.com/in/kasaini/

Abstract. In this extended abstract, we provide a brief synopsis of our published, ongoing and future work on data driven exploration of perceptual properties of 3D shapes. We focus on research problems related to 3D data in massive online libraries which are difficult to search, organise and reuse. Our approach involves the use of crowdsourcing and machine learning methods to mitigate the challenges. We collect a considerable amount of human judgements on the style matching and aesthetics of 3D shapes by leveraging crowdsourcing platforms. Our method demonstrates that machine learning and crowdsourcing are suitable techniques to build data-driven models of visual perceptual attributes of 3D shapes. We build data-driven models of two perceptual characteristics, namely style similarity and aesthetics. As demonstrated by our results, the learned models can be employed in various 3D graphics applications such as search, organisation, scene composition, and visualisation of 3D shape data.

Keywords: Computer graphics · Perception · Machine learning

1 Introduction

The recent increase in 3D content production has resulted in the emergence of massive online libraries [1]. ShapeNet is an example of such an online 3D shape library which contains 55 common object categories with about 51,300 unique 3D models. Design and implementation of computer graphics applications relying on search, organisation, and reuse of 3D data in such massive online repositories face significant challenges. To tackle such challenges, we investigate the use of crowdsourcing and machine learning methods that focus on the visual perceptual characteristics of 3D shapes. The primary focus of our research is on two distinct perceptual properties of 3D shapes, style similarity and aesthetics. Specifically, for style similarity, we learn to predict how similar in style a given 3D shape is to any other 3D shapes. And for aesthetics, for a given 3D shape we learn to predict it's aesthetics score (i.e. a numeric value).

Our goal is to study the visual perceptual properties of 3D shapes from the visual computing perspective and demonstrate how machine learning and crowdsourcing techniques can be utilised to construct data-driven models that

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2026
I. Potapov et al. (Eds.): RTC-Digital 2023, CCIS 2647, pp. 20–25, 2026.
https://doi.org/10.1007/978-3-032-04731-1_3

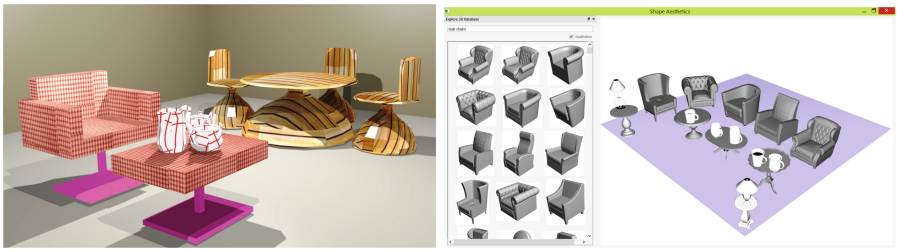


Fig. 1. Example applications of our work in 3D shape search and scene composition. Left: example scene composed using our style similarity metric, depicting two groups of 3D shapes having similar styles. Right: the screenshot shows our tool for searching 3D shapes (left panel) and composing scenes (right panel) based on perceptual aesthetics. In the left panel, 3D shapes can be ranked according to their aesthetics scores computed using our technique.

accurately represent these visual perceptual properties. These models can then be implemented to enhance the search, organisation, scene composition, and visualisation of 3D shape data in online libraries that are constantly growing.

To solve these problems, we first collect a considerable amount of human judgements on these two attributes of 3D shapes by leveraging crowdsourcing platforms. By utilising this data, we model or create metrics (i.e. style similarity and aesthetics) that accurately measure style matching and aesthetics. We give more details of our approach in the sections below.

2 Modeling Style and Aesthetics of 3D Shapes

In the sections below, we provide more details on how we learn to match shapes based on their style and predict their aesthetics scores.

2.1 Style Similarity

We develop a metric that can accurately match the styles of 3D shapes based on their shape, colour, and texture. Specifically, we learn a distance function that assigns a real value as a measure of style distance between two given input 3D shape descriptors. The learned metric uses human style preference data collected using crowdsourcing platforms (Fig. 2).

Our novelty is in expanding existing feature-based style metric learning techniques [6,8]. Specifically, we learn our style similarity metric using a shape descriptor that combines geometric, colour, and textural information for metric learning. The results of our empirical research demonstrate that this approach is effective for developing texture- and colour-aware shape style similarity metrics.

Our method involves two steps [2]. Firstly, we use crowdsourcing to obtain a significant amount of style preference data, which allows us to define styles based on examples rather than pre-defined rules. To collect style similarity data, we

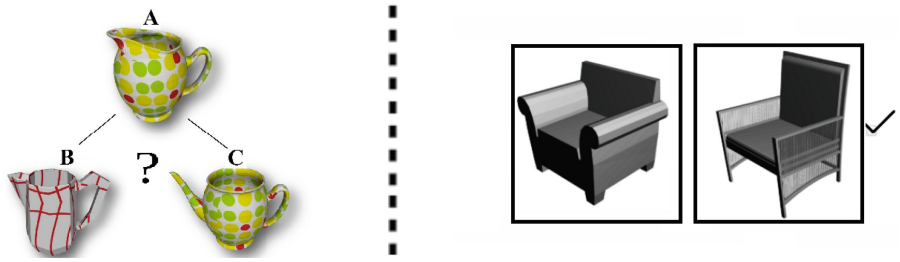


Fig. 2. Example data collection tasks. Left: users are asked to choose shape B or C based on perceptual style similarity with shape A. Right: users are asked to choose a shape they perceive as more aesthetic in each pair.

present participants with three shapes: a source shape (A) and two target shapes (B and C) in a relative comparison "triplet" format (Fig. 2: left image). Secondly, we use a pairwise metric learning method to compute a style distance function based on the crowdsourced style preference data. This approach is derived from existing methods for learning distance metrics. To compute the style similarity metric, our algorithm uses features computed on 3D meshes and their texture and colour attributes, and the triplet responses received from the crowdsourcing platform. Our algorithm also allows learning of personalised style metrics. We implement this feature by letting users interact with the application and provide their style preferences, which can be used to personalise the objective style metric. An application of our approach is demonstrated in Fig. 1.

2.2 Aesthetics

We leverage deep neural networks and crowdsourcing to construct a data-driven model capable of predicting the perceptual aesthetics of 3D shapes [3]. Our learning algorithm takes two inputs: human aesthetics judgements (Fig. 2:right) and multi-view rendered images of 3D shapes (Fig. 3). In comparison to existing approaches [7], our algorithm automatically identifies the relevant features or shape descriptors to learn the aesthetic function. Specifically, unlike prior studies that use predefined features, such as curvature and symmetry, to explore shape aesthetics, we approach visual aesthetics as a perceptual concept and gather a significant number of human aesthetics judgements using crowdsourcing to automatically learn a model capable of predicting the aesthetic score of an input 3D shape.

Overall, our approach offers a way to learn a measure of the visual aesthetics of 3D shapes that can be extended to include other properties of 3D shapes. This model can be applied to a variety of graphics applications (Fig. 1).

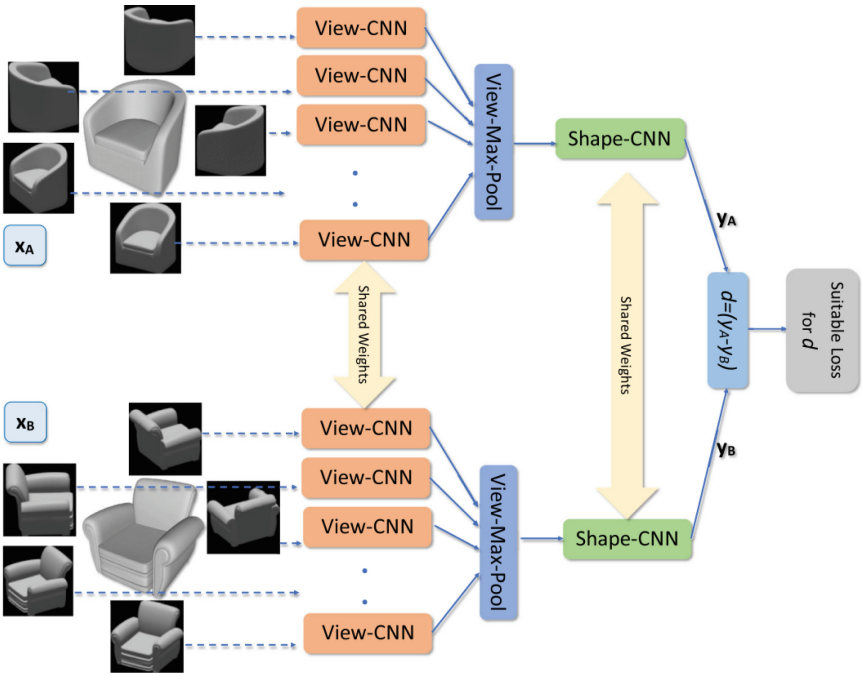


Fig. 3. Our multiview deep ranking architecture for shape aesthetics is illustrated here. The architecture involves two CNNs: View-CNN, which processes rendered views as its input, and Shape-CNN, which receives the max-pooled output of View-CNN as its input. The objective is to handle pairs of shapes (x_A, x_B) with specified inequality constraints (i.e. x_A is rated higher than x_B). In this scenario, the output y_A is expected to possess a higher value than y_B. To achieve this, the difference between y_A and y_B is used in the loss function. Please refer the original paper for more details.

3 Perception of 3D Shape Aesthetics

We expand our work on perception of aesthetics [5] and submit to a computer graphics journal [4]. The expanded work focuses on studying the impact of different shape representations on the perceptual aesthetics of 3D shapes in pairs. For example, we pair 3D shapes in different representation to collect human aesthetics preferences (Fig. 4). We use voxels, points, wireframe, and polygon as four different 3D representations. On analysing the collected human preference data we found that relatively low-resolution representation (e.g. points or voxels) are comparable to those on high resolution representation (i.e. polygon meshes). This result is significant as it suggests that humans do not need very detailed shape representation to make aesthetic decisions, since limited number of points or a low resolution voxel representation are sufficient for humans to discern aesthetics preferences. Additionally, we propose that humans may unconsciously engage in smoothing or interpolation to visualise a smoothed surface rather than the

jaggedness of voxels. Our results have implications for the studies using pairwise aesthetics data.

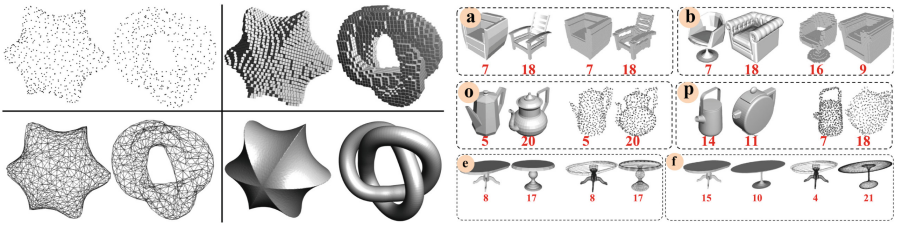


Fig. 4. Left: Illustration of a 3-D shape pair portrayed through four representations: point cloud (500 points), voxels ($32 \times 32 \times 32$ resolution), wireframe mesh, and polygon mesh. Right: (a, b): comparison of polygon mesh with voxels ($32 \times 32 \times 32$ resolution), (o, p): point clouds (500 points), and (e, f): wireframe mesh. User responses (A,B) below shapes shows the number of user who prefer that shape compared to the other in a pair. Left columns (i.e. a, o, e) show shape pairs with consistent responses for both polygon mesh and the respective comparison. Right columns (i.e. b, p, f) display shape pairs with divergent responses between polygon mesh and the comparison

4 Future Work

Through our research, we have developed distinct measures for assessing 3D shape style similarity and aesthetics, which can prove valuable for scene composition and other related applications in computer graphics. However, the decision of whether to prioritise style or aesthetics when selecting new shapes to add to a 3D scene, either manually or through automation, warrants exploration. Some users may place more emphasis on matching the existing shapes' style, while others may prioritise selecting more visually appealing shapes. By comprehending the relative significance of style and aesthetics in constructing 3D scenes, we can design data-driven models that optimise these properties, resulting in scenes that are more visually attractive and stylish.

We argue that additional research is necessary to understand how humans compare shapes to determine which is more or less aesthetically pleasing in a given pair. It is still unclear, from a perceptual standpoint, whether people quickly evaluate the overall shape or engage in a more localised or part-level comparison before providing their judgements.

When using shape representations in machine learning, there are two primary approaches: image-based and object-based representations. Image-based representations consist of depth-images or images generated from multiple camera angles, while object-based representations use voxelized volumes or point clouds. However, given the significant time and memory constraints of deep learning algorithms, the choice of shape representation can significantly impact

these factors. Consequently, we maintain that a comprehensive exploration of different shape representations as inputs to deep networks is essential to understand their relationship with time and memory requirements.

Our research setup can be extended to problems associated with the perception of 3D shapes. The computer graphics research community continues to confront a significant obstacle in creating user-friendly tools that novices can use with ease to generate 3D content. To overcome these challenge, one approach is to employ data-driven models capable of automatically synthesising new shapes by incorporating their perceptual properties. These models can learn from a set of examples and generate novel shape models. By applying our shape aesthetics measure to these generative models, visually more appealing shapes can be automatically generated.

Finally, our approach showcases the utilisation of machine learning and crowdsourcing techniques to develop data-driven models of the visual perceptual characteristics of 3D shapes. These models can be employed in a variety of applications, including search, organisation, scene composition, and visualisation of 3D shape data available in the ever-growing online 3D shape content libraries. Our exploration of the perceptual features of 3D shapes is likely to stimulate further research into other critical perceptual properties associated with our visual system.

Acknowledgments. The authors would like to thank the volunteers on Facebook for providing their responses to the shape aesthetics study. They would also like to thank Microsoft Research for supporting this research.

References

1. Chang, A.X., et al.: ShapeNet: an information-rich 3D model repository. arXiv preprint arXiv:1512.03012 (2015)
2. Dev, K., Kim, K., Villar, N., Lau, M.: Improving style similarity metrics of 3D shapes. In: Proceedings of the 42nd Graphics Interface Conference, pp. 175–182 (2016)
3. Dev, K., Lau, M.: Learning perceptual aesthetics of 3-D shapes from multiple views. IEEE Comput. Graph. Appl. **42**(1), 20–31 (2020)
4. Dev, K., Palumbo, L., Lau, M.: Comparing shape representations for the aesthetic judgment of 3-D shape pairs. IEEE Comput. Graph. Appl. **43**(6), 33–45 (2023). https://doi.org/10.1109/MCG.2023.3286047
5. Dev, K., Villar, N., Lau, M.: Polygons, points, or voxels? Stimuli selection for crowdsourcing aesthetics preferences of 3D shape pairs. In: Proceedings of the Symposium on Computational Aesthetics, pp. 1–7 (2017)
6. Liu, T., Hertzmann, A., Li, W., Funkhouser, T.: Style compatibility for 3D furniture models. ACM Trans. Graph. (TOG) **34**(4), 85 (2015)
7. Lu, X., Lin, Z., Jin, H., Yang, J., Wang, J.Z.: Rating image aesthetics using deep learning. IEEE Trans. Multimedia **17**(11), 2021–2034 (2015)
8. Lun, Z., Kalogerakis, E., Sheffer, A.: Elements of style: learning perceptual shape style similarity. ACM Trans. Graph. (TOG) **34**(4), 84 (2015)

Application Surrogate Models to Study Stress State of Dispersion Strengthened Composite

Ruslan Babudzhan, Mariia Shapovalova(✉), and Oleksii Vodka

National Technical University Kharkiv Polytechnic Institute, Kyrpychova 2 str., Kharkiv 61002, Ukraine
ruslan.babudzhan@infiz.khpi.edu.ua,
mariia.shapovalova@khpi.edu.ua,
oleksii.vodka@khpi.edu.ua
https://web.kpi.kharkov.ua/dpm/uk/shapovalova-mariya-igorivna/,
https://web.kpi.kharkov.ua/dpm/uk/vodka-oleksij-oleksandrovich/

Abstract. This work deals with the creation of a surrogate model for approximating the finite element solution of the problem of dispersion-strengthened composite plane sample deformation. The authors propose an algorithm for constructing a parametric two-dimensional model of a composite and use the ANSYS Mechanical program to create the calculation model. The stress-strain state of the material microstructure is processed using a convolutional neural network based on the U-Net architecture of the encoder-decoder type. The model loss function is defined as the root mean square error, and optimisation of the loss function is performed using the ADAM optimisation method. The surrogate model is significantly ahead of the FEM and is used to speed up calculations and determine the overall quality of the approximation of problems of mechanics of this type.

Keywords: Surrogate Models · Stress State · Dispersion Strengthened Composite · Neural Networks

1 Introduction

Industries are increasingly relying on machine learning for various tasks, including pattern and image recognition, natural language processing, optimisation of operations, data mining, and knowledge discovery. Machine learning algorithms are also being utilised to approximate solutions to complex mathematical models, such as differential or variational equations. However, traditional simplified solutions for determining the physical properties of structures have limitations in terms of accuracy and configuration, leading to more conservative design choices.

Fortunately, surrogate models based on intelligent approximation methods using data have the potential to overcome these limitations. A surrogate model in computational and applied mathematics is the approximate model which is used

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2026
I. Potapov et al. (Eds.): RTC-Digital 2023, CCIS 2647, pp. 26–38, 2026.
https://doi.org/10.1007/978-3-032-04731-1_4

to mimic the behaviour of a complex system more efficiently than the detailed physics based model. These models are frequently used in place of the original model when it is impossible to do a direct numerical simulation due to time or computational costs. Surrogate models often work by exploiting a training dataset consisting of sampled data from the original model to learn underlying patterns and dependencies. The goal is to strike a balance between precision and computational proficiency, and facilitating quicker evaluations in situations such as optimisation, sensitivity study, and real-time decision making. At its core, surrogate models are a convenient and economical way to represent the behaviour of a complicated system with a much reduced computational overhead. This paper focuses on the development of a surrogate model utilising a convolutional neural network. The process involves creating a dataset through finite element analysis, enabled by automating the ANSYS software package. This approach not only facilitates data generation but also ensures the integration of comprehensive finite element method (FEM) insights into the training process.

Further, the paper delves into the detailed architecture of the model, its training process, and a thorough analysis of the results. The neural network construction took place utilising the Python programming language, harnessing the functionalities of the PyTorch deep learning framework. This methodology ensures that the surrogate model, with its optimised similarity function between the approximated solution and the original mathematical model, can accurately replace traditional methods like FEM. With sufficient training data, these surrogate models demonstrate an ability to achieve accuracy comparable to FEM, while offering enhanced flexibility and potential for handling complex approximations.

2 Literature Review

The surrogate model that improves the accuracy of macroscale finite element models for composite materials is developed in [18]. This model uses data from smaller, mesoscale unit cells, enhancing prediction accuracy in larger-scale applications. Complementing this, the study in [11] introduces a surrogate model adaptable to different fibre volumes and properties in composite materials, offering versatility in handling varying compositions within composites.

A comprehensive overview of surrogate modelling in finite element analysis is presented in [8]. This review includes discussions on the validation of these models and their various types, thereby expanding comprehension of the applications of surrogate models. The research in [2] describes a unique surrogate model based on a U-Net convolutional neural network, specifically designed for analysing stress-strain states in material microstructures. Additionally, the utilisation of U-Net deep learning models for predicting stress in composite materials is examined in [15], presenting these models as effective alternatives to the conventional FEM.

The main focus of the authors in work [17], is on the applications of machine learning in biomechanics, emphasising the growing significance of machine learning as a potent substitute for finite element analysis in biomechanical systems. A

method for reducing data dimensions in composite microstructures through the use of deep-learned surrogate models is introduced in [7]. This approach enhances computational efficiency and accuracy in handling complex data. The study in [3] discusses the development of a surrogate model for nonlinear, stochastic multiscale analysis, specifically addressing how microstructural variability impacts material response, aiding in more comprehensive virtual testing.

An inventive boundary-oriented graph embedding approach for finite element analysis is presented in [4]. This method expedites design and prototyping processes, with a specific focus on stress field prediction and topological optimisation. A user-friendly surrogate model tailored for complex engineering systems is elucidated in [1], streamlining the exploration of high-dimensional parameter spaces with diminished computational demands. Additionally, a special issue, as outlined in [10], provides a comprehensive summary of the crucial role played by finite element modelling in establishing the link between microstructure and macroscopic behaviour in composites. This underscores the method's significance in virtual testing and prototyping.

A deep learning approach for deriving mechanical properties from microstructural images of composite hydrogels is proposed in [5], which demonstrates high accuracy in predicting Young's modulus and Poisson's ratio, employing a combination of finite element analysis and deep neural networks. The application of deep learning surrogate models in solid mechanics is investigated in [9]. The exploration aims to proficiently predict structural behaviour under diverse loads, providing a time-efficient alternative to conventional finite element simulations. The application of OpenCV computer vision technology, as detailed in [13], is employed for the analysis of cast iron images containing graphite inclusions. This utilisation aids in material classification and enhances understanding of the distribution of inclusions based on their concentration.

A convolutional neural network tailored for the analysis of steel microstructure images is introduced in the work referenced as [14]. This network predicts material compositions with over 83% accuracy, showcasing the potential of machine learning in microstructure analysis. Implementing Physics-Informed Neural Networks (PINN), the research focuses on resolving both linear and nonlinear solid mechanics problems, offering a sturdy framework for inversion and surrogate modelling within this domain, as outlined in [6]. Conclusively, the introduction of Generalisable Surrogate Models (GSMs) represents a noteworthy advancement in stress approximation for 3D trusses [12]. This is achieved by consolidating diverse structural data to improve the accuracy of stress predictions. The utilisation of Autoencoders and Long Short-Term Memory (LSTM) Networks in aerodynamic modelling, showcases a notable advancement in forecasting aerodynamic behaviour [16]. This marks a convergence of machine learning principles with engineering applications.

The extensive review of existing literature underscores the evolving landscape of surrogate modeling in the realm of material science and engineering. Pioneering studies like [11,18], and [8] have laid a foundational understanding of the capabilities and versatility of surrogate models in approximating finite

element analyses and managing complex microstructural data. Innovations in neural network applications, as seen in [2] and [15], further demonstrate the integration of machine learning with traditional engineering methods. This research aims to build upon these advancements by introducing a novel approach that utilises U-Net based convolutional neural networks for more efficient and accurate stress-strain state predictions in composite materials. The proposed model seeks to address existing gaps in computational efficiency and scalability, aiming to provide a more robust and versatile tool for material analysis and design. In doing so, it not only aligns with the trajectory set by recent studies but also contributes to the broader scientific problem of optimising material property prediction through advanced computational techniques.

3 Objectives of the Research

The primary goal of this study is to refine and advance the field of surrogate modelling for composite materials, specifically focusing on samples with chaotically arranged inclusions. This objective stems from the literature review, which highlighted the potential and challenges of surrogate models in approximating finite element solutions in diverse material configurations. The approach is detailed as follows:

1. The objective is to construct a computational deformation model for a representative sample of a dispersion-reinforced composite material. This model will simulate the deformed state under varying conditions, as visualised in Fig. 1, which depicts the sample setup.
2. A diverse dataset of stress field solutions for various configurations of composite material samples will be gathered. This dataset, crucial for training the surrogate model, reflects the variability in inclusion arrangements as described in Table 1, showing material parameters.
3. The architecture of a neural network-based surrogate model will be designed to approximate the equivalent stress distribution. This model will leverage the finite element method's computational rigour, addressing the research gap identified in the literature.
4. The neural network will undergo training to refine the surrogate model, aiming to approximate the FE results accurately.
5. Ultimately, the evaluation of the surrogate model's accuracy involves comparing its solutions with those obtained through the finite element method. This process includes assessing error margins and evaluating overall performance.

This research addresses the need for efficient and accurate surrogate models in handling the complexities of composite materials with chaotic inclusions, as underscored by the reviewed literature. By integrating computational models with neural network architectures, this study aims to contribute significantly to the optimisation of material property predictions in composite materials. This study investigates a dispersion-reinforced composite material comprising an isotropic matrix and orthotropic inclusions. The parameters of the matrix and inclusions materials (in GPa) provides in Table 1.

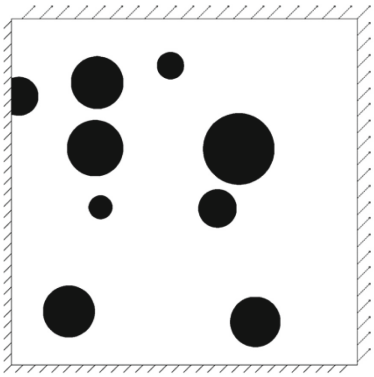


Fig. 1. Typical composite section

Table 1. Elastic constants of matrix and inclusions

E_m	ν_m	C_{11}, C_{22}	C_{12}	C_{13}, C_{23}	C_{33}	C_{44}, C_{55}	C_{66}
180	0.35	1060	290	109	46.6	2.3	385

4 Creating a Dataset

This section details the dataset creation process essential for training the convolutional neural network (CNN)-based surrogate model. The process of generating a dataset is outlined in Fig. 2, illustrating its logical structure.

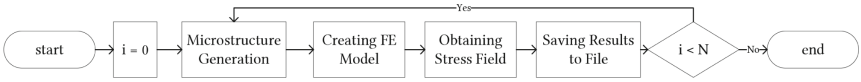


Fig. 2. Logical scheme of dataset creation

Recognising that CNNs are a form of supervised learning, they necessitate an appropriate training dataset. For the intended purposes, this dataset consists of a set of representative examples, particularly addressing solved stress-strain problems. The surrogate model's parameters are finely tuned to ensure that its output closely aligns with these pre-solved examples. Through this process, the model acts as an effective approximator for the solver.

Finite Element Analysis (FEA) is employed as the primary solver. This involves generating a collection of geometric representations of composite volumes. These geometries are then processed using the ANSYS software, where the stress distribution across the mesh nodes is calculated and recorded. In the training dataset, these geometries and their corresponding loads form the input

features, while the computed stresses at the nodes serve as the target outputs for each example.

As the formation of the stiffness matrix is the most costly action in the finite element method, generating 1,000 different samples provides sufficient samples for analysis. To construct the composite geometry, the manually developed software solution for generating a statistically equivalent artificial microstructure of cast iron is used. This approach assumes that the size distribution of inclusions follows the normal law, and the placement of inclusions on the plane occurs randomly based on a uniform distribution of the value. This methodology produces enough samples for analysis. ANSYS software is used to create the data set, with a displacement vector formed and applied to the corresponding finite element nodes at each overload step. The initial bias is obtained from the uniform distribution, and the stress state calculation is simultaneously performed for all load steps for one generated sample geometry. The stresses inside the finite elements are calculated from the obtained displacements of each node of the structure after the automatic solution of the equation system.

The Python programming language is used to form the algorithm for constructing a computational model and subsequent processing of the results, which provides software tools for analysing data. The convolutional neural network algorithm accepts structured matrices as input, necessitating further processing of stress data imported into Python by cubic spline interpolation. At the output, the stresses obtained are at the centers of the pixels. Rasterising the geometry of each sample is necessary to work with neural networks, resulting in a binary mask applied to the neural network input. The load vector data array corresponding to each load distribution is concatenated with the latent space of the model.

5 Neural Network Architecture

This section explores the creation and training of a neural network predicting equivalent von Mises stresses in a two-dimensional composite sample under tension/compression. The model, implemented in Python using TensorFlow, adopts an encoder-decoder architecture. The encoder compresses the input data array into a dimensionally restricted latent space, while the decoder reveals the compressed feature vector, producing a modified data array at the output. Sequentially connecting and simultaneously optimising both networks' neurone weights with a chosen algorithm yields a neural network serving as a surrogate model for determining the composite sample's stress-strain state.

The architecture employed in this research is based on the U-NET model, specifically designed for the segmentation of biomedical images. A distinctive aspect of U-NET involves replicating the feature map within each layer block. One duplicate advances 'deeper' through the network, while the other integrates with the symmetrical decoder block, circumventing deeper network layers. Due to challenges in concatenating these loads into extensive feature maps without incurring a significant increase in computational complexity, this feature has

been omitted. As a result, a direct sequence of layers has been adopted from the U-NET architecture. Notable adaptations include modifying convolutional layer types to improve training speed and stability.

The respective architectures of the encoder and decoder, showcasing a sequential interconnection of blocks in the network structure, illustrate in Figs. 3 and 4. Each block contains the following layers:

1. Convolution (Conv2D, SeparableConv2D, Conv2DTranspose). These layers are different implementations of the idea of convolutional layers. Conv2D is a regular convolutional layer, SeparableConv2D is a layer that also implements the convolution function but using two 1D convolution kernels instead of one 2D kernel. Conv2DTranspose implements 'unfolding', that is, convolution with a dilation factor greater than 1;
2. Batch normalisation (BatchNormalization);
3. Activation (Activation). The block applies the activation function.
4. Subsampling (MaxPooling2D, UpSampling2D). Layers that implement scaling of feature maps (reducing by half in the case of MaxPooling2D and increasing with UpSampling2D).

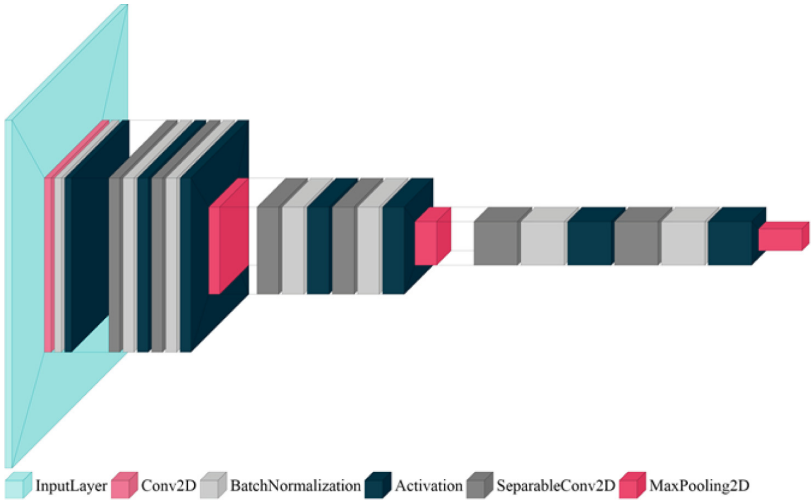


Fig. 3. Encoder architecture

As previously indicated, the neural network's input comprises a geometry-representing mask for the composite sample. In Fig. 3, the input layer is denoted as InputLayer, with an encoder input dimension of (256, 256, 1), signifying acceptance of a 256 × 256 pixel image with a single channel. The decoder's output dimension also remains (256, 256, 1), generating a matrix of equivalent stresses in a 256 × 256 format. The final architectural component is the latent space,

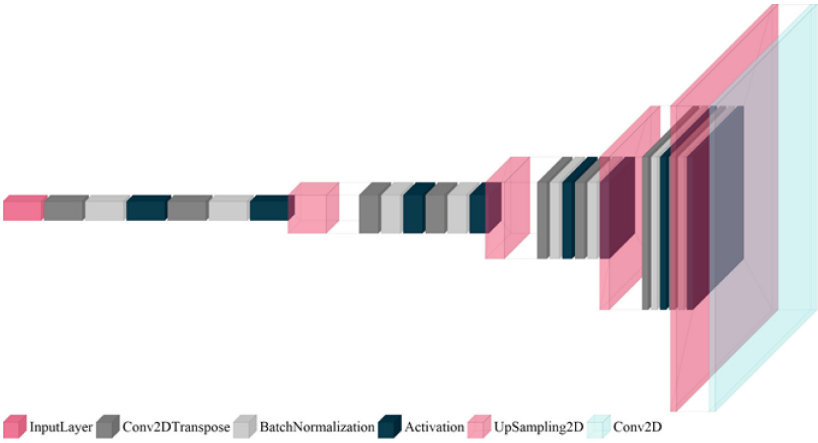


Fig. 4. Decoder architecture

bridging the encoder and decoder, incorporating information about the applied load—specifically, a vector denoting displacements of the sample's edge nodes under load, with dimensions (2, 1). The encoder's output, akin to the decoder's input, has dimensions (16, 16, 256), constituting 256 feature maps sized 16 × 16.

In the latent space segment (Fig. 5), Dense, Reshape, and Concatenate layers coexist. The Dense layer, configured with a (2, 1) vector input, yields an output of dimensions (256, 1). Reshape transforms spatial dimensions to (16, 16, 1) without arithmetic operations, unfolding the input vector into a single feature map. Subsequently, this block combines with a previously described convolutional block (without subsampling). The output of this block's last layer, with dimensions (16, 16, 32), concatenates with the encoder's output, resulting

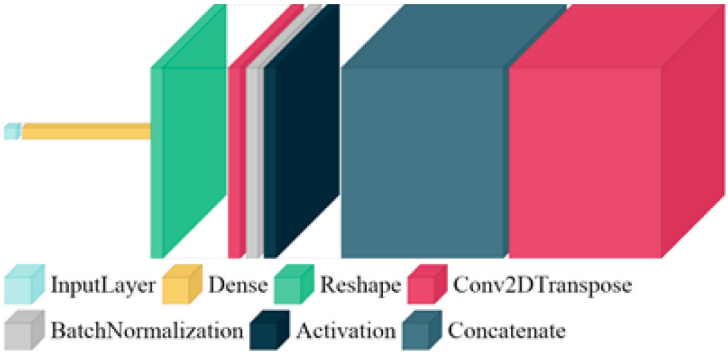


Fig. 5. Latent space architecture

in dimensions of (16, 16, 288). Convolution via Conv2DTranspose on these layers produces the latent space, encoding the input data.

The loss function chosen is the root mean square error across all points of the output matrix:

$$Loss = \sqrt{\frac{1}{256 \times 256} \sum_{i=1}^{256} \sum_{j=1}^{256} (y_{ij} - \hat{y}_{ij})^2}. \quad (1)$$

6 Training and Validation of the Neural Network

The neural network has been trained on a dataset comprising 10,000 samples, with 80% of the data used for training and 20% for validation. The optimisation process has been conducted using the Adam optimisation algorithm with a learning rate of 0.0001. The utilised loss function is mean squared error (MSE).

This process is depicted through various graphs that illustrate the relationships between loss functions and supplementary metrics, as referenced in Figs. 6, 7, and 8. A notable observation is the alignment of these indicators across both the training and validation datasets. In the context of deep learning theory, such an alignment signifies the model's ability to generalise. Consequently, the model demonstrates equivalent predictive performance on both familiar data (the training set) and unfamiliar data (the validation set). The results shown in Fig. 6 reveal a steady reduction in the loss function, suggesting that the gradient descent method has not yet converged to a local minimum. This implies that further optimisation of the network's weights could potentially enhance the model's performance. Typically, reaching a local minimum is indicated by a slowed rate of change in the loss function graph concerning the training dataset. The absence of this trend necessitates an increase in the number of training epochs. At each iteration of network training, the values of the functions are also calculated: the root-mean-square difference ($RMSE_{max}$) of the maxima of equivalent stresses (2) and the root-mean-square error at 80% ($RMSE_{80\%}$):

$$RMSE_{\max} = \sqrt{(y_{\max} - \hat{y}_{\max})^2}, \quad (2)$$

$$RMSE_{80\%} = \sqrt{\frac{1}{256 \times 256} \sum_{i=1}^{256} \sum_{j=1}^{256} [(y_{ij} - \bar{y}_j)^2 \times \theta(y_{ij} - Q_{80}(y_j))]}, \quad (3)$$

The predicted field of equivalent stresses on the training set serves a dual purpose. Firstly, it assesses the model's effectiveness in addressing the primary objective of the research. In the second place, through a comparison of the output and metric values of the model with established standards, it is possible to determine the adequacy of the model. Results of NN output with comparison to FEM solution is shown on Fig. 9. The figure shows the comparison between the finite element analysis (FEA) solution and the surrogate model prediction for a dispersion-strengthened composite material's stress state under load. The

Fig. 6. The loss function on the training and validation samples

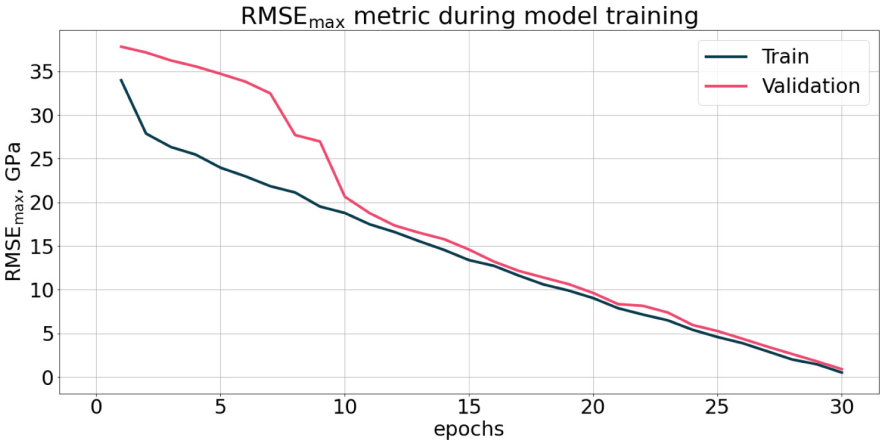


Fig. 7. The loss function on the training and validation samples

left column depicts the sample geometry with inclusions and the load direction, the middle column displays the stress distribution obtained via FEA, and the right column showcases the neural network's prediction of stress distribution. The colour spectrum from blue to red indicates the stress magnitude, with red being the highest stress region.

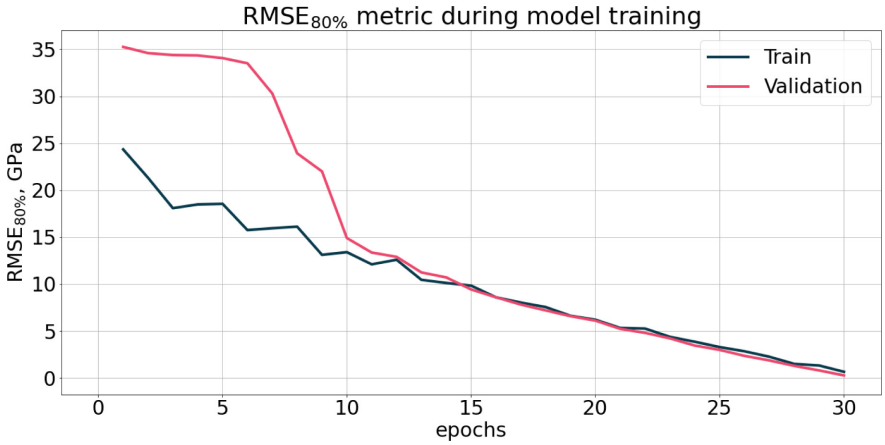


Fig. 8. The loss function on the training and validation samples

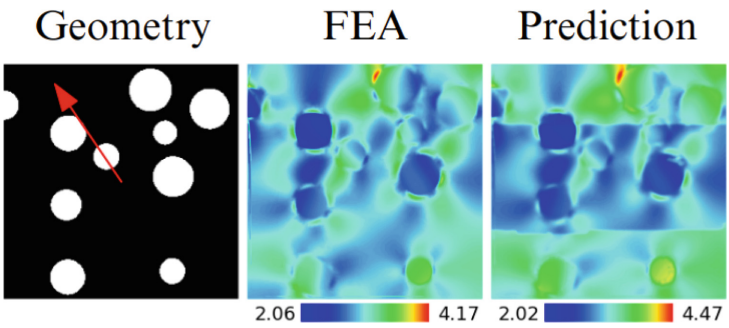


Fig. 9. Sample geometry, FE solution, and surrogate model output. Validation sample, GPa

7 Conclusions

This study involved the development of a surrogate model utilising a convolutional neural network to approximate finite element solutions for the deformation of dispersion-strengthened composite samples. This model aims to speed up stress-strain calculations in material microstructures. The network, trained on 10,000 stress state variants of a composite material sample, is based on the U-Net encoder-decoder architecture. It predicts equivalent stress distributions based on sample geometry and load values. The model's performance showed a mean squared error (MSE) of approximately 540 MPa for maximum stresses and 412 MPa on average across the stress distribution. The neural network demonstrated a significant time-saving advantage compared to the finite element method (FEM). The FEM analysis of 10,000 configurations took 430 min on a standard PC, with each loading step taking 19 s. In contrast, training the neural network took 135 min, and the trained model could calculate equivalent

stress matrices for 32 samples in just 9 s, demonstrating a time efficiency up to 70 times faster than FEM.

Acknowledgments. This work has been supported by the Ministry of Education and Science of Ukraine in the framework of the realisation of the research project "Algorithms, models, and tools of artificial intelligence for two-level modelling of complex materials behaviour for dual-use technologies" (State Reg. Num. 0124U000450).

References

1. Arndt, C., Crusenberry, C., Heng, B., Butler, R., TerMaath, S.: Reduced-dimension surrogate modeling to characterize the damage tolerance of composite/metal structures. Modelling **4**(4), 485–514 (2023). https://doi.org/10.3390/modelling4040028
2. Babudzhan, R., Vodka, O., Shapovalova, M., et al.: Application of computational intelligence methods for the heterogeneous material stress state evaluation. Herald Adv. Inf. Technol. **3**(5), 198–209 (2022)
3. Calleja Vázquez, J.M., Wu, L., Nguyen, V., Noels, L.: A micromechanical mean-field homogenization surrogate for the stochastic multiscale analysis of composite materials failure. Int. J. Numer. Meth. Eng. **124**(23), 5200–5262 (2023). https://doi.org/10.1002/nme.7344
4. Fu, X., Zhou, F., Peddireddy, D., Kang, Z., Jun, M.B.G., Aggarwal, V.: An finite element analysis surrogate model with boundary oriented graph embedding approach for rapid design. J. Comput. Des. Eng. **10**(3), 1026–1046 (2023). https://doi.org/10.1093/jcde/qwad025
5. Gholami, K., Ege, F., Barzegar, R.: Prediction of composite mechanical properties: integration of deep neural network methods and finite element analysis. J. Compos. Sci. **7**(2), 54 (2023). https://doi.org/10.3390/jcs7020054
6. Haghighat, E., Raissi, M., Moure, A., Gomez, H., Juanes, R.: A physics-informed deep learning framework for inversion and surrogate modeling in solid mechanics. Comput. Meth. Appl. Mech. Eng. **379**, 113741 (2021). https://doi.org/10.1016/j.cma.2021.113741. https://www.sciencedirect.com/science/article/pii/S0045782521000773
7. Kim, H., Jeong, I., Cho, H., Cho, M.: Surrogate model based on data-driven model reduction for inelastic behavior of composite microstructure. Int. J. Aeronaut. Space Sci. **24**(3), 732–752 (2022). https://doi.org/10.1007/s42405-022-00547-3
8. Kudela, J., Matousek, R.: Recent advances and applications of surrogate models for finite element method computations: a review. Soft. Comput. **26**(24), 13709–13733 (2022). https://doi.org/10.1007/s00500-022-07362-8
9. Li, H., Wang, Q.: A surrogate model for finite element simulation of elastic wave propagation in composite materials. SSRN (2023). https://doi.org/10.2139/ssrn.4425267
10. Luo, Y.: Finite element modeling of microstructures in composite materials: a special issue in materials. Materials **16**(15), 5332 (2023). https://doi.org/10.3390/ma16155332
11. Nakka, R., Harursampath, D., Ponnusami, S.A.: A generalised deep learning-based surrogate model for homogenisation utilising material property encoding and physics-based bounds. Sci. Rep. **13**(1) (2023). https://doi.org/10.1038/s41598-023-34823-3

12. Nourbakhsh, M., Irizarry, J., Haymaker, J.: Generalizable surrogate model features to approximate stress in 3D trusses. Eng. Appl. Artif. Intell. **71**, 15–27 (2018). https://doi.org/10.1016/j.engappai.2018.01.006. https://www.sciencedirect.com/science/article/pii/S095219761830006X
13. Shapovalova, M., Vodka, O.: Computer methods for constructing parametric statistically equivalent models of high-strength cast iron microstructure to analyze its elastic characteristics. Notes of the Tavrida National Univ. VI Vernadsky. Series: Technical Sciences **6**(1), 179–187 (2019)
14. Shapovalova, M., Vodka, O.: Image microstructure estimation algorithm of heterogeneous materials for identification their chemical composition. In: 2019 IEEE 2nd Ukraine Conference on Electrical and Computer Engineering (UKRCON), July 2019, pp. 975–979. IEEE (2019). https://doi.org/10.1109/ukrcon.2019.8879861
15. Shokrollahi, Y., Nikahd, M.M., Gholami, K., Azamirad, G.: Deep learning techniques for predicting stress fields in composite materials: a superior alternative to finite element analysis. J. Compos. Sci. **7**(8), 311 (2023). https://doi.org/10.3390/jcs7080311
16. Tekaslan, H.E., Demiroglu, Y., Nikbay, M.: Surrogate unsteady aerodynamic modeling with autoencoders and LSTM. Networks (2021). https://doi.org/10.2514/6.2022-0508
17. Vurtur Badarinath, P., Chierichetti, M., Davoudi Kakhki, F.: A machine learning approach as a surrogate for a finite element analysis: status of research and application to one dimensional systems. Sensors **21**(5), 1654 (2021). https://doi.org/10.3390/s21051654
18. Yan, S., Zou, X., Ilkhani, M., Jones, A.: An efficient multiscale surrogate modelling framework for composite materials considering progressive damage based on artificial neural networks. Compos. B Eng. **194**, 108014 (2020). https://doi.org/10.1016/j.compositesb.2020.108014

Deep Learning for Space Applications

Serhii Khoroshylov(✉) and Mykhailo Redka

Institute of Technical Mechanics of NASU and SSAU, 15 Leshko-Popelya Street, Dnipro 49005, Ukraine
skh@ukr.net

Abstract. The advances in deep learning have revolutionized the field of artificial intelligence, demonstrating the ability to create autonomous systems with a high level of understanding of the environments where they operate. These advances, as well as new tasks and requirements in space exploration, have led to an increased interest in these deep learning methods among space scientists and practitioners. The problems of controlling the attitude and relative motion of spacecraft are considered for both traditional and new missions, such as contactless space debris removal. Both supervised and reinforcement learning is used to solve such problems based on various architectures of artificial neural networks, including convolutional ones. The possibility of using deep learning together with methods of control theory is analyzed to solve the considered problems more efficiently. The difficulties that limit the application of these methods for space applications are highlighted. The necessary research directions for solving these problems are indicated.

Keywords: reinforcement learning · convolutional neural network · space debris · spacecraft

1 Introduction

At this time, artificial intelligence methods attract a great interest of researches and practitioners all over the world, which is largely because of the impressive results obtained using deep learning (DL) techniques. DL has rapidly evolved and showed promising results in solving complex tasks, finding non-trivial solutions of existing problems [1].

Meanwhile, space control systems have been designed mostly based on classical methods, for example. However, the developers face the problems of adaptability, robustness and autonomy when they attack new problems of space exploration using conventional techniques. Complex tasks of an orbital service, such as releasing a payload or capturing non-interacting targets, is accompanied by rapid changes in attitude and mass parameters of the spacecraft (SC), which can lead to unstable motion and tumbling of the satellite. In such conditions, the driving modes and mass characteristics are unpredictable. In these cases, conventional attitude control techniques cannot always stabilize the satellite, since they depend on the mass parameters of the plant.

Different approaches to space debris removal have been introduced recently [2]. Among those approaches, the ion-beam shepherd (IBS) concept can be selected because

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2026
I. Potapov et al. (Eds.): RTC-Digital 2023, CCIS 2647, pp. 39–46, 2026.
https://doi.org/10.1007/978-3-032-04731-1_5

it assumes the contactless removal of a space debris object (SDO). The main idea of this concept is to use the momentum from an ion thruster plume to transmit the deorbiting momentum to the SDO. To implement the IBS concept, it is necessary to have an efficient technique to determine the force that transmits the ion thruster (IT) to the SDO because these values are required for mission planning and relative control tasks [3]. The conventional approach for this task results in computationally complex algorithms since they are based on integration of the elementary forces over the SDO surface.

This work demonstrates that these tasks can be addressed using supervised [4, 5] and reinforcement learning (RL) [6] techniques.

2 Determination of the Ion Beam Impact on a Space Debris Object Using Convolutional Neural Networks

2.1 Problem Statement

Using the IBS concept, we consider the deorbiting of an SDO from a low Earth orbit. According to this concept, a shepherd satellite (SS) is equipped with an impulse transfer thruster (ITT) and an impulse compensation thruster (ICT). The ion plume from the ITT is pointed towards the SDO and used for transferring the deorbiting momentum. The nozzle of the ICT is pointed in the opposite direction to compensate for the reaction force created by the ITT. For efficient contactless de-orbiting, the SS has to fly at a sufficiently small distance in front of the SDO of the order of a few SDO diameters.

We assume that the SS has visual camera that can make the SDO images. The study aims to develop a neural net model, which can map an SDO image to the force transmitted by an IT plume to this object, and estimate the accuracy of such models.

2.2 Methods

We propose to use a convolutional neural network (CNN) [7] to determine the force transmitted by the IT plume to an SDO using visual images as the input. The input for the CNN is an SDO's image. The force projections are the outputs of the network. The hidden layers include two convolution layers with ReLU activation and two max-pool layers to reduce the dimensions of the features. Four consecutive fully-connected layers with ReLU activations follow the last max-pool layer. The output layer consists of three neurons corresponding to three force projections.

The mean squared error between the predicted and ground truth force values is used as the loss function to train the CNN.

In order to calculate the ground truth values of the ion beam force, we use a model of the ion beam interaction with SDO [8]. For all the experiments, we use the following ion thruster parameters: the initial ion thruster radius is 0.0805 m; the ion mass (xenon) is $2.18\ 10^{-25}$ kg; the initial plasma density is $4.13 \cdot 10^{15}$ m^3; the initial axial velocity of ions is 71580 m/s; the divergence angle is 7 deg; the initial electron temperature is 0.001 eV.

The upper stage of the Cyclone-3 launch vehicle is considered as the SDO, which is approximated by a cylinder with a height of 2.6 m and a base diameter of 2.2 m.

The datasets contain the images of the SDO as the input and the corresponding force vector as the output. The ground-truth forces are calculated using the function, which

is designed using the methodology from Sub-Sect. 3.2. The inputs for this function are generated randomly using a continuous uniform distribution and variation ranges specifically for each input parameter. Then, we use these input parameters to generate the SDO's synthetic images. The images are rendered using Blender open-source software. A camera focal length of 25 and black background are used for rendering grayscale images with a size of 200 × 200. We did not use any preprocessing techniques for input data, but we normalized outputs to have the ground-truth values in a range of [–1.0, 1.0].

Three different approaches for the end-to-end training of the CNN were investigated. The first model, called CNN1, uses a single CNN, which is trained on the dataset consisting of 10000 SDO images. In this case, the relative position inputs vary in ranges of [–1.0, 1.0] m and [5.0, 9.0] m for x, y and z coordinates, respectively. The input orientation angles

ψ, φ, θ, , vary in a range of [1.57, 1.57] rad. The second model, called CNN2, is an ensemble network. It consists of four sub-models Q1 - Q4. Each particular model is trained with its own dataset containing 10000 images. In this case, different models are used to determine the force depending on the relative distances to the SDO. A classifier is used to pick the specific model. To train the classifier, the output regression layer of CNN1 was replaced with a classification layer to perform four-class classification for mapping each classifier output to the specific model. According to this approach, the CNN2 model uses the classifier at first to determine the specific model, and then the specified model performs regression using the same image as the input. The last model, called CNN3, is very similar to the CNN1. However, it is trained on the extended dataset consisting of all images used to train the CNN2 model. Thus, the CNN3 is trained on the dataset containing 40000 images. All thee models have the same CNN architecture, which is shown in Fig. 1.

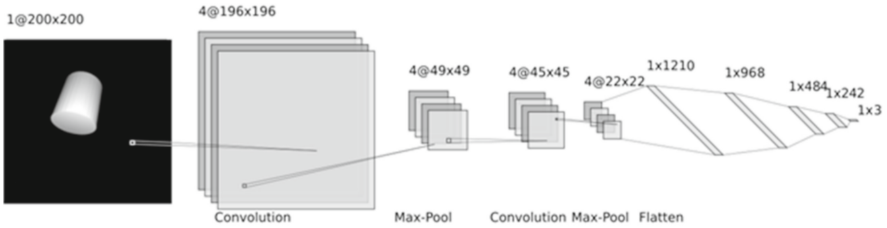


Fig. 1. CNN architecture.

2.3 Results and Discussion

Each particular sub-model of the ensemble model CNN2 is trained independently using the specific dataset. The training results for sub-models used to determine the force are given in Table 1.

We tested our trained CNN models considering different positions and orientations of the SDO relative to the ITT. Testing results show that CNN3 is the most accurate model, with an average normalized error of 7.689%. All models demonstrated the best accuracy for the same case with normalized errors of 2.863% ($1.821 \cdot 10^{-5}$ N), 1.555% ($9.888 \cdot 10^{-6}$ N) and 4.131% ($1.326 \cdot 10^{-3}$ N) for the CNN1, CNN2, CNN3, respectively.

Testing case 15 is the worst for the CNN1 model, where a relative error of 27.492% and an absolute error of 1.754·10–4 N were registered. CNN2 demonstrates the worst performance with a relative error of 16.889% and an absolute error of $1.070 \cdot 10^{-4}$. For the CNN3 model, a maximum relative error of 15.323% and an absolute error of $9.76 \cdot 10^{-5}$ N are the less accurate results. In this regard, CNN3 is similar to CNN1.

Table 1. Average loss for CNN2 sub-models.

Sub-model	Training loss	Validation loss
Q1	4.4·10–4	$4.9 \cdot 10^{-4}$
Q2	9.8·10–4	$1.19 \cdot 10^{-3}$
Q3	5.5·10–4	$6.4 \cdot 10^{-4}$
Q4	6.0·10–4	$7.2 \cdot 10^{-4}$

Although some relative errors may seem pretty significant, their impact on the SDO is minimal. For example, a relative error of 27.492% for CNN1 corresponds to an absolute error of 1.754·10–4 N, which is only 0.5% from the nominal thrust of the ITT (0.031 N). Another conclusion that can be drawn from the testing results is that the ion force prediction's accuracy goes up with the size of the dataset used for training.

The computation time required to determine the ion beam force by the ground truth function (traditional method) and CNN models is presented in Table 2 (Table 3).

Table 2. Computation time (s).

Ground truth	CNN1	CNN2	CNN3
19.99	8.49	8.50	8.65

Table 3. Control performance.

Case	$\tilde{L}_1/\tilde{L}_2$	Firing number	Total momentum, s	x-error	y-error	Mean-error
1	PWM	25	1397	8.50e-03	4.20e-03	6.35e-03
1	5e-5 / 0	13	2262	7.80e-02	5.80e-03	4.19e-02
2	PWM	37	334	4.30e-02	3.60e-02	3.95e-02
2	5e-5 /5e-5	98	839	8.50e-03	1.70e-03	5.10e-03

This table shows that every trained CNN model can determine the ion beam force significantly faster than the traditional method. Although there is little difference in time performance between the CNN models, they are all at least twice as fast as the conventional method. These results demonstrate that the CNN model is an alternative technique for determining the ion beam force, providing admissible accuracy and a shorter computation time than the conventional approach.

3 Spacecraft Attitude and Relative Control via Reinforcement Learning

3.1 Problem Statement

It is assumed that at the control system design stage an approximate (nominal) model of the spacecraft dynamics is known. This model differs from the real one both by its parameters and unmodeled dynamics. This nominal model is used to synthesize a basic control algorithm sufficient for the SC to perform some initial simple tasks. Next, the SC begins operating in orbit using this control algorithm. Then, the intelligent control system sequentially performs the following actions:

1. Collect data about peculiarities of the SC dynamics;
2. Learn model of SC dynamics using the obtained data;
3. Improve of the SC control algorithm using the updated model.

These actions are repeated until a required control performance is achieved. In the end, the control algorithm should be obtained that is as close as possible in terms of performance to the optimal control synthesized using an accurate mathematical model of the plant.

3.2 RL Algorithm

Neural networks have great potential for describing various processes based on experimental data. Unfortunately, this approach requires a very large amount of data to provide high-quality results, which makes it difficult to use them in the SC cases. Considering this issue, we use a statistical model based on the concept of Gaussian processes. This approach allows us to obtain the posterior distribution of the function based on the available data using non-parametric Bayesian regression. Using this model and the actor critic architecture an efficient RL-based algorithm can be presented as follows:

1. Select a deterministic model of the SC dynamics $h(X_k, U_k)$;
2. Design an initial controller $\pi(X_k)$ using the model $h(X_k, U_k)$;
3. Train actor $\pi_\theta(X)$;
4. Train critic $V_\eta^\pi(X)$;

Loop start

5. Collect data on the peculiarities of the SC dynamics using control $\pi_\theta(X)$.
6. Update the statistical model $g(X_k, U_k)$.

Loop start

7. Improve control $\pi_\theta(X)$.
8. Update the critic $V_\eta^\pi(X)$.

Loop end

Loop end

3.3 On-off Relative Control

Thrusters are usually used control SC motion. Such a thruster is a significantly nonlinear actuator, which have only two states: on or off. This property significantly complicates the direct synthesis of control laws [1, 2]. To overcome this issue, a linear control law is usually first synthesized taking into account the minimization of the selected performance criterion, and then a pulse width modulator (PWM) is used [9]. This approach does not allow explicitly optimizing the number of thruster firings and control performance degrades if the thrust resolution is large relative to the controller sample period. RL-based controller design can mitigate these drawbacks.

We use the following cost function:

$$C_k = Q^T X_k Q + {L_k}^T R^T U_k L_k R$$

This function is similar to the usual LQR criterion with constant weights Q and R, but an additional variable weight L_k allows us optimize control policy more flexibly, for example to encourage the agent to use wider pulses.

The agent receives as input a state vector and information about the control action on the previous control step as follows $\left[X_k, u_{k-1}\right]^T$. The actor and critic are implemented as fully connected neural networks with three hidden layers. Most layers use the Relu activation functions, only the actor output uses the SoftMax function.

The agent is trained using the PPO algorithm [10] and the following system data: sample period –200 s; thruster resolution –10 s; thrust - 2 mN; Q = diag(0.625, 6.4, 0.0002, 0.001); R = 0.16. Two ranges of the state vector variation are considered, such as Case 1: $x_m = 250$ m, $y_m = 800$ m, $\dot{x}_m$=0.4 m/s, $\dot{y}_m$=1 m/s; Case 2: $x_m = 12.5$ m, $y_m = 40$ m, $\dot{x}_m$=0.4 m/s, $\dot{y}_m$=1 m/s.

We use both constant action weights ${L_k}^2 = \tilde{L}$ and variables ones formed as follows ${L_k}^2 = \tilde{L}_1$, if $u_k \neq u_{k-1}$ and ${L_k}^2 = \tilde{L}_2$ if $u_k = u_{k-1}$. Figures 2 and 3 show the variations of the normalized state vector (a) and the thrust (b) for those cases. Control performance indicators are presented in the Table. 3.

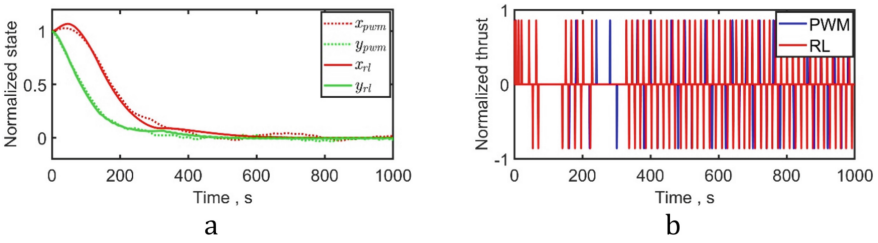


Fig. 2. Normalized state vector (a) and thrust (b) for case 2 and $\tilde{L} = 0.00005$.

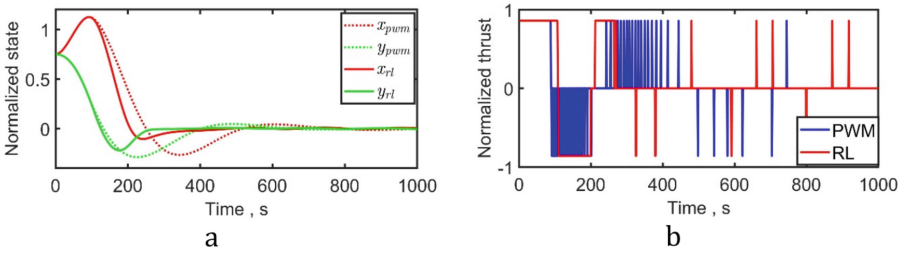


Fig. 3. Normalized state vector (a) and thrust (b) for case 1 and $\tilde{L}_1 = 0.00005$ та $\tilde{L}_2 = 0$

Simulation results demonstrate that this RL-based algorithm can outperform conventional control laws in terms of accuracy, propellant consumption, thruster firings.

4 Conclusions

This work demonstrates how supervised and RL techniques can be used for space applications. CNNs are utilized to determine the force impact of the IT plume on the SDO. The CNN models can determine the force impact without prior information about the SDO's relative position and orientation using only the SDO's images as the input. Moreover, CNNs make it possible to obtain the results significantly faster in comparison with the methods used before. Although the CNN models turn out less accurate than the traditional method, their errors are insignificant for practical applications. All these features allow us to state that the CNN model is a promising technique for both spacecraft onboard algorithms and the simulation of contactless space debris removal missions.

Spacecraft control performance can be improved the during its operation using RL. To increase the effectiveness of training, a model-based concept with Gaussian processes was used. The proposed approach makes it possible to develop control systems that can improve their performance as data accumulate during the operation of a specific object. The methodology makes it possible to reduce requirements for control system's elements (sensors, actuators), avoid using special bench equipment, and reduce development time and cost.

References

1. Khoroshylov, S.V., Redka, M.O.: Deep learning for space guidance, navigation, and control. Space Sci. Technol. **27**(6), 38–52 (2021). https://doi.org/10.15407/knit2021.06.038
2. Hakima, H., Reza Emami, M.: Assessment of active methods for removal of LEO debris. Acta Astronaut. **144**, 225–243 (2018). https://doi.org/10.1016/j.actaastro.2017.12.036
3. Bombardelli, C., Peláez, J.: Ion beam shepherd for contactless space debris removal. J. Guid. Control. Dyn. **34**(3), 916–920 (2011). https://doi.org/10.2514/1.51832
4. Redka, M.O., Khoroshylov, S.V.: Determination of the force impact of an ion thruster plume on an orbital object via deep learning. Space Sci. Technol. **28**(5), 15–26 (2022). https://doi.org/10.15407/knit2022.05.015
5. Redka, M.O., Khoroshylov, S.V.: Convolutional neural networks for determining the ion beam impact on a space debris object. Sci. Innov. **19**(6), 19–30 (2023). https://doi.org/10.15407/scine19.06.019

6. Oestreich, C., Linaresy, R., Gondhalekarz, R.: Autonomous six-degree-of-freedom spacecraft docking maneuvers via reinforcement learning. J. Aerospace Inform. Syst. **18**(7), 417–428 (2021). https://doi.org/10.2514/1.I010914
7. LeCun, Y., Boser, B., Denker, J.S., et al.: Backpropagation applied to handwritten zip code recognition. Neural Comput. **1**(4), 541–551 (1989). https://doi.org/10.1162/neco.1989.1.4.541
8. Alpatov, A., Cichocki, F., Fokov, A., et al.: Determination of the force transmitted by an ion thruster plasma plume to an orbital object. Acta Astronaut. **119**, 241–251 (2016). https://doi.org/10.1016/j.actaastro.2015.11.020
9. Alpatov, A., Khoroshylov, S., Lapkhanov, E.: Synthesizing an algorithm to control the angular motion of spacecraft equipped with an aeromagnetic deorbiting system. Eastern-Eur. J. Enterpr. Technol. **1**(5–103), 37–46 (2020). https://doi.org/10.15587/1729-4061.2020.192813
10. Schulman, J., Wolski, F., Dhariwal, P., et al.: Proximal Policy Optimization Algorithms, [Cs] (2017). https://arxiv.org/abs/1707.06347

A Framework for Decision-Support in Landmine Surveying: Spatio-Temporal Analysis of Land Cover Change Using Sentinel-2

Alaa AlZoubi[1(✉)] and Nasir Ibrahim[2]

[1] School of Computing, University of Derby, Derby DE22 3AW, UK
a.alzoubi@derby.ac.uk
[2] School of Computing, The University of Buckingham, Buckingham MK18 1EG, UK
nasir.ibrahim@buckingham.ac.uk

Abstract. Casualties from mines and explosive remnants of war (ERW) continue to be significant. Landmine monitoring group recorded more than 4,000 cases of injuries and fatalities because of ERW in 2022. In addition to suffering casualties, ERW results in significant impact on social and economic development. To mitigate the impact of mines, technical and non-technical surveys are employed to identify, clear and release potential hazardous regions. Currently, the land release process is effective but incur inefficiencies in time and cost, hence, the need to provide technical solution to increase the efficiency of the process. We extend our previous approach by analysing larger dataset collected over five years. In particular, we explore a framework for land clearance prioritisation using land-cover change analysis to support decision-making in landmine surveying. We present analysis using remote sensing images from sentinel-2 to estimate the changes in the land cover over time and provide results to assist in the clearance priority of different hazardous regions. A case study in the Kingdom of Cambodia is presented with several observations of satellite images for the years 2019 and 2023. Our study demonstrates that analysing the changes in land-cover is a promising direction to help in the non-technical survey process and increasing the productivity of the land release.

Keywords: Land Cover Analysis · Non-Technical Survey · Satellite Imagery · Sentinel-2 Data

1 Introduction

Unexploded landmines and other explosive devices (both anti-vehicle and anti-personnel) that are buried during conflict or war present effects that are felt long-term after the conflict or war – and the immediate inhabitants are hit harder by the devastation. These weapons can be found on roads, footpaths, agricultural fields, deserts, forests, and surrounding houses and school. They

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2026
I. Potapov et al. (Eds.): RTC-Digital 2023, CCIS 2647, pp. 47–53, 2026.
https://doi.org/10.1007/978-3-032-04731-1_6

deny access to water, food and hinders the movement and other basic activities of people [2]. In 2022, Landmine and cluster munition monitor [5] recorded more than 4,000 casualties (injuries and deaths) of mines and ERW. Of the casualties, 85% were civilians and 49% children [5]. Land Release (LR) process reduces the risk from landmines, sub-munition and Explosive Ordnance (EO) to a level where people can live safely. Land release process consists of establishing and improving the definition of where EO are to be found (and where they are not) through the application of all reasonable effort, until it can be shown with justifiable confidence that EO are either not present in an area or, if they were found to be present, have all been destroyed or removed from that area. The process has three main stages: 1) Non-Technical Survey, 2) Technical Survey and 3) Clearance.

Non-Technical Survey (NTS) is an essential step before the technical survey and mine clearance processes to identify the locations (and boundaries) of hazardous areas and gather information about the land-cover usage. It involves collecting information from different sources such as speaking to the populations in suspected areas, reviewing maps and field inspections. Understanding the changes in the land-cover before deploying the technical survey provides valuable information on the land usage, the scale of the problem, the resources needed to clear the field, and estimate the clearance time. Such analysis can help in identifying minefield related indicators such as changes in buildings, roads and vegetations. It also ensures that most hazardous areas are identified and prioritised. Usually, teams are deployed in potential areas, and they conduct interviews asking residents about the land use. However, this process is time and effort-consuming and costly.

This paper extends our prior work [1,4] in proposing a new framework to improve the efficiency of non-technical survey process by identifying key indicators for effective and precise technical survey. Our framework uses remote sensing images from Sentinel-2 to estimate the changes in key classes (vegetation and non-vegetation) in different suspected areas. We extend our prior work by analysing changes in 5 years period and providing insights and recommendations on suspected hazardous areas in Cambodia qualified for technical survey and clearance.

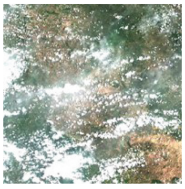
2 Data, Methods and Analysis

469 suspected hazardous areas (or polygons) in Cambodia were collected by landmine operators and the remote sensed data were collected from Copernicus Sentinel-2 [3]. The polygons size ranges from $1,200\,\mathrm{m}^2$ to $519,267\,\mathrm{m}^2$, spanning over 5 communes and 29 villages in Cambodia. For this study, 157 polygons with large size ($80,000\,\mathrm{m}^2$) were selected and analysed. Figure 1 shows example polygons overlaid on Google Earth.

We collected 54 observations (one observation every month) over five years (January 2019–June 2023). However, due to the cloud cover in the observations,

Fig. 1. Example Polygons Projected on Google Earth.

only 23 observations were suitable for analysis. Figure 2 shows images with different levels of cloud cover and Table 1 shows the selected observations. The observations selected have either low or no cloud cover.



(a) High Cloud Cover

(b) Medium Cloud Cover

(c) Low Cloud Cover

(d) No Cloud Cover

Fig. 2. Example Images with Different Levels of Cloud Cover.

Table 1. Selected Observations.

Year/Month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sept	Oct	Nov	Dec
2023	x	x	x	x		x						
2022	x	x	x								x	x
2021	x	x	x									
2020	x	x	x								x	x
2019	x	x	x								x	x

The analysis framework [1,4] comprises of four components:

1. `Segmentation`: The Sentinel-2 images were segmented using the Scene Classification algorithm [3] into different classes – vegetation, non-vegetation – soils/desert cloud, cloud shadow, snow and water. Only vegetation and non-vegetation classes were used in the analysis.

2. `Change Detection`: The segmented outputs are used for consecutive observations change detection. The method outputs a binary change mask - which represents the change locations and the change type.
3. `Change Description`: For each polygon, we extract and estimate the amount and number of land cover changes by estimating the changes in every two consecutive observations. The change detection method employed allows granular detailed information to be derived such as, From-To changes (the class change from one observation to the next observation) and the amount (in m^2) of change per class.
4. `Change Analysis and Recommendation`: Following the change description, significant changes between and within polygons are estimated over the observation period. After the change estimation, the decision makers (e.g., landmine operators) can consider the findings which will assist in prioritising regions (and sub-regions) for technical survey and clearing.

3 Change Analysis and Recommendation

The change analysis and recommendation component provide two levels of analyses: 1) polygon-based change analysis; and 2) polygon sub-region change analysis to help in identifying the hazardous areas (or polygons) and support the NTS threat classification process.

Polygons Change Analysis: The polygon-based change analysis estimates significant changes between consecutive observations in a polygon over time. First, the amount of changes between observations is computed (see Table 1 for an example polygon). Second, the number of significant changes is determined - a change is considered significant if the amount is greater than a threshold (2000 m^2, in this study). Table 2 shows the number of significant changes for one of the polygons.

Table 2. Example Polygon Changes between Consecutive Observations from Jan 2019–Dec 2020.

Change Detection	Jan.19 – Feb.19	Feb. 19 – Mar. 19	Mar. 19 – Nov. 19	Nov. 19 – Dec. 19	Dec. 19 – Jan. 20	Jan. 20 – Feb. 20	Feb. 20 – Mar. 20	Mar. 20 – Nov. 20	Nov. 20 – Dec. 20
Amount of Changes (m^2)	54,000	70,400	33,600	12,800	79,600	40,800	43,200	114,800	9600

In the case of the example polygon (Table 2), the number of significant changes is 9. Further, Table 3 highlights proportions of significant changes over multiple time periods (over 1 year, 2 years, 3 years and 5 years) for all 157 polygons. Figure 3 shows annual significant changes of several areas (polygons) over 5-year observation period.

Table 3. Proportion of Significant Changes in the Analysed Polygons in all Analysis Periods.

Analysis Periods	Num. of Observ.	Number of Significant Changes										
		0	1	2	3	4	5	6	7	8	9	10+
2023	5	0.00%	0.00%	1.91%	28.03%	70.06%	-	-	-	-	-	-
2019–2020	10	0.00%	0.00%	0.00%	0.00%	0.64%	1.91%	14.01%	22.93%	19.75%	40.76%	-
2021–2023	13	0.00%	0.00%	0.00%	0.00%	0.00%	0.64%	0.64%	1.91%	2.55%	10.19%	84.08%
2019–2023	23	**0–5 Changes**			**6–10 Changes**		**11–15 Changes**		**16–20 Changes**		**>20 Changes**	
		0.00%			0.00%		3.18%		62.42%		34.39%	

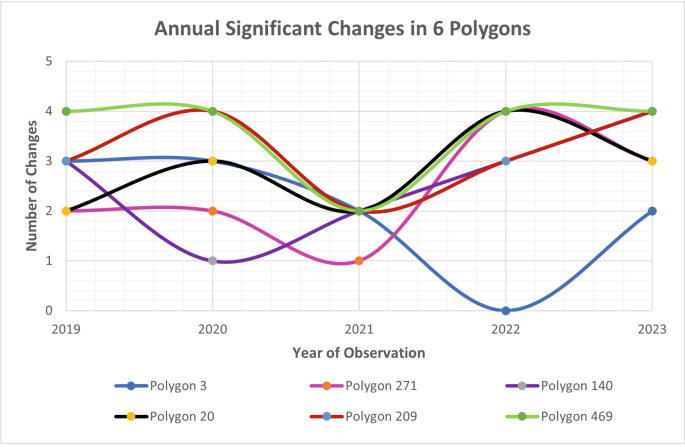


Fig. 3. Examples of Annual Significant Changes in 6 Polygons over 5-Years.

Polygon Sub-region Change Analyses: In addition to analysing the polygon significant changes over consecutive observations, each polygon sub-region representing a 400 m^2 (20 m × 20 m) is analysed and the changes on that sub-region over time is estimated. In this case, a change is considered if the land cover type changes between observations. For instance, if a region has vegetation type in observation one and non-vegetation in observation two, that will be considered a change otherwise no change. This number is determined for all observations in the sub-region. Figure 4 shows a visual output of the sub-region change analysis.

For the 157 polygons analysed, Table 4 highlights the proportion of polygons with land cover changes in the sub-regions over multiple analysis periods (over 1 year, 2 years, 3 years and 5 years).

These analyses can help the landmine surveyors to understand the overall usage of areas and decide which polygon (area) or sub-polygon area to survey first – priority could be given to polygon areas with small changes – indicating less usage. Further, in conjunction with our results, the final recommendation of the area could be decided according to the NTS Threat Classification process.

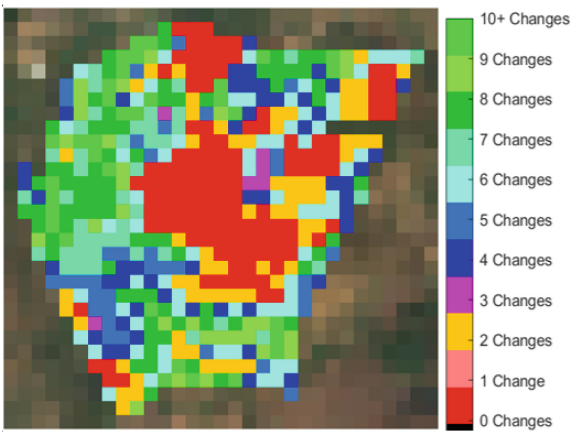


Fig. 4. Example Polygon Sub-Region Change Analysis.

Table 4. Proportion of Polygons with Sub-region Having the Represented Number of Changes.

AnalysisPeriod	0 Change	1 Change	2 Changes	3 Changes	4 Changes	5 Changes	6 Changes	7 Changes	8 Changes	9 Changes	10+ Changes
2023	83.44%	98.73%	100.00%	79.62%	23.57%	-	-	-	-	-	-
2019–2020	48.41%	40.76%	85.35%	100.00%	100.00%	91.08%	57.96%	15.92%	5.10%	0.00%	-
2021–2023	40.76%	37.58%	87.90%	93.63%	99.36%	99.36%	96.82%	83.44%	71.97%	28.03%	12.74%
2019–2023	18.47%	3.82%	45.22%	36.94%	70.06%	78.98%	92.99%	98.73%	100.00%	98.73%	97.45%

4 Conclusions and Future Work

In this paper, we have presented an extended analysis on our proposed framework for land-cover analysis to help improve the efficiency in land release process. Our analysis shows promise in analysing and estimating changes in suspected hazardous areas with publicly available data (Sentinel-2). We presented analysis over 5 years period providing insights and recommendations to assist decision makers in prioritising areas qualified for clearance. The Sentinel-2 data suffers from cloud and shadow occlusion which limited the classes of interest to analyse. Hence, pre-processing (cloud and shadow correction) needs to be applied for more complete analysis. Further, segmenting and analysing Sentinel-2 with resolution $20\,\mathrm{m} \times 20\,\mathrm{m}$ showed potential in detecting vegetation and non-vegetation areas as indicators of mines presence and absence. However, to create adequate land cover segmentations map and identify small objects, high-resolution images are needed. Encouraged by our results, we plan to extend our work to further support ongoing research into land cover change analysis by analysing other indicators such as changes in roads, buildings and water using high resolution remote sensing images.

Acknowledgments. The authors wish to thank Mr H.E Suy Chiheang, Vice president of Cambodian Mine Action and Victim Assistance Authority (CMAA), for providing the polygon dataset for the research.

References

1. AlZoubi, A., Ibrahim, N.: Land cover analysis using Sentinel-2 for humanitarian mine action and ERW survey. In: Poster, Digital Theme UK-Ukraine Research Twinning Conference (2023)
2. Cormack, D.: Landmines and unexploded ordnance will endanger the lives of Ukrainian people for years to come [online] (2023). https://www.maginternational.org/whats-happening/CEO-op-ed-Ukraine-year-anniversary/
3. European Space Agency: Sentinel overview: Sentinel missions [online] (2023). https://sentinels.copernicus.eu/web/sentinel/missions
4. Ibrahim, N., Fahs, S., AlZoubi, A.: Land cover analysis using satellite imagery for humanitarian mine action and ERW survey. In: Agaian, S.S., Asari, V.K., DelMarco, S.P., Jassim, S.A. (eds.) Multimodal Image Exploitation and Learning 2021. Society of Photo-Optical Instrumentation Engineers (SPIE) Conference Series, April 2021, vol. 11734, p. 1173402 (2021). https://doi.org/10.1117/12.2589792
5. Landmine Monitor: Landmine monitor 2022 [online] (2022). http://the-monitor.org/en-gb/reports/2022/landmine-monitor-2022/major-findings.aspx

Pipeline for Automatic Segmentation of Epithelial Cells in Buccal Swabs Slides with Custom Cellpose-Based Model

Dmytro Horyslavets[1], Oleksandr Skorobohatov[1(✉)], Yaroslav Ryndyk[1], Pavlo Areshkov[1], Olena Romantsova[1], Charlotte Oliver[2], Anna Paterson[3], Wei Cope[3], Emily Joslin[3], Emily O'Dea[4], Jon Teague[4], Peter Clapham[4], Moritz Przybilla[4], Mykhailo Tukalo[1], James McCafferty[4], Inigo Martincorena[4], and Alina Frolova[1]

[1] Institute of Molecular Biology and Genetics, Zabolotnogo 150, Kyiv 03143, Ukraine
mtukalo@imbg.org.ua, skorobogatov@gmail.com
[2] North West Anglia NHS Foundation Trust, 169 London Rd., Old Fletton, Peterborough PE2 9DS, UK
coliver3@nhs.net
[3] Cambridge University Hospitals NHS Foundation Trust, Hills Rd, CB2 0QQ Cambridge, UK
{anna.paterson,wei.cope1,emily.joslin}@nhs.net
[4] Wellcome Sanger Institute, Wellcome Trust Genome Campus, Hinxton, CB10 1RQ Saffron Walden, UK
emily.odea1@nhs.net, {jwt,pc7,mp34,jdmc,inigo.martincorena}@sanger.ac.uk

Abstract. Somatic mutations are known to be the root cause of cancer development and progression. However, detecting mutations in single cells or clones remains a challenging task due to the high error rates and restrictions on specific cell types. With the recent development of such protocol as NanoSeq it is now possible to reliably detect mutations in single DNA molecules, which could transform understanding of somatic mutagenesis and enable non-invasive studies on large-scale cohorts. Thus, we are in urgent need of large-scale studies of human cohorts of different ages, occupations, lifestyles, and health statuses. One of the promising non-invasive approaches is collecting buccal swab samples to study epithelial cell mutations. However, it is complicated by the need to establish a robust quality control pipeline of microscopy image data for cell count estimation in a given sample to avoid possible contamination. Such pipeline should be able to process hundreds of images in a reasonable time and accurately detect cells of a certain type automatically. We present the Automated Buccal Swabs Cells Recognition (ABSCR) tool that allows for the automated segmentation of human epithelial cells from buccal swabs microscopy image data using GPU-enabled Deep Learning methods and integration with the OMERO platform, which is widely used for managing, visualising and analysing microscopy images and associated metadata.

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2026
I. Potapov et al. (Eds.): RTC-Digital 2023, CCIS 2647, pp. 54–59, 2026.
https://doi.org/10.1007/978-3-032-04731-1_7

Keywords: ABSCR · Buccal Swabs · Cell Segmentation · Image Processing

1 Introduction

It is known that somatic mutations drive the development of cancer, but despite their importance, the difficulty of detecting mutations that are only present in single cells or small clones has limited our knowledge of somatic mutagenesis to a minority of tissues. The error rates of single-cell approaches have been too high [3] and other approaches are limited to mitotically active cell types. Thus, a research team from Wellcome Sanger Institute has previously developed nanorate sequencing (NanoSeq), a duplex sequencing protocol with error rates of less than five errors per billion base pairs in single DNA molecules from cell populations [1]. Results obtained by this protocol suggest that mutational processes that are independent of cell division play a significant role in somatic mutagenesis.

It is hard to study somatic mutation rate in humans in a non-invasive way, so collecting buccal swab samples is a promising approach. At the same time, buccal swab microscopy image data can be used during the quality control step to assess possible contamination of different cell types. As the amount of data may approach hundreds of high-resolution slides, the need for an automated epithelial cell segmentation and classification pipeline arises. To address this need we established a collaboration between Welcome Sanger Institute, UK, and the Institute of Molecular Biology and Genetics in Ukraine within The Cancer, Ageing, and Somatic Mutation Programme at Sanger. This program pioneers data aggregation and informatics innovation, develops high-throughput cellular models of cancer for genome-wide functional screens and drug testing, and explores basic scientific questions about the role somatic mutation plays in clonal evolution, aging, and development. As a result of collaboration, we developed the Automated Buccal Swabs Cells Recognition (ABSCR) tool, an efficient and scalable Python library that fulfils these purposes with the help of GPU-powered Sanger computational infrastructure.

ABSCR addresses several problems, including the need to reliably determine the characterisation and count of epithelial cells in the sample, the need to scale up the procedure to much higher volumes, including automation of manual pathology work such as cell detection, a lack of well-established and systematised capture, management and storage of spatial data with the required metadata, as well as the need to make the solution, data and associated tools widely available, putting power in researchers' hands.

2 Methods

During the model selection step, we evaluated three popular cell segmentation algorithms: ilastik [2], StarDist [4], and Cellpose[5]. After careful consideration, we decided to use the Cellpose (v.2.0) - generalist algorithm for cell and nucleus segmentation that can be optimized for your data and which is based on U-Net convolution neural network architecture.

2.1 Manual Epithelial Cell Annotation

30 out of 142 Sanger-provided buccal swab slides were chosen for manual annotations. After visual examination in NDP.view2 (v.2.9.29), each slide was divided into at least 5 regions 6,000 × 6,000 pixels in size in QuPath (v.0.4.2), which was further used for epithelial cell manual annotation. Contours of individual epithelial cells were outlined using the annotation tools available in QuPath, followed by filling cell interiors. The obtained annotations were exported as 8-bit binary TIF images, and the slides' regions were exported as 16-bit PNG images. The resulting dataset consisted of 165 pairs of raw slides' regions and corresponding ground truth masks, each containing, on average, 14 epithelial cell masks.

2.2 Image Preprocessing

The ABSCR library provides the option to crop images in order to get rid of obstructing artifacts. ABSCR supports all the popular slide formats such as Hamamatsu (.vms, .vmu, .ndpi) or Philips (.tiff) since it employs the TiffSlide library to read microscopy images. One can also use .png or .jpg images. To make the segmentation process time-efficient, where possible we used the metadata of the slides to downscale the images. We tested segmentation for the optimal downscale factor, and we also calculated cell diameter in pixels, important for the segmentation steps.

2.3 Model Training

The dataset was randomly split into 90% training and 10% validation sets. We trained 3 different models: Cellpose-derived cyto and cyto2, and a model trained from scratch. The mean training mask diameter was automatically calculated by the Cellpose algorithm when pre-trained models were used at the start of the training, or set to 200 otherwise. We used the default values for other training parameters.

2.4 Whole Slide Segmentation

The average resolution of the provided slide images was 200,000 × 100,000 pixels, so we used a downscale factor of 16, resulting in images of 10000 pixels in height. The overall segmentation time of the scaled images, with the help of Sanger GPU-powered computational facilities, took approximately 24 h, which is about 11 min per slide.

2.5 OMERO Integration

We implemented integration with the OMERO platform since it is widely used for microscopy data handling (https://www.openmicroscopy.org/omero/). In particular, ABSCR utilizes the OMERO python API to allow for downloading data and uploading the cell masks and metadata, obtained after the segmentation step, to the OMERO server.

2.6 GUI Tool

ABSCR provides the Python Bokeh-based GUI for manual removal and adjustment of the masks obtained after the segmentation step, which is necessary for custom model training or deleting unpleasant artefacts from the original image. It is capable of generating a scatter plot with linked histograms on both axes. It uses an image file and a mask file to calculate a set of properties of the shapes defined by the masks. Thus, the library comprises a complete set of instruments needed for efficient cell segmentation.

3 Results

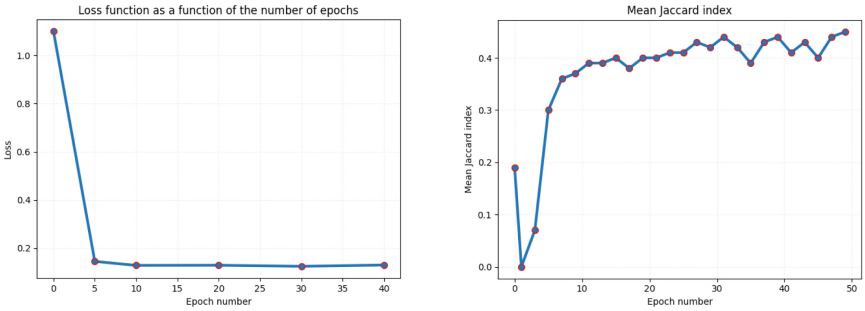
All models were trained for 50 epochs, and model parameters were saved every 2 epochs. A single model's training, with the help of Sanger GPU-powered computational facilities, lasted for approximately 2 h and required 100 GB on 4 GPU A100 NVIDIA nodes.

The obtained model, trained from scratch using the Cellpose algorithm, along with the cyto and cyto2 models, both native and trained additionally on our data, were evaluated on the validation data. The intersection-over-union, also known as the Jaccard Index, was chosen as the primary metric for segmentation accuracy evaluation. We identified that the ABSCR model based on the pretrained cyto2 model, which was additionally trained on our data, showed the best results indicated in Table 1. The loss function stabilised after the 5th epoch, while the mean Jaccard Index calculated on the validation set exhibited plato-like behaviour after the 10th epoch for the winning ABSCR model. The plot of the change of both metrics during the training process is given in Fig. 1.

Table 1. Comparison of models' performance on validation data.

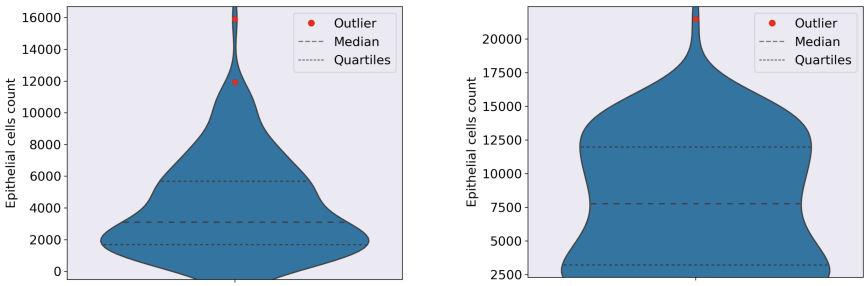
Model name	Mean Jaccard Index	Number of segmented images
ABSCR	0.45	17/17
cyto2	0.33	16/17
cyto	0.32	16/17
scratch	0.15	13/17

We ran the ABSCR model on 142 whole slide buccal swab images provided by the Wellcome Sanger Institute to count the number of epithelial cells. The resulting distribution of the cell counts is shown in Fig. 2. The overall segmentation time took approximately 24 h on a single GPU A100 NVIDIA node, which is about 11 min for the whole slide with an average resolution of $200{,}000 \times 100{,}000$ pixels, which were previously downscaled by the factor of 4. The segmentation accuracy example is illustrated in Fig. 3.



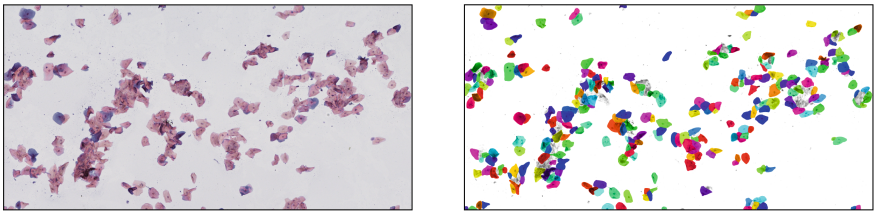
(a) Model training loss (b) Mean Jaccard index

Fig. 1. Cellpose cyto2-based custom model training using annotated buccal swabs whole-slide image data.



(a) Buccal swab slides provided by Sanger (b) Randomly generated data

Fig. 2. Epithelial cell count distribution. We compared the distribution of cell counts on slides with a discrete uniform distribution to ensure that the model accurately captures objects rather than image noise.



(a) Original image region (b) Cellpose segmentation performance

Fig. 3. Cellpose's cyto model epithelial cell segmentation performance on a buccal swab whole slide image.

References

1. Abascal, F., et al.: Somatic mutation landscapes at single-molecule resolution. Nature **593**(7859), 405–410 (2021). https://doi.org/10.1038/s41586-021-03477-4
2. Berg, S., et al.: ilastik: interactive machine learning for (bio)image analysis. Nat. Methods (2019). https://doi.org/10.1038/s41592-019-0582-9
3. Petljak, M., et al.: Characterizing mutational signatures in human cancer cell lines reveals episodic APOBEC mutagenesis. Cell **176**(6), 1282–1294.e20 (2019). https://doi.org/10.1016/j.cell.2019.02.012
4. Schmidt, U., Weigert, M., Broaddus, C., Myers, G.: Cell detection with star-convex polygons. In: Medical Image Computing and Computer Assisted Intervention – MICCAI 2018 - 21st International Conference, Granada, Spain, September 16-20, 2018, Proceedings, Part II, pp. 265–273 (2018). https://doi.org/10.1007/978-3-030-00934-2_30
5. Stringer, C., Wang, T., Michaelos, M., Pachitariu, M.: Cellpose: a generalist algorithm for cellular segmentation. Nat. Meth. **18**(1), 100–106 (2021)

Improving Short Utterance Dialogue Act Classification Using a Transformer Ensemble

Harry Maltby[1(✉)], Julie Wall[1], Tracy Gooodluck Constance[1], Mansour Moniri[1], Cornelius Glackin[2], Marvin Rajwadi[2], and Nigel Cannings[2]

[1] University of East London, London, UK
u1606661@uel.ac.uk
[2] Intelligent Voice Ltd., London, UK
https://intelligentvoice.com/

Abstract. An influx of digital assistant adoption and reliance is demonstrating the significance of reliable and robust dialogue act classification techniques. In the literature, there is an over-representation of purely lexical-based dialogue act classification methods. A weakness of this approach is the lack of context when classifying short utterances. We improve upon a purely lexical approach by incorporating a state-of-the-art acoustic model in a lexical-acoustic transformer ensemble, with improved results when classifying dialogue acts in the MRDA corpus. Additionally, we further investigate the performance on an utterance word count basis, showing classification accuracy increases with utterance word count. Furthermore, the performance of the lexical model increases with utterance word length and the acoustic model performance decreases with utterance word count, showing the models complement each other for different utterance lengths.

Keywords: Dialogue Act Classification · Ensemble · NLP · Transformers · Lexico-Acoustic Models

1 Introduction

There is a surge in the use of digital assistants in our homes. As device hardware and capabilities have improved in recent years, it has resulted in people, especially the elderly and disabled, becoming more reliant on them for everyday purposes. An essential requirement for an AI agent is to reliably understand a user's intended request and distinguish between a command and a question. This task is known as Dialogue Act (DA) classification and is essential to comprehending communication as it exposes the intent of the requests [7]. Detecting questions from speech is difficult for machines as automatic speaker recognition (ASR) transcripts lack punctuation, relying solely on lexical information, leading

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2026
I. Potapov et al. (Eds.): RTC-Digital 2023, CCIS 2647, pp. 60–66, 2026.
https://doi.org/10.1007/978-3-032-04731-1_8

to ambiguity. Acoustic information can aid in resolving ambiguity, as demonstrated by Ortega [8]. For example, in the case of a Declarative Question in the text without question marks, such as 'this is your car(?)', determining whether it is a statement, or a question requires additional contextual information. Our study shows that combining lexical and acoustic information can enhance DA classification. We use predictions from both a lexical and an acoustic model to improve overall performance. Our proposed transformer ensemble, which consists of BERT, Wav2Vec2, and a multi-layer perception (MLP), fuses lexical and acoustic features in a cascaded process for DA classification. By combining the strengths of both models and addressing their weaknesses, our ensemble approach achieves better results than either single model alone, as demonstrated on the Meeting Recorder Dialogue Act (MRDA) corpus.

2 Related Work

Dialogue Act (DA) classification is the task of classifying an utterance to capture the intent of the communication, i.e. whether it is a question, statement, etc. [2]. The most used publicly available benchmark data sets for this task are the Switchboard Dialogue Act (SwDA) and the (MRDA) corpus. The MRDA corpus was first described by Shriberg in 2004 [10]. The corpus was later revised and relabeled into 5 classes, as described by Ang et al. in 2005 [1], where the data set was separated into the labels: *backchannel*, *disruption*, *floorgrabber*, *question* and *statement*. It was noted that researchers at that time encountered many challenges due to the data set containing multiple speakers, frequent speaker overlap and high rates of speaker interruption. These challenges hindered the effectiveness of traditional techniques of the time to perform tasks such as DA classification and speaker diarisation. There have been several approaches to DA classification. Li et al. [7] utilize neural network models to classify DAs using a recurrent neural network. Raheja et al. [9] propose a context-aware self-attention mechanism with a hierarchical recurrent neural network. The work of Ortega et al. is among the few that have explored the effectiveness of acoustic features and found that they can be effective when lexical information is insufficient or strong lexical indicators are absent. However, their analysis also revealed the heavy reliance on question marks for question classification, demonstrating the need for acoustic information in the classification process. Recent works propose neural and attention-based models for DA classification in spoken dialogues, however, only a few have explored the effectiveness of acoustic information. Research for this task appears to be converging and maturing into the use of hierarchical techniques to capture context correlation data within the lexical data of the data set. We build upon this and further explore the efficacy of using the acoustic portion of the data set to augment the performance of mature lexical techniques for DA classification.

3 Methodology

3.1 Data Set and Rationale

We performed our experiments on the MRDA corpus, the 5-class version of the MRDA corpus was used: Statement (S), Question (Q), Floorgrabber (F), Backchannel (B), and Disruption (D). The MRDA corpus is an audio data set consisting of 75 meetings (each roughly an hour in duration) with 53 unique speakers (an average of about 6 speakers per meeting), mainly male speakers (40 male and 13 female speakers). We used the processing utilities from [6] and followed the guidelines outlined by the authors in [10], these utilities formatted the utterances in the format "Speaker | Utterance Text | Basic DA Tag | General DA Tag | Full DA Tag", removed non-verbal utterances such as laughter or throat clearing, removed unlabelled data and removed disfluency annotations. The train, validation and test splits are split meeting-wise according to these guidelines. The statistics of our data split are displayed in Table 1. Of the 75 meetings, the guidelines suggest the omission of the last 2 meetings which has been followed.

Table 1. Training, validation, and testing indicate the number of utterances in their respective splits.

Set	Floorgrabber	Statement	Disruption	Backchannel	Question	Total
Training	5,324	45,099	9,740	10,265	4,640	75,068
Validation	1,039	8,939	2,313	2,049	1,043	15,383
Testing	1,313	8,569	2,107	1,960	1,115	15,064
Whole	7,676	62,607	14,160	14,274	6,798	105,515

Our study aims to leverage the acoustic aspect of the MRDA dataset for performance enhancement. State-of-the-art techniques, such as BERT, rely on context information for classification, but they struggle with short, contextually poor utterances, leading to classification ambiguity as shown in Table 2. For example, single-word utterances like **'yeah'** can belong to multiple classes, with **'yeah'** specifically being able to belong to any class. We hypothesize that lexical models benefit from longer utterances, as they provide more context for classification, while acoustic models may perform worse on lengthy utterances. Our research seeks to validate these hypotheses.

3.2 Lexical and Acoustic Models

For the lexical model, we used the Hugging Face transformers package [11] and used PyTorch to implement a BERT model to classify text transcripts. BERT is a state-of-the-art Natural language Processing method developed by the Google AI research team in 2018, they trained BERT on the English Wikipedia (2,500M

Table 2. MRDA data samples showing noise in the lexical modality with unpunctuated textual data.

Sample ID	Unpunctuated	DA Class
Bed006-c2_2288240_2288760	yeah	Question
Bmr028-c4_3230200_3230480	yeah	Backchannel
Bmr028-c0_3106910_3110880	yeah	Statement
Bed006-c2_0564787_0565647	yeah	Disruption
Bmr010-c2_0397975_0398275	yeah	Floorgrabber

words) and Books corpus (800M words) [5]. BERT has proven useful and efficient for a vast array of language tasks such as text summarization and sentiment analysis. The transcript of the MRDA corpus was used to fine-tune the BERT pre-trained model. To explore the efficacy of using the acoustic data from the MRDA data set to predict DA classes, we chose to use a Wav2Vec2 [3] model to classify the 5 classes. The model used was Facebook's XLRS Large 960 self-model which comes pre-trained and fine-tuned on 960 h of the Libri-Light and Librispeech data sets using a self-training objective [12]. This model was chosen due to providing performance greater than the best semi-supervised methods while using 100 times less labelled data. This was of great consideration due to the unbalanced nature of the data set as shown in Table 1, as the data set classes follow the expected distribution of dialogue acts in natural spoken language. From the literature, it is evident that utilisation of attention for DA classification has shown great success with much of the SOTA using some form of attention mechanism.

3.3 Model Architecture

In our transformer ensemble approach, we implemented two models, a lexical BERT model and an acoustic Wav2Vec2 model. Utilising the output probabilities from both models concatenated as input features we trained an MLP classifier. Using the output probabilities of both models resulted in 10 input features as each model provided a class-wise probability providing 5 probabilities from each model. To process the concatenated probability features from our transformer models we utilised an MLP classifier. Our MLP classifier has 2 layers with a hidden dimension size of 16, we discovered a hidden layer size of 16 was the most effective through a grid parameter search. To ensure the greatest level of generalisation we included dropout and regularisation layers with a dropout ratio of 0.2 between the layers. In Fig. 1 the BERT, Wav2Vec2 and MLP ensemble is shown.

4 Experiments and Results

Our models' performance, as shown in Table 3, revealed the BERT model achieving 89.44% accuracy and 0.87 F1 score, while the Wav2Vec2 model achieved

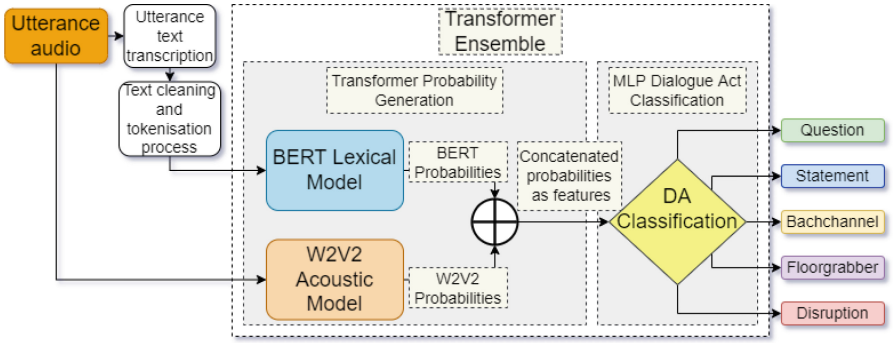


Fig. 1. Transformer ensemble classification architecture.

Table 3. Model performance metrics, Statement (S), Question (Q), Floorgrabber (F), Backchannel (B), and Disruption (D).

Technique	F (F1)	S (F1)	D (F1)	B (F1)	Q (F1)	Macro F1	Accuracy %
BERT	0.82	0.93	0.91	0.75	0.96	0.8740	89.44
Wav2Vec2	0.61	0.80	0.34	0.54	0.39	0.5354	67.88
Ensemble	0.82	0.94	0.91	0.76	0.96	0.8765	90.18
Baseline	0.57	0.92	0.82	0.69	0.71	0.6904	80.05

Table 4. Performance comparison table

Author	Accuracy	Features	Technique
Chapuis et al. (2021) [4]	92.4%	Lexical	Bespoke hierarchical transformer encoder
Li et al. (2019) [7]	92.2%	Lexical	CRF
Raheja et al. (2019) [9]	91.1%	Lexical	Hierarchical Dual-Attention RNN CRF
Ortega et al. (2018) [8]	84.7%	Lexico-Acoustic	CNN LSTM + CNN
Ortega et al. (2018) [8]	84.1%	Lexical	CNN LSTM
Ortega et al. (2018) [8]	67.8%	Acoustic	CNN
Our system	90.18%	Lexico-Acoustic	Transformer Ensemble
Our system	89.44%	Lexical	BERT
Our system	67.88%	Acoustic	Wav2Vec2

67.88% accuracy and 0.54 F1 score. The MLP ensemble outperformed Ortega et al. [8], with 90.18% accuracy and an F1 score of 0.8765. When analyzing utterance word count and plotting a grouped bar chart (see Fig. 2), we observed a positive correlation in the F1 score for the BERT classifier with increasing word count and a negative trend for the Wav2Vec2 model, aligning with our hypothesis. The MLP ensemble, particularly excelling in classifying short utterances, demonstrated the advantages of combining lexical and acoustic features (Fig. 4).

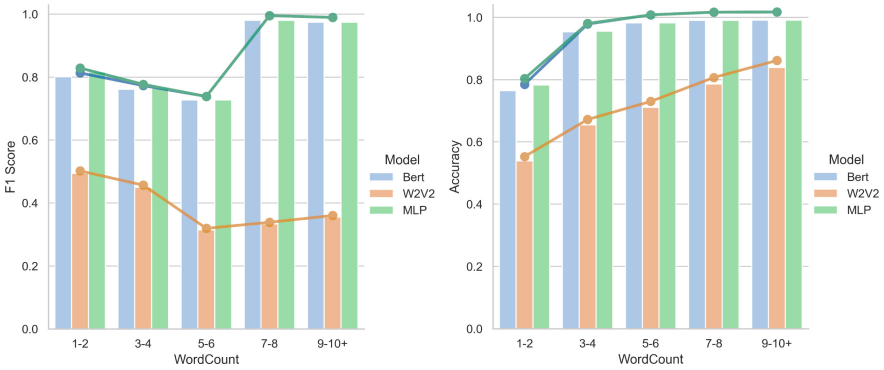


Fig. 2. Per-model accuracy and F-score on sets of utterances grouped by word count.

5 Conclusions

To summarise, the analysis of the performance of BERT, Wav2Vec2, and an MLP ensemble for the task of DA classification using both lexical and acoustic features showed that there is potential for performance improvement by exploiting both features. The results showed that BERT performs better on longer utterances, while Wav2Vec2 performs better on shorter ones. Our analysis of utterance word count shows that general accuracy increases as word count increases for all models, but this does not accurately describe the situation as F1 score plotted against word count shows that BERT performance declines from 1–2 words to 5–6 words but then greatly increases when utterances are of 7–8 words or longer. The MLP ensemble performed better than any single model, especially for short utterances. The negative correlation between the output of BERT and Wav2Vec2 suggests that they complement each other as an ensemble as one model can pick up where another model fails. Overall, the results suggest that there is room for further research to fully exploit both lexical and acoustic features in this task, large language models are especially of note showing great promise for improving dialogue act classification in the future for both modalities, offering the opportunity for more accurate and efficient analysis of conversations by way of their scale at the cost of more computation. Exploring their application and fine-tuning is essential for significant progress in this area.

References

1. Ang, J., Liu, Y., Shriberg, E.: Automatic dialog act segmentation and classification in multiparty meetings. In: Proceedings of the IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP), vol. 1, p. I–1061 (2005)
2. Austin, J.: How to Do Things with Words. Oxford University Press (1975)
3. Baevski, A., Zhou, Y., Mohamed, A., Auli, M.: wav2vec 2.0: a framework for self-supervised learning of speech representations. Adv. Neural Inf. Process. Syst. **33**, 12449–12460 (2020)

4. Chapuis, E., Colombo, P., Manica, M., Labeau, M., Clavel, C.: Hierarchical pre-training for sequence labelling in spoken dialog. arXiv preprint arXiv:2009.11152 (2020)
5. Devlin, J., Chang, M.W., Lee, K., Toutanova, K.: BERT: pre-training of deep bidirectional transformers for language understanding. arXiv preprint arXiv:1810.04805 (2018)
6. Duran, N.: Utilities for Processing the Meeting Recorder Dialogue Act Corpus (2018). https://github.com/NathanDuran/MRDA-Corpus
7. Li, R., Lin, C., Collinson, M., Li, X., Chen, G.: A dual-attention hierarchical recurrent neural network for dialogue act classification. arXiv preprint arXiv:1810.09154 (2018)
8. Ortega, D., Vu, N.T.: Lexico-acoustic neural-based models for dialog act classification. In: 2018 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), pp. 6194–6198 (2018)
9. Raheja, V., Tetreault, J.: Dialogue act classification with context-aware self-attention. arXiv preprint arXiv:1904.02594 (2019)
10. Shriberg, E., Dhillon, R., Bhagat, S., Ang, J., Carvey, H.: The ICSI meeting recorder dialog act (MRDA) corpus. Technical report, International Computer Science Inst. Berkeley CA (2004)
11. Wolf, T., et al.: Transformers: state-of-the-art natural language processing. In: Proceedings of the Conference on Empirical Methods in Natural Language Processing: System Demonstrations, pp. 38–45. Association for Computational Linguistics, October 2020. https://www.aclweb.org/anthology/2020.emnlp-demos.6
12. Xu, Q., et al.: Self-training and pre-training are complementary for speech recognition. In: 2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), ICASSP 2021, pp. 3030–3034 (2021). https://doi.org/10.1109/ICASSP39728.2021.9414641

Mathematical Foundations for Digital Technologies

The Exploration of Elastic SH Waves and Their Second Harmonics in Anisotropic Waveguide in the Form of a Layer on a Halfspace

Nadia Zhogoleva(✉)

Institute of Applied Mathematics and Mechanics NAS of Ukraine, 19 Batiuka Street, Sloviansk, Ukraine
zhogoleva.nadia@gmail.com
http://iamm.in.ua/

Abstract. In this article, using the methods of computer algebra we investigate the small nonlinear wave effects of second harmonics resulting from the propagation of elastic SH waves in a composite crystalline waveguide. The waveguides under consideration comprise a layer located on a half-space, with varied boundary conditions on the outer surface. The materials constituting the waveguide include diverse anisotropic crystals of the m3m class cubic system. The research is grounded in a model incorporating both geometric and physical nonlinearity of dynamic deformation processes. The approach involves decomposing the system into terms using a small parameter. Applying this approach in a special symbolic-numeric computing environment, it becomes possible to determine analytical representations of the displacement vector components of the second harmonics of elastic shear waves. Numerical results are presented for a waveguide composed of a NaCl layer on a Si half-space.

Keywords: computer algebra methods · anisotropy m3m class · nonlinear second harmonics · elastic surface waves

1 Problem Statement

The waveguide is associated with the rectangular coordinate system $O\widetilde{x}_1\widetilde{x}_2\widetilde{x}_3$ and comprises the layer V_1 situated on the half-space V_2. The layer occupies the region $V_1 = \{-\infty < \widetilde{x}_1,\ \widetilde{x}_2 < \infty,\ -h \le \widetilde{x}_3 \le 0\}$, while the half-space occupies $V_2 = \{-\infty < \widetilde{x}_1,\ \widetilde{x}_2 < \infty,\ 0 < \widetilde{x}_3 < \infty\}$.

The materials forming the waveguide consist of anisotropic cubic crystals of the $m3m$ class, characterised by three independent second-order matrix constants $\widetilde{c}_{11}^{(p)}$, $\widetilde{c}_{12}^{(p)}$, $\widetilde{c}_{44}^{(p)}$, five independent third-order matrix constants $\widetilde{c}_{111}^{(p)}$, $\widetilde{c}_{112}^{(p)}$, $\widetilde{c}_{114}^{(p)}$, $\widetilde{c}_{155}^{(p)}$, $\widetilde{c}_{123}^{(p)}$, $\widetilde{c}_{456}^{(p)}$, and the density $\widetilde{\rho}^{(p)}$. The superscript p denotes that the

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2026
I. Potapov et al. (Eds.): RTC-Digital 2023, CCIS 2647, pp. 69–75, 2026.
https://doi.org/10.1007/978-3-032-04731-1_9

corresponding characteristic pertains to the waveguide component V_p in the following ratios. The crystallographic directions of the layer and half-space are collinear.

The transition to dimensionless normalised characteristics is used. The components $\widetilde{u}_j^{(p)}$ of the vector of elastic wave displacements are normalised by the amplitude parameter $u_* = max_{x_1,x_2,x_3,t,j}|\widetilde{u}_j(x_1,x_2,x_3,t)|$. Mechanical stresses and elastic constants are related to the normalising parameter c_*. The value $R_* = h$ is employed as a normalising parameter for coordinate variables $\widetilde{x}_j$. Thus, $x_j = \widetilde{x}_j R_*^{(-1)}$, $u_j = \widetilde{u}_j u_*^{(-1)}$, $c_{ij}^{(p)} = \widetilde{c}_{ij}^{(p)}/c_*$, $c_{ijl}^{(p)} = \widetilde{c}_{ijl}^{(p)}/c_*$.

This work uses a physical and geometric nonlinear deformation model based on elastic potential U with quadratic and cubic terms of finite deformations:

$$U = \frac{1}{2}c_{jqrk}\varepsilon_{jq}\varepsilon_{rk} + \frac{1}{6}c_{jqrklm}\varepsilon_{jq}\varepsilon_{rk}\varepsilon_{lm} \qquad (j,q,r,k,l,m = \overline{1,3}). \tag{1}$$

Additionally, it relies on nonlinear representations of the components of the mechanical deformation tensor:

$$\varepsilon_{jk} = \frac{1}{2}(u_{l,k} + u_{k,j} + u_{l,j}u_{l,k}), \tag{2}$$

where $u_{r,k} = \partial u_r/\partial x_k$ and u_r represents the components of the wave elastic displacement vector. The components of the mechanical stress tensor σ_{jd} that are related to the choice of elastic potential above are represented as the sum of linear and nonlinear components

$$\sigma_{jd} = \sigma_{jd}^{(l)} + \sigma_{jd}^{(n)}, \tag{3}$$

where

$$\sigma_{jd}^{(l)} = c_{jdrk}u_{r,k}, \quad \sigma_{jd}^{(n)} = \frac{1}{2}c_{jdrk}u_{l,r}u_{l,k} + c_{pdrk}u_{j,p}u_{r,k} + \frac{1}{2}c_{jdrklm}u_{r,k}u_{l,m}. \tag{4}$$

The equation of motion for the elastic medium can be represented in a tensor form:

$$\rho\ddot{u}_j^{(p)} - \sigma_{jd,d}^{(p,l)} = \sigma_{jd,d}^{(p,n)}, \quad (j = \overline{1,3}). \tag{5}$$

The research of small nonlinear wave effects is based on small parameter method, where the normalised complex wave displacement functions $u_j^{(p)}$ $(j = \overline{1,3})$ are sought as the representation $u_j = u_j^{(l)} + \delta u_j^{(n)}$. The value $\delta = u_*/R_* << 1$ for elastic waves of low intensity is implied to be a small parameter.

2 Problem Solution

In the considered problem of generalised linear shear waves spreading in an m3m class cubic system monocrystal layer on a half-space of the same anisotropic

class, the linear components of the investigated wave field are determined from a homogeneous spectral boundary value problem:

$$\sigma_{2j,j}^{(p,l)} - \rho_p \ddot{u}_2^{(p)} = 0, \quad (p = 1, 3) \quad (j = \overline{1,3}). \tag{6}$$

The condition of ideal mechanical contact between the layer and the half-space is given by:

$$(u_2^{(1,l)})_{x_3=0} = (u_2^{(2,l)})_{x_3=0}, \quad \sigma_{32}^{(1,l)})_{x_3=0} = (\sigma_{32}^{(2,l)})_{x_3=0}. \tag{7}$$

For the outer face, three variants of boundaries will be considered:

stress-free face:

$$(\sigma_{32}^{(1,l)})_{x_3=-1} = 0; \tag{8}$$

Rigid fastening:

$$(u_j^{(1,l)})_{x_3=-1} = 0 \quad (j = \overline{1,3}); \tag{9}$$

Membrane coating:

$$(u_1^{(1,l)})_{x_3=-1} = (u_2^{(1,l)})_{x_3=-1} = 0, \quad (\sigma_{33}^{(1,l)})_{x_3=-1} = 0 \quad (j = \overline{1,3}). \tag{10}$$

The solution to the first approximation problem (6) of the linear surface SH waves $(u_2^{(1,l)})$ for all of boundary conditions is as follows:

$$u_2^{(l,1)} = u_2^{(0)}(cos(\alpha^{(1)}x_3) - c_{44}^{(2)}\alpha^{(2)}sin(\alpha^{(1)}x_3))/(c_{44}^{(1)}\alpha^{(1)})E_1, \quad x_3 \in [-1; 0];$$

$$u_2^{(2,l)} = u_2^{(0)}e^{-\alpha^{(2)}x_3}E_1, \quad x_3 \in [0; \infty). \tag{11}$$

Here, $u_2^{(0)}$ is the normalised amplitude parameter, ω is the circular frequency, $\alpha^{(1)} = ((\Omega_1^2/c_{44}^{(1)})^{1/2} - k^2)^{1/2}$, $\alpha^{(2)} = (k^2 - (\Omega_2^2/c_{44}^{(2)})^{1/2})^{1/2}$, $\Omega_p^2 = \rho_p\omega^2 R_*^2/c_*$ and $E_1 = e^{-i(\omega t - kx_1)}$. The dispersion relation that determines the family of dependencies of ω and wave number k for the case of waveguide upper surface free from disturbances (6), (7), (8) is

$$\tan(k_n(\Omega_1^2/k_n^2 - 1)^{1/2}) = c_{44}^{(2)}(1 - \Omega_2^2/k_n^2)^{1/2}/(c_{44}^{(1)}(\Omega_1^2/k_n^2 - 1)^{1/2}). \tag{12}$$

The dispersion relation for generalised linear SH waves of mode n in a waveguide of type (6), (7), (9) and (6), (7), (10) has the form:

$$\tan(k_n(\Omega_1^2/k_n^2 - 1)^{1/2}) = -c_{44}^{(1)}(\Omega_2^2/k_n^2 - 1)^{1/2}/(c_{44}^{(2)}(1 - \Omega_1^2/k_n^2)^{1/2}). \tag{13}$$

The structures (11), (12) and (11),(13) will be used to define relations for the problem of finding appropriate nonlinear anharmonic perturbations for the surface SH waves. The relations of the second approximation have the following form:

$$(\sigma_{ij.j}^{(p,l)})_{\boldsymbol{u}^{(p)}=\boldsymbol{u}^{(p,n)}} - \rho_p \ddot{u}_i^{(p,n)} = -(\sigma_{ij.j}^{(p,n)})_{\boldsymbol{u}^{(p)}=\boldsymbol{u}^{(p,l)}}. \tag{14}$$

The boundary conditions on the contacting border of the layer and halfspace of the considered waveguides are presented:

$$u_1^{(2,n)} = u_1^{(1,n)}, \quad u_3^{(2,n)} = u_3^{(1,n)} \text{ where } x_3 = 0,$$
$$(\sigma_{3i}^{(1,l)})_{\boldsymbol{u}^{(1)}=\boldsymbol{u}^{(1,n)}} + (\sigma_{3i}^{(1,n)})_{\boldsymbol{u}^{(1)}=\boldsymbol{u}^{(1,l)}} = 0, \quad (i = 1, 2) \tag{15}$$
$$(\sigma_{33}^{(1,l)})_{\boldsymbol{u}^{(1)}=\boldsymbol{u}^{(1,n)}} + (\sigma_{33}^{(1,n)})_{\boldsymbol{u}^{(1)}=\boldsymbol{u}^{(1,l)}} = (\sigma_{33}^{(2,l)})_{\boldsymbol{u}^{(2)}=\boldsymbol{u}^{(2,n)}} + (\sigma_{33}^{(2,n)})_{\boldsymbol{u}^{(2)}=\boldsymbol{u}^{(2,l)}}.$$

The boundary conditions on the top of the layer $x_3 = -1$ are the following: For the stress-free case:

$$(\sigma_{ij}^{(1,n)})_{\boldsymbol{u}^1=\boldsymbol{u}^{(1,l)}} + (\sigma_{ij}^{(1,l)})_{\boldsymbol{u}^1=\boldsymbol{u}^{(1,n)}} = 0; \tag{16}$$

For the case of rigid fixing:

$$u_i^{(1,n)} = 0 \quad (i = 1; 3); \tag{17}$$

For the membrane layers top:

$$u_1^{(1,n)} = 0, \quad (\sigma_{33}^{(1,n)})_{\boldsymbol{u}^1=\boldsymbol{u}^{(1,l)}} + (\sigma_{33}^{(1,l)})_{\boldsymbol{u}^1=\boldsymbol{u}^{(1,n)}} = 0. \tag{18}$$

The components $u_i^{(p,n)}$ of the complex stress vector of the second harmonics for each case of boundary conditions are determined from the relations of the boundary value problems (14–16); (14), (15), (17); (14), (15), (18) in analytic form, using computer algebra methods. In this case, the second harmonics for the layer and surface are represented as the sum of partial and general solutions of the corresponding inhomogeneous boundary value problem. From the analysis of the structure of the boundary value problem, it becomes apparent that the second harmonics of the discovered linear surface waves are waves of P-SV type. The structure of the analytical solution of the problem of the search for elastic wave displacements $u_j^{(n,p)} \quad (j = 1, j = 3)$ in the nonlinear second harmonics of investigated waves can be written in a brief form:

$$u_1^{(1,n)} = (u_2^{(0)})^2(\tilde{\lambda}_{11}cos(\zeta_1^{(1)}x_3)+\tilde{\lambda}_{12}cos(\zeta_2^{(1)}x_3))+\tilde{\mu}_{11}\sin(\zeta_1^{(1)}x_3)+\tilde{\mu}_{12}\sin(\zeta_2^{(1)}x_3)$$
$$+\tilde{\nu}_1+\tilde{\chi}_1\cos(2\alpha^{(1)}x_3)+\tilde{\xi}_1\sin(2\alpha^{(1)}x_3))exp(-2i(\omega t-kx_1)),$$
$$u_3^{(1,n)} = (u_2^{(0)})^2(\tilde{\lambda}_{31}\sin(\zeta_1^{(1)}x_3)+\tilde{\lambda}_{32}\sin(\zeta_2^{(1)}x_3)+\tilde{\mu}_{31}\cos(\zeta_1^{(1)}x_3)+\tilde{\mu}_{32}\cos(\zeta_2^{(1)}x_3)$$
$$+\tilde{\nu}_3+\tilde{\chi}_3\sin(2\alpha^{(1)}x_3)+\tilde{\xi}_3\cos(2\alpha^{(1)}x_3))exp(-2i(\omega t-kx_1)), \tag{19}$$
$$u_1^{(2,n)} = (u_2^{(0)})^2(\tilde{\beta}_{11}^{(2)}exp(\zeta_1^{(2)}x_3)+\tilde{\beta}_{12}^{(2)}exp(\zeta_2^{(2)}x_3))exp(-2i(\omega t-kx_1)),$$
$$u_3^{(2,n)} = (u_2^{(0)})^2(\tilde{\beta}_{31}^{(2)}exp(\zeta_1^{(2)}x_3)+\tilde{\beta}_{32}^{(2)}exp(\zeta_2^{(2)}x_3))exp(-2i(\omega t-kx_1)).$$

3 Numerical Results

Numerical investigations of the kinematic characteristics of the studied surface waves and their nonlinear second harmonics for the waveguide in the form of a V_1 layer from a sodium chloride monocrystal, located on the V_2 halfspace from a silicon monocrystal. The waveguide components are characterised by independent physical-mechanical constants for the NaCl crystal in the first row of the Table 1 below and for Si in the second row, respectively. The parameter values c_*, ρ_* are $c_* = 10^{10}$ (N/m^2), $\rho_* = 10^3$ (kg/m^3).

Table 1. Values of Elastic Constants and Density for the Waveguide Materials

Material	c_{11}	c_{12}	c_{44}	ρ
NaCl	$4,958c_*$	$1,306c_*$	$1,279c_*$	$2,1678\rho_*$
Si	$16,7c_*$	$7,9c_*$	$6,5c_*$	$2,3283\rho_*$

Material	c_{111}	c_{112}	c_{123}	c_{144}	c_{456}	c_{155}
NaCl	$-86,36c_*$	$-4,96c_*$	$0,93c_*$	$1,32c_*$	$0,71c_*$	$-5,87c_*$
Si	$-82,5c_*$	$-45,1c_*$	$-6,4c_*$	$1,2c_*$	$-6,4c_*$	$-31,0c_*$

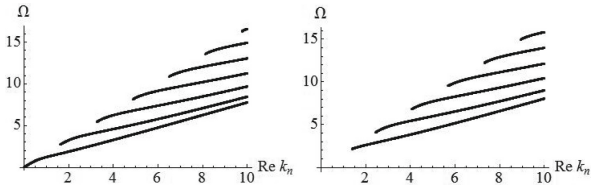


Fig. 1. Dispersion relations that determine the dependence of Ω and k

Numerical studies were carried out for the Ω and k ratios taken from the two lower branches of Fig. 1, representing mode 0 and mode 1, respectively, reflecting the admissible families of solutions. The first image illustrates the dependency of Ω and k taken from (12) for the case of the upper surface of the waveguide free from disturbances. The second image illustrates the dependency from (13) for the cases of rigid fastening and membrane coating, respectively. Interestingly, solving the last mentioned two problems in the first stage, when determining the type of $|u_2^{(l)}|/u_2^{(0)}$, leads to the observation that, in the first approximation, it is the same problem. Only in the second approximation, when determining the nonlinear wave additions $|u_1^{(n)}|/(u_2^{(0)})^2$ and $|u_3^{(n)}|/(u_2^{(0)})^2$, does the difference in behaviour for these wave propagations become apparent (Fig. 3).

For a comparative frequency-parametric analysis of the identified nonlinear wave effects, we conducted calculations for dimensionless normalised amplitudes

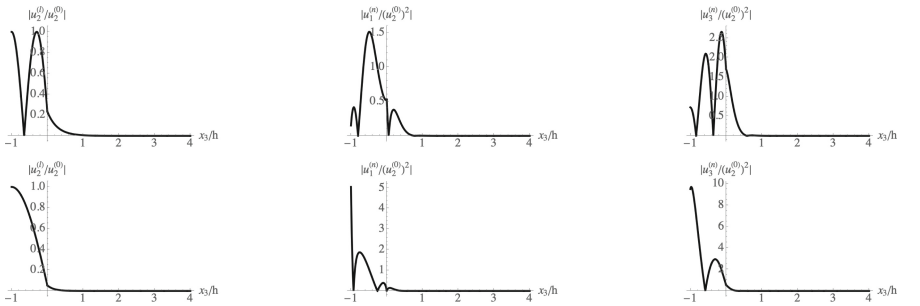


Fig. 2. Spreading of normalised values $|u_2^{(l)}|/u_2^{(0)}$, $|u_1^{(n)}|/(u_2^{(0)})^2$, $|u_3^{(n)}|/(u_2^{(0)})^2$ with $\Omega_0 = 5$ for the upper row and $\Omega_1 = 5$ –lower row, for the case of stress-free of the layers outer boundary

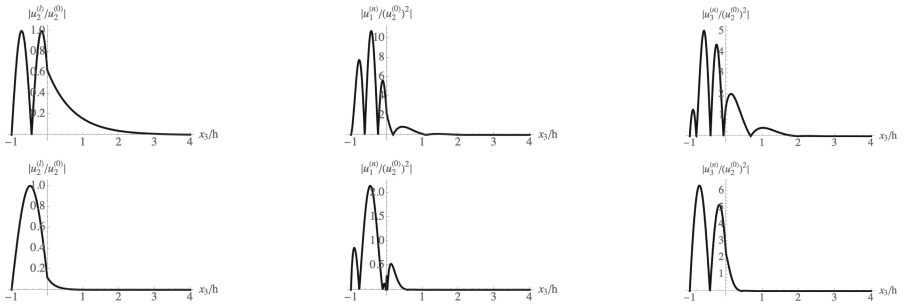


Fig. 3. Spreading of normalised values $|u_2^{(l)}|/u_2^{(0)}$, $|u_1^{(n)}|/(u_2^{(0)})^2$, $|u_3^{(n)}|/(u_2^{(0)})^2$ with $\Omega_0 = 5$ for the upper row and $\Omega_1 = 5$ –lower row, for the case of rigid fixation of the layers outer boundary

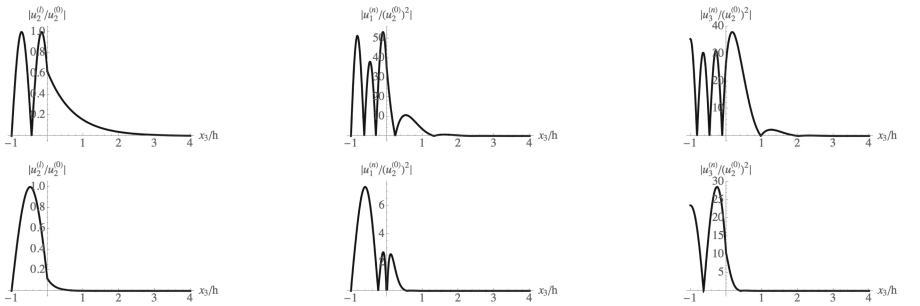


Fig. 4. Spreading of normalised values $|u_2^{(l)}|/u_2^{(0)}$, $|u_1^{(n)}|/(u_2^{(0)})^2$, $|u_3^{(n)}|/(u_2^{(0)})^2$ with $\Omega_0 = 5$ for the upper row and $\Omega_1 = 5$ –lower row, for the case of membrane coating of the layers outer boundary

of elastic lateral shifts in linear waves $|u_2^{(l)}|/u_2^{(0)}$, along with their corresponding characteristics $|u_1^{(n)}|/(u_2^{(0)})^2$ and $|u_3^{(n)}|/(u_2^{(0)})^2$ in their second harmonics. The

analysis covers the region along the thickness of the waveguide, comprising the layer area $x_3 \in [-1;0]$ and the region $x_3/h \in (0;4)$ in the half-space. The distributions of $|u_1^{(n)}|/(u_2^{(0)})^2$ and $|u_3^{(n)}|/(u_2^{(0)})^2$ are illustrated in Fig. 2 to Fig. 4, representing various boundary conditions: the stress-free face of the waveguide, rigid fastening, and membrane coating, for the frequency value $\Omega(k) = 5.0$.

Acknowledgments. This work was partially supported by a grant from the Simons Foundation (Award 1160640, Presidential Discretionary-Ukraine Support Grants, Nadia Zhogoleva).

References

1. Blistanov, A.A., Bondarenko, V.S., Chkalova, V.V.: Pod redaktsiey M.P.Shashkol'skoy. Akusticheskiye kristally. M.: Nauka, 632 p. (1982). (in Russian)
2. Shevchenko V.P., Zhogoleva N.V.: Nonlinear second harmonics of localized shear waves in an anisotropic layer between anisotropic half spaces under the conditions of imperfect contact. J. Math. Sci. **236**, 199–211 (2019)

Development of Computational Algorithm of Investigation Self-heating Processes in Elastomers

Yuliia Viazovychenko(✉) and Oleksiy Larin

Kharkiv Polytechnic Institute, National Technical University, Kyrpychova Str. 2, Kharkiv, Ukraine
yuliia.viazovychenko@khpi.edu.ua
oleksiy.larin@khpi.edu.ua
https://web.kpi.kharkov.ua/dpm/uk/vyazovichenko-yuliya-andriyivna/,
https://web.kpi.kharkov.ua/dpm/uk/larin-oleksiy-oleksandrovych

Abstract. The work deals with development of computational algorithms for modeling complex processes using the example of self-heating processes in elastomer composites. The calculation approach was proposed and applied for the calculation of the heat process of the tire in operation. The approach deals with computational modelling of stress-strain state processes and solving thermal conductivity problem. It includes obtaining data from one calculation, its processing and implementation to another. The several integrated macros codes and specific routines has been written in frame of work.

Keywords: Algorithm · Data processing · Self-heating · Fourier series · FE-method · Strain cycles

1 Introduction

Elastomers and their composites are essential in diverse industries like automotive, aviation, energy, construction, and technology. These materials exhibit characteristics such as hyperelasticity, viscoelasticity, moderate deformations, and temperature sensitivity.

Analysing their applications reveals significant cyclic dynamic loads, forming hysteresis loops and causing self-heating phenomena. Elevated temperatures lead to changes in mechanical properties [1], reducing strength [2] and accelerating aging and degradation [3], impacting fatigue processes [4]. This necessitates development of effective calculation methods to study cyclic deformation and thermal processes in such materials. Additionally, it is crucial to assess their influence on thermal stress in structural elements and evaluating reliability under varying operational conditions.

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2026
I. Potapov et al. (Eds.): RTC-Digital 2023, CCIS 2647, pp. 76–83, 2026.
https://doi.org/10.1007/978-3-032-04731-1_10

2 General Idea

The heat generation process in elastomers is contingent on the heat released per deformation cycle. The current work proposes a numerical algorithm to model this process, and involves three main stages:

FE-simulation of static tire-road interaction: The outcomes include deformation cycles.

Heat Generation Rate Computation: Computing the rate of heat generation in each element of model. This includes determining the energy released during one deformation cycle, with previously established material parameters [5].

Temperature Field Determination: The final stage centres on the stationary thermal conductivity problem. Each element is loaded with the heat generation rate function derived in the preceding stage.

The general block scheme, illustrated in Fig. 1, outlines the fundamental steps involved in identifying internal heat resources. The algorithm is applied for analysis of pneumatic tire self-heating.

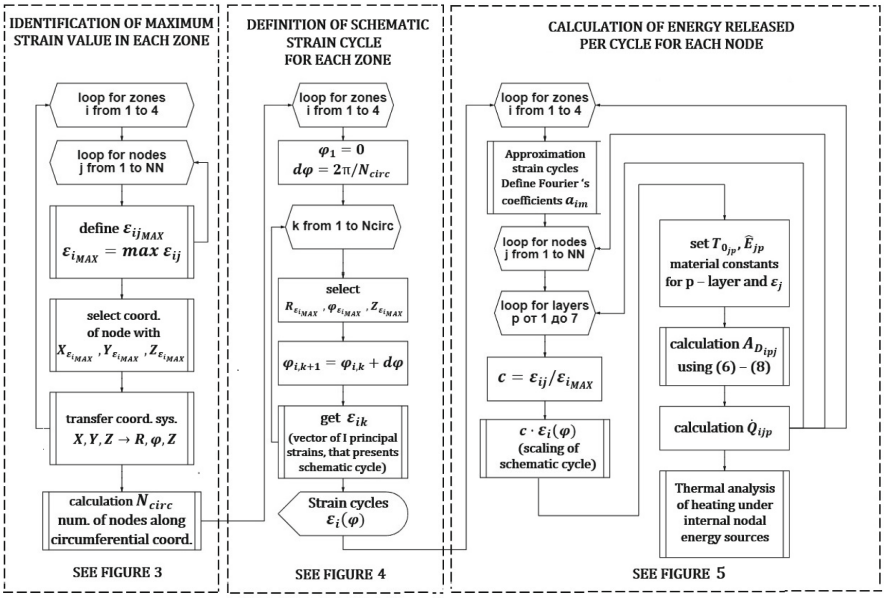


Fig. 1. General block-scheme of heat internal source calculation (stage 2 and 3)

3 Strain Cycles Determination

To ensure a precise identification of stress and strain levels, a multi-scale approach, outlined in [6], was implemented. The main tire response on the static

deformation was performed finally on the sector model represented on Fig. 2, that was cut out from whole tire model handling the intricate multilayer structure and curved shape of the studied object.

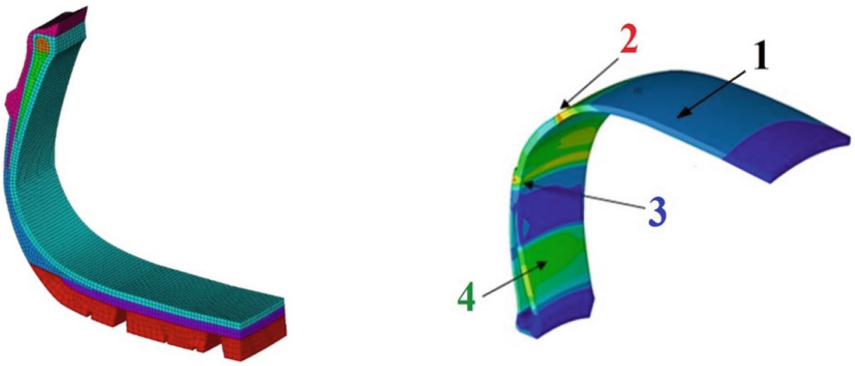


Fig. 2. Tire sector; a–FE-model, b–Strains zones

The computation of the deformed state was used for producing results aligned with wheel rotation, as an approximation of stationary periodic deformation.

Detailed analysis of the deformed state of the tire elements facilitated the identification of 4 zones (see Fig. 3) with the highest strains in the cord layers, corresponding to the shoulder zone (carcass and breaker), bead (carcass) and the location where the carcass is doubled.

The identification of the deformation cycles was held as follows. Identification of FE-node with maximum strains value for each zone. Then, the coordinates of those nodes were changed to cylindrical system, i.e. (X, Y, Z) -¿ (R, Phi, Z) and extended as a cycle the tire moved on with rotation around Z axis. The detail description is shown on Fig. 1. It allows us define strains in node with corresponding circumferential coordinate by changing it cyclically keeping fixed two others. The information was saved to the external file. Figure 3 represents cycles of the tire carcass deformed state in the control zones. The entire model was conventionally divided into the four zones mentioned above, and it was considered that the general nature of the deformation cycle for all nodes within one zone is the same [6]. The deformation values in the cycle were scaled relative to value in each specific node.

4 Calculation of Heat (source) Generation

The process of calculating heat generation functions comprises multiple steps.

Strain Cycle Import: In the initial step, strain cycles are imported from the preceding stage. Cycle Approximation: The next phase involves obtaining

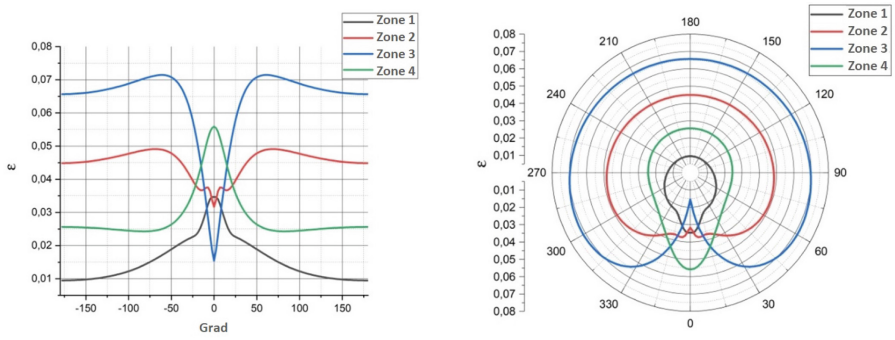


Fig. 3. Cycle of carcass deformed state for the pressure value 1.58 MPa and vertical load 1.2 t: a – in the Cartesian coordinate system, b – in the polar coordinate system

a fitting approximation of these cycles, potentially utilising piecewise quadratic functions. Refinement via Fourier's Transformation: Subsequently, the obtained approximations undergo refinement by applying Fourier's transformation to establish the cyclic nature of the strain-time relationship.

Calculation of Heat Generation Rate: The final step focuses on computing the heat generation rate. This calculation is based on a 2-parameters viscoelasticity model, incorporating previously obtained real operation cycles.

For specific cycles, frequency components with a more pronounced contribution in the Fourier series (Fig. 4) compared to the first three terms are selectively chosen.

4.1 Determination of the Heat Generation Function

The heat generation rate is determined per unit volume as follows

$$\dot{Q} = \int_0^t \sigma_{ij}(\tau) \frac{d\epsilon_{\mathbf{ij}}(\tau)}{dt}, \tag{1}$$

where $\varepsilon_{ij}(\tau)$ – strain cycles, $\sigma_{ij}(\tau)$ – strain tensor, which depends on the material parameters.

The stress patterns over time, taking into consideration viscoelastic behaviour, display a distinct structure. It's crucial to highlight that when performing steady-state calculations, this effect is overlooked, and only the amplitude strain values for each cycle are provided:

$$\sigma(t) = \int_{-\infty}^{t} G(t-\tau) \frac{\partial \varepsilon}{\partial \tau} d\tau, \varepsilon(t) = \varepsilon_0 + \sum_{i=1}^{N} \varepsilon_i cos\omega_i t, \tag{2}$$

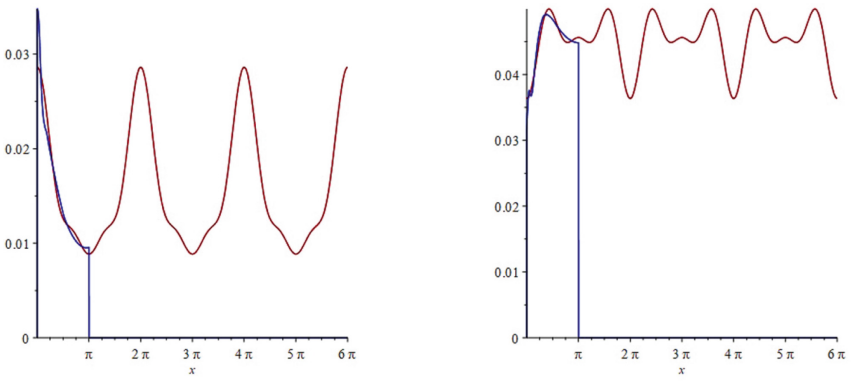
where $G(t)$ is the relaxation kernel; ε_0 and ε_i are decomposition harmonics.

The work was limited to the two-parameter model of linear viscoelasticity [7]:

$$G(t) = E + \hat{E}e^{-\frac{\hat{E}}{\eta}t} = E + \hat{E}e^{\frac{t}{T_0}}, \tag{3}$$

where $\hat{E}, \eta$– are material constants obtained from the experiment [8]; T_0 – relaxation time ($T_0 = \frac{\eta}{\hat{E}}$).

On the basis of (2) and (3) the stress is determined from (4) and will be written as follows:



a) b)

Fig. 4. Visualization of strain cycles approximation for zone 1 (a) and 2 (b) (pressure value – 158 kPa, weight – 1.2 t.

$$\sigma = Ae^{-i\lambda t} + B\sum_{i=1}^{N} b_i cos\omega_i t + C\sum_{i=1}^{N} c_i sin\omega_i t + D, \tag{4}$$

where A, B, C, D, b_i, c_i are integration constants that depend on the material characteristics $\hat{E}, T_0$, as well as the strain amplitudes values. They can be found analytically and have a complex algebraic form.

Then, for the stationary case, the released energy amount is equal to

$$A_D = \int_0^T \sigma \frac{d\varepsilon}{dt} dt = A_D\left(\hat{E}, T_0, \omega_i, \varepsilon_i\right), \tag{5}$$

where ω_i is the load frequency,T_0– period of one deformation cycle.

5 Solving of Stationary Heat Conduction Problem

This problem was also solved with usage of FE method. The matrix equation of steady thermal analysis can be represented as

$$[P]\,\Theta = \dot{Q}, \tag{6}$$

where $[P]$ - thermal conductivity matrix, Θ – nodal temperature value, internal heat source $\dot{Q}$.

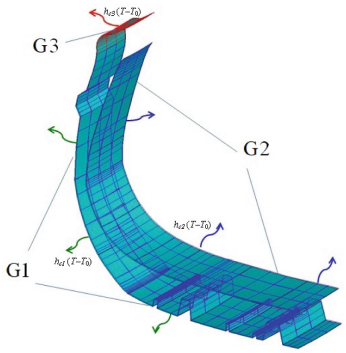


Fig. 5. Boundary conditions in thermal analysis.

Thus, for each specific zone, we integrated Fourier fitting of deformation cycles and strains values ε_i were assigned to each node within that zone, taking into account the respective layers. Subsequently, the cycle for this zone was adjusted in relation to the obtained $\varepsilon_i(\tau)$, resulting in an individualised strain cycle for each node in the sector model. Then we define A_D – amplitude of released energy amount, with respect to ε_i amplitude value using $\hat{E}_{ip}, T_{0ip}$– heat dissipation material parameters (strain amplitude dependent value).

The FE model was built to solve the problem of heat conduction problem. Figure 5 illustrated surfaces with different boundary conditions. The temperature (T_0, Co) and the convective coefficient $(h_c, W/m^2Co)$ are set as boundary conditions on the external (G1), internal (G2) surface and the point of interaction with the tire rim (G3) [9]. As a load in each node, the function of the heat generation rate $\dot{Q}$, is given, which is calculated by formula (6) for each zone according to the deformation cycle. Figure 6 shows the temperature distribution formed during rolling of a tire at a speed of 50 km/h loaded with an internal pressure of 158 kPa and with a vehicle weight of 1.2 t. According to results, the carcass zone is the most heated, due to the presence of larger deformations in it, compared to other layers. The obtained temperatures correspond to the actual heating levels of the rolling tire at the given operational modes, which proves the feasibility of using the proposed algorithm.

6 Conclusions

In summary, this study presents a robust algorithm for investigating complex processes, focusing on the example of self-heating phenomena in elastomers. The

introduced data processing approach enhances the usability of deformed state calculation results for solving heat conductivity problems.

The algorithm's practical application was demonstrated through the calculation of the tire's heat state during operation, considering the impact of self-heating processes. Key aspects, such as tire-road interaction, calculation of heat generation in tire elements, and determination of the temperature field in structural elements, have been considered. Identifying zones with the highest deformations in the cords allows for optimising computational resources by considering the general nature of the deformation cycle for all nodes in one zone. The model accurately represented the spatial aspects of the tire layers, and the results demonstrated temperature distribution during tire rolling under specific operational conditions. The obtained temperature distribution along the tire profile exhibited a noteworthy correlation with the actual temperature distribution observed during real-world tire operation. This successful implementation underscores the algorithm's effectiveness in comprehensively capturing and predicting the thermal behaviour of elastomers.

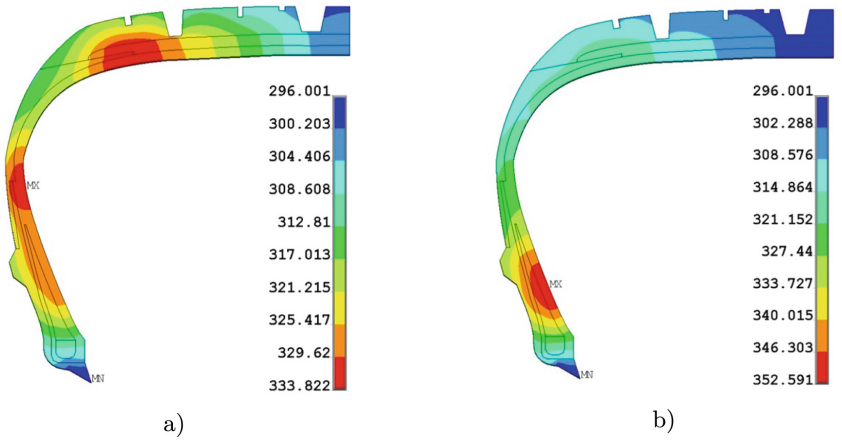


Fig. 6. Temperature distribution field along the tire profile at an internal pressure of 158 kPa and a vertical load of 1.2 t (a), 2.1 t (b)

References

1. Li, Z., Wan, J., Zhang, L., et al.: Effects of heat and moisture on characteristics, tensile properties of RFL-coated rayon cords, and their adhesion with NR/SBR matrix. J. Appl. Polym. Sci. **134**(48), 147–165 (2017)
2. Shim, W., Jang, J., Choi, J.H., et al.: Simulating rate- and temperature-dependent behaviors of adhesives using a nonlinear viscoelastic model. Mech. Mater. **147**, 103–446 (2020)

3. Kondé, A., Rosu, I., Lebon, F., et al.: Thermomechanical analysis of an aircraft tire in cornering using coupled ale and Lagrangian formulations. Open Eng. **3**(2), 191–205 (2013)
4. Luo, W., Li, M., Huang, Y., et al.: Effect of temperature on the tear fracture and fatigue life of carbon-black-filled rubber. Polymers **11**(5), 201–214 (2019)
5. Larin, A.A., Vyazovichenko, Y.A., Barkanov, E., Itskov, M.: Investigation of viscoelastic characteristics of rubber-cord composites considering the process of their self-heating. Strength Mater. **50**, 841–851 (2018)
6. Larin A.A., Petrova Y.A.: Determination of stress cycles of pneumatic tire elements using the submodeling procedure. NTU "KhPI" Bull. Ser. Dyn. Strength Mach. **57**, 37-49 (2014)
7. Haupt P.: Continuum Mechanics and Theory of Materials, p. 643. Springer, Heidelberg (2002)
8. Viazovychenko, Y., Larin, O.: Stochastic optimization algorithms for data processing in experimental self-heating process. In: Integrated Computer Technologies in Mechanical Engineering, ICTM 2020. LNNS, vol. 188, pp. 644–653. Springer (2020)
9. Cho, J.R., Lee, H.W., Jeong, W.B., et al.: Numerical estimation of rolling resistance and temperature distribution of 3-D periodic patterned tire. Int. J. Solids Struct. **50**(1), 86–96 (2013)

Application of the Discrete Models of the Topologies on a Finite Set to the Enumeration and Calculation of All Labeled T_0-Topologies

Anna Skryabina(✉) and Polina Stegantseva

Zaporizhzhya National University, Zhukovskogo Str. 66, Building 1, Office 21-A, Zaporizhzhya, Ukraine
anna_29_95@ukr.net

Abstract. The tasks of studying topologies on a finite set are closely related to the tasks of studying bijunctive Boolean functions, partial orders, and graphs. This explains the possibility of using combinatorial methods for their research, and also makes such topologies important mathematical objects. The problem of enumeration of homotopy types of finite topological spaces is as important as the problem of enumeration of homotopy types of finite simplicial complexes. These tasks are related to the digital processing of images based on finite sets of observations, that is, trying to understand the content of the image based on the concept of proximity of points. The problem of computing topologies on a finite set remains unsolved at the moment, so interest in it remains, publications by researchers from different countries with new results have appeared. This paper uses two models to study topology on a finite set. In the first model, each topology is assigned a nondecreasing sequence of non-negative integers (topology vector), in the second - the conjunctive normal form of the Boolean function.

Keywords: discrete model · topology weight · topology vector · minimum element neighborhood · T_0-topologies · consistent topologies

1 Introduction

A topological structure is often the base on which other mathematical structures are built. Topological spaces and their continuous mappings appear in many branches of mathematics. Studies of topology on the finite sets have numerical applications and are parallel to studies of partially ordered sets, finite graphs, and Boolean functions. Topologies on a finite set play a key role in the theory of pattern recognition [1, 2], the theory of molecular structures [3, 4], geometries on finite sets [5].

The first research results of topologies on finite sets appeared in the 1960s and 1970s [6, 7]. T_0-topologies play an important role in the study of topologies on a finite set. In the article by J.W. Evans., F. Harary, M.S. Lynn [6] a formula for the relationship

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2026
I. Potapov et al. (Eds.): RTC-Digital 2023, CCIS 2647, pp. 84–89, 2026.
https://doi.org/10.1007/978-3-032-04731-1_11

between the number $T(n)$ of all topologies on an n-element set and the number $\tilde{T}(m)$ of all T_0-topologies on its m-element subsets is given:

$$T(n) = \sum_{m=1}^{n} S(n, m) \cdot \tilde{T}(m) \quad (1)$$

where $S(n, m)$ are Stirling numbers of the second kind.

Most early and modern publications describe the structure of topologies on a finite set and perform direct calculations of the number of topologies. There were also attempts to find patterns of transition from topologies on $(n-1)$-element to topologies on n -element set.

2 Study of T_0-Topologies with Weight $2^{n-2} < m \leq 2^{n-1}$ Using the Topology Vector

A topology on an arbitrary n-element set X is said to belong to the m -class of topologies (or has weight m) if it consists of m elements ($m = 2,3, \ldots, 2^n$).

Let X be a finite set and let τ_X be a topology on it. A set A from τ_X is called maximal in τ_X if A is not contained in any other set from τ_X except the set X itself. The set X in this case is called ambient.

Assume that $A, X, A \subset X$ are given sets and τ_A is a given topology. We will restore all such topologies on the set X, for which the set A is maximal and which induce the topology τ_A on the set A (in this case the topology τ_A and the corresponding topologies on X will be called consistent).

The intersection of all neighborhoods of a point $x_i \in X$ is called the minimum neighborhood M_i of this point. The index of a point a in the topology is a number equal to the number of points different from a in its minimal neighborhood M_a [8].

Topologies with weight $m > 2^{n-1}$ are called close to discrete topology, they are fully investigated in [8] using the concept of a topology vector – a nondecreasing sequence of nonnegative integers that determine the minimal neighborhoods of the elements of a given finite set. In particular, it is shown that all of them are T_0-topologies with topology vectors of three types:

a) $v(\tau) = (0, \ldots, 0, \alpha_n)$, $1 \leq \alpha_n \leq n-1$, $|\tau| = 2^{n-1} + 2^{n-1-\alpha_n}$.

b) $v(\tau) = \left(\underbrace{0, \ldots, 0}_{k}, 1, \ldots, 1\right)$, $1 \leq k \leq n-2$ i $\bigcap_{m=k+1}^{n} M_m = \{y\}$, $|\tau| = 2^{n-1} + 2^{k-1}$.

c) $v(\tau) = (0, ..., 0,1, 1)$ i $M_{n-1} \cap M_n = \varnothing$, $|\tau| = 2^{n-1} + 2^{n-4}$, where M_i - the minimal neighborhood of the element $x_i \in X$.

The sequence of the weights of the topologies close to the discrete topology on an n-element set has the form

$$\left(2^{n-1} + 2^0\right), \left(2^{n-1} + 2^1\right), \underbrace{\ldots}_{2^1-1}\left(2^{n-1} + 2^2\right), \ldots, 5 \cdot 2^{n-3}, \underbrace{\ldots}_{2^{n-3}-1}, 3 \cdot 2^{n-2}, \underbrace{\ldots}_{2^{n-2}-1}, 2^n$$

where the numbers of classes that do not have any topology are indicated by curly brackets.

There are also published results about topologies with weight $2^{n-2} < m \leq 2^{n-1}$. We note the works of R. Stanley [9] and Kolli [10, 11]. In [12] T_0-topologies with weight $2^{n-2} < m \leq 2^{n-1}$ on an n-element set, consistent with topologies close to discrete on an $(n-1)$-element set with vectors of types a) and c) were found. Here we will add information about T_0-topologies, which are consistent with close to discrete topology of type b). They can be described by vectors $\left(\underbrace{0,\ldots,0}_{k},\ \underbrace{1,\ldots,1,}_{n-k-1}\ \alpha_n\right)$, where $1 \leq k \leq n-3$, $2 \leq \alpha_n \leq n-1$ and $\bigcap_{m=k+1}^{n-1} M_m = \{y\}$. The non-homeomorphic topologies among them have different forms of the minimal neighborhood of the element x_n. For example, if the minimal neighborhood of an element x_n. Has the form $M_n = \left\{\underbrace{x_1,\ldots,y,\ldots,x_d,x_{n-(\alpha_n-d)},\ldots,x_{n-1},x_n}_{\alpha_n-d}\right\}$, i.e. $\bigcap_{m=k+1}^{n-1} M_m \bigcap M_n = \{y\}$, then the weight of such topologies is calculated by the formula $|\tau| = 2^{n-2} + 2^{k-1} + 2^{k-d} + 2^{k-d}(2^{n-k-(\alpha_n-d+1)} - 1)$. If the minimal neighborhood of an element x_n has the form $M_n = \{x_1,\ldots,x_{\alpha_n+1},x_n\}$ and contains no element y, i.e. $\bigcap_{m=k+1}^{n-1} M_m \bigcap M_n = \varnothing$, then the weight of such topologies is calculated by the formula $|\tau| = 2^{n-2} + 2^{k-1} + 2^{k-\alpha_n} + 2^{k-(\alpha_n+1)}(2^{n-k-1} - 1)$. These are all possible cases in which T_0-topologies have weight $|\tau| \in [5 \cdot 2^{n-4}, 2^{n-1}]$.

Let us now consider some classes of T_0-topologies on an n-element set with weight $2^{n-2} < m \leq 2^{n-1}$, which are not consistent with close to discrete topology on an $(n-1)$-element set.

In work [12] it was shown that in classes of topologies with weight $|\tau| \in [13 \cdot 2^{n-5}, 2^{n-1}]$ all T_0-topologies are consistent with close to discrete or are dual to them. Among the classes of T_0-topologies with weight $|\tau| \in [5 \cdot 2^{n-4}, 13 \cdot 2^{n-5})$ there are such classes that are not exhausted by T_0-topologies consistent with close to discrete and dual to them. This is proven by comparing the total number of T_0-topologies in these classes with the number of T_0-topologies in them that are consistent with close to discrete and dual to them. There was also an example of a class in which there is not a single T_0-topology consistent with close to discrete.

Now we continue the list of T_0-topologies with weight $|\tau| \in (2^{n-2}, 2^{n-1}]$, which are not topologies consistent with close to discrete.

Consider the T_0-topologies with vectors $(0,\ldots,0,1,3,3)$, $(0,\ldots,0,1,2,\alpha_n)$, where $2 \leq \alpha_n \leq n-1$, $(0,\ldots,0,1,1,1,2,2)$, $(0,\ldots,0,1,1,1,2,3)$, $(0,\ldots,0,1,1,1,1,2,2)$, $(0,\ldots,0,1,2,2,2)$, $(0,\ldots,0,2,\ldots,2)$. Obviously, they are not topologies consistent with close to discrete. Let's find the conditions on the minimal boundaries of elements under which the weight of such T_0-topologies belongs to the interval $\left(2^{n-2}, 2^{n-1}\right]$.

For example, let us analyze T_0-topologies with vector $(0,\ldots,0,1,3,3)$ for $n \geq 5$. It is sufficient to consider only the minimal neighborhoods of elements whose index in the topology vector is not equal to zero. In this case, these are M_{n-2}, M_{n-1}, and M_n, where M_{n-2} is a two-element set, and M_{n-1} and M_n are four-element sets. The

weight of the topology will depend on the pairwise intersections of the specified minimal neighborhoods. We divided all possible cases into three classes, describing the powers of pairwise intersections of minimal neighborhoods. Let us formulate the obtained results.

1) If $|M_{n-2} \cap M_{n-1}| = 2$ and the power of at least one of the other two intersections is zero, then the weight of the topology is less than or equal to 2^{n-2}, that is, it does not belong to the set $\left(2^{n-2}, 2^{n-1}\right]$;
2) If $|M_{n-2} \cap M_{n-1}| = 1$ and at least one of the other two intersections is an empty set, then the weight of the topology does not belong to the set $\left(2^{n-2}, 2^{n-1}\right]$;
3) If $|M_{n-2} \cap M_{n-1}| = 0$, then the weight of the topology does not belong to the set $\left(2^{n-2}, 2^{n-1}\right]$.

In all other cases, T_0-topologies with vector $(0, \ldots, 0{,}1, 3{,}3)$ for $n \geq 5$ have weight $|\tau| \in (2^{n-2}, 2^{n-1}]$.

In the case of T_0-topologies with the vector $(0, \ldots, 0{,}1, 2, \alpha_n)$, where $2 \leq \alpha_n \leq n-1$, the analysis of the intersections of the minimal neighborhoods M_{n-2}, M_{n-1}, and M_n has made it possible to obtain the following conclusions:

1) If $|M_{n-2} \cap M_{n-1}| = 2$ or $|M_{n-2} \cap M_{n-1}| = 1$, then the weight of the topology belongs to the set $\left(2^{n-2}, 2^{n-1}\right]$;
2) If $|M_{n-2} \cap M_{n-1}| = 0$, then the weight of the topology does not belong to the set $\left(2^{n-2}, 2^{n-1}\right]$.

For the rest of the vectors, a similar analysis was carried out, conditions on minimal neighborhoods were found, and it was shown that among the corresponding T_0-topologies there are those that have a weight from the set $\left(2^{n-2}, 2^{n-1}\right]$.

3 Application of 2-CNF Boolean Functions for the Study of T_0-Topologies

When modeling topologies with Boolean functions, a single conjunctive normal form of a certain form (maximal 2-CNF) is matched to each T_0-topology [13]. This tool turned out to be quite convenient for describing mutually dual and self-dual T_0-topologies.

We will show the application of this model for enumerating and counting all T_0-topologies with weight $25 \cdot 2^{n-6}$.

In work [10], table 6 indicates that there are 7 non-homeomorphic T_0-topologies in the class $25 \cdot 2^{n-6}$, and in work [11] it is shown that the number of all T_0-topologies of this class for $n > 6$ is equal to $\frac{n+14}{24} \cdot (n)_6 + \frac{1}{24} \cdot (n)_7$.

Theorem. A T_0-topology has the weight $25 \cdot 2^{n-6}$ if and only if its vector (under the specified conditions for the minimal neighborhoods of the elements) has one of the following forms:

1) $(0, \ldots, 0{,}1, 5)$, whenever either a) $M_{n-1} \subset M_n$ or b) $M_{n-1} \cap M_n$ is a one-element set;
2) $(0, \ldots, 0{,}2, 2)$, if $M_{n-1} \cap M_n = \varnothing$,
3) $(0, \ldots, 0{,}1, 1{,}1, 1{,}2)$, if $\bigcap_{m=n-4}^{n-1} M_m = \{y\}$ is one-element set and either a) $M_{n-1} \subset M_n$ or b) $\bigcap_{m=n-4}^{n-1} M_m = \{y\}$;

4) $(0, \ldots, 0,1, 1,2)$, if the intersection $M_{n-2} \cap M_{n-1}$ is a one-element set and $\bigcap_{m=n-2}^{n} M_m = \varnothing$;
5) $(0, \ldots, 0,1, 1,1, 1)$, if $M_{n-3} \cap M_{n-2} = \{y\}$ and $M_{n-1} \cap M_n = \{z\}$ are one-element sets with $y \neq z$.

With the help of 2-CNF, we showed the following:

- T_0-topology τ with vector $(0, \ldots, 0,1, 1,2)$ under conditions $M_{n-2} \cap M_{n-1} = \{x_i\}$ and $\bigcap_{m=n-2}^{n} M_m = \varnothing$ is self-dual;
- T_0-topologies with vectors $(0, \ldots, 0,1, 5)$ and $(0, \ldots, 0,1, 1,1, 1,2)$, $(0, \ldots, 0,2, 2)$ and $(0, \ldots, 0,1, 1,1, 1)$ are mutually dual.

Using the vector of T_0-topology in work [13], the number of all T_0-topologies with the specified vectors was calculated, and it was proved that this number coincides with the one obtained in Kolli's work, that is. So there are no other T_0-topologies with such a weight.

Another example of effective application of the topology model in the form of a maximal 2-CNF is the problem of enumerating all non-homeomorphic T_0-topologies on 4-element and 5-element sets. This problem is reduced to listing the maximal 2-CNFs of all non-homeomorphic T_0-topologies in each class.

Let's stop only at the case of a 4-element set. To solve the problem, we used the connection between the 2-CNF of the topology and 2-CNF of dual T_0-topology and the properties of the 2-CNF of the self-dual T_0-topology [13]:

- If some 2-CNF defines the topology τ, then the 2-CNF obtained from it by replacing $x_i \to$, defines a dual topology τ;
- Two 2-CNF f_1 and f_2 of the topology on the set X are called equivalent if there exists a bijection $\varphi : X \to X$ such that $f_2 = \varphi(f_1)$. If f_τ and $f_{C\tau}$ are equivalent, then each of them will be called a self-dual 2-CNF, and the corresponding homeomorphic topologies will be called self-dual.

In the following list, each maximal 2-CNF specifies one of two mutually dual T_0-topologies or a self-dual T_0-topology. The complete list of maximal 2-CNF of all non-homeomorphic T_0-topologies on a 4-element set can be obtained by adding to each non-self-dual maximal 2-CNF the maximal 2-CNF that is dual to it.

List of maximal 2-CNF T_0-topologies on a 4-element set:
5- class: $(x_1 \vee \overline{x}_2)(x_1 \vee \overline{x}_3)(x_1 \vee \overline{x}_4)(x_2 \vee \overline{x}_3)(x_2 \vee \overline{x}_4)(x_3 \vee \overline{x}_4)$ – self-dual;
6- class: $(x_1 \vee \overline{x}_2)(x_1 \vee \overline{x}_3)(x_1 \vee \overline{x}_4)(x_2 \vee \overline{x}_3)(x_2 \vee \overline{x}_4)(x_3 \vee \overline{x}_4)$;
$(x_1 \vee \overline{x}_2)(x_1 \vee \overline{x}_3)(x_1 \vee \overline{x}_4)(x_2 \vee \overline{x}_4)(x_3 \vee \overline{x}_4)$ – self-dual;
7- class: $(x_1 \vee \overline{x}_3)(x_1 \vee \overline{x}_4)(x_2 \vee \overline{x}_4)(x_3 \vee \overline{x}_4)$;
$(x_1 \vee \overline{x}_3)(x_1 \vee \overline{x}_4)(x_2 \vee \overline{x}_3)(x_2 \vee \overline{x}_4)$– self-dual;
8- class: $(x_1 \vee \overline{x}_3)(x_1 \vee \overline{x}_4)(x_2 \vee \overline{x}_3)$ – self-dual;
$(x_1 \vee \overline{x}_3)(x_1 \vee \overline{x}_4)(x_3 \vee \overline{x}_4)$– self-dual;
9- class: $(x_1 \vee \overline{x}_2)(x_1 \vee \overline{x}_3)(x_1 \vee \overline{x}_4)$;
$(x_1 \vee \overline{x}_3)(x_2 \vee \overline{x}_4)$– self-dual;
10- class: $(x_1 \vee \overline{x}_3)(x_1 \vee \overline{x}_4)$;

12- class: $(x_1 \vee \overline{x}_4)$ – self-dual;
16- class: $f(x_1, x_2, x_3, x_4) \equiv 1$.

4 Conclusions

In this work, topologies on a finite set were modeled by a non-decreasing sequence of non-negative integers - a topology vector, as well as a conjunctive normal form of a certain form - a maximal 2-CNF. The results of the application of these models were obtained. Vectors of T_0-topologies with weight $2^{n-2} < m \leq 2^{n-1}$ on the n-element set are found, which are consistent with topologies close to discrete topology on the $(n-1)$-element set. Some classes of T_0-topologies with weight $2^{n-2} < m \leq 2^{n-1}$ are distinguished, which are not consistent with topologies close to discrete topology. The application of both models for enumeration and counting of all T_0-topologies with a weight of $25 \cdot 2^{n-6}$ is shown. The 2-CNFs of all non-homeomorphic T_0-topologies on the 4-element set are listed. There is an expectation that the study of the regularities of the transition from topologies on the $(n-1)$-element set to topologies on the n-element set can lead to qualitative results.

References

1. Kovalevsky, V.A.: Finite topology as applied to image analysis. Comput. Vis. Graph. Image Process. **46**, 141–161 (1989)
2. Kovalevsky, V.A.: Finite topology and image analysis. Adv. Electr. Electron Phys. **84**, 197–259 (1992)
3. Merrifield, R., Simmons, H.: The structures of molecular topological spaces. Theoretica Chimica Acta. **55**, 55–75 (1980)
4. Merrifield, R., Simmons, H.: Topological Methods in Chemistry, p. 230. Wiley, New York (1989)
5. Rosenfeld, A., Kong, T., Wu, A.: Graphical models and image processing. Digit. Surf. **53**, 305–312 (1991)
6. Evans, J.W., Harary, F., Lynn, M.S.: On the computer enumeration of finite topologies. Commun. ACM **10**(5), 295–297 (1967)
7. Sharp, J.H.: Cardinality of finite topologies. J. Comb. Theory **5**, 82–86 (1968)
8. Velichko, I.G., Stegantseva, P.G., Bashova, N.P.: Enumeration of topologies close to discrete on finite sets. Izv. Vyssh. Uchebn. Zaved., Ser. Mat. **11**, 23–31 (2015)
9. Stanley, R.P.: On the number of open sets of finite topologies. J. Comb. Theory **10**, 74–79 (1971)
10. Kolli, M.: On the Cardinality of the T_0-topologies on a finite set. Int. J. Comb. 798074, 1–7 (2014)
11. Kolli, M.: Direct and elementary approach to enumerate topologies on a finite set. J. Integer Sequences **10**, 07.3.1, 1–11 (2007)
12. Stegantseva, P.G., Skryabina, A.V.: Topologies on the n-element set consistent with topologies close to the discrete on an (n-1)-element set. Ukrainian Math. J. **2**(73), 276–288 (2021). https://doi.org/10.37863/umzh.v73i2.6174
13. Skryabina, A., Stegantseva, P., Bashova, N.: The properties of 2-CNF of the mutually dual and self-dual T0-topologies on the finite set and the calculation of T0-topologies of a certain weight. In: Proceedings of the International Geometry Center, no. 1, vol. 15, pp. 75–85 (2022), https://doi.org/10.15673/tmgc.v15i1.2084

Crystal Structure Prediction, A Challenging and Fundamental Problem

Duncan Adamson(✉)

Leverhulme Research Centre for Functional Materials Design, University of Liverpool, Liverpool, UK
d.a.adamson@liverpool.ac.uk
https://duncanadamson.github.io

Abstract. In this papers, we highlight recent work done on one of the most fundamental problems in computational chemistry, *Crystal Structure Prediction*. In CSP, the goal is to find a configuration of ions in 3D space that yields the lowest (i.e. the greatest negative) pairwise potential energy. This problem has only recently been formalised as a mathematical problem, leaving many interesting open problems. This abstract will highlight the recent work between computer science and chemistry, as well as presenting several interesting open problems. In particular, we show that this problem is undecidable in the general case and NP-hard when the size of the period defining the crystal is given. Further, we give an overview of the combinatorial results on these structures that may provide future tools for solving this problem in some restricted cases. Finally, we provide an integer programming model for solving this problem in the case of a fixed-size period.

Keywords: Crystal Structure Prediction · NP-hardness · Combinatorics on words

1 Introduction

New materials are needed to tackle the challenges of the 21st century, providing new structures that are stronger, lighter, or more flexible than currently available. *Crystals* are a fundamental form of matter, defined by a repeating *unit cell*, a 3-dimensional box that is repeated across the structure acting as the period of the structure. Each unit cell contains a set of *ions*, charged atoms defining the properties of these structures. Each ion belongs to a class known as a species, which defines the pairwise interaction via an *energy function*, taking as input a pair of species and the distance between the ions. A positive interaction represents the repulsion of these ions, while a negative value is the attraction.

The *Crystal Structure Prediction problem* (CSP) asks, given a set of *ion species*, classes of ions, what is the "best" placement of these ions within a charged balanced unit cell [11]. The quality of a solution here is defined by the sum of the pairwise interaction functions between each ion in the unit cell with

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2026
I. Potapov et al. (Eds.): RTC-Digital 2023, CCIS 2647, pp. 90–96, 2026.
https://doi.org/10.1007/978-3-032-04731-1_12

every other ion in the *global* structure, with the aim being to find the lowest (greatest negative) total interaction [14,16].

Current techniques of CSP rely upon heuristic techniques [9,10,13,17], which have shown some promise, however they still lack the ability to guarantee optimality and are computationally very demanding. Until recently, the computational hardness of CSP was conjectured [12], without any formal proof. Our recent work has shown the this problem is not only NP-hard, but undecidable in the most general case [3,4]. On the other hand recent work has applied techniques from theoretical computer science to CSP [2,4,6,15], providing a basis for future research.

2 Hardness of CSP

Perhaps the most important of our work on CSP has been on formally establishing the computational hardness of CSP. This not only justifies the application of heuristic techniques but also highlights where the challenges of CSP originate.

In [3], we consider the problem of *Crystal Structure Prediction by Ion Removal* (CSP-IR). In this variation of CSP, we are given as input an initially densely packed unit cell and tasked with removing ions in order to minimise the total energy of the unit cell. We first show that this problem is NP-complete in the general case via a reduction from the clique problem. This is strengthened by showing that CSP-IR can not be approximated within any constant factor. We note that this result requires the number of ion species to be part of the input.

We further show that CSP-IR is NP-complete when the number of ion species is restricted to only two, the minimal number required to ensue charge neutrality. This is done via a novel reduction from the *Independent Set Problem on Penny Graphs*. A penny graph is a graph that can be represented as the adjacency graph of unit-sized, non-overlapping circles, known as pennies. This reduction is done by representing each penny as a pair of ions, with the energy function giving a positive cost penalty for adjacent pairs of ions and a negative total energy for the removal of ions leaving an independent set. A representation of this reduction is given in Fig. 1.

This work is strengthened in [4] by showing that the more general problem of *Crystal Structure Prediction by Ion Insertion*, CSP-II, (formally the *average pairwise energy minimisation problem*) is NP-Hard when the size of the unit cell is fixed, and undecidable in the general case when the unit cell may be chosen arbitrarily. Both results are proven by a reduction from the *periodic tiling problem* via a series of intermediary problems. The periodic tiling problem asks if a tiling of the infinite grid can be produced from an input set of tiles. An overview of the series of reductions is provided in Fig. 2.

3 Algorithmic and Combinatorial Results

Alongside the hardness of this problem, we also have a set of positive algorithmic and combinatorial results. The key tool we use for these results are *necklaces* (also

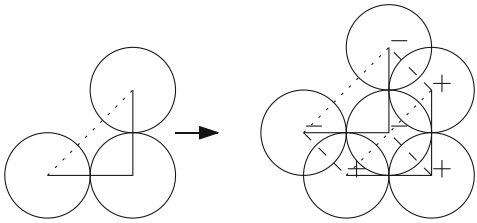


Fig. 1. An overview of the reduction from the independent set problem on Penny Graphs to CSP-IR. Left is a penny graph, while the right shows a representation of the crystal structure. The crystal structure corresponds to a two-layer system, with each layer corresponding to a drawing of the penny graph with an ion placed in the centre of each penny. The layers are stacked on top of each other, with one layer containing negatively charged ions and the other positively charged ions.

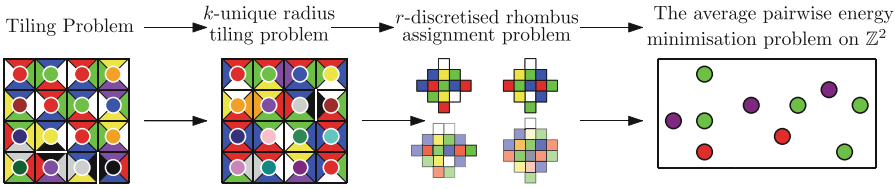


Fig. 2. A sketch of the set of problems used to show the undecidability of CSP-II. We start with the periodic tiling problem, asking if a tiling of the infinite grid exists from an input set of tiles. This is reduced to the *k-unique radius periodic tiling problem*, a variation of the classic tiling problem where only one copy of each tile is allowed within a radius of k. This is reduced to the *r-discretised rhombus assignment problem*, a problem taking as input a set of discretised rhombuses with radius r over the integer grid, with each square on the gird coloured from some set $\mathcal{C}$. The goal in this problem is to tile the grid using these rhombuses such that any pair of rhombuses can overlap if and only if they colour each square with the same colours. Finally, this problem is reduced to CSP-II.

known as cyclic words), representations of infinite words with a finite period. This follows the model given in [7] of crystals as a 1-dimensional structure.

Our main result is based on the idea of sampling the space of potential crystal structures. Our approach is to model the space of crystal structures as a set of necklaces. The goal is to select a *representative set* of the space, defined under a distance as a set of necklaces, minimising the distance between the further necklaces from the sample under this metric. This model of sampling is provided in [2], where we provide strong approximation results for the general problem of sampling necklaces, with an approximation factor of $1+\frac{\log_q(k\cdot n)}{n-\log_q(k\cdot n)}-\frac{\log_q^2(k\cdot n)}{2n(n-\log_q(k\cdot n))}$, where k is the size of the set and n the length of the necklace. We show that this result can be generalised to *multidimensional necklaces* with the same approximation factor.

Following the same approach as [2], we provide tools allowing for further classes of cyclic words to be sampled in the same manner. In doing so, we provide more accurate representations for crystalline structures, capturing more sophisticated symmetries. In [5], we show that the same techniques can be applied to the class of *bracelets*, equivalence classes of words under both the cyclic shift and reflection operations. Similarly, in [1], we provide the same tools for the class of *unlabelled necklaces*, equivalence classes under both the cyclic shift and the *relabelling* operations. A relabelling of a word is a bijective mapping of each symbol from the alphabet back to the alphabet.

3.1 Integer Programming

Since the original presentation of this abstract, we have provided a significant new publication [8] regarding the use of *integer programming* to find optimal solutions to CSP-II. Integer programming provides a well-studied and powerful tool for solving NP-hard problems efficiently. In [8], we provide an integer programming model for solving CSP-II optimally, as well as several experimental results. We will provide a brief summary of the model here, referring the reader to [8] for the full results.

Model. Formally, we use *Integer Quadratic Binary programming*, that is, a set of constraints in the form of quadratic inequalities where each variable has a value of 1 or 0, and a quadratic optimisation equation. For a unit-cell defined on the integer grid $n_1 \times n_2 \times n_3$, with a set of σ types of atom, we have the set of variables X_{t,p_1,p_2,p_3} where:

- $X_{t,p_1,p_2,p_3} = 1$ if there is an atom of type t at position (p_1, p_2, p_3) on the integer grid,
- $X_{t,p_1,p_2,p_3} = 0$ if there is not an atom of type t at position (p_1, p_2, p_3) on the integer grid.

Alongside the values of n_1, n_2, n_3 and σ, we assume that we are given a *explicit* chemical formula for the unit cell, i.e. a set of σ values $c_1, c_2, \ldots, c_\sigma \in \mathbb{N}$ such that c_i represents the number of copies of atom type i in the target unit-cell.

We introduce three types of constraints. First, we have that each position can be occupied by *at most* 1 atoms. We formalise this as $\sum_{t \in 1,2,\ldots,\sigma} X_{t,p_1,p_2,p_3}$, for every position $(p_1, p_2, p_3) \in n_1 \times n_2 \times n_3$. Second, we have that the number of copies of each type of atom is exactly equal to the number given in the input formula. Formally, $\sum_{(p_1,p_2,p_3) \in n_1 \times n_2 \times n_3} X_{t,p_1,p_2,p_3} = c_t$, for every type of atom $t \in 1, 2, \ldots, \sigma$. Finally, we introduce a special combinatorial constraint to help reduce the search space. Assuming that $c_1 \leq c_2 \leq \cdots \leq c_\sigma$, the first position $(1, 1, 1)$ is fixed to containing some atom of type 1, formally $X_{1,1,1,1} = 1$. This is done to reduce the number of identical solutions that can be found by simply translating the unit cell, moving every atom by (i_1, i_2, i_3) modulo (n_1, n_2, n_3).

To define our optimisation function, we need the final input, the energy function $E : t_1, t_2 \in 1, 2, \ldots, \sigma, (p_1, p_2, p_3), (q_1, q_2, q_3) \in n_1 \times n_2 \times n_3 \mapsto \mathbb{R}$.

We treat E as a precomputed table where $E[t_1, (p_1, p_2, p_3), t_2, (q_1, q_2, q_3)]$ is the pairwise energy between some atom of type t_1 at position (p_1, p_2, p_3), and some atom of type t_2 at position (q_1, q_2, q_3). Observe that the binary equation $E[t_1, (p_1, p_2, p_3), t_2, (q_1, q_2, q_3)] \cdot X_{t_1,p_1,p_2,p_3} \cdot X_{t_2,q_1,q_2,q_3}$ returns $E[t_1, (p_1, p_2, p_3), t_2, (q_1, q_2, q_3)]$ if there is an atom of type t_1 at (p_1, p_2, p_3), and an atom of type t_2 at (q_1, q_2, q_3), and 0 otherwise. Using this observation, and the notation $\mathbf{p} = (p_1, p_2, p_3)$, $\mathbf{q} = (q_1, q_2, q_3)$, we formulate our objective function as:

$$\sum_{t_1,t_2 \in 1,2,\ldots,\sigma} \sum_{\mathbf{p},\mathbf{q} \in n_1 \times n_2 \times n_3} E[t_1, \mathbf{p}, t_2, \mathbf{q}] \cdot X_{t_1,\mathbf{p}} \cdot X_{t_2,\mathbf{q}}.$$

4 Open Problems

Crystal structure prediction remains a new and relatively unstudied field with many clear research directions. There are several clear directions in which to extend this work. The hardness results of CSP-II show this is a highly challenging problem, making finding an optimal algorithm for the fixed period case highly unlikely and impossible in the general case. Despite this, some clear directions exist for both positive and hardness results. The first directions we consider are *parameterised algorithms* and *approximation algorithm* relative to the number of ion species in the unit cell. The key motivation for this direction is that the number of species is, in most cases, fixed as a relatively small constant, often 5 or fewer. As such, even an algorithm with a runtime of the order of $O(N^{\sigma^c}$ for some constant c may be acceptable within the context of CSP-II, a problem that currently attracts a great deal of computing power for the use of heuristic techniques. At the same time, determining if either case is non-viable via a hardness result would present clear guidance in the future of the field, in particular, that the focus on tools such as integer programming and heuristic techniques are the best ways to proceed.

Problem 1. Fixed Parameter Tractability for CSP-II

Our existing hardness results for CSP-II require the number of species to be part of the input. However, in real-world applications, the number of species is often restricted to a small constant, such as 3 or 4. This leads to the following question,

Question: Can CSP-II be solved in time $O(N^{f(\sigma)})$ time for a unit cell with N position and $\sigma - 1$ species of atoms?

While either a parameterised algorithm or a hardness result for parameterised tractability would be an interesting result, we conjecture that such an algorithm exists. Our intuition behind this conjecture is that the encoding needed for the hardness result of CSP-II requires multiple species to be added to the instance for every tile in the original tiling instance. Further, the hardness result for CSP-IR regarding a fixed number of ion species requires a precise structure that would not be replicated in CSP-II.

Problem 2. Approximation for CSP-II

Question: Can CSP-II be approximated within some factor of $O(g(\sigma))$ in $O(N^{f(\sigma)})$ time for a unit cell with N positions and $\sigma - 1$ species of atoms?

In the same direction as fixed-parameter tractability, a natural question to ask is if there exists an efficient approximation algorithm with time complexity either parameterised by the number of ion species or as some polynomial of the number of positions and ion species.

We conjecture that both problems can be answered in the affirmative. Our main intuition behind this is that our current hardness results are based on the use of a large number of ion species. Indeed, there is no obvious method of converting our current results to using only a fixed number of ion species.

4.1 Quantum Computing

Beyond theoretical computer science, we are also interested in extending our optimisation approaches to new forms of computing. In particular, we are interested in utilising *quantum computing* as a means to speed up the optimisation problem formulated in our integer programming model. In our initial work implementing the integer programming model [8], we explored the use of the commercially available D-Wave quantum annealer. While this showed some initial successes, there are clear limitations. First, as is the case with all existing quantum systems, there is a lack in the scalability of these models due to the challenges of entangling many atoms. Secondly, and more specific to the D-Wave machine, we have only explored the use of quantum annealers, which are, by definition, heuristic methods. While these are powerful and allow many thousands of runs of the model in a fraction of the time it would take to make even a single run on a classical machine, it still leaves open the possibility of missing the optimal solution.

Problem 3. Quantum Algorithms for CSP-II

Question: Can a quantum algorithm be formulated for CSP-II that guarantees optimality while maintaining a speedup over classical methods?

We mean, by this question, if we can formulate an algorithm for our CSP-II model that can be solved by a quantum computer optimally, while taking advantage of the so-called "quantum-advantage", such as the improvement of Shor's algorithm in reducing the time to factorise the integer n from $O(n)$ time, to $\mathrm{PolyLog}\, n$. We conjecture that such an algorithm is likely, or at least some algorithm with running time parameterised by the number of ion species.

References

1. Adamson, D.: Ranking binary unlabelled necklaces in polynomial time (2022)
2. Adamson, D., Deligkas, A., Gusev, V.V., Potapov, I.: The k-centre problem for necklaces. arXiv preprint arXiv:2005.10095 (2020)
3. Adamson, D., Deligkas, A., Gusev, V.V., Potapov, I.: On the hardness of energy minimisation for crystal structure prediction. Fundam. Informaticae **184**(3), 181–203 (2021). https://doi.org/10.3233/FI-2021-2096
4. Adamson, D., Deligkas, A., Gusev, V.V., Potapov, I.: The complexity of periodic energy minimisation. In: Szeider, S., Ganian, R., Silva, A. (eds.) 47th International

Symposium on Mathematical Foundations of Computer Science, MFCS 2022, 22–26 August 2022, Vienna, Austria. LIPIcs, vol. 241, pp. 8:1–8:15. Schloss Dagstuhl - Leibniz-Zentrum für Informatik (2022). https://doi.org/10.4230/LIPIcs.MFCS.2022.8
5. Adamson, D., Gusev, V.V., Potapov, I., Deligkas, A.: Ranking bracelets in polynomial time. In: Gawrychowski, P., Starikovskaya, T. (eds.) 32nd Annual Symposium on Combinatorial Pattern Matching, CPM 2021, July 5-7, 2021, Wrocław, Poland. LIPIcs, vol. 191, pp. 4:1–4:17. Schloss Dagstuhl - Leibniz-Zentrum für Informatik (2021). https://doi.org/10.4230/LIPIcs.CPM.2021.4
6. Antypov, D., Deligkas, A., Gusev, V., Rosseinsky, M.J., Spirakis, P.G., Theofilatos, M.: Crystal structure prediction via oblivious local search. arXiv preprint arXiv:2003.12442 (2020)
7. Collins, C., Darling, G.R., Rosseinsky, M.: The Flexible Unit Structure Engine (FUSE) for probe structure-based composition prediction. Faraday Discuss. **211**, 117–131 (2018)
8. Gusev, V.V., et al.: Optimality guarantees for crystal structure prediction. Nature **619**(7968), 68–72 (2023)
9. Lyakhov, A.O., Oganov, A.R., Valle, M.: How to predict very large and complex crystal structures. Comput. Phys. Commun. **181**(9), 1623–1632 (2010)
10. Mellot-Draznieks, C., Girard, S., Férey, G., Schön, J.C., Cancarevic, Z., Jansen, M.: Computational design and prediction of interesting not-yet-synthesized structures of inorganic materials by using building unit concepts. Chem. Eur. J. **8**(18), 4102–4113 (2002)
11. Oganov, A.: Modern Methods of Crystal Structure Prediction. Wiley–VCH (2010)
12. Oganov, A.R.: Crystal structure prediction: reflections on present status and challenges. Faraday Discuss. **211**, 643–660 (2018)
13. Oganov, A.R., Glass, C.W.: Crystal structure prediction using ab initio evolutionary techniques: principles and applications. J. Chem. Phys. **124**(24) (2006)
14. de Souza, V.K., Wales, D.J.: The potential energy landscape for crystallisation of a Lennard-Jones fluid. J. Stat. Mech. Theor. Exp. **2016**(7), 074001 (2016). https://doi.org/10.1088/1742-5468/2016/07/074001
15. Theofilatos, M.: Stability and Structure-Formation Problems for Dynamic and Chemical Systems. Ph.D. thesis, University of Liverpool (2021)
16. Wales, D.J.: Exploring energy landscapes. Ann. Rev. Phys. Chem. **69**(1), 401–425 (2018). pMID: 29677468. https://doi.org/10.1146/annurev-physchem-050317-021219
17. Wang, Y., et al.: Materials discovery via CALYPSO methodology. J. Phys. Condens. Matter Instit. Phys. J. **27**, 203203 (2015)

Parallel Methods for Composing Difference Approximations in Simulation Dynamic Processes

Olga Dmytriyeva[1,2(✉)] and Vira Huskova[1]

[1] Institute of Applied System Analysis, National Technical University of Ukraine, Beresteyska av. 37, Kyiv 03056, Ukraine
huskova.vira@lll.kpi.ua

[2] Institute for Modelling Hydraulic and Environmental Systems, University of Stuttgart, Pfaffenwaldring 5a, Stuttgart 70049, Germany
dmytriyeva.olga@lll.kpi.ua

Abstract. This paper investigates methods for composing difference approximations to form one-step and many-step parallel difference schemes of a given order, with a focus on numerically solving the Cauchy problem for both ordinary differential equations and evolutionary partial differential equations. The proposed discrete approximations allow for varying the order of error while moving away from the maximum possible on a fixed set of knots, while ensuring the absolute or A-α stability of the numerical solutions. It is shown that one-step schemes of the maximum approximation order remain absolutely stable as the dimensionality of computational blocks increases. Fulfilment of the condition of absolute or A-α stability for multistep methods is provided by reducing the maximum possible order of approximation.

Keywords: Parallel methods · Difference approximations · Absolute stability · Simulation · Cauchy problem

1 Introduction

Modern physical problems often lead to mathematical models that lack advanced methods for their solution [11]. These models typically have an evolutionary character and include both ordinary and partial differential equations [9]. Additionally, dynamic processes that have unique physical features are often described by the same equations or their systems [2,9]. The same mathematical model can be used to solve different physical problems, while the model itself may be transformed multiple times during numerical experiments, which requires improvement of the constructed algorithm [2,4,9]. Therefore, it is of utmost importance to develop computational algorithms that can efficiently perform numerical experiments, which is a modern direction in mathematical modelling contributing to the development of new numerical methods. Another aspect to consider is the effective organisation and realisation of calculations. The high

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2026
I. Potapov et al. (Eds.): RTC-Digital 2023, CCIS 2647, pp. 97–103, 2026.
https://doi.org/10.1007/978-3-032-04731-1_13

dimensional of systems of difference equations obtained as a result of discretisation is the main feature. Therefore, it is necessary to take into account the computational efficiency related to the possibility of parallelising the constructed algorithms, the number of processors involved, and the peculiarities of topological characteristics of processor fields.

2 Analysis of Modern Numerical Approaches in Mathematical Modelling of Dynamic Processes

The process of constructing new numerical algorithms is usually based on the implementation of the two most common sequential steps [9], which consist of composing discrete (difference) approximations of equations and verifying the a priori qualitative characteristics of these approximations, such as approximation errors, stability, convergence, and accuracy of the obtained difference schemes. The problems associated with the implementation of the first stage are discussed in detail and substantiated in [5,10]. Paper [1] proposes difference schemes of Bicart-type methods, which also provide the ability to control the integration step. Also, in [3], the use of collocation difference schemes in the numerical solution of evolutionary partial differential equations is considered. However, it should be noted that a promising difference scheme for approximating the original equation does not guarantee obtaining high-quality approximations. A significant problem may be the weak stability and/or small order of approximation of the composite scheme. Issues related to the analysis of the accuracy and stability of the resulting difference scheme are not only of fundamental theoretical importance but are also key factors in obtaining effective numerical approximations based on reducing the approximation error. The main result of the studies conducted in this area is the identification of a subset of stable (absolutely or A-α) difference schemes of the primary family. Another task, far from trivial, is to estimate the approximation errors. Within the found subset of stable schemes, schemes with a given accuracy, computational complexity, and other desirable properties and parameters are determined [9]. The main tasks that arise at this stage include determining the achievable order of accuracy of a difference scheme for different classes of problems, developing schemes for solving a wide class of problems with a certain guaranteed accuracy, constructing schemes of increased accuracy in narrower classes of problems, and formulating general principles for constructing stable difference schemes with minimal implementation complexity. In recent years, there has been a significant increase in the number of papers published on the estimation of error orders and the study of stability in numerical schemes. Spectral approaches are commonly used in these works, but they often come with restrictions on the linearity of the difference operators, the dimensionality and types of equations [8,12], and the location of collocation points [7]. Despite the large number of studies in these areas, the development of difference schemes that provide stable and accurate numerical solutions remains a relevant and challenging research topic.

3 Methods for Composing Difference Approximations

Let us consider the Cauchy problem for the ordinary differential equation (1), which could be initially set or reduced to it by the initial partial differential equation, for example, by the method of lines.

$$x^{'}(t) = f(t, x(t)),\ x(t_0) = x_0,\ t \in [t_0, T], \tag{1}$$

Let $u_{n,0}$ denote the approximate value of the solution to the Cauchy Problem (1) at the starting point $t_{n,0}$ of the block to be processed. It is assumed that within the same block the grid points are at equal distances τ

$$t_{n,i} = t_{n,0} + i\tau, i = 1, 2, ..., s.$$

For one-step methods, the difference equations are as follows

$$u_{n,i} = u_{n,0} + i\tau \left(d_i F_{n,0} + \sum_{j=1}^{s} a_{i,j} F_{n,j} \right), i = 1, 2, ..., s, n = 1, 2, ..., N, \tag{2}$$

where $F_{n,j} = f(t_{n,0} + j\tau, u_{n,j})$,

N – is the number of calculation blocks,
s – is the block dimension (number of design points),
$d_i, a_{i,j}$ – are the coefficients of the calculation schemes, $i = 1, 2, ..., s, j = 1, 2, ..., s$.

We obtain the coefficients of the equations that allow us to represent general one-step multipoint methods in the canonical form. Each i-th difference equation (2) contains $s+1$ unknown coefficients. To determine them, a system of equations of the appropriate dimension should be drawn up and solved to ensure that the approximation conditions are met. The approximation error is of order p if the conditions are met:

$$d_i + \sum_{j=1}^{s} a_{i,j} = 1,\ \sum_{j=1}^{s} j^{l-1} a_{i,j} = \frac{i^{l-1}}{l}, i = 1, 2, ..., s, l = 2, ..., p. \tag{3}$$

System (3) for each fixed i contains p equations and $s + 1$ unknowns. If we require that $p = s + 1$, then from (3) we can determine all the unknown coefficients d_i and $a_{i,j}\, i, j = 1, 2, ..., s$, which we will further consider as elements of the vector D and matrix A, respectively. The requirement of coincidence $p = s + 1$ provides the appropriate order of approximation of the one-step s-point block method. The method error is $O(\tau^{(s+1)})$.

The difference equations of multi-step multi-point methods on a template with a calculation block containing s points and a reference block containing m points will be as follows

$$u_{n,i} = u_{n,0} + \tau \left(\sum_{j=1}^{m} b_{i,j} F_{n,j-m} + \sum_{j=0}^{s} a_{i,j} F_{n,j} \right), i = 1, 2, ..., s, n = 1, 2, ..., N, \tag{4}$$

where the notations $F_{n,j}, N, s, a_{i,j}$ coincide with the descriptions (2),

m is the dimension of the reference block (number of reference points).
$b_{i,j}$ are the coefficients of the calculation scheme, where $i = 1, 2, \ldots, s$, $j = 1, 2, \ldots, m$.

The approximation error is of order p if the following conditions are met

$$\sum_{j=1}^{m} b_{i,j} + \sum_{j=1}^{s} a_{i,j} = 1, i = 1, 2, ..., s, \tag{5}$$

$$\sum_{j=1}^{s} a_{i,j} j^{l-1} - \sum_{j=1}^{m} b_{i,j} (j-m)^{l-1} = \frac{i^{l-1}}{l}, i = 1, 2, ..., s, l = 2, ..., p.$$

The system of Eqs. (5) for each fixed i contains p equations and $m + s$ unknowns $b_{i,1}, b_{i,2}, ..., b_{i,m}, a_{i,1}, a_{i,2}, ..., a_{i,s}$. When $p = m + s$, all unknown coefficients from the system can be determined. The obtained coefficients will allow us to form the corresponding matrices A and B. The highest order of approximation of the multistep s-point block method is $m + s$.

4 Investigation of the Stability of One and Multi-step Block Methods

The integro-interpolation approach to the formation of the calculated difference coefficients considered in [3] allows us to obtain the desired values while maintaining the highest order of approximation, i.e., of order $O(\tau^{s+1})$ for single-step schemes and of order $O(\tau^{s+m})$ for multi-step schemes, where s and m are the number of calculation and reference points, respectively. At the same time, it has been proven [5] that single-step schemes of the highest order of accuracy are absolutely stable, while multi-step schemes of the highest order of approximation are unsuitable for numerical calculations because they are unstable. Therefore, when forming the calculation coefficients, it is necessary to artificially reduce the accuracy order of the method in order to obtain absolutely stable difference schemes. Let us consider some variants of such formation of calculation schemes, which, on the one hand, have a sufficiently high order of accuracy, and, on the other hand, remain absolutely stable both in terms of the initial data and the right-hand side.

The characteristic equation of the transition matrix of the one-step method for a homogeneous case has the form

$$\lambda^{s-1}(1 - \lambda) = 0,$$

the roots of this equation are $\lambda_1 = 1, \lambda_i = 0, i = 2, 3, ..., s$, so Dahlquist's conditions [5,6] are met, and the one-step multistep method is stable with the initial data. The matrix S for the multi-point method (4) also has the form with the same characteristic equation and roots, and, therefore, the multi-step multi-point method is also always stable according to the initial data.

To study the stability of the difference method on the right-hand side, we use [6,10] the model one-dimensional equation of the form

$$x^{'}(t) = \lambda x(t), x(t_0) = x_0, t \in [t_0, T], \tag{6}$$

where λ - is a complex number ($\lambda < 0$).

A method is considered A-stable if the boundary of the method's stability region in complex space is the imaginary axis, and $A-(\alpha)$ is stable if its stability region contains the angle $|arg(-\mu)| < \alpha$, $\mu = \lambda\tau$ [10].

In Figs. 1 and 2 show hodographs that determine the stability limits of one-step methods of the maximum order of accuracy for each design point in a block of the corresponding dimension and demonstrate the absolute stability (A-stability) of one-step methods. This property of the methods is preserved for any block dimension. The A-stable method is absolutely stable at any value of the integration step τ, provided that the solution of the initial differential equation (system of equations) is stable [3]. Unfortunately, multistep methods of maximum order approximation lack this feature.

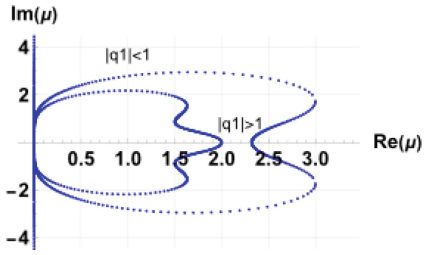


Fig. 1. Stability region of the one-step method, $s = 3$ with the highest order of approximation $O(\tau^{s+1}) = O(\tau^4)$

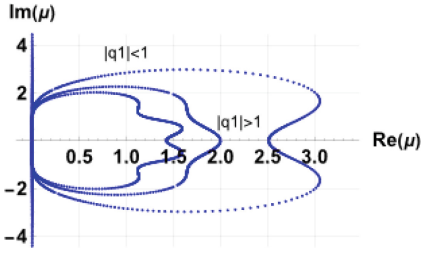


Fig. 2. Stability region of the one-step method, $s = 4$ with the highest order of approximation $O(\tau^{s+1}) = O(\tau^5)$

To ensure the absolute stability (A) or stability $A-\alpha$ of the multistep methods (4), it is necessary to reduce the order of accuracy while maintaining the entire set of reference and design points. Figures 3 and 4 show the stability histograms of the difference schemes when the dimensions of the blocks do not coincide, $m < s$. It can be seen that when the maximum possible order of approximation is reduced, the difference scheme becomes absolutely stable.

5 Conclusions

In this paper, we investigate the formation of single- and multi-step difference schemes of a given order of accuracy for numerical solutions of the Cauchy problem for both ordinary differential equations and partial differential equations, which can be reduced to solving ODEs by the method of lines. The proposed difference schemes are designed to be used in parallel computing systems and provide the ability to simultaneously determine values in a set of calculation points of a block, provided there are a sufficient number of processors. The

construction of discrete approximations is carried out by limiting the order of approximation of the difference method, taking into account the dimensions of the calculation and/or reference blocks.

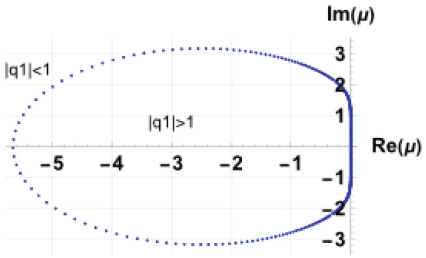


Fig. 3. Stability region of the multistep method, $s = 3, m = 2$ with the highest order of approximation $O(\tau^{s+m}) = O(\tau^5)$

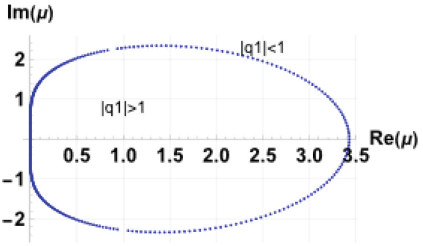


Fig. 4. Stability region of the multistep method, $s = 3, m = 2$ with the order of approximation $O(\tau^{s+m-1}) = O(\tau^4)$

The scientific novelty of this work lies in the proposed approaches to the construction of methods for the composition of difference approximations that provide a given order of approximation in the numerical solution of the Cauchy problem for both ordinary differential equations and evolutionary partial differential equations. The study of the properties of the proposed transition matrices, which connect the reference and design points, allows for determining the nature of the stability of the generated difference schemes according to the initial data and right-hand sides. The practical value of the work is in the creation of a software system that generates difference relations for given block dimensions, determines the approximation orders at the design points, and evaluates the stability of the obtained schemes. Automatic generation of stable difference schemes takes into account the topology of the available processor field, thus maximising the use of the available computing resource.

Acknowledgments. The authors express their deep gratitude to the Philipp Schwartz Initiative of the Alexander von Humboldt Foundation for the opportunity to conduct research.

References

1. Bashkov, Y., Dmitrieva, O., Huskova, N.: Parallel multiple blocked methods of Bickart type. Proc. SPIE **11176**(1) (2019). https://doi.org/10.1117/12.2536798
2. Butcher, J.: Methods for Ordinary Differential Equations. Wiley, Chichester (2016)
3. Dmitrieva, O., Huskova, V.: Generation of parallel colocation difference schemes using the integro-interpolation approach. Inf. Sci. Model. **1**(7), 95–108 (2022). https://doi.org/10.20998/2411-0558.2022.02.09
4. Dongarra, J., Hammarling, S., Higham, N.J., Relton, S.D., Valero-Lara, P., Zounon, M.: The design and performance of batched BLAS on modern high-performance computing systems. Procedia Comput. Sci. vol. 108, pp. 495–504 (2017)

5. Feldman, L., Dmitrieva, O.: Development and research of parallel collocation block methods. Inf. Cybern. Comput. Technol. **16**(1), 28–35 (2012)
6. Hairer, E., Wanner, G.: Solving Ordinary Differential Equations II: Stiff and Differential-Algebraic Problems. Springer Verlag (2004)
7. Jameson, A.: A proof of the stability of the spectral difference method for all orders of accuracy. J. Sci. Comput. **5**(1), 348–358 (2010). https://doi.org/10.1007/s10915-009-9339-4
8. Ma, Y., Chen, L.: Error bounds of a finite difference/spectral method for the generalized time fractional cable equation. Fractal Fractional **6**(88) (2022). https://doi.org/10.3390/fractalfract6080439
9. Samarskii, A.: The Theory of Difference Schemes. CRC Press, Boca Raton (2001)
10. Samarskii, A., Gulin, A.: Numerical method. Science (1989)
11. Zhu, Q., Fusco, G.: Complex dynamic system modelling, identification and control. Entropy **24**(3), 380–389 (2022). https://doi.org/10.3390/e24030380
12. Zlotnik, A., Chetverushkin, B.: Spectral stability conditions for an explicit three-level finite-difference scheme for a multidimensional transport equation with perturbations. Differ. Eq. **57**(7), 891–900 (2021). https://doi.org/10.1134/s0012266121070065

Classification of Transformations of Equivalent Kernels of Some Determinantal Point Processes

Harry Sapranidis Mantelos(✉)

University of Warwick, Coventry CV4 7AL, UK
h.sapranidis-mantelos@warwick.ac.uk
https://warwick.ac.uk/fac/sci/statistics/staff/research/students/mantelos

Abstract. Determinantal point processes (abbr., DPPs) are randomly arranged points whose distribution is characterised via determinants of matrices. The entries of these matrices are given by one fixed function of two variables, $K(x, y)$, called the kernel of the DPP. It is well-known that the kernel K is not unique and that there exist various other functions of two variables that are valid kernels of the same DPP. We refer to such kernels as *equivalent kernels to* K. It was recently shown by Stevens in [Random Matrices: Theory and Applications, 10(03):2150027, 2021] that, restricting to the case of symmetric kernels, all equivalent kernels of some DPP can be transformed into one another by conjugation transformations. This partially solves a conjecture of Bufetov from 2017 which states that all equivalent kernels of some DPP can be transformed into one another by conjugation and transposition transformations. In this work, we completely relax the symmetry assumptions on the kernels. We go through why this conjecture cannot hold in this general setting, but that under some surprisingly simple and natural conditions on the kernel it does.

Keywords: Determinantal point processes · Equivalent correlation kernels · Cycle graphs · Cocycle functions

1 Introduction and Motivation

Point processes are probabilistic models for random scatterings of points in some mathematical space. These processes are characterised by their correlation function. If a point process is determinantal (a.k.a., the point process is a DPP) then its correlation function is of a determinantal form (see, e.g., [1,2,4]).

More precisely, if we denote by ρ_n the n^{th} correlation function of such a point process on some measure space Λ, then there exists a function $K : \Lambda^2 \to \mathbb{F}$ such that for any $n > 0$ and any tuple $(x_1, \ldots, x_n) \in \Lambda^n$, we have

$$\rho_n(x_1, \ldots, x_n) = \det\left(K(x_i, x_j)\right)_{i,j=1}^{n},$$

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2026
I. Potapov et al. (Eds.): RTC-Digital 2023, CCIS 2647, pp. 104–112, 2026.
https://doi.org/10.1007/978-3-032-04731-1_14

where $\mathbb{F}$ is a suitable field - which in most (if not all) cases is either $\mathbb{R}$ or $\mathbb{C}$ (and so this is what we also take $\mathbb{F}$ to be in our work). We then call K a **correlation kernel** of the DPP.

As in [5] (the paper we build on), for our research problem, we neglect the measure space structure of Λ and just consider it as a set. Hence, the kernel K can be regarded as a mere function. If there exists another function $Q : \Lambda^2 \to \mathbb{F}$ such that

$$\det(Q(x_i, x_j))_{i,j=1}^n = \det(K(x_i, x_j))_{i,j=1}^n \qquad \forall x_1, \ldots, x_n \in \Lambda \ \forall n \in \mathbb{N}, \tag{1}$$

then we call K and Q **equivalent kernels** ($K \equiv Q$ for short).

Remark 1. If Q is of the form

$$Q(x, y) = g(x)g(y)^{-1}K(x, y), \qquad x, y \in \Lambda,$$

for some non-zero function $g : \Lambda \to \mathbb{F}$, we say that Q is a **conjugation transformation** of K with **conjugation function** g. It is then easy to check that $Q \equiv K$.

Remark 2. It is not difficult to see that if $Q(x, y) = K(y, x)$ for all $x, y \in \Lambda$, then $Q \equiv K$. In this case we say that Q is a **transposition transformation** of K.

Although the problem of classifying transformations that yield equivalent kernels of DPPs - in the above framework - originates (and is of theoretical interest) in the theory of determinantal point processes, it essentially asks a basic question about a pair of functions of two variables: given that function K is related to Q via Eq. (1) - and thus $K \equiv Q$ - what are all the possible transformations that transform K into Q? In this setup, it was established in [5] that, restricting to the case of symmetric kernels (a function h of two variables is **symmetric** if $h(x, y) = h(y, x)$ for every x, y), K must be a conjugation transformation of Q. More precisely, the following result was established in [5].

Theorem 1. ***(Theorem 1.5 from** [5]. Suppose that Λ is a set, let $\mathbb{F}$ be a field and let $K, Q : \Lambda^2 \to \mathbb{F}$ be symmetric kernels. If K and Q are equivalent (i.e., Eq. (1) holds), then it must be the case that Q and K are conjugation transformations of one another, i.e., there exists a non-zero function $g : \Lambda \to \mathbb{F}$ such that for every $x, y \in \Lambda$,*

$$Q(x, y) = g(x)g(y)^{-1}K(x, y). \tag{2}$$

Of course, in the (symmetric) setting of the above theorem, a transposition transformation would have just been the trivial identity transformation. Thus, [5] partially solves the following conjecture of Bufetov, in the case where **both** equivalent kernels K and Q are symmetric.

Conjecture 1. (Conj. 1.4 from [5]. If K and Q are equivalent kernels (i.e., (1) is satisfied), then they can be transformed into one another by transposition and conjugation transformations.

It was then left as an open problem to solve Bufetov's conjecture in the general setting without the symmetry assumptions on the kernels. It is imperative, of course, to first determine whether the conjecture does indeed hold in that general setting. In the next section we show that, in fact, it is possible to find counterexamples to the conjecture. However, the nature of such counterexamples, as we shall see, is rather exceptional. Fortunately, our research findings show that it is possible to impose some simple and natural conditions on the kernels that elegantly rule out such counterexamples. The proof of the conjecture (under these conditions) - briefly outlined in this paper - is essentially revolved around the recurrent use of an elementary lemma (Lemma 1) and some basic graph theory and combinatorics.

Hereafter we summarize the research conducted in [arXiv:2302.02471].

2 Counterexamples of Bufetov's Conjecture

Firstly, let us consider the case where Λ from Sect. 1 is any discrete set of cardinality $n \geq 4$. For simplicity, assume $\Lambda = \{1, 2, \ldots, n\}$. Then, consider the kernels $K : \Lambda^2 \to \mathbb{F}$ and $Q : \Lambda^2 \to \mathbb{F}$ defined by

$$(K(x,y))_{1\leq x,y\leq n} = \left[\begin{array}{c|c} \mathrm{C} & 0 \\ \hline 0 & \mathrm{D} \end{array}\right], \qquad (Q(x,y))_{1\leq x,y\leq n} = \left[\begin{array}{c|c} C^T & 0 \\ \hline 0 & D \end{array}\right],$$

where C and D are some non-symmetric square matrices of the same dimension. In this case $K \equiv Q$ but K is neither a transposition nor a conjugation transformation of Q. Thus, Bufetov's conjectured classification of transformations does not hold: we have just identified one other type of transformation - other than the two stated in the conjecture - which yields equivalent kernels. Perhaps a suitable name for it would be a"**partial transposition transformation**", since only *part* of the matrix is being transposed.

Interestingly, by the block determinant formula, even if the null sub-matrices of the above pair of matrices were replaced by matrices of ones, it would still be the case that $K \equiv Q$. However, for the same reason as in the previous paragraph, Bufetov's conjecture would not hold.

For reasons that will be explained in the next section, our analysis is restricted to non-zero kernels (i.e., kernels K such that $K(x,y) \neq 0$ for every $x, y \in \Lambda$). To this end, let us further analyse the counterexample specified in the preceding paragraph with the additional condition that the matrices C and D are entrywise non-zero. We seek for a condition on K that would make kernels K and Q of the previously described form a non-admissible pair satisfying (1). We first need to deal with the case when $n = 4$, in which case kernels K and Q from before satisfy

$$(K(x,y))_{1\leq x,y\leq n} = \begin{pmatrix} a & b & 1 & 1 \\ c & d & 1 & 1 \\ 1 & 1 & e & f \\ 1 & 1 & g & h \end{pmatrix}, \qquad (Q(x,y))_{1\leq x,y\leq n} = \begin{pmatrix} a & c & 1 & 1 \\ b & d & 1 & 1 \\ 1 & 1 & e & f \\ 1 & 1 & g & h \end{pmatrix}, \qquad (3)$$

for some $a, b, c, d, e, f, g, h \in \mathbb{F} \setminus \{0\}$ such that $c \neq b$ and $f \neq g$.

Clearly, the sub-matrices of ones in the upper-right and lower-left block regions of the above two block matrices are the ones that bring about the block structure that enables this "partial transposition" transformation discussed previously to be a viable transformation that transforms equivalent kernels (of a special form) into one another. And so by far the most intuitive condition to impose on our pair of matrices that we would hope would be enough to make this type of transformation non-permissible would be to require non-zero determinants in the upper-right and lower-left 2×2 sub-matrix regions, that is, to require

$$\begin{vmatrix} K(1,3) & K(1,4) \\ K(2,3) & K(2,4) \end{vmatrix}, \quad \begin{vmatrix} K(3,1) & K(3,2) \\ K(4,1) & K(4,2) \end{vmatrix} \neq 0,$$

which - in the case of $n = 4$ - is the same as requiring for every $x, y, z, w \in \Lambda$ distinct,

$$\begin{vmatrix} K(x,y) & K(x,w) \\ K(z,y) & K(z,w) \end{vmatrix} \neq 0.$$

Our research findings not only show that this condition, in fact, is sufficient in solving Bufetov's conjecture in the above setting with $\Lambda = \{1, 2, \ldots, n\}$ and $n = 4$ (or for any discrete set Λ of any cardinality, for that matter); but even for a **general** abstract set Λ (of any kind of cardinality).

3 Main Result

Precisely, our main result in [3] reads as follows.

Theorem 2. *Suppose that Λ is a set, let $\mathbb{F}$ be a field and let $K, Q : \Lambda^2 \to \mathbb{F}$ be non-zero (not necessarily symmetric) kernels such that for every $x, y, z, w \in \Lambda$ distinct,*

$$\begin{vmatrix} K(x,y) & K(x,w) \\ K(z,y) & K(z,w) \end{vmatrix} \neq 0. \tag{4}$$

*If K and Q are equivalent (i.e., Eq. (1) holds - in which case, Eq. (4) with K replaced by Q also holds), then the following two transformations are the **only** possible transformations that transform K into Q:*

- *Conjugation transformations, i.e., there exists a non-zero function $g : \Lambda \to \mathbb{F}$ such that (2) holds for every $x, y \in \Lambda$.*
- *Transposition transformations followed by conjugation transformations, i.e., there exists a non-zero function $g : \Lambda \to \mathbb{F}$ such that for every $x, y \in \Lambda$,*

$$Q(x,y) = g(x)g(y)^{-1}K(y,x). \tag{5}$$

In the last section, the significance of the (4) condition from Theorem 2 was explained and illustrated; but the reason why we require non-zero kernels was not. The reason we insist on the latter is because, had it not been so, then we may

have been presented with quite a few problems in solving Bufetov's conjecture in this general (not necessarily symmetric) setting. Indeed, Eq. (1) with $n = 1$ and $n = 2$ can easily be seen to yield the identity

$$K(x,y)K(y,x) = Q(x,y)Q(y,x), \qquad x,y \in \Lambda. \tag{6}$$

This identity, in the symmetric setting of [5] (where both K and Q are symmetric), implies that for every $x, y \in \Lambda$,

$$K(x,y) = 0 \text{ if and only if } Q(x,y) = 0. \tag{7}$$

However, when we drop this symmetry assumption on both kernels, we do not have the luxury of concluding the statement in (7). On the contrary, there could very well exist $\bar{x}, \bar{y}, \hat{x}, \hat{y} \in \Lambda$ such that, for example, $K(\bar{x},\bar{y}) = 0$, $K(\bar{y},\bar{x}) \neq 0$, $Q(\bar{x},\bar{y}) \neq 0$, $Q(\bar{y},\bar{x}) = 0$, $K(\hat{x},\hat{y}) = 0$, $K(\hat{y},\hat{x}) \neq 0$, $Q(\hat{x},\hat{y}) = 0$ and $Q(\hat{y},\hat{x}) \neq 0$. In this case, both the pair $\bar{x}, \bar{y}$ and the pair $\hat{x}, \hat{y}$ satisfy Eq. (6) - so there's no contradiction to the (1) equation. However, the pair $\bar{x}, \bar{y}$ could not possibly satisfy Eq. (2) for some non-zero function g; nor could the pair $\hat{x}, \hat{y}$ satisfy Eq. (5) for some non-zero function g. Therefore, K and Q in this case could not possibly be transformed into one another through conjugation and transposition transformations: neither Eq. (2) nor Eq. 5) is satisfied **for every** $x, y \in \Lambda$.

4 Idea and Outline of the Proof of Theorem 2

The main tools we used to prove Theorem 2 were the classic Leibniz formula for determinants, some (perhaps known) results from graph theory regarding cycles of lengths 2, 3 and 4, the concept of a cocycle function (see [5] for more details), and some basic combinatorics. We provide a brief outline of our proof below. We begin by laying some necessary groundwork and introducing some (perhaps non-standard) terminology.

Definition 1. *We call an $(n+1)$-tuple $(x_0, x_1, \ldots, x_n) \in \Lambda^{n+1}$ such that $x_n = x_0$ a **cycle of length** n. Moreover, if all the x_i's in the tuple are distinct (except for x_0 and x_n - which by definition are equal), then we call $(x_0, x_1, \ldots, x_n)$ a **simple cycle**.*

Definition 2. *We call a function $c : \Lambda^2 \to \mathbb{F}$ a **cocycle function** if for every $r \geq 1$ and for every tuple $(z_1, \ldots, z_r) \in \Lambda^r$,*

$$c(z_1,z_2)c(z_2,z_3)\cdots c(z_{r-1},z_r)c(z_r,z_1) = 1. \tag{8}$$

Definition 3. *If $Q : \Lambda \to \mathbb{F}$ and $K : \Lambda \to \mathbb{F}$ are functions that satisfy*

$$Q(x,y) = c(x,y)K(x,y) \qquad \forall\ x,y \in \Lambda$$

*for some cocycle function $c : \Lambda^2 \to \mathbb{F}$, then we say that Q is a **cocycle transformation** of K.*

One can easily verify that a cocycle transformation of a kernel yields an equivalent kernel. It is also a straightforward exercise to show that, under our paper's non-zero kernel assumption, a kernel Q is a cocycle transformation of a kernel K if and only if Q is a conjugation transformation of K. This is a critical observation which simplifies our work, since establishing that some function satisfies the cocycle property (8) is objectively easier than constructing some conjugation function from scratch. In particular, the previous observation means that if we are able to prove that either the function

$$S : \Lambda^2 \to \mathbb{F}, \;\; S(x,y) = \frac{Q(x,y)}{K(x,y)},$$

or the function

$$\tilde{S} : \Lambda^2 \to \mathbb{F}, \;\; \tilde{S}(x,y) = \frac{Q(x,y)}{K(y,x)}$$

is a cocycle function, then, since for every $x, y \in \Lambda$,

$$Q(x,y) = S(x,y)K(x,y) \text{ and } Q(x,y) = \tilde{S}(x,y)K(y,x),$$

it would follow that Q must be in one of the two forms from the conclusion of Theorem 2 - which is exactly the statement of Bufetov's conjecture. And so the proof of Theorem 2 boils down to proving that one of the functions S or $\tilde{S}$ must be a cocycle function.

For non-zero functions of two variables (which, under our paper's non-zero kernel assumption, both S and $\tilde{S}$ are), there is a shortcut to establishing the cocycle property (8); we state it below.

Proposition 1. *If a non-zero function $c : \Lambda^2 \to \mathbb{F}$ satisfies*

1. $c(x,x) = 1$ *for every* $x \in \Lambda$;
2. $c(x,y)c(y,x) = 1$ *for every* $x, y \in \Lambda$;
3. $c(x,y)c(y,z)c(z,x) = 1$ *for every* $x, y, z \in \Lambda$,

then c is a cocycle function.

In other words, c satisfying the cocycle property for every cycle of length 1, 2 and 3 is both a necessary and sufficient condition for c to be a (full) cocycle function, that is, for it to satisfy the cocycle property for any cycle of any length. This result simplifies/shortens our analysis in that instead of having to prove the cocycle property (8) for every cycle in Λ (of every possible length) for either the function S or $\tilde{S}$; proving it for cycles of lengths 1, 2 and 3 is sufficient.

It is immediate from Eq. (1) with $n = 1$ and Eq. (6) that both S and $\tilde{S}$ automatically satisfy the cocycle property for any cycle of length 1 and 2 in Λ. Thus, by Proposition 1, it suffices to show that either S or $\tilde{S}$ satisfies the cocycle property for all cycles of length 3 in Λ. This task, however, is not at all trivial in this general setting. In particular, it is no longer enough to just use Eq. (1) with $n = 3$ in conjunction with the Leibniz formula for determinants - as was

the case in [5]. The reason being that in this general setting the aforementioned step will merely yield, for any simple cycle $p = (p_i)_{i=0}^3$ of length 3 in Λ,

$$\overbrace{K(p_0,p_1)K(p_1,p_2)K(p_2,p_0)}^{=:\,K[p]} + \overbrace{K(p_0,p_2)K(p_2,p_1)K(p_1,p_0)}^{=:\,K'[p]}$$

$$= \tag{9}$$

$$\underbrace{Q(p_0,p_1)Q(p_1,p_2)Q(p_2,p_0)}_{Q[p]\,:=} + \underbrace{Q(p_0,p_2)Q(p_2,p_1)Q(p_1,p_0)}_{Q'[p]\,:=}.$$

Note that, had both K and Q been symmetric, Eq. (9) would simplify to $2K[p] = 2Q[p]$, which proves the cocycle property for all cycles of length 3 in Λ for the function S from before. In contrast, to be able to proceed with Eq. (9) in the non-symmetric setting, one needs to make the additional observation that $K[p]K'[p] = Q[p]Q'[p]$ also holds. This is an immediate consequence of Eq. (6). Thereafter the following elementary lemma can be applied.

Lemma 1. *Let $a, b, a', b' \in \mathbb{F}$ be constants satisfying*

$$a + b = a' + b' \qquad \text{and} \qquad ab = a'b'. \tag{10}$$

Then, it is either the case that $a = a'$ and $b = b'$; or, $a = b'$ and $b = a'$.

This simple lemma is the key ingredient of our proof, as it is what first introduces the "*dichotomy*" that enables us to eventually reach our desired conclusion about S and $\tilde{S}$ - and is repeatedly invoked.

Thereafter we need to extract a bit more information about the relationship between K and Q. For this we make use of Eq. (1) with $n = 4$ in conjunction with the Leibniz formula for determinants. As a result cycles of length 4 also enter the scene; and so then our task is to find a way to link cycles of lengths 2, 3 and 4 together in some suitable way. In Figs. 1, 2 and 3 below we represent graphically the three lemmas from graph theory we used to achieve this. In very broad terms, the way in which we exploited these lemmas was via the following graphical identification we made for functions of two variables evaluated on cycles: for an arbitrary function $h : \Lambda^2 \to \mathbb{F}$, we identify (graphically) the product $h[p] := \prod_{i=1}^n h(p_{i-1}, p_i)$, where $p := (p_i)_{i=0}^n \in \Lambda^{n+1}$ is a cycle of length n in Λ, by precisely the (drawn) cycle graph of p.

5 Open Problems

As we saw in Sect. 2, when we put no constraint on the structure of K, there are potentially many more types of transformations that yield equivalent ker-

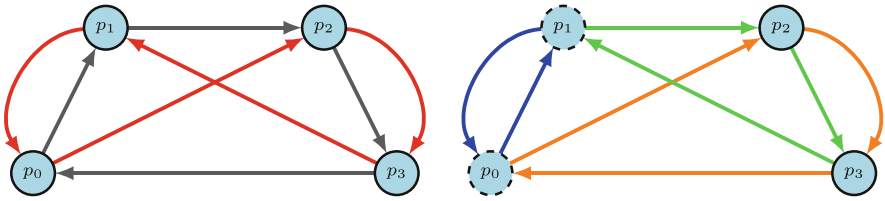


Fig. 1. In this figure we seek to write the disjoint union of two distinct simple cycles of length 4, $p = (p_0, p_1, p_2, p_3, p_0)$ and $q = (p_0, p_2, p_3, p_1, p_0)$, drawn in black and red, respectively, in the LHS directed graph, as the disjoint union of distinct simple cycles of lengths 2 and 3. We do this in the RHS directed graph by considering the 3-cycle $s^{(1)} = (p_2, p_3, p_1, p_2)$, the 3-cycle $s^{(2)} = (p_2, p_3, p_0, p_2)$ and the 2-cycle $r = (p_0, p_1, p_0)$. (Color figure online)

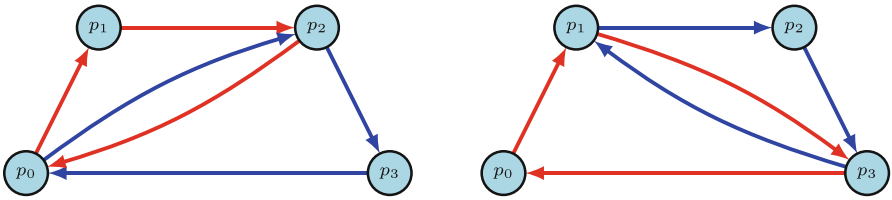


Fig. 2. In this figure we decompose a simple cycle of length 4, $p = (p_0, p_1, p_2, p_3, p_0)$, into two simple cycles of length 3. There are two possible such decompositions: on the LHS graph, in red is the 3-cycle $s^{(1)} = (p_0, p_1, p_2, p_0)$, and in blue is the 3-cycle $s^{(2)} = (p_0, p_2, p_3, p_0)$; on the RHS graph, in red is the 3-cycle $r^{(1)} = (p_0, p_1, p_3, p_0)$, and in blue is the 3-cycle $r^{(2)} = (p_1, p_2, p_3, p_1)$. (Color figure online)

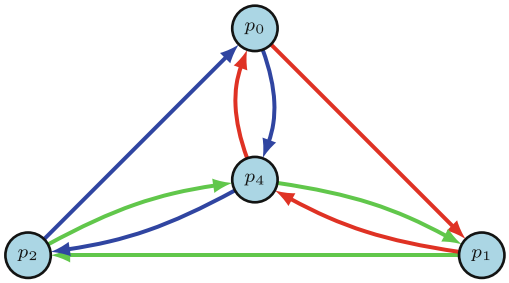


Fig. 3. In this figure we decompose a 3-cycle $p = (p_0, p_1, p_2, p_0)$ into three distinct simple cycles of length 3 that each contains the vertex p_4, where $p_4 \neq p_i$ for all $i \in \{0, 1, 2\}$. In red is the 3-cycle $s^{(1)} = (p_0, p_1, p_4, p_0)$, in green is the 3-cycle $s^{(2)} = (p_1, p_2, p_4, p_1)$, and in blue is the 3-cycle $s^{(3)} = (p_2, p_0, p_4, p_2)$. (Color figure online)

nels other than the ones stated in Bufetov's conjecture. Thus, another (rather ambitious) research problem that one could attempt is to classify **every** possible type of transformation.

Another interesting question to ask is the following. Given that we start with a non-symmetric kernel, say K, of a DPP, does there necessarily exist an equivalent kernel of K, say Q, which is symmetric?

Finally, one could also attempt to extend the results from [5] and [3] by taking potential measure space structures of Λ into consideration.

Acknowledgments.. Harry Sapranidis Mantelos is supported by the Warwick Mathematics Institute Centre for Doctoral Training, and gratefully acknowledges funding from the University of Warwick.

References

1. Hough, J., Krishnapur, M., Peres, Y., Virág, B.: Zeros of Gaussian Analytic Functions and Determinantal Point Processes. University Lecture Series, American Mathematical Society (2009). https://books.google.co.uk/books?id=KCuCAwAAQBAJ
2. Kulesza, A., Taskar, B.: Determinantal Point Processes for Machine Learning. Foundations and Trends in Machine Learning. World Scientific (2012). https://books.google.gr/books?id=FY-6lAEACAAJ
3. Mantelos, C.S.: Classification of transformations of equivalent kernels of some determinantal point processes (2023). https://doi.org/10.48550/ARXIV.2302.02471
4. Soshnikov, A.: Determinantal random point fields. Russ. Math. Surv. **55**(5), 923 (2000)
5. Stevens, M.: Equivalent symmetric kernels of determinantal point processes. Random Matrices Theor. Appl. **10**(03), 2150027 (2021)

On the Computational Complexity of the Interval Dependence Problem in Credal Networks

Marco de Angelis[1(✉)], Hector Estrada Lugo[2], Scott Ferson[3], and Edoardo Patelli[1]

[1] University of Strathclyde, 75 Montrose St., Glasgow G1 1XJ, UK
marco.de-angelis@strath.ac.uk
[2] Technological University Dublin, Grangegorman Lower, Dublin D07 ADY7, Ireland
[3] University of Liverpool, Brodie Tower, Liverpool L69 3GB, UK

Abstract. Credal networks are Bayesian networks with interval conditional probability tables CPTs. Even though algorithms that trade complexity in exchange for approximation have been developed to compute with credal networks in applications, the complexity of the interval dependence problem is still not very well understood. We investigate such computational complexity on the variable elimination algorithm on small-sized two-state credal networks by means of computer arithmetic. Automatic differentiation is coupled to interval arithmetic in a strategic attempt to isolate monotonic trends for a particular inferential or diagnostic query. The projection of interval uncertainty through the partial derivatives with respect to the parameters proves instrumental to reduce the dimensionality of the dependency problem. In fact, the parameters that are monotonic with respect to a particular query can then be collapsed to a singleton reducing the number of intervals to be propagated to yield the tightest bounds. The dependency problem of interval computation has at worst exponential complexity, so reducing the size of this problem can be vital to compute with credal networks. The considerations drawn in this work on a small network, can be readily generalised to larger networks, whilst the qualitative account of the parameter space can be utilised to build more elaborated algorithms.

Keywords: Credal networks · Computer arithmetic · Automatic differentiation · Verified computing

1 Introduction

The inference computation is itself an NP-hard task for both exact and approximate inference algorithms even in the precise probability domain of Bayesian networks [5,9]. In the case of credal networks upper and lower bounds of the query node must be found which increases the complexity of the inference computation. The so called exact inference methods can be reformulated to find lower

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2026
I. Potapov et al. (Eds.): RTC-Digital 2023, CCIS 2647, pp. 113–118, 2026.
https://doi.org/10.1007/978-3-032-04731-1_15

and upper bounds of the queried variables when interval conditional probabilities are at play. The complexity of credal networks has been studied by multiple scholars [4,7]. In the case of big networks, the inference problem is often recast in an optimisation fashion, because rigorous outer approximation is not affordable, especially for large interval uncertainty. The idea of using optimization to solve credal networks has been extensively explored by de Campos et al. and Antonucci et al. [1,3]. The literature regarding imprecise probability in graphical models is growing due to the fact that both concepts (imprecise probabilities and graphical models) have only recently been related. Tessem [10] introduced the idea of interval propagation for Bayesian networks, with an algorithm called annihilation/reinforcement (A/R), with approximate solutions but running in polynomial time. He adopted the idea of Kyburg [6] about representing uncertainty with convex sets instead of probability points and combines it with the causal inference of Judea Pearl [9]. The idea of representing uncertainty with sets of probability measures has been extensively studied by scholars in the imprecise probability community [2]. In this work, we draw a strategy towards the development of a more general algorithm for computation in credal networks. We briefly outline the dependency problem of plain interval arithmetic and ways to overcome it using the tools of interval analysis. The aim of this work is to compute best 'possible bounds' on any particular inferential or diagnostic query.

2 Inference in Credal Networks

Probabilistic inference consists in the computation of a posterior probability of a node of interest, $P(X_q)$. This process is carried out adopting the information available given by the prior probability $P(X_e|X)P(X)$, and the evidence $P(X_e)$. *Predictive reasoning* refers to the computation of a posterior probability containing new information about the effects of a given event. In the Bayesian case, predicting the posterior of a variable x_q means marginalising out the variables that are not x_q. This is represented by the sum $\sum_{x_1,\ldots,x_m \setminus x_q}$ in (1), where $(x_1, \ldots, x_m \setminus x_q)$ is the set of variables not involved in the query x_q. When interval probabilities are present the joint probability must be minimised and maximised to attempt to bound the posterior. For instance, the lower bound of the marginal probability is

$$\underline{P}(x_q) = \min_{P(X)\in K(X)} P(x_q) = \min_{P(X_i|\pi_i)\in K(X_i|\pi_i)} \sum_{x_1,\ldots,x_n \setminus x_q} \prod_{i=1}^{n} P(x_i|\pi_i), \qquad (1)$$

where $K(X_i|\pi_i)$ is the interval CPT. In *Diagnostic reasoning* the analysis is done to find the cause of an outcome (sometimes called symptoms [5]). This task consists in applying Bayes' rule, $P(x_q|x_e) = P(x_e|x_q)P(x_q)/P(x_e)$, to obtain the posterior probability of the queried variable (with $P(x_e) > 0$),

$$\underline{P}(x_q|x_e) = \min_{P(X_i|\pi_i)\in K(X_i|\pi_i)} \frac{\sum_{i=0}^{n} P(x_i, x_e)}{P(x_e)}. \qquad (2)$$

The computational effort to compute exact inference for the precise case increases exponentially with the number of nodes in the network. Additionally for the imprecise case, the inference computation is complicated further due to the combinatorial problem.

3 Proposed Strategy

3.1 Interval Analysis and Repeated Variables

Interval computations in credal networks comprise addition, multiplication, subtraction, and division between intervals in $[0,1]$, $\boldsymbol{a} = \{a \in [0,1] : \underline{a} < a < \overline{a}\}$

$$\begin{aligned} \boldsymbol{a} + \boldsymbol{b} &= [\underline{a} + \underline{b}, \overline{a} + \overline{b}] \\ \boldsymbol{a} * \boldsymbol{b} &= [\min\{\cdot\}, \max\{\cdot\}] \\ \boldsymbol{a} - \boldsymbol{b} &= [\underline{a} - \overline{b}, \overline{a} - \underline{b}] \\ \boldsymbol{a}/\boldsymbol{b} &= [\underline{a}/\overline{b}, \overline{a}/\underline{b}], \text{ with } 0 < \boldsymbol{a},\ \boldsymbol{b} < 1 \end{aligned} \tag{3}$$

where $\{\cdot\} = \{\underline{a}\underline{b},\ \underline{a}\overline{b},\ \overline{a}\underline{b},\ \overline{a}\overline{b}\}$. Note that operations between intervals generalise the case of precise numbers. The use of interval analysis is often discouraged by its inclination to produce *inflated* uncertainty. Interval analysis produces results that are 'best possible', i.e. without *inflated* uncertainty, when the variables appear only once in their explicit expressions. For example, every quadratic polynomial can be reduced to its *SUE* (single use expression) by completing the square: $\boldsymbol{x}^2 + \boldsymbol{x} = \left(\boldsymbol{x} + \frac{1}{2}\right)^2 - \frac{1}{4}$. Being able to reduce expressions to *SUE* can be crucial when the expression has only a few interval variables, like in the case of polynomials. For more about interval analysis the reader is referred to the seminal work of Moore [8].

4 Results

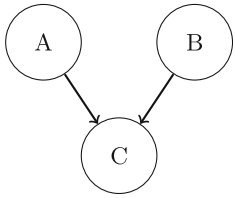
We aim to obtain an outer approximation of the inference probability of variable C for the three-node network shown in Fig. 1. In order to show the computations involved, suppose that the probability of the variable C being on state c is queried, i.e., $P(c)$. By marginalising the joint distribution the expression for $P(c)$ can be obtained as follows

$$\begin{aligned} &P(c) = \textstyle\sum_{A,B} P(A)P(B)P(c|A,B) \\ &= \textstyle\sum_{A} P(A) \sum_{B} P(B)P(c|A,B) \\ &= a\ b\ P(c|a,b) + (1-a)\ b\ P(c|\neg a, b) \\ &+ a\ (1-b)P(c|a,\neg b) + (1-a)(1-b)P(c|\neg a, \neg b) \end{aligned} \tag{4}$$

Equivalently, on the same network, we can do a diagnostic analysis and query the probability of A in state $\neg a$, given $C = \neg c$. The states of these variables are denoted as $1-a$ and $1-c$, for calculation purposes, i.e., $P(1-a|1-c)$. We study the derivatives of $P(a|c)$ with respect to a, b, and $c_{1,2,3,4}$. The interval derivatives show that, for this network, the inference posterior probabilities are monotone

A	a	$\neg a$
	a	$1-a$

B	b	$\neg b$
	b	$1-b$



A	a		$\neg a$	
B	b	$\neg b$	b	$\neg b$
c	c_1	c_2	c_3	c_4
$\neg c$	$1-c_1$	$1-c_2$	$1-c_3$	$1-c_4$

Fig. 1. Credal network with three node variables and related interval conditional probability tables.

in a and $c_{1,2,3,4}$. Therefore, we can use a combination of *endpoints analysis* and single use expressions to compute the best possible rigorous bounds with respect to these parameters. Thanks to monotonicity we can reduce the problem to a one variable interval propagation problem. For example, the inference interval probability $[P(a|c)]$ can be computed as follows

$$[P(a|c)] = \underline{P(a|c)}^{\boldsymbol{b}} \cup \overline{P(a|c)}^{\boldsymbol{b}} \tag{5}$$

where

$$\underline{P(a|c)}^{\boldsymbol{b}} = \frac{\underline{a}\ \boldsymbol{b}\ \underline{c_1} + \underline{a}\ (1-\boldsymbol{b})\ \underline{c_2}}{\underline{a}\ \boldsymbol{b}\ \underline{c_1} + (1-\underline{a})\ \boldsymbol{b}\ \overline{c_3} + \underline{a}\ (1-\boldsymbol{b})\ \underline{c_2} + (1-\underline{a})(1-\boldsymbol{b})\ \overline{c_4}} \tag{6}$$

and

$$\overline{P(a|c)}^{\boldsymbol{b}} = \frac{\overline{a}\ \boldsymbol{b}\ \overline{c_1} + \overline{a}\ (1-\boldsymbol{b})\ \overline{c_2}}{\overline{a}\ \boldsymbol{b}\ \overline{c_1} + (1-\overline{a})\ \boldsymbol{b}\ \underline{c_3} + \overline{a}\ (1-\boldsymbol{b})\ \overline{c_2} + (1-\overline{a})(1-\boldsymbol{b})\ \underline{c_4}} \tag{7}$$

The value of the first interval derivative straddling zero also shows that the inference probability is not monotone with respect to b, which is why we are going to need the *reducing template* for b or its *centred forms*. We can now compute the expression of Eq. 6 and Eq. 7, using the *SUE* of Eq. 9

$$P(a|c)^{\boldsymbol{b}} = \frac{a\ c_2\ +\ a\ (c_1 - c_2)\ \boldsymbol{b}}{a\ c_2 + (1-a)\ c_4\ +\ (a\ (c_1 - c_2) + (1-a)\ (c_3 - c_4))\ \boldsymbol{b}} = \frac{s}{u}\ -\ \left(\frac{s\ t}{u^2} - \frac{r}{u}\right) \Big/ \left(\frac{t}{u} + \boldsymbol{b}\right) \tag{8}$$

where, $r = a\ c_2$, $s = a\ (c_1 - c_2)$, $t = a\ c_2 + (1-a)\ c_4$, and $u = a\ (c_1 - c_2) + (1-a)\ (c_3 - c_4)$. The *SUE* of (8) can be used to obtain the *best possible* bounds on $P(a|c)^{\boldsymbol{b}}$ by means of interval arithmetic. Similarly, we can obtain *SUE* in b for the remaining combination of states: $P(\neg a|c)^{\boldsymbol{b}}$, $P(a|\neg c)^{\boldsymbol{b}}$, $P(\neg a|\neg c)^{\boldsymbol{b}}$. These results corresponding to all diagnostic queries involving a and c are summarised in Table 1. These expressions are reported in the original form (before rearrangement) for simplicity.

4.1 Reducing Template for Inference

In Bayesian inference, posterior probabilities are given in terms of rational explicit expressions. Thus, the examination of each interval variable node individually can often lead to an inference expression to the form of (9), which luckily comes with its respective *SUE*,

$$P_{\text{posterior}} = \frac{r + s\ \boldsymbol{x}}{t + u\ \boldsymbol{x}} = \frac{s}{u} - \frac{st/u^2 - r/u}{t/u + \boldsymbol{x}}. \tag{9}$$

Where $\boldsymbol{x}$ is the interval variable, and r, s, t, and u are coefficients that group all the remaining variables. The first derivative with respect to x is

$$\frac{d}{dx}\left(\frac{s}{u} - \frac{st/u^2 - r/u}{t/u + \boldsymbol{x}}\right) = \frac{st/u^2 - r/u}{(t/u + \boldsymbol{x})^2}. \tag{10}$$

Using automatic differentiation on the variable elimination algorithm we can obtain the gradient with respect to all variables without knowing the closed-form expression (9).

5 Conclusions

This paper has served as a demonstrator for the use of interval computer arithmetic to investigate the computational complexity of the dependency problem for the variable elimination algorithm in credal networks. The propagation of intervals through the partial derivatives computed with automatic differentiation has provided insights into the complexity of the dependency problem and has showed that such problem is manageable on small networks. Future studies will focus on connecting this strategy with approximate inference algorithms, which will enable larger networks. The appealing feature of this strategy in connection with

Table 1. Summary of explicit inference forms in single use expression for $\boldsymbol{b}$

Query	Single use expression for $\frac{r+s\,\boldsymbol{b}}{t+u\,\boldsymbol{b}} \rightarrow \frac{s}{u} - \left(\frac{s\,t}{u^2} - \frac{r}{u}\right) \Big/ \left(\frac{t}{u} + \boldsymbol{b}\right)$
$P(a\|c)$	$\dfrac{a\ c_2 + a\ (c_1 - c_2)\ \boldsymbol{b}}{a\ c_2 + (1-a)\ c_4 + (a\ (c_1 - c_2) + (1-a)\ (c_3 - c_4))\ \boldsymbol{b}}$
$P(a\|\neg c)$	$\dfrac{a\ (1-c_2) + a\ ((1-c_1) - (1-c_2))\ \boldsymbol{b}}{a\ (1-c_2) + (1-a)\ (1-c_4) + (a\ ((1-c_1) - (1-c_2)) + (1-a)\ ((1-c_3) - (1-c_4)))\ \boldsymbol{b}}$
$P(\neg a\|c)$	$\dfrac{(1-a)\ c_4 + (1-a)\ (c_3 - c_4)\ \boldsymbol{b}}{a\ c_2 + (1-a)\ c_4 + (a\ (c_1 - c_2) + (1-a)\ (c_3 - c_4))\ \boldsymbol{b}}$
$P(\neg a\|\neg c)$	$\dfrac{(1-a)\ (1-c_4) + (1-a)\ ((1-c_3) - (1-c_4))\ \boldsymbol{b}}{a\ (1-c_2) + (1-a)\ (1-c_4) + (a\ ((1-c_1) - (1-c_2)) + (1-a)\ ((1-c_3) - (1-c_4)))\ \boldsymbol{b}}$

approximate inference is its outward character capable of rigorously bounding the uncertainty due to imprecision, effectively confining the uncertainty within the Bayesian counterpart of inference process. The extension of this work to examples involving a bigger number of nodes and exploring the case of multiple states and n directly connected nodes is also envisioned.

Acknowledgments. We would like to acknowledge the support of EPSRC under grant number EP/R006768/1.

References

1. Antonucci, A., de Campos, C.P., Huber, D., Zaffalon, M.: Approximate credal network updating by linear programming with applications to decision making. Int. J. Approx. Reason. **58**, 25–38 (2015)
2. Augustin, T., Coolen, F.P., De Cooman, G., Troffaes, M.C.: Introduction to Imprecise Probabilities. Wiley (2014)
3. de Campos, C.P., Cozman, F.G.: Inference in credal networks using multilinear programming. In: Proceedings of the Second Starting AI Researcher Symposium, pp. 50–61 (2004)
4. de Campos, C.P., Cozman, F.G.: The inferential complexity of Bayesian and credal networks. In: IJCAI, vol. 5, pp. 1313–1318. Citeseer (2005)
5. Kjaerulff, U.B., Madsen, A.L.: Bayesian networks and influence diagrams **200**, 114 (2008)
6. Kyburg, H.E.: Bayesian and non-Bayesian evidential updating. Artif. Intell. **31**, 241–288 (1984)
7. Mauá, D.D., de Campos, C.P., Benavoli, A., Antonucci, A.: Probabilistic inference in credal networks: new complexity results. J. Artif. Intell. Res. **50**, 603–637 (2014)
8. Moore, R.E.: Interval Analysis. Oxford University Press (1966)
9. Pearl, J.: Probabilistic Reasoning in Intelligent Systems: Networks of Plausible Inference. Morgan Kaufmann (1988)
10. Tessem, B.: Interval probability propagation. Int. J. Approx. Reason. **7**(3), 95–120 (1992)

Homogenization as a Method for Modeling Processes in Photonic Crystals

Gennadiy V. Sandrakov[1,2(✉)]

[1] Taras Shevchenko National University of Kyiv, 64,Volodymyrska Street, Kyiv 01601, Ukraine
gsandrako@gmail.com
[2] Pukhov Institute for Modelling in Energy Engineering of NAS of Ukraine,
15 G. Naumov Street, Kyiv 03164, Ukraine

Abstract. The homogenizations of initial boundary value problems for wave equations with asymptotically degenerate rapidly oscillating periodic coefficients are considered. Such problems model wave processes in heterogeneous periodic media like photonic crystals. The homogenized problems have the form of initial boundary value problems for integro-differential equations in convolutions and with external forces, even if in the original problem the forces were zero. The appearance of problems with convolutions indicates the possible presence of gaps in the spectrum of the relevant model, as is the case with semiconductors. Statements on the approximation of solutions to the original problem by solutions of the homogenized problem in various cases are discussed.

Keywords: Wave processes · Homogenized problems · Periodic media

1 Homogenization Methods

Understanding the processes in crystals is useful, for example, for designing computers. Such processes are modeled on the basis of the Schrodinger equation. As a rule, crystals have a periodic structure, which is characterized by the periodicity of the Schrodinger equation potential. By analogy with such crystals, physicists are developing photonic crystals, where light waves propagate. There are books where it is proposed to model processes in photonic crystals based on the Schrodinger equation. This approach is not entirely suitable, since light wave processes are modeled by Maxwell's equations, which form a system and can sometimes be reduced to a wave equation in contrast to the Schrodinger equation. The speed of such waves is much higher than the speed of waves in conventional crystals, and this may be useful for some applications. The periodicity of photonic crystals is also natural technologically. Having chosen a cell of periodicity, one can continue the cell in a periodic way to obtain a photonic crystal, similar to how walls and squares are covered with tiles. Usually there are a lot of such cells and the crystal has two scales, which are characterized by its size and the number of cells. Thus, composite continuous two-scale media arise with a periodic structure, which are similar to composite materials with microscale. And modeling, for example, electromagnetic and wave processes in such continuous media becomes relevant and natural.

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2026
I. Potapov et al. (Eds.): RTC-Digital 2023, CCIS 2647, pp. 119–126, 2026.
https://doi.org/10.1007/978-3-032-04731-1_16

Homogenization theory was developed by I. Babuska, N. S. Bakhvalov, A. Bensoussan, J.-L. Lions, and G. Papanicolau to simplify computer simulations for composite material models (references can be found, for example, in [1, 2]). Accordance to the method the simulation of a process in a composite with a periodic structure can be replaced by the simulation of some process for a homogeneous (homogenized) material with a suitable approximation. Appropriate theorems on the approximation were proved, for example, in [1, 2]. Artificial composites have proven their usefulness and application in many fields, and perhaps the creation of artificial photonic crystals will also be useful for applied purposes. However, it is unnatural to create artificial composites from materials with similar properties. But, if the properties of the components of the composite differ greatly, then this difference should be compared with the microscale parameter, since the models can be completely different, which will be checked further. Thus, when composite materials composed of materials with very different properties, additional small and large parameters may appear in models. Homogenization for some initial boundary value problems with several parameters was considered, for example, in [3–5], where homogenized problems with convolutions are obtained and theorems on approximation were proved. The presence of convolutions in models is called the memory effect [6]. Note that the presence of convolutions in such models is far from trivial, since the spectrum of operators has gaps like those of semiconductors. Accordingly, in the absence of convolutions, such gaps do not arise. More detailed information and precise statements can be found in [5]. Here we consider some applications of homogenization methods for modeling wave processes in composite materials, since some photonic crystals are designed as composite materials with a periodic structure. The exact formulation of such a model as an initial boundary value problem for a wave equation with several parameters is given in the next section.

2 Models of Wave Processes for Composites with a Periodic Structure

The purpose here is to investigate models of wave processes in a composite with a periodic structure, which occupies a bounded domain $\Omega \subset \mathbb{R}^n$ and is composed of two materials with very different properties. It is assumed that the domain Ω is represented as a union of two domains $\Omega_1^\varepsilon = F_1^\varepsilon \cap \Omega$ and $\Omega_0^\varepsilon = F_0^\varepsilon \cap \Omega$ with a periodic structure determined by a small positive parameter ε and periodic $F_1^\varepsilon, F_0^\varepsilon$. The domains correspond to two materials with very different properties. Their density and conductivity coefficients are defined by the formulas

$$\lambda_\sigma^\varepsilon = \sigma\, \lambda_0 \quad in \quad \Omega_0^\varepsilon \quad and \quad \lambda_\sigma^\varepsilon = \lambda_1 \quad in \quad \Omega_1^\varepsilon,$$

$$m_\mu^\varepsilon = \mu\, m_0, \quad s_\mu^\varepsilon = \mu\, s_0 \quad in \quad \Omega_0^\varepsilon \quad and \quad m_\mu^\varepsilon = m_1, \quad s_\mu^\varepsilon = s_1 \quad in \quad \Omega_1^\varepsilon,$$

where m_0, s_0, m_1, s_1 are positive constants and the constant matrices λ_0 and λ_0 are symmetric and elliptic. Here, it is assumed that $n \geq 2$. Moreover, positive ε is small parameter, σ and μ are finite parameters (where $0 < \sigma \leq 1$ and $0 < \mu \leq 1$ for precise determinations).

Define $u = u(t, x)$ as a solution to the following initial boundary value problem

$$m_\mu^\varepsilon u'_{tt} + s_\mu^\varepsilon u'_t - div\, \lambda_\sigma^\varepsilon(\nabla u) = 0 \quad in \quad \Omega \times (0, T), \tag{1}$$

$$u|_{t=0} = \rho^\varepsilon u_0, \quad u'_t|_{t=0} = \varphi^\varepsilon u_1 \quad in \quad \Omega, \quad u = 0 \quad on \quad \partial\Omega \times (0, T),$$

where $\rho^\varepsilon = \rho(x/\varepsilon)$, $\varphi^\varepsilon = \varphi(x/\varepsilon)$ and it is assumed that $\rho(y)$ and $\varphi(y)$ are 1-periodic regular functions. The initial data $u_0 = u_0(x)$ and $u_1 = u_1(x)$ are also regular and compactly supported functions by assumption. Moreover, $F_1^\varepsilon = \varepsilon F_1$ and $F_0^\varepsilon = \varepsilon F_0$ by definition, where the set F_1 is an open connected 1-periodic (periodic with period 1 for each $x_1, \ldots, x_n$) subset of $\mathbb{R}^n$ with a locally Lipschitz boundary and $F_0 = \mathbb{R}^n \backslash \overline{F}_1$ is a set with a locally Lipschitz boundary.

Thus, $Y = (0, 1)^n$ is a periodicity cell and the sets F_1 and F_0 with the common boundary ∂F_1 are completely determined by the sets $Y_1 = F_1 \cap Y$ and $Y_0 = F_0 \cap Y$ with the boundary $\Gamma = \partial F_1 \cap Y$. The sets Y_1 and Y_0 split the periodicity cell Y into two sets corresponding to different materials of the composite media that are separated by a common boundary Γ.

Following [2], the problem (1) can be rewritten in an equivalent form as an initial boundary value problem with conjugation conditions at the interface of the domains:

$$\mu m_0 u'_{tt} + \mu s_0 u'_t - \sigma\, div\, \lambda_0(\nabla u) = 0 \quad in \quad \Omega_0^\varepsilon \times (0, T),$$

$$m_1 u'_{tt} + s_1 u'_t - div\, \lambda_1(\nabla u) = 0 \quad in \quad \Omega_1^\varepsilon \times (0, T), \tag{2}$$

$$u|_{t=0} = \rho^\varepsilon u_0, \quad u'_t|_{t=0} = \varphi^\varepsilon u_1 \quad in \quad \Omega, \quad u = 0 \quad on \quad \partial\Omega \times (0, T),$$

$$\Upsilon_\varepsilon \cdot (A_1^\varepsilon \nabla u) = \sigma\, \Upsilon_\varepsilon \cdot (A_0^\varepsilon \nabla u) \quad on \quad \Gamma_\varepsilon \times (0, T),$$

and $u \in H_0^1(\Omega)$, where the last equality means the continuity of flows on the internal boundary (contact surfaces of the considered media Γ_ε) is satisfied for almost all $t \in (0, T)$. Here and in what follows Υ_ε denotes the outer normal to the locally Lipschitz boundary Γ_ε.

The equivalent form (2) for problem (1) clarifies exactly that one can consider media with small coefficients of density and conductivity on the domain Ω_0^ε for small μ and σ. It is clear that one can multiply the equation in problem (2) by σ^{-1} and consider the case of large coefficients of density and conductivity on the domain Ω_1^ε for small σ and $\mu = \sigma$. Accordingly, by multiplying these equations by $\sigma^{-1/2}$, one can consider the case of small coefficients on Ω_0^ε and large ones on Ω_1^ε for small σ and $\mu = \sigma$. Other options are possible when multiplying by $\mu^{-\kappa}$ for a fixed positive κ. Of course, such multiplications do not affect the final form of approximations for solutions to problem (1) for sufficiently small ε and possibly σ.

3 Regular Homogenized Solution Approximations

For fixed parameters μ, σ and ε, problem (1) has a unique solution, for example, in accordance with [6]. Thus, one can try to construct an approximation for the solution. The equation of problem (1) is hyperbolic for fixed parameters. But, for very small μ or

σ, this hyperbolicity may degenerate. Hence, to begin with, we will consider the simpler case $\sigma = 1$, which is regular in some sense. For a precise formulation of homogenized problems, additional definitions are needed. Let us define constants and matrices with constant components by

$$m = \int\limits_Y m_\mu^1(y)\,dy, \quad s = \int\limits_Y s_\mu^1\,dy, \quad m_\rho = \int\limits_Y m_\mu^1\,\rho^1 dy, \tag{3}$$

$$m_s = \int\limits_Y s_\mu^1 \rho^1 dy, \quad m_\varphi = \int\limits_Y m_\mu^1 \varphi^1 dy, \quad \lambda = \int\limits_Y (\lambda_1^1 + \lambda_1^1 \nabla_y N)\,dy.$$

where the vector function $N = N(y)$ is a 1-periodic solution of the following problem $-\,div_y\,\lambda_1^1(\nabla_y N\,) = div_y\,\lambda_1^1$, which is usually called *a cell problem.*

In the case, the homogenized initial boundary value problem for the function $v = v(t, x)$ is

$$m\,v'_{tt} \;+\; s\,v'_t \;-\; div\,\lambda\,(\nabla\,v\,) \;=\; 0 \quad in \quad \Omega \times (0, T), \tag{4}$$

$$v|_{t=0} = m^{-1} m_\rho\,u_0, \quad v'_t|_{t=0} = m^{-1} m_\varphi\,u_1 + m^{-1} m_{s\rho} u_0 \quad in \quad \Omega, \quad v = 0 \quad on \quad \partial\Omega \times (0, T),$$

where the coefficient $m_{s\rho} = m_s - s\,m^{-1} m_\rho$ is equal to zero if $\rho^\varepsilon = 1$ in problem (1). What was unexpected in this problem was the appearance of an additional initial velocity.

It is known [2] that the matrix λ is elliptic and a unique solution of problem (4) exists and is regular. Consequently, following [3, 4], it is possible to check that the approximation accuracy

$$\|\,u - v\,\|^2_{C^0([0,T];L^2(\Omega_1^\varepsilon))} \;+\; \mu\;\|\,u - v\,\|^2_{C^0([0,T];L^2(\Omega_0^\varepsilon))} \;\le\; C\,\varepsilon, \tag{5}$$

is valid, where the constant C is independent of μ and ε for $0 < \mu \le 1, 0 < \varepsilon \le \varepsilon_0$ with relevant ε_0. This means that estimate (5) is uniform with respect to μ and therefore also holds for small μ. In the latter case, one can write $(\varepsilon + \mu^2)$ in the right side of (5).

Thus, for modeling wave processes, instead of solving problem (1) (for example, numerical), one can solve the homogenized problem (4) with a suitable approximation for small ε. Naturally, the numerical solution of problem (1) is much more difficult than the solution of problem (4), since a very fine grid is required to take into account the complex geometry of the donains Ω_1^ε and Ω_0^ε. Thus, a rather complex periodic medium can be replaced by a homogeneous medium with guaranteed accuracy, which is essential for modeling.

4 Homogenized Solution Approximations with Oscillations

Let us further assume that σ is a small parameter in problem (1). In this case, the hyperbolicity of problem (1) is lost on the domain Ω_0^ε and additional definitions are needed. To present an approximation of the solution, we define constants $\widehat{m}$, $\widehat{s}$, $\underset{\rho}{\widehat{m}}$, $\underset{s}{\widehat{m}}$, $\underset{\varphi}{\widehat{m}}$, $\underset{s\rho}{\widehat{m}} = \underset{s}{\widehat{m}} - \widehat{s}\,m^{-1}\,\underset{\rho}{\widehat{m}}$, and a constant matrix $\widehat{\lambda}$ using the equalities (3), where integration

over Y is replaced by integration over Y_1 and the vector function $N(y)$ is a 1-periodic solution of the following problem: $div_y(\lambda_1 \nabla_y N) = 0$ *in* Y_1, $-\Upsilon \cdot (\lambda_1 \nabla_y N) = \Upsilon \cdot \lambda_1$ *on* ∂Y_1, where Υ is the outward normal to ∂Y_1 and the dot denotes the inner product of vector functions.

In addition, let the functions $p_m = p_m(t, y)$, $p_\rho = p_\rho(t, y)$, $p_\varphi = p_\varphi(t, y)$, $p_s = p_s(t, y)$, and $p_{ss} = p_{ss}(t, y)$ be 1-periodic solutions of the initial boundary value problems on Y_0:

$$\mu\, m_0(p_m)'_{tt} + \mu\, s_0(p_m)'_t - \vartheta\, div_y(A_0 \nabla_y p_m) = 0 \quad in \quad Y_0 \times (0, \infty),$$

$$p_m|_{t=0} = 0, \quad (p_m)'_t|_{t=0} = 1 \quad in \quad Y_0, \quad p_m = 0 \quad on \quad \overline{Y}_1 \times (0, \infty), \tag{6}$$

$$\mu\, m_0(p_\rho)'_{tt} + \mu\, s_0(p_\rho)'_t - \vartheta\, div_y(A_0 \nabla_y p_\rho) = 0 \quad in \quad Y_0 \times (0, \infty),$$

$$p_\rho|_{t=0} = 0, \quad (p_\rho)'_t|_{t=0} = \rho(y) \quad in \quad Y_0, \quad p_\rho = 0 \quad on \quad \overline{Y}_1 \times (0, \infty)$$

and similar boundary value problems for the remaining functions, where $\vartheta = \sigma/\varepsilon^2$ is considered as a parameter. It is well known [6] that the relevant solutions of problems (6) exist and are uniquely defined. Following [5], one can check that the functions $q_m = q_m(t)$, $q_\rho = q_\rho(t)$, …, $q_{ss} = q_{ss}(t)$ are well defined as functions of time by the formulas

$$q_m = \int_{Y_0} m_0(p_m)'_{tt} dy + \int_{Y_0} s_0(p_m)'_t dy, \quad \dots, \quad q_{ss} = \int_{Y_0} m_0(p_{ss})'_{tt} dy + \int_{Y_0} s_0(p_{ss})'_t dy,$$

In the case, the homogenized initial boundary value problem for the function $v = v(t, x)$ is

$$\widehat{m}\, v'_{tt} + s\, v'_t - q_m * (v'_{tt}) - q_s * (v'_t) - div\, \lambda\, (\nabla\, v) = (\widehat{\underset{\rho}{q}})'_t u_0 + \varphi t\, u_1 + \widehat{\underset{s\rho}{q}}\, t\, u_0, \tag{7}$$
$$v|_{t=0} = m^{-1} m_\rho\, u_0, \quad v'_t|_{t=0} = m^{-1} m_\varphi\, u_1 + m^{-1} m_{s\rho} u_0 \quad in \quad \Omega, \quad v = 0 \quad on \quad \partial\Omega \times (0, T),$$

where $\widehat{\underset{\rho}{q}} = q_\rho - \widehat{\underset{\rho}{m}}\, \widehat{m}^{-1} q_m$, $\widehat{\underset{\varphi}{q}} = q_\varphi - m\widehat{\underset{\varphi}{}}\, \widehat{m}^{-1} q_m$, $\widehat{\underset{s\rho}{q}} = q_{ss} - \widehat{\underset{s\rho}{m}}\, \widehat{m}^{-1} q_m - \widehat{\underset{\rho}{m}}\, \widehat{m}^{-1} q_s$ are zero functions if $\rho = 1$ and $\varphi = 1$ in problem (1). Here the star $*$ denotes the convolution operator with respect to the variable t, for example, by definition $q_m * (v''_t) = \int_0^t q_m(t - \tau)(v''_{\tau\tau}(\tau, x))\, d\tau$.

In comparison with homogenized problem (4), the homogenized problem may contain convolutions and non-trivial external forces. In terms of elasticity, the latter means that the presence of microstresses in the composite can lead to the action of external forces on it, the so-called effect of residual stresses. According to (4), the effect can be observed only for composites with contrast coefficients for the models under consideration. Following [5], one can verify that the unique solution v to (7) exists and the approximation for solution to (1) is

$$v_a = v + (\tilde{p}^\varepsilon_\rho)'_t u_0 + (\tilde{p}^\varepsilon_\varphi)\, u_1 + (\tilde{p}^\varepsilon_{s\rho})\, u_0 - p^\varepsilon_s * (v'_t) - p^\varepsilon_m * (v'_{tt}), \tag{8}$$

where $\tilde{p}^{\varepsilon}_{\rho} = p^{\varepsilon}_{\rho} - \widehat{\underset{\rho}{m}}^{-1} p^{\varepsilon}_{m}, \tilde{p}^{\varepsilon}_{\varphi} = p^{\varepsilon}_{\varphi} - \widehat{\underset{\varphi}{m}}^{-1} p^{\varepsilon}_{m}, \tilde{p}^{\varepsilon}_{s\rho} = p^{\varepsilon}_{ss} - \widehat{\underset{s\rho}{m}}^{-1} p^{\varepsilon}_{m} - \widehat{\underset{\varphi}{m}}^{-1} p^{\varepsilon}_{s}$, $p^{\varepsilon}_{\rho} = p_{\rho}(t, x/\varepsilon)$, $p^{\varepsilon}_{s} = p_{s}(t, x/\varepsilon)$, $p^{\varepsilon}_{m} = p_{m}(t, x/\varepsilon)$ are defined by (6) and are zero on Ω^{ε}_{1}.

Consequently, following [4, 5], it is possible to check that the approximation accuracy

$$\| u - v \|^{2}_{C^{0}([0,T];L^{2}(\Omega^{\varepsilon}_{1}))} \quad \leq \quad C\ (\varepsilon + \sigma) \tag{9}$$

is valid, where the constant C is independent of μ, σ and ε for $0 < \mu \leq 1$, $0 < \sigma \leq \sigma_0 0 < \varepsilon \leq \varepsilon_0$ with relevant σ_0, ε_0. This means that estimate (9) is uniform with respect to μ and so also holds for small μ. In the latter case, one can write $(\varepsilon + \sigma + \mu^2)$ in the right side of (9). However, the relevant accuracy estimate on the set Ω^{ε}_{0} is much more complicated, which is associated with the specific regularity of solutions to problems (6).

Therefore, by definition v_a, the approximation of solutions to problem (1) contains five rapidly oscillating terms on the set Ω^{ε}_{0}, and these terms are not trivial, since they are determined by nontrivial problems (6). In addition, the last term is determined by the second derivative of the homogenized solution. The representation of wave processes is quite complex, but it can be simulated by numerical methods, where instead of solving problem (1) (for example, numerically), it is possible to solve the homogenized problem (7) with relevant approximation for small ε and σ. The appearance of strongly oscillating terms in the approximation of solution (8) also characterizes the energy loss on a set with low conductivity and can be used to design wave-absorbing composite materials. In fact, from results [5] it follows that the absorption properties depend on the parameter $\vartheta_{\mu} = \sigma/\varepsilon^{2}\mu$, on which problems (6) actually depend. And as proven in [5], there may be gaps in the absorption of the width ϑ_{μ}, which may have practical significance. Thus, homogenization methods can be useful for modeling wave processes in composites similar to photonic crystals. The methods are especially relevant for composites with very different properties. Undoubtedly, the transfer of the methods to the system of Maxwell's equations is very topical and perspective.

5 Computational Aspects of Such Modeling by Homogenization

Let's explain the advantages of this approach to studying problem (1). If the assumptions of Sect. 3 are met, then one can solve computationally with guaranteed accuracy by virtue of (5), instead of problem (1), problem (4) for an equation with constant coefficients, which is much simpler. Moreover, by separating the variables, it is possible to present the exact solution to problem (4) in the form of a series through the eigenfunctions of the operator in (4). For some domains Ω, such eigenfunctions are well known, otherwise they can be found computationally.

In turn, if the assumptions of Sect. 4 are met, then the computational situation is much more complicated. In fact, for very small σ one cannot find a computational solution to problem (1) with guaranteed accuracy, since standard accuracy estimates contain σ^{-1} as a factor and it is necessary to choose a grid step to solve problem (1) significantly smaller than σ^{-1}. In addition, this grid step must be significantly smaller than ε in order to take into account the geometry of the sets Ω^{ε}_{0} and Ω^{ε}_{1}. Consequently, even by choosing

a very small grid step, one cannot guarantee the accuracy of the numerical calculations, since accuracy estimates strongly depend on the geometry of these sets. This is actually confirmed by expansion (8), which is a good approximation to the solution of problem (1) due to estimate (9) and is quite complexly oscillating on the set Ω_0^ε, which indicates the absorption of energy on the set. Following [5], the magnitude of such absorption can be calculated, which may be actually for applications.

Thus, it remains to explain the computational aspects for problem (7), the solution of which approximates the solution of problem (1) due to (9), at least on the set Ω_1^ε. Note that a similar estimate has been carried out on the set Ω_0^ε, but the formulation of such an estimate is much cumbersome. Details for related problems can be found in [4] and [5]. Next, we remark that problem (7) has constant coefficients and the only unusual thing is the presence of convolutions with respect to the time variable. It is known [5] that these convolutions are correctly defined and are expressed through solutions to problems (6). Computational solution of problems (6) is standard. Moreover, these solutions can be represented as series through the eigenfunctions of the operator in (6). Therefore, it is possible to solve problem (7) computationally in parallel with problems (6) or using series to represent solutions (6). This solution is standard and has guaranteed accuracy, at least for a fixed parameter ϑ/μ in (6).

Naturally, in a specific problem the parameter ϑ/μ is fixed. Moreover, if in the problem the parameter μ is small, then it is easy to check that the influence of convolutions is of the order of μ and therefore these convolutions can be considered equal to zero in (7). Consequently, problem (7) for small μ, like problem (4), has constant coefficients and can be solved by standard computational methods with guaranteed accuracy. In addition, the oscillating terms in expansion (8) are not significant for small μ, which can be verified by following [5].

In conclusion, we note that the results presented for problem (1) are new. Typically, this problem was studied for external forces with homogeneous initial conditions and for $s_\mu^\varepsilon = 0$. But, for homogeneous initial conditions without external forces, the solution to problem (1) is trivial. Whereas, as noted for problem (7), inhomogeneous initial conditions can induce external forces in a homogenized problem, but the converse is not true, as is well known. Note also that when applied to photonic crystals, it is necessary to simulate the passage of waves through them. Such waves, of course, do not satisfy homogeneous initial conditions. Thus, the cases considered here are of greatest interest for such wave propagation modeling.

Acknowledgments. This work has been partially supported by the Ministry of Education and Science of Ukraine Grant for perspective development of a scientific direction "Mathematical sciences and natural sciences" at Taras Shevchenko National University of Kyiv.

References

1. Bensoussan, A., Lions, J.-L., Papanicolau, G.: Asymptotic Analysis for Periodic Structures. North-Holland, Amsterdam, New York, Oxford (1978)
2. Bakhvalov, N.S., Panasenko, G.P.: Homogenization: Averaging Processes in Periodic Media. Kluwer, Dordrecht (1989)

3. Sandrakov, G.V.: The homogenization of nonstationary equations with contrast coefficients. Dokl. Math. **56**(1), 586–589 (1997)
4. Sandrakov, G.V.: Homogenization of nonstationary problems of the theory of strong nonhomogeneous elastic media. Dokl. Akad. Nauk SSSR **358**(3), 308–311 (1998)
5. Sandrakov, G.V.: Multiphase homogenized diffusion models for problems with several parameters. Izvestiya Mathematics **71**(6), 1193–1252 (2007). https://doi.org/10.1070/IM2007v071n06ABEH002387
6. Duvaut, G., Lions, J.-L.: Les Inequations en Mecanique et en Physique. Dunod, Paris (1972)

New Cellular Automata. Models, Algorithms, Applications and Anticipation Accounting

Aleksander Makarenko(✉)

Institute of Applied System Analysis at National Technical University of Ukraine (I. Sykorski KPI), Prospect Pobedy 37, Building 35, Kyiv 03056, Ukraine
makalex51@gmail.com

Abstract. New generalizations of cellular automata CA are described. This survey contains recent results published in research journals, conference proceedings etc. Cellular automata had been considered in the restricted regions of space. In this work, we propose the concepts for modelling logical gates by propagation of the gliders of cellular automata in a bounded domain and consider special roles for collisions of gliders with walls and obstacles. Also, the general formulations and properties of cellular automata with the strong anticipatory property of cells (introduced by D. Dubois) are considered. Multivalued behaviour (hyper-incursion) of solutions of such CA is described. This poses new research problems of computation theory related to the presumable multivaluedness of cellular automata with strong anticipation properties. Extending of classical automata, Turing machine and algorithms had been proposed. Also, some relations of such cellular automata and quantum mechanics are discussed. Applications of cellular automata can remind well-known processes in real life like football games; migration in science and higher education; epidemic spreading; and artificial life.

Keywords: Cellular automata · new models · applications · anticipation

1 Introduction

Cellular automata (CAs) are a very interesting class of models, both in terms of orthographic and applications (see [1–6] and many others). The idea of classical cellular automata is roughly as follows:

- the space is divided into identical elements (for example, as in the classic game “life” - into square cells on the plane);
- each element has a finite number of states (in the game of life, there are only two states);
- classical CA models are considered at discrete moments of time;
- dynamic laws for changing the states of the cell from the previous moment to the next moment are formulated;
- dynamic laws for changing the states of each cell are constructed in such a way that their formulas depend only on the values of cell states in the local neighbourhood of the cell (for example, in the game “life”, a neighbourhood of 4 cells (Neumann neighbourhood), or a neighbourhood of 8 cells (Moore neighbourhood)).

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2026
I. Potapov et al. (Eds.): RTC-Digital 2023, CCIS 2647, pp. 127–134, 2026.
https://doi.org/10.1007/978-3-032-04731-1_17

All these settings hold for classical cellular automata which have a large number of both theoretical studies and applied applications. However, non-classical CAs are also being developed in parallel. For example, CAs with fuzzy states of elements, CAs with memory, CAs with special geometry of the underlying space, including those on graphs, probabilistic CAs, and many others.

The proposed article describes the results of constructing generalized models of cellular automata, studying the properties of such CAs and their applications. The following are mentioned among such CAs: Non-classical cellular automata; Elements constructed from CA for emulation of logical operation; Non-determinism in CA; Demography and migration; Crowd movement; Football modelling; Epidemic modelling.

Particularly interesting are cellular automata with anticipation (accounting of the future), which have the possibility of multiple states of cellular automata (i.e., many states of one cell element at a fixed time). The article concludes with several directions for further research.

2 Short Description on Investigations on CA

2.1 Some Theoretical Investigations on CA

Here we pose the short titles and references on some of our theoretical research related to the cellular automata: non-regular elements in CA [7]; probability in local withinities [8]; multilevels CA [9]; modeling of logical gates by CA in restricted domain [10]; CA with the anticipation property [11]; artificial Life in cellular space [12]; CA of game "Life" in restricted domain [13]; identification problems for CA rules [14].

2.2 Some New Applications of Cellular Automata

Here we pose the short titles and references on some of our research related to the cellular automata applications: prediction of electoral processes with cellular automata [15]; prediction of pedestrian crowds movement [16]; football modeling by cellular automata [17]; epidemic modeling [18]; demography modeling [19]; mobility of students and researches [20].

2.3 List of Multivalued Cellular Automata Investigations

Here we pose the short titles and references on some our theoretical research related to the cellular automata in multivalued case, especially of.CA with strong anticipation: cellular automata with strong anticipation [21]; multivalued cellular automata [22]; CA and non-determinism [23]; CA in quantum mechanic [24]; CA and theory of computation [25]; complex-valued CA with branching [26].

3 The Three Most Interesting Research Results

3.1 Multivalued Cellular Automata with Strong Anticipation

In this paragraph, we give some results on multivalued CA with strong anticipation research. Considered in literature until now cellular automata (CA) had dynamic laws in which the state on the next temporal step (with discrete time) relied only on the state of elements in the local neighbouring in the previous moments of time (see for example [1–3]). Consideration of the possibility of foresight in the dynamics of elements leads to equations with anticipation.

The term 'anticipation' has been introduced in active use in biology and applied mathematics by Robert Rosen [27]. So this name is well known for common researchers. Less known is the contribution by Daniel Dubois to strong anticipation. Since the beginning of 90-th in the works by D.Dubois – see [28, 29] the idea of strong anticipation had been introduced: "Definition of an incursive discrete strong anticipatory system …: an incursive discrete system is a system which computes its current state at the time, as a function of its states at past times , ..., $t-3, t-2, t-1$ present time, t, and even its states at future times $t+1, t+2, t+3$, ...

$$x(t+1) = A(..., x(t-2), x(t-1), x(t), x(t+1), x(t+2), ..., p) \tag{1}$$

where the variable x at future times $t+1$, $t+2$, $t+3$, ... is computed in using the equation itself".

In the general case it follows formal dynamic equations (for the case of discrete-time of t = 0, 1…, n…) (see for example [21–24, 28, 29]):

$$s_i(t+1) = G_i(\{s_i(t)\}, ..., \{s_i(t+g(i))\}, R), \tag{2}$$

where the R is the set of external parameters (environmental, control etc.) is, {s(t)} - the state of the system in the moment of t, g(I) - horizon of foresight, {G} – a collection of nonlinear functions for the calculation of change of the state of elements. These equations are poorly explored even in mathematics. Therefore here we will pose only the pair of the simplest examples necessary for the exposition the ideas of the article.

One of the simplest examples of cellular automata is the well-known game «Life» by J.Conway ([1–3]). In the «classic" game "Life» by Conway every cell can have only two states (we will designate them 0 and 1, S = {0,1}); neighbouring, which is used for the record of local rules of change –Moor's neighbouring, consisting of eight neighbouring cells:

In most descriptions of the game «Life» following biological associations are proposed: two states of the cell 0 or 1 are interpreted as «dead" or «living» cells. Then the changes of the state of cells can be described by the following rule:

Let in the moment of t some subset of cells is made by living cells. In the moment of time of t + 1 those cells which in the moment of t fulfilled conditions will be living:

1. If a living cell has two or three living neighbors in its neighboring, it remains living in the next moment of time. Otherwise, the cell dies.
2. If a dead cell has exactly three living neighbors in neighboring, then she becomes «living» in a next moment of time.

All «births» and «deaths» take place simultaneously.

Rules for calculation of the states in the next moment of time in the game «Life» can be written down how some functions of transition are, the representing states in the local neighboring of cell in the state of the cell in the next moment of time. At the consideration of foresight (in the simplest variant) this state relies also on the possible states of cells in the local neighboring also and in the next moment of time.

These results were described in a separate publication (see for example [20–24])), but one of the most fundamental results of accounting anticipation which is important for us is the possibility of the existence of many branches of solutions (possibility of a few states of every cell in a cellular automaton in a given moment of time). Here in the picture, we pose one of the examples of such solutions (Fig. 1):

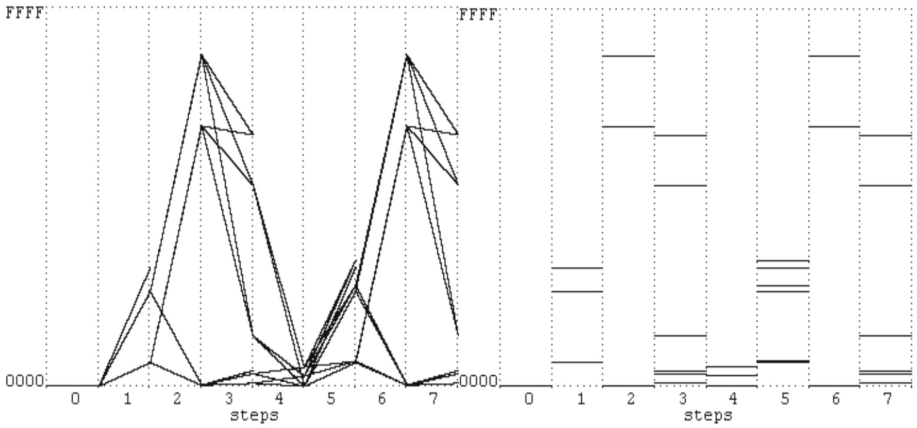


Fig. 1. Solution of CA with strong anticipation with a few possible states (on abscise axis the steps of discrete time are postponed, and on a y-axis the number of different system configurations is represented (each configuration is the collection of the single–valued states of all cells of the system)).

In the given example we also display the number of the possible states. Further possibilities, as well as details, have been suggested in other publications. So, the basic distinctive feature is the multiplicity of solutions and presumable configurations at every instant. Possible consequences of these facts and the problems of reconstruction of automata theory may be considered.

3.2 Modelling of Logical Gates by CA in a Restricted Domain

As far as we are concerned, the implementation of logic gates we have made is based on three points. First of all the information is no longer represented by a glider beam as described above but by a single glider. We no longer need to resort to glider guns, which is a big novelty. Then we set up a particular network with “walls”. Each logical gate is a particular configuration of space, a feature that is exploited by bouncing the gliders wisely as described in part 4. Finally, the method relies on collisions between gliders. And especially the two collisions that we analyzed in part 2.

Generally, the logic gates set up have two ducts at the top (representing the two inputs). A glider in the conduit means that the entry is at 1 otherwise it is at 0 (see Fig. 2). And a conduit down (representing the exit). The particular configuration of the rest of the network will allow or not to obtain a glider in the lower conduit depending on the nature of the logic gate.

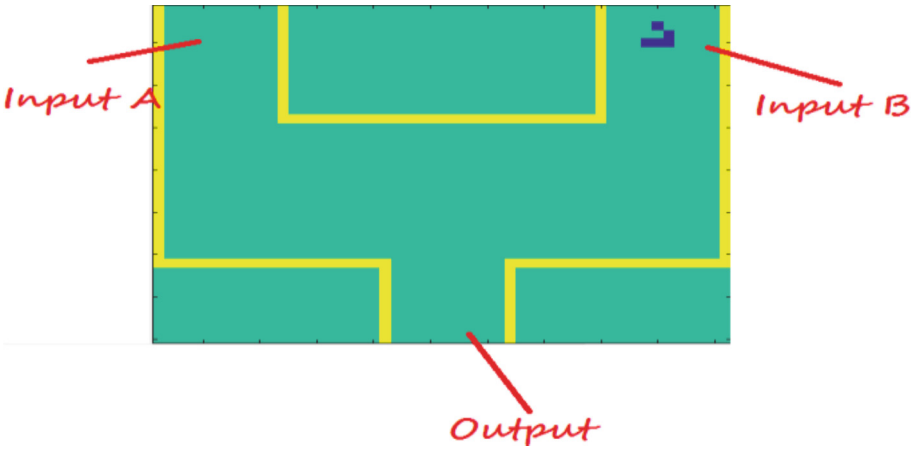


Fig. 2. Logic gate with two inputs $A = 0$ and $B = 1$.

Each of the four logic gates has been implemented. Namely: the NOT gate, the OR gate, the AND gate and the XOR gate.

3.3 Mobility of Students and Researchers

The problem of migration of researchers and students is very relevant, especially for Ukraine in the current situation. A method of forecasting this process is proposed. Computer programs for cellular automata models are proposed and implemented. A flexible structure of computer programs is developed and the structure of software modules and their interaction is proposed. The migration calculation was made for a server-based solution based on Guscuit for an unlimited number of users with the ability to access the application from any mobile device. The implementation used the Model-View-Controller concept. The model provides data and responds to controller commands. The view is responsible for displaying the model data. This technology allows you to modify each component independently of each other (Fig. 3).

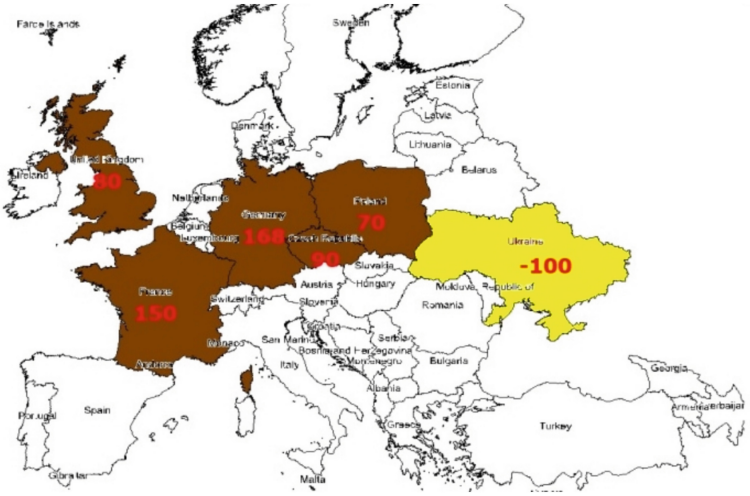


Fig. 3. Final distribution of Ukrainian students in Western Europe (one of the scenarios).

Acknowledgments. The authors would like to thank their colleagues, PhD students and students for helping in the research, especially for D. Krushinski, J. Brajon, L. Facchetti, A. Popov, B. Goldengorin.

References

1. Wolfram, S.: New kind of science. Wolfram Media Inc. (2023)
2. http://en.wikipedia.org/wiki/Cellular_Automaton
3. Illiachinski, A.A.: Cellular Automata. A Discrete Universe. World Scientific Publishing, Singapoure (2001)
4. Liu, Y.: Modelling Urban Development With Geographical Information Systems and Llular Automata. CRC Press (2009)
5. Chopard, B., Droz, M.: Cellular Automata Modeling of Physical Systems. Cambridge Univ. Press, Cambridge (1998)
6. Barto, F.: Tagliabae Jacopo Cellular Automata . The Stanford Encyclopedia of Phylosophy (Spring 2022 Edition). Edward N. Zalta (ed.) (https://plato.stanford.edu/archives/spr2022/entries/cellular_automata/)
7. t'Hooft The Cellular Automation Interpretation of Quantum Mechanics ArXiv: [qiant-ph] 1405.1548, v3, 259 (2015)
8. Grossing, G., Zellinger, A.: Structures in quantum cellular automata physica. B **151**, 366–370 (1988)
9. Facchett, L., Makarenko, A.: 'Game of life' with modifications: non-regular space, different rules and many hyerarchical levels. Int. J. Inf. Content Process. **4**(1), 21–50 (2017)
10. Makarenko, A., Brajon, J.: Logical gates on gliders in restricted space domain cellular automata springer nature switzerland AG. In: 2021 Gwizdałła, T.M., et al. (eds.): ACRI 2020, LNCS 12599, pp. 1–4 (2021). https://doi.org/10.1007/978-3-030-69480-7_22
11. Makarenko, A.: Multivaluedness in cellular automata with strong anticipation and some prospects for computation theory. WSEAS Trans. Inf. Sci. Appl. **17**, 69–78 (2020). https://doi.org/10.37394/23209.2020.17.8

12. Zavertaniy, V., Makarenko, A.: Elements of simulation of problems of artificial life. monograph. NSC IPSA at NTUU KPI named after Igor Sikorsky, Kyiv, p. 132 (2021). ISBN 978-617-7021-84-0
13. Ostrovskiy, Z., Makarenko, A.: Cellular automata in regions with obstacles as the models and tools for analysis and modeling. In: Proceedings 2019 IEEE 2nd Ukraine Conf. on Electrical and Computer Engineering (UKRCON), p. 5 (2019). https://doi.org/10.1109/UKRCON.2019.887998
14. Popov, A., Makarenko, A.: Presumable applications of deep learning for cellular automata identification In: Book Recent Advances in Big Data and Deep Learning (2020). https://doi.org/10.1007/978-3-030-16841-4_13
15. Terpil, E., Makarenko, A.: Simulation of public opinion with ideas of cellular automata. In: Conference: International Conference on Cellular Automata, LNCS, vol. 8751, pp. 518 – 525 (2014). https://doi.org/10.1007/978-3-319-11520-7_54
16. Goldengorin, B., Krushinski, D., Makarenko, A.: Synchronization of movement for large – scale crowd. In: Kyamakya, K., Halang, W.A., Unger, H., Chedjou, J.C., Rulkov, N.F.. Li, Z. (eds.) Recent Advances in Nonlinear Dynamics and Synchronization: Theory and Applications, pp. 277 – 303. Springer, Berlin/Heidelberg (2009)
17. Makarenko, A., Krushinski, D., Musienko, A., Goldengorin, B.: Towards cellular automata football models with mentality accounting. In: Bandini, S., Manzoni, S., Umeo, H., Vizzari, G. (eds.) Cellular Automata. ACRI 2010. Lecture Notes in Computer Science, vol. 6350. Springer, Berlin, Heidelberg (2010). https://doi.org/10.1007/978-3-642-15979-4_16
18. Popova, A.B., Makarenko, A.: Modeling the spread of epidemics using cellular automata. In: Abstracts of Reports of the 2nd Science-practice. Conference with International Participation "Mathematical and Simulation Modeling of Systems" MODS'2007, p. 4. Kyiv, June, (2007)
19. Makarenko, O.S., Samorodov, E.L.: To the statement of tasks of research of migration processes in education and science. In: International Scientific and Practical Conference «Intellectual Systems and Information Technologies»: Conference Proceedings/Odessa State Environmental University. Odessa, pp. 238–243 (2021). ISBN 978-617-7711-43-7
20. Krushinskiy, D., Makarenko, A.: Cellular automata with anticipation. examples and presumable applications. In: Dubois, D.M., (ed.) AIP Conference Proceedings, vol. 1303, pp. 246–254. USA (2010)
21. Makarenko, A.: Cellular automata with strong anticipation property of elements. In: Proceedings 19th International Workshop on Cellular Automata and discrete Complex Systems (AUTOMATA- 2013) - Exploratory Papers (Gissen, Germany, 17–19 September 2013) . IFIG Research Report 1302, pp. 49–56 (2013)
22. Makarenko, A.: Cellular automata and some problems in concepts reconstruction. Herald of Chmelnickiy State University (Ukraine), **2**, 248-251 (2007). (in Russian)
23. Makarenko, A., Popov., A., Pribega, V.V.: (Idea of cellular automata application in two-slit experiments. In: Chopard, B., Bandini, S., Dennunzio, A., Arabi Haddad, M. (eds) Cellular Automata. ACRI 2022. Lecture Notes in Computer Science, vol. 13402, p. 10. Springer, Cham. https://doi.org/10.1007/978-3-031-14926-9_18
24. Makarenko, A.: Presumable applications of cellular automates with strong anticipation in quantum physics J. Phys.: Conf. Ser. **1251**, 012031 (2019). https://iopscience.iop.org/article/https://doi.org/10.1088/1742-6596/1251/1/012031/pdf
25. Makarenko, A.: One method of modeling turing machines. In: Proceedings of the conference RNAOMP-2020, p. 2 Lutsk (2020)
26. Makarenko, A.: Cellular automata models with complex valued transition functions. Syst. Res. Inf. Technol. **4**, 141–147 (2020). https://doi.org/10.20535/SRIT.2308-8893.2020.4.11
27. Rosen, R.: Anticipatory Systems. Pergamon Press (1985)

28. Dubois, D.: Generation of fractals from incursive automata, digital diffusion and wave equation systems. BioSystems **43**, 97–114 (1997)
29. Dubois, D.: Incursive and hyperincursive systems, fractal machine and anticipatory logic. In: AIP Conference Proceedings, vol. 573, pp. 437–451. American Institute of Physics (2001)

The Maximum Cover for Passive Localisation

Igor Potapov, Jason F. Ralph, and Theofilos Triommatis(✉)

School of Electrical Engineering and Electronics and Computer Science, University of Liverpool, Liverpool L69-3BX, UK
{potapov,jfralph,Theofilos.Triommatis}@liverpool.ac.uk

Abstract. Imagine a polygon-shaped platform P and only one static spotlight outside P; which direction should the spotlight face to light most of P? This problem occurs in maximising the visibility, as well as in limiting the uncertainty in localisation problems. More formally, we define the following maximum cover problem: "Given a convex polygon P and a Field Of View (FOV) with a given centre and inner angle ϕ; find the direction (an angle of rotation θ) of the FOV such that the intersection between the FOV and P has the maximum area". In this paper, we provide the theoretical foundation for the analysis of the maximum cover with a rotating field of view. Since the optimal solution is a real number, we develop an algorithm that approximates the direction of the field of view, with precision ε, and complexity $\mathcal{O}(n(\log n + (\log \varepsilon)/\phi))$.

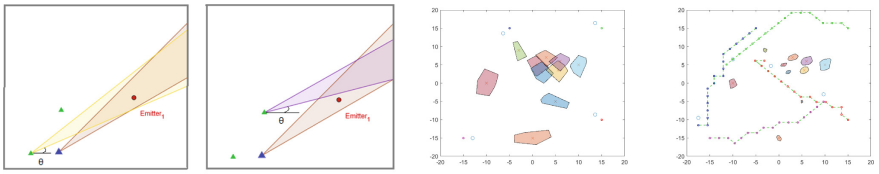
Keywords: Computational Geometry · Area Optimisation · Rotated Field Of View · Maximum Cover

1 Introduction

The use of antennas, sensors and cameras in "smart" or autonomous systems motivates the study of various visibility problems [8,12,13] with applications in computer graphics, motion planning, and other areas. The most known visibility problems are the art gallery problem, region visibility, point or edge visibility, viewshed, see [1,4–6,14]. Point or edge visibility is the decision problem of checking whether these objects are visible from a viewpoint in a context of a given set of obstacles. In the art gallery problem, the objective is to find the minimal number of locations to place guards (with restricted or unrestricted Field of View, FOV) within a polygon room to observe the room's whole area [2,9]. In this paper, we study the problem of finding the maximal visibility area from a viewpoint with a rotating FOV. Imagine a polygon-shaped platform P and only one static spotlight outside of P. Which direction should the spotlight face to light most of P? More formally, we define the following problem: "Given a polygon P and a Field Of View (FOV) with a given centre and inner angle ϕ; find the direction (as an angle θ) of the FOV such that the intersection between the FOV and P has the maximum area". This problem occurs in maximising

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2026
I. Potapov et al. (Eds.): RTC-Digital 2023, CCIS 2647, pp. 135–145, 2026.
https://doi.org/10.1007/978-3-032-04731-1_18

the visibility, as well as in limiting the uncertainty in localisation problems. The occurrence of this problem in the former setting is straightforward to understand. However, its occurrence in the latter is more subtle because we assume inside the polygon an object which we need to detect by the maximising probability of detection in the following scan without prior knowledge of its position. In [17], the geometric approach for passive localisation of static emitters is based on the problem of finding the maximum intersection of a polygon and a rotating FOV. For a passive sensor, a measurement is an angle with an error that points to the direction of a transmission's origin point. The angle with its angular error creates a cone of possible locations for the emitter. After consecutive iterations, a sensor computes a polygon by intersecting multiple measurements from different positions. A sensor needs to make a decision to move to its next position from a given finite set. The choice is made by evaluating all the available positions according to an objective function. In a myopic (greedy) decision-making strategy, a sensor moves by minimising the maximum uncertainty on its subsequent measurement, achieved by evaluating the maximal intersection of polygons that contain the emitters' position and FOVs with centres that represent sensors' available positions. Experimental results in [17] were based on a heuristic to estimate the intersection. Here we provide an algorithm with proven guarantee and precision (Fig, 1).



(a) (b) (c) (d)

Fig. 1. In (a) and (b), the sensor (blue triangle) calculates the worst-case uncertainty, which corresponds to the maximum cover, for two positions of "East" and "North" (green triangles) to select the one which minimises it, which in this case is "North". In (c) and (d), the sensors explore by minimising the maximum uncertainty in each step which makes the areas of uncertainty (the polygons) smaller. (Color figure online)

In this paper, we design an algorithm with a mathematical guarantee and provide the theoretical foundation for analysis of the maximum cover with a rotating field of view. We show an alternative way to express the maximum cover by various compositions of a function $A_\theta(\phi)$ (with a variable inner angle ϕ and a fixed direction θ) that has an analytical solution. The core component of the solution is to find the maximal intersection of a fixed sector (field of view with infinite radius) and a rotated one under a restricted rotation angle since the area of intersection of two sectors without restrictions can be infinite. Surprisingly, the function of the area, even in such a restricted case, is non-monotonic. Nonetheless, it is possible to find the maximal intersection as shown

in Sect. 3 by using functions that calculate the area with a fixed rotation angle and the inner angle as a variable. Later, we show how to express more complex shapes of the intersection of a polygon and a rotated sector as a combination of multiple two-sector intersections. Finally, we complete the solution by identifying how an infinite number of intersections can be decomposed into a finite number of equivalence classes and propose at the same time a partitioning algorithm as well as a solution for each equivalence class. Since the optimal solution is a real-value number, we develop an algorithm that approximates the direction of the field of view, with precision ε, and complexity $\mathcal{O}(n(\log n + (\log \varepsilon)/\phi))$.

2 The Maximum Intersection Problem

To begin with we introduce the notations we will use throughout the paper. Let P and Q be two points, we will notate with $\hat{PQ}$ the slope's angle of the line that P and Q define. In other words the slope of the line that P and Q define is $\tan\left(\hat{PQ}\right)$. Throughout this paper, when we mention angles we mean the positive (counterclockwise) angles and we will use the notation $A\hat{B}C$ to denote the positive angle with apex B. Moreover, denote a convex polygon $\mathcal{P} = (P_1, \ldots, P_n)$ as the list of its vertices $P_i \in \mathbb{R}^2$, $i \in \{1, \ldots, n\}$, in counter-clockwise order. A field of view in the 3D space is in essence a cone, and we assume that its height tends to infinity. Since we study the problem on a 2D plane, the field of view is actually a sector of a circle with a radius that tends to infinity.

Definition 1. *A* ***sector*** $S[C, \varepsilon_r, \varepsilon_\ell]$ *is the set of points that lie inside an angle* $0 < R\hat{C}L < \pi$ *that is formed by two half lines* ε_r*, and* ε_ℓ*,* $R \in \varepsilon_r$ *and* $L \in \varepsilon_\ell$*, that share a common endpoint* C*, called the centre of the sector. We will call* ε_r*, and* ε_ℓ *the* ***right*** *and the* ***left semi-line*** *of the sector respectively.*

As we are interested in studying the sector under rotation we introduce an alternative definition that is based on the angles of the arrays' slopes.

Definition 2. *A* ***sector*** $S[C, \varepsilon_r, \varepsilon_\ell]$ *defined by two semi-lines* $\varepsilon_r, \varepsilon_\ell$ *with gradients* θ*,* $\theta + \phi$ *and a common endpoint* C *can be represented by another triplet* $S(C, \theta, \phi)$*, where the angle* ϕ *is the* ***inner angle of the sector*** *and the angle* θ *is the* ***direction of the sector****.*

Note that ε_r is a half line that extends from C, and the direction $\theta \in [0, 2\pi]$ corresponds to exactly one semi-line because if x is the horizontal line that passes through C, then $\theta = X\hat{C}R$ where $X \in x$ and $R \in \varepsilon_r$. Now we are ready to formulate the problem properly.

Problem 1. Given a convex polygon $\mathcal{P} = (P_1, \ldots, P_n)$, a point $C = (x_0, y_0)$ outside of the polygon on the Euclidean plane and $0 < \phi < \pi$, find the direction $\theta \in [0, 2\pi]$ such that the intersection $S(C, \theta, \phi) \cap \mathcal{P}$ has the maximum area.

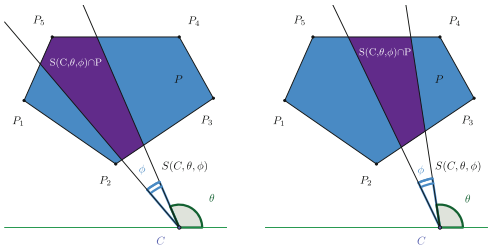


Fig. 2. The intersection between a polygon P and a sector $S(C, \theta, \phi)$ with centre C, inner angle ϕ and direction θ. The intersection becomes a quadrilateral from a pentagon as S is rotated clockwise.

Let $\mathcal{P}$ be a convex set, and a sector $S(C, \theta, \phi)$ with its centre C outside of $\mathcal{P}$. We will say that the sector $S(C, \theta, \phi)$ **contains** $\mathcal{P}$ if $\mathcal{P} \cap S(C, \theta, \phi) = \mathcal{P}$; **fully intersects** $\mathcal{P}$ if both semi-lines of $S(C, \theta, \phi)$ intersect an edge or a vertex of $\mathcal{P}$; **partially intersects** $\mathcal{P}$ if only one of the two semi-lines of $S(C, \theta, \phi)$ intersects an edge or a vertex of $\mathcal{P}$; **does not intersect** $\mathcal{P}$ if none of the two semi-lines of $S(C, \theta, \phi)$ intersect an edge or a vertex of $\mathcal{P}$. We present an algorithm that guarantees to find the maximum intersection by obtaining a maximal intersection in each equivalence class.

3 Studying the Area of Intersection

The Intersection Area Function: The intersection of a fixed sector and a rotating one, when the rotating sector fully intersects the other one, is given in the following theorem.

Theorem 1. *Let two sectors on the plane, $S(C, \theta, \phi)$, $S(K, \theta_K, \phi_K)$ with $C \notin S(K, \theta_K, \phi_K)$, and $\mathcal{R} \subseteq [0, 2\pi] \times (0, \pi)$. The area of the bounded intersection $S(C, \theta, \phi) \cap S(K, \theta_K, \phi_K)$ is*

$$A(\theta, \phi) = \frac{d_1 \sin\phi \ \cos^2(\theta_K + \phi_K)}{2\sin(\theta + \phi - \theta_K - \phi_K)\sin(\theta - \theta_K - \phi_K)} - \frac{d_2 \sin\phi \ \cos^2(\theta_K)}{2\sin(\theta + \phi - \theta_K)\sin(\theta - \theta_K)} \quad (1)$$

for every $(\theta, \phi) \in \mathcal{R}$, where $\phi_K, \theta_K \in (-\pi/2, \pi/2)$, and d_1,$d_2 \in \mathbb{R}$, are constants representing distances (as in Fig. 3(a)).

If two parallel lines intersect with a sector, then Eq. (1) holds for $\phi_K = 0$ and $\mathcal{R} = (\theta_k, \pi + \theta_k - \phi) \times (0, \pi)$ that is (Fig. 3)

$$A(\theta, \phi) = \frac{(d_1 - d_2)\sin\phi \ \cos^2(\theta_K)}{2\sin(\theta + \phi - \theta_K)\sin(\theta - \theta_K)} \quad (2)$$

The proof of Theorem 1 and other omitted proofs can be found in the paper's full version [16]. The original function $A_\phi(\theta)$ could be standardised and expressed

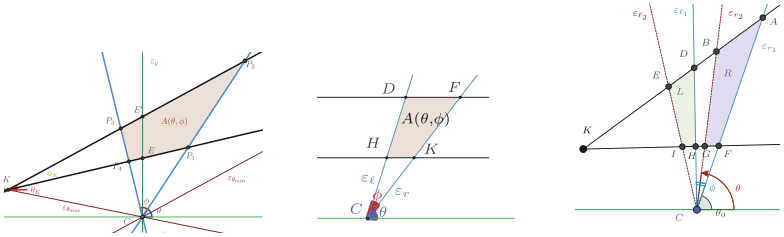


Fig. 3. (a) The area $A(\theta, \phi)$ of intersection of two sectors $S(C, \theta, \phi) \cap S(K, \theta_K, \phi_K)$. The function $A(\theta, \phi)$ is defined between the lines $\varepsilon_{\theta_{max}}$, and $\varepsilon_{\theta_{min}}$. Finally, if ε_y is a vertical line that passes through C, then $E' = \varepsilon_y \cap KP_2$, $E = \varepsilon_y \cap KP_1$ and $d_1 = |CE'|^2$, $d_2 = |CE|^2$. (b) The intersection of a rotating sector with two parallel lines, which can be considered a special case of Theorem 1, where Eq. (1) hols. (c) The sector $S(C, \theta_0, \phi)$ has as borders the blue lines ε_{ℓ_1}, and ε_{r_1} while the sector $S(C, \theta, \phi)$ has as borders the red ones ε_{ℓ_2}, and ε_{r_2}. The line ε_{ℓ_1} partitions $S(C, \theta, \phi) \cap S(K, \theta_K, \phi_K)$ into two quadrilaterals.

as an exponential polynomial function (polynomials with non-integer powers). However, the direct maximisation of these non-convex functions is difficult, see [10]. The main difficulty is that there are no constructive criteria to check the number of possible solutions of $dA_\phi/d\theta = 0$, which means there is no guarantee of finding the global maximum value within a given precision and computation time by applying naively general numerical methods. The domain of function $A(\theta, \phi)$ is $\mathcal{R} = \mathcal{D} \times \mathcal{I} \subseteq [0, 2\pi] \times (0, \pi)$. We will denote the restriction of $A(\theta, \phi)$ at $\mathcal{D}$ and $\mathcal{I}$ respectively, as $A_\phi : \mathcal{D} \to \mathbb{R}$, and $A_\theta : \mathcal{I} \to \mathbb{R}$. The function $A_\phi(\theta)$ is a rational function of the form $P(x)/Q(x)$, where $P(x)$, and $Q(x)$ are exponential polynomials (polynomials with non-integer powers). Even though it is hard to find the local extreme points of $A_\phi(\theta)$, we found a way to identify its global maximum indirectly following the analysis of the function $A_\theta(\phi)$. The function $A_\theta(\phi)$ has a symmetry that allows the cancellation of terms and gives us the possibility to calculate the explicit analytical form of its extreme points.

The function $A_\theta(\phi)$ has at most two local extreme points in $\mathcal{I} \subseteq (0, \pi)$, and they can be explicitly calculated.

This allows us to calculate the roots of the equation $dA_\theta/d\phi = 0$, $\phi_1 < \phi_2$. Now in the interval (ϕ_1, ϕ_2) function $dA_\theta/d\phi$ is either positive or negative, which means that $A_\theta(\phi)$ is strictly increasing or decreasing. In the following section, we express the area of intersection as a linear combination of $A_\theta(\phi)$ functions. Using the local extreme points of each $A_\theta(\phi)$, we can identify the intervals where there is at most one solution of the linear combination, which leads to an effective method of finding a global maximum for the original $A_\phi(\theta)$ function (see Lemma 1).

Approximating the Maximum Area Under Restricted Rotations: The objective of maximising the area of the intersection of two sectors without restricting the domain of rotations is an ill-posed optimisation problem because

there are unbounded intersections which means that the maximum is infinity. Even if we disregard those, the natural domain of rotations where Eq. (1) is well-defined is an open set $(\theta_{min}, \theta_{max})$, which means that we can create a strictly increasing sequence of $A_\phi(\theta_i)$ which tends to infinity as θ_i tends to either θ_{min} or θ_{max}. For these reasons, we study the maximisation of $A(\theta, \phi)$ under restricted rotations, i.e. we consider that θ belongs in a closed and bounded subset of $\mathbb{R}$, which guarantees the existence of a maximum.

Problem 2. Given an interval $[a, b]$, a fixed sector $S(K, \theta_K, \phi_K)$ and a sector $S(C, \theta, \phi)$ with the centre $C \notin S(K, \theta_K, \phi_K)$, calculate the area of the intersection when $\theta \in [a, b]$, and $S(C, \theta, \phi)$ fully intersects $S(K, \theta_K, \phi_K)$.

From Eq. (1), one can verify that $A_\phi(\theta)$ is not a convex function which means that this function may have multiple extreme points inside a given interval $[a, b]$. Since finding the local extreme points of $A_\phi(\theta)$ analytically is a non-trivial problem, we first conceptualise the change in direction as an increase or a decrease of two different sectors to express $A_\phi(\theta_i)$ as a combination of $A_\theta(\phi)$ functions. Secondly, we apply numerical methods to approximate the solutions of equations. The method taken into consideration is the Newton Raphson method [3,7,15]. It is easy to modify Newton Raphson to return a value equivalent to a negative value if it does not converge after a constant number of iterations. Keep in mind that if Newton Raphson converges, then the time complexity needed to approximate the solution of an equation $f(x) = 0$ up to $\varepsilon > 1$ accuracy is $F(\varepsilon) \cdot \log \varepsilon$, where $F(\varepsilon)$ is the complexity of computing f'/f, up to ε precision, that is $|apx - opt| < 10^{-\varepsilon}$.

We initially focus on the case where the angle of the rotation θ is bounded by the inner angle ϕ, i.e. $\theta \in [\theta_0 - \phi, \theta_0 + \phi]$ because it allows us to express $A_\phi(\theta)$ with only two functions A_θ and a constant $A(\theta_0, \phi)$, see Lemma 1 and Fig. 3.(c). In the following lemma not only do we express the function A_ϕ as a summation of A_θ but we also divide the domain $[\theta_0, \theta_0 + \phi]$ into a finite number of intervals where in each there exists at most one point that is a root of the first derivative of the function of the area to identify every possible local maximum. In the end, we obtain the maximum by selecting the maximum out of all local maximums.

Lemma 1. *For $\phi \in (0, \pi)$, and $\theta \in [\theta_0, \theta_0 + \phi]$, the function $A_\phi(\theta)$ is expressed as:*

$$A_\phi(\theta) = A(\theta_0, \phi) + A^L_\theta(\theta - \theta_0) - A^R_\theta(\theta - \theta_0), \tag{3}$$

where $A(\theta_0, \phi)$ is constant. The maximum value of $A_\phi(\theta)$ can be approximated with precision $\varepsilon > 1$, in time $\mathcal{O}(\log \varepsilon)$.

$\mathcal{LMR}$ Intersection and the Global Objective Function: We can decompose the area of intersection of a polygon $\mathcal{P}$ and a sector $S[C, \varepsilon_r, \varepsilon_\ell]$ as a summation of multiple areas of intersections of two sectors. Notice that if $\mathcal{P} \cap S[C, \varepsilon_r, \varepsilon_\ell]$ does not contain any vertices of $\mathcal{P}$, then this case is identical to the intersection of two sectors.

In the case that $\mathcal{P} \cap S[C, \varepsilon_r, \varepsilon_\ell]$ contains one vertex $P_0 \in \mathcal{P}$ or it contains two vertices colinear with C, then we can express the area as $Area(\mathcal{P} \cap S[C, \varepsilon_r, \varepsilon_\ell])$ as $Area(\mathcal{P} \cap S[C, \varepsilon_r, CP_0]) + Area(\mathcal{P} \cap S[C, CP_0, \varepsilon_\ell])$, see Fig. 4(a). We will refer to $Area(\mathcal{P} \cap S[C, \varepsilon_r, CP_0])$, $Area(\mathcal{P} \cap S[C, CP_0, \varepsilon_\ell])$ as right and left respectively. Now we can use Eq. (1), so $Area(\mathcal{P} \cap S[C, \varepsilon_r, \varepsilon_\ell]) = A^L(\theta_1, \phi_1) + A^R(\theta_2, \phi_2)$.

If $\mathcal{P} \cap S[C, \varepsilon_r, \varepsilon_\ell]$ contains two non colinear vertices $P_1, P_2 \in \mathcal{P}$, then using the lines CP_1 and CP_2 we can express the area of intersection of $Area(\mathcal{P} \cap S[C, \varepsilon_r, \varepsilon_\ell])$ as $Area(\mathcal{P} \cap S[C, \varepsilon_r, CP_2]) + Area(\mathcal{P} \cap S[C, CP_1, \varepsilon_\ell]) + Area(\mathcal{P} \cap S[C, CP_1, CP_2])$ (see Fig. 4(b)). The area of $S[C, CP_1, CP_2]$ remains constant for certain rotations and will call it the middle area. Similarly using Eq. (1), the area of intersection is $Area(\mathcal{P} \cap S[C, \varepsilon_r, \varepsilon_\ell]) = A^L(\theta_1, \phi_1) + A^R(\theta_2, \phi_2) + A^M$ where A^M is a constant unless the number of the vertices that $\mathcal{P} \cap S[C, \varepsilon_r, \varepsilon_\ell]$ contains, changes. The same argument can be said in the case that $\mathcal{P} \cap S[C, \varepsilon_r, \varepsilon_\ell]$ contains $P_1, \ldots, P_k$ vertices $k > 2$. The only difference is that the middle area will be $\sum_{i=1}^{k-1} Area(S[C, P_i, P_{i+1}])$. This is a different decomposition from the one we presented in the previous section, which enables the identification of the extreme points.

In the next section, we will partition the domain of directions $\mathcal{D}_G \subseteq [0, 2\pi]$ into intervals $[\theta_i, \theta_{i+1}]$, where only the left, and right area change for every rotation $\theta \in [\theta_i, \theta_{i+1}]$. This means that the area of the intersection $S(C, \theta, \phi) \cap \mathcal{P}$ in each interval is

$$\begin{aligned} Area(S(C, \theta, \phi) \cap \mathcal{P}) &= f_i(\theta) = Area(L) + Area(M) + Area(R) \\ &= A^L_\phi(\theta) + A^R_\phi(\theta) + Area(M), \quad \theta \in [\theta_i, \theta_{i+1}] \end{aligned}$$

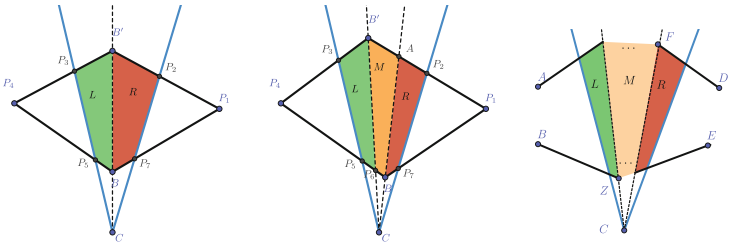
Where $Area(M)$ is a constant, and it can be calculated either using the shoelace formula [11] or as the sum of its quadrilateral sections. Now we can define properly the objective function of the area $f : \mathcal{D}_G \to \mathbb{R}$ as

$$f(\theta) = f_i(\theta) = A^{L_i}_\phi(\theta) + A^{R_i}_\phi(\theta) + Area(M_i) \qquad \theta \in [\theta_i, \theta_{i+1}] \tag{4}$$

4 Partitioning and Maximum Intersection Algorithm

Let us now explain how to decompose an infinite set of intersections into a finite number of independent sub-problems. First, we partition the polygon into quadrilateral sections by a set of lines from a point C to polygon vertices. Then every intersection $S(C, \theta, \phi) \cap \mathcal{P}$ can be written as a union of three unique convex sets L,M, and R (Left, Middle, Right), where L and R are subsets of the polygon's sections. By defining an equivalence relation on the L, M, R sets, we are partitioning all intersections $S(C, \theta, \phi) \cap \mathcal{P}$ into finite families. So, we can obtain the maximal intersection by selecting the maximum overall maximums in the equivalent classes. More formally:

Definition 3. *A **partition** of a set S is a collection of nonempty subsets of S such that every element of S is in exactly one of the subsets. The subsets are the **cells** of the partition.*



(a) (b) (c)

Fig. 4. The case of intersection $\mathcal{P} \cap S[C, \varepsilon_r, \varepsilon_\ell]$ has two colinear vertices with C, has two non-colinear vertices and has more than two vertices.

Notation 2 (Counterclockwise Vertices' Angular Ordering). *Let*

$$\mathcal{P} = (P_1, \ldots, P_n)$$

be a polygon with n vertices, a point C outside of $\mathcal{P}$, and the semi-lines of CP_i that extend from C. We will denote with $\{P_{k_i}\}_{i=1}^{n}$ the strictly increasing sequence of the vertices such that $P_{k_i} < P_{k_{i+1}}$ if $\hat{CP_i} < \hat{CP_{i+1}}$ (see Fig. 5(a)).

Definition 4. *Let*

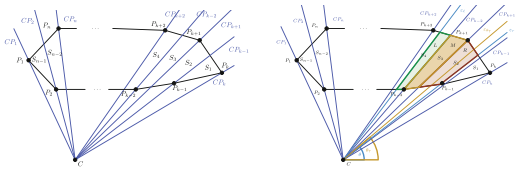
$$\mathcal{P} = (P_1, \ldots, P_n)$$

*be a polygon with n vertices, a point C outside of $\mathcal{P}$, and the semi-lines of CP_i that extend from C. We will call the sequence $\{S_i\}_{i=1}^{m}$, $m \leq n-1$, **vertex partitioning of** $\mathcal{P}$ **from** C and a set S_i a **section** of $\mathcal{P}$ where: $S_i = S[C, CP_{k_{i+1}}, CP_{k_i}] \cap \mathcal{P}$.*

Notice that the sequence S_i partitions the polygon $\mathcal{P}$ using the angular position of the vertices of $\mathcal{P}$ from point C. Sorting the semi-lines CP_i in a strictly increasing sequence means that we exclude semi-lines where CP_i coincides with CP_{i+1}, which means the number of sections is at most $n-1$. We can partition the intersection into three sets, Left, Middle, and Right; where during a "small rotation" only the Left and Right are quadrilaterals and change while the middle remains constant. These can be expressed uniquely if defined as in Definition 5 and also illustrate an example in Fig. 4.

Definition 5. *Let $S(C, \theta, \phi)$ be a sector that either fully or partially intersects a polygon $\mathcal{P}$, $K = \mathcal{P} \cap S(C, \theta, \phi)$, and $\{S_i\}_{i=1}^{m}$ is the vertex partitioning of $\mathcal{P}$ from C. Let us define three sets L, M, R (Left, Middle, Right) for the intersection K: $L = \emptyset$ or $L = K \cap S_i$ for $i = \max\{q \in \{1, \ldots, m\} : S_q \cap K \neq \emptyset\}$ or $L = K$ if $K \subseteq S_i$, $i \in \{1, \ldots, m\}$; $R = \emptyset$ or $R = K \cap S_j$, for $i \neq j = \min\{q \in \{1, \ldots, m\} : S_q \cap K \neq \emptyset\}$; $M = K \setminus (L \cup R)$.*

Every intersection $K = \mathcal{P} \cap S(C, \theta, \phi)$ is a union of three unique sets L, M, R as defined in Definition 5. That means that there is a bijection between an intersection $\mathcal{P} \cap S(C, \theta, \phi)$ and its decomposition L, M, R, so an equivalence



(a) (b)

Fig. 5. (a) The vertex partitioning of $\mathcal{P}$ from C. The vertices of $\mathcal{P}$ are sorted counterclockwise, and the sequence $S_1, \ldots, S_{n-1}$ is sorted from right to left. (b) The LMR partition with $L \subseteq S_4$, $R \subseteq S_2$, and $M = S_3$ is valid for every $\theta \in [\hat{CP}_{k-1}, \theta_7]$, where $\theta_7 = \hat{CP}_{k+2} - \phi$. When $\theta = \theta_7$ then ε_r will cross CP_{k+2}.

relation on these sets not only partitions them but partitions the intersections as well. The intersections expressed their decomposition L,M,R share the same branch of Eq. (4) if both their left sets are subsets of the same section and at the same time, both their right sets are subsets of the same section.

Definition 6. *Let two intersections* $K_1 = S(C, \theta_1, \phi) \cap \mathcal{P} = L_1 \cup M_1 \cup R_1$, $K_2 = S(C, \theta_2, \phi) \cap \mathcal{P} = L_2 \cup M_2 \cup R_2$, *and* $\{S_i\}_{i=1}^{m}$ *be the vertex partitioning of* $\mathcal{P}$ *from* C. *We define the relation* $\mathcal{LMR}$, *and we will say that* K_1 *and* K_2 *are* $\mathcal{LMR}$ ***related*** *if and only if the following statements are both true, for* $j < i \in \{1, \ldots, n\}$*:* $L_1 \subseteq L_2 \subseteq S_i$ *or* $L_2 \subseteq L_1 \subseteq S_i$ *or* $L_1 = L_2 = \emptyset$*;* $R_1 \subseteq R_2 \subseteq S_j$ *or* $R_2 \subseteq R_1 \subseteq S_j$ *or* $R_1 = R_2 = \emptyset$

The intersections $S(C, \theta_1, \phi) \cap \mathcal{P}$, and $S(C, \theta_2, \phi) \cap \mathcal{P}$ are not $\mathcal{LMR}$ related if during the rotation from θ_1 to θ_2 either the left or the right semi-line of the sector $S(C, \theta, \phi)$ crosses one of the CP_i lines of polygon $\mathcal{P}$, $i \in \{1. \ldots, n\}$. In other words, if in an interval $[\theta_s, \theta_f]$ contains an angle of rotation θ of the form $CP_i - \phi$ or CP_i then the intersections $S(C, \theta_s, \phi) \cap \mathcal{P}$, and $S(C, \theta_f, \phi) \cap \mathcal{P}$ are not $\mathcal{LMR}$ related.

The equivalence relation $\mathcal{LMR}$ partitions naturally the domain of rotations into intervals $[\theta_i, \theta_{i+1}]$, where the sequence $\{\theta_i\}_{i=1}^{Q}$ is the merged sorted list of the two strictly increasing sequences of angles $\{\hat{CP}_{k_i}\}_{i=1}^{n}$, and $\{\hat{CP}_{k_i} - \phi\}_{i=1}^{n}$. The number of $\mathcal{LMR}$ cells Q is at most $2n$. If a sector $S(C, \theta_0, \phi)$ intersects either fully or partially a polygon $\mathcal{P} = (P_1, \ldots, P_n)$ then there exists a partition L,M,R of $\mathcal{P} \cap S(C, \theta_0, \phi)$. The partition of L,M,R belongs to a unique $\mathcal{LMR}$ cell over the interval of rotations $[\theta_i.\theta_{i+1}]$, $i \in \{1, ..., Q-1\}$, and the change of the intersection area is equal to the change in the sum of the two quadrilaterals L and R for every rotation $\theta \in [\theta_i.\theta_{i+1}]$ as the area M remains constant.

The algorithm starts with computing the sections of the polygon $\mathcal{P}$. We can do that using a rotational sweep on the vertices of $\mathcal{P}$ from the centre of the sector C, to compute all the CP_i lines $i \in \{1, \ldots, n\}$, and their derivatives $\hat{CP}_i$. Then we can compute also compute $\hat{CP}_i - \phi$ and merge them with the ordered list $\hat{CP}_i$ to create the sequence of $\{\theta_i\}_{i=1}^{Q}$ that make up the intervals $[\theta_i, \theta_{i+1}]$ of the independent problems.

If we consider that the intersection $K = \mathcal{P} \cap S(C, \theta, \phi)$ is a quadrilateral then we need to know the edges of the polygon that contribute to K to be able to evaluate Eq. (1).

Algorithm 1. *Identifying the Lower and Upper Edge of Each Section*

Input: A polygon $(P_i)_{i=1}^n$, and a rearrangement $(P_{k_i})_{i=1}^n$ ordered by their $C\hat{P}_{k_i}$ values
Output: Two lists of the upper and lower edge of each section of the polygon.
1: **Set** upper_Edges = $\{P_{k_1}, P_{k_1+1}\}$, lower_Edges = $\{P_{k_1}, P_{k_1-1}\}$
2: **Set** cu = $k_1 + 1$ //The index of the left vertex of the current Upper edge
3: **for** $j = 2$ to n **do**
4: //Check if the change is in the upper or lower edge
5: **if** P_{k_j} **coincides with** P_{cu} **then**
6: **Set** upper_Edges = upper_Edges $\cup\{P_{k_j+1}\}$, cu = $k_j + 1$
7: **else**
8: **Set** lower_Edges = lower_Edges $\cup\{P_{k_j-1}\}$
9: **return** upper_Edges **and** lower_Edges

We can identify the upper and lower edges of each section by using an algorithm that goes through the upper and lower hull of $\mathcal{P}$ from centre C using the counter-clockwise order $(P_1, \ldots, P_n)$ and the order $(P_{k_1}, \ldots, P_{k_n})$. Now we are ready to present an algorithm that ε approximates the maximum intersection with accuracy $\varepsilon > 1$, $|apx - opt| < 10^{-\varepsilon}$.

Algorithm 2. *Maximum Intersection of a Polygon and a Sector*

Input: The set P of the polygon's points, the centre $C \in \mathbb{R}^2$, the angle of the sector $\phi \in (0, \pi)$
Output: The direction θ_{max} that yields to the intersection with the maximum area
1: **Compute** $\hat{CP}_{k_1}, \ldots, \hat{CP}_{k_n}$ by using $(P_1, \ldots, P_n, C)$
2: **Compute** $\theta_1, \ldots, \theta_{2k_n}$ by sorting the vlaues $(\hat{CP}_{k_1}, \ldots, \hat{CP}_{k_n}, \hat{CP}_{k_1} - \phi, \ldots, \hat{CP}_{k_n} - \phi)$
3: **Compute** the upper and lower edges for each LMR using **Algorithm 1**
4: **for** $i = 1$ to $2k_n - 1$ **do**
5: //For every partition LMR find the points of the local maximum area from Eq. 4
6: **Find** $r_i = \text{argmax}_{r \in [0, \theta_{i+1} - \theta_i]} f_i(r)$
7: $local_max_i = \theta_i + r_i$
8: $\theta_{max} = \max(local_max_i)$
9: **return** θ_{max}

To find the maximum in line 6 then we examine all the local extreme points where $df_i/d\theta = 0$ plus the values $f(\theta_i)$, and $f(\theta_{i+1})$.

Theorem 2. *Given a convex polygon* $\mathcal{P} = (P_1, \ldots, P_n)$ *with* n *vertices, and a sector* $S(C, \theta, \phi)$ *where* $\theta \in [0, 2\pi]$*, then Algorithm 2 approximates up to* ε *accuracy the direction* θ_{max} *such that the area of* $S(C, \theta_{max}, \phi) \cap \mathcal{P}$ *is maximised, in time* $\mathcal{O}(n(\log n + \log \varepsilon/\phi))$.

References

1. Abrahamsen, M.: Covering polygons is even harder. In: 2021 IEEE 62nd Annual Symposium on Foundations of Computer Science (FOCS), pp. 375–386 (2022)
2. Abrahamsen, M., Adamaszek, A., Miltzow, T.: The art gallery problem is ∃ ℝ-complete. J. ACM **69**(1) (2021)
3. Argyros, I.K.: Convergence and Applications of Newton-type Iterations. Springer (2008)
4. Asano, T., Guibas, L., Hershberger, J., Imai, H.: Visibility of disjoint polygons. Algorithmica **1**(1), 49–63 (1986). https://doi.org/10.1007/BF01840436
5. Barequet, G., Goryachev, A.: Offset polygon and annulus placement problems. Comput. Geom. **47**(3, Part A), 407–434 (2014)
6. Bose, P., Lubiw, A., Ian Munro, J.: Efficient visibility queries in simple polygons. Comput. Geom. **23**(3), 313–335 (2002)
7. Engquist, B. (ed.): Encyclopedia of Applied and Computational Mathematics. Springer, Heidelberg (2015). https://doi.org/10.1007/978-3-540-70529-1
8. Gadiraju, V., et al.: Novel sensor/access-point coverage-area maximization for arbitrary indoor polygonal geometries. IEEE Wirel. Commun. Lett. **10**(12), 2767–2771 (2021)
9. Kamaliah, S., Md Said, A., Ismail, I.: Optimal camera placement for 3D environment. In: Software Engineering and Computer Systems, pp. 448–459. Springer, Heidelberg (2011)
10. Lax, P.D.: The quotient of exponential polynomials. Duke Math. J. **15**(4), 967–970 (1948)
11. Lim, W.: Shoelace formula: connecting the area of a polygon and vector cross product. Math. Teach. **110**m 631–636 (2017)
12. Sitaraman, R., Marengoni, M., Hanson, A.: A system to place observers on a polyhedral terrain in polynomial time. Image Vis. Comput. **18**(10), 773–780 (2000)
13. Murat, U., et al.: Automated camera layout to satisfy task-specific and floor plan-specific coverage requirements. Comput. Vis. Image Underst. **103**(3), 156–169 (2006)
14. Otfried, C., Alon, E., Har-Peled, S.: Finding a guard that sees most and a shop that sells most. Discrete Comput. Geom. **37**(4), 545–563 (2007)
15. Polyak, R.A.: Introduction to Continuous Optimization. Springer Optimization and its Applications, vol. 172. Springer, Switzerland (2021)
16. Potapov, I., Ralph, J.F., Triommatis, T.: The maximum cover with rotating field of view. CoRR, abs/2309.15573 (2023). https://doi.org/10.48550/arXiv.2309.15573. arXiv arXiv:2309.15573
17. Triommatis, T., et al.: A geometric approach to passive localisation. In: 25th International Conference on Information Fusion (FUSION), pp. 1–8 (2022)

Mathematical Formalisms for Relational (Table) Databases

Iryna Lysenko(✉)

Nizhyn Mykola Gogol State University, Grafska str., 2, Nizhyn 16600, Ukraine
iryna.glushko@ndu.edu.ua
https://cutt.ly/1wYSPJYm

Abstract. In this paper, we develop theoretical foundations for relational (table) databases. Two mathematical formalisms such as table algebra of infinite tables and multiset table algebra are considered. The question of the relationship between table algebra of infinite tables and multiset table algebra is also considered. As the main result, we show the equivalence of table algebra of infinite tables and corresponding relational calculi. This result generalises the classical result about the equivalence of Codd's relational algebra and tuple (domain) relation calculus.

Keywords: Relational (table) databases · Table algebra · Multiset table algebra · Relational calculi

1 Introduction

The development of database technology is largely based on the relational data model proposed by E. F. Codd in 1970 [2]. Since then, the set of relational algebra operations proposed by Codd has been expanded according to the needs of query languages (A. Klug, P. Gray, G. Gardarin, P. Valduriez, R. Elmasri, S. Navathe, R. Riordan, and others). Tuple calculus (E. F. Codd) and domain calculus (M. Lacroix, A. Pirotte) have been developed. Scientists L. Libkin, L. Wong, G. Lamperti, M. Melchiori, M. Zanella, G. Garcia-Molina, J.D. Ullman, J. Widom, K. Ross, J. Stoyanovich, A. Silbeschatz, and others paid attention to the issue of expanding the capabilities of the database due to the use of multisets. Among the scientists engaged in research in the field of relational databases, it is also worth mentioning D. Maier, C.J. Date, T.M. Connolly, K.E. Begg, H. Garcia-Molina, A. Silberschatz, and others.

E. F. Codd introduced Relational Algebra as a set of operators on the relations. In the monograph [7] authors V. Redko, J. Brona, D. Buy, S. Poliakov specified a table (relation) in terms of nominal sets. Traditionally the finite set of tuples is understood under the table and the authors take it into account. However, as a rule, mathematical statements about standard properties of relation operations remain true for infinite relations. At the same time, the question arises about the equivalence of table algebra and the corresponding relational calculi. This paper contains the results, which continue to develop the current problem of the development of the theoretical basis of the table databases.

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2026
I. Potapov et al. (Eds.): RTC-Digital 2023, CCIS 2647, pp. 146–152, 2026.
https://doi.org/10.1007/978-3-032-04731-1_19

2 Definition of Two Mathematical Formalisms

Let's introduce the basic concepts of table algebra of infinite tables and multiset table algebra in terms of monographs [1] and [7]. Let $\boldsymbol{A}$ be the set of attributes and $\boldsymbol{D}$ be the universal domain. An arbitrary finite set of attributes $R \subseteq \boldsymbol{A}$ is called a scheme. The tuple of scheme R is a nominal set on the pair R, $\boldsymbol{D}$, where the projection of this nominal set for the first component is equal to R, i.e. $s : R \to \boldsymbol{D}$.

Under table algebra of infinite tables is understood an algebra

$$\langle \boldsymbol{T}, \Omega_{P,\Xi} \rangle,$$

where $\boldsymbol{T}$ is the set of all tables,

$\Omega_{P,\Xi} = \{\bigcup_R, \bigcap_R, \setminus_R, \sigma_{p,R}, \pi_{X,R}, \underset{R_1,R_2}{\otimes}, \div_{R_2}^{R_1}, Rt_{\xi,R}, \sim_R\}_{X,R,R_1,R_2 \subseteq \boldsymbol{A}}^{p \in P, \xi \in \Xi}$ is the signature,

P, Ξ are the sets of parameters.

A table of the scheme R ($R \subseteq \boldsymbol{A}$) is a pair

$$\langle t, R \rangle,$$

where t is a set (in particular infinite) of tuples of the fixed scheme R.

The signature $\Omega_{P,\Xi}$ of table algebra of infinite tables contains set-theoretical operations (union, intersection, and difference), selection, projection, join, division, renaming, and active complement. Table algebra of infinite tables is also filled up with additional operations such as inner join (Cartesian join, natural join, join using attributes and join on predicate), outer join (outer left join, outer right join, outer full join and union join), semijoin, outer set operations (outer left join, outer right join, outer full join and union join) and aggregate operations (*Sum*, *Avg*, *Max*, *Min*, *Count*). The special element NULL is inserted in the universal domain to define outer join and outer set operations. The operations of signature $\Omega_{P,\Xi}$ and the formal mathematical semantics of additional operations are defined in [1].

Theorem 1. *Any expression of the table algebra of infinite tables can be replaced by an equivalent to him expression which uses only operations of selection, join, projection, union, difference, and renaming.*

Under multiset table algebra is understood an algebra

$$\langle \Psi, \Omega_{P,\Xi} \rangle,$$

where Ψ is the set of all tables,

$\Omega_{P,\Xi} = \{\bigcup_{All}^R, \bigcap_{All}^R, \setminus_{All}^R, \sigma_{p,R}, \pi_{X,R}, \underset{R_1,R_2}{\otimes}, Rt_{\xi,R}, \sim_R\}_{X,R,R_1,R_2 \subseteq \boldsymbol{A}}^{p \in P, \xi \in \Xi}$ is the signature,

P, Ξ are the sets of parameters.

A table of the scheme R ($R \subseteq \boldsymbol{A}$) is a pair

$$\langle \psi, R \rangle,$$

where the first component ψ is an arbitrary multiset, the basis of which $\Theta(\psi)$ is the set of tuples of the same scheme and the second component R is a scheme of the table. The notation $Occ(s, \psi)$ denotes the number of duplicate tuple s in the multiset ψ. Let's agree to write down a multiset as

$$\{s_1^{n_1}, ..., s_k^{n_k}\},$$

where $n_i = Occ(s_i, \psi)$, $i = 1, ..., k$, and $\Theta(\psi) = \{s_1, ..., s_k\}$ is a basis of the multiset ψ.

The signature $\Omega_{P,\Xi}$ of multiset table algebra contains set-theoretical operations (union, intersection, and difference), selection, projection, join, renaming, and active complement. The signature is also filled up with additional operations such as inner join (Cartesian join, natural join, join using attributes and join on predicate), outer join (outer left join, outer right join, outer full join and union join), semijoin and aggregate operations (*Sum*, *Avg*, *Max*, *Min*, *Count*). The special element NULL is inserted in the universal domain to define outer join and outer set operations. The operations of signature $\Omega_{P,\Xi}$ and the formal mathematical semantics of additional operations are defined in [1].

Theorem 2. *The following statements hold:*

1. *Any expression of the multiset table algebra can be replaced by an equivalent to him expression which uses only operations of selection, join, projection, union, difference, and renaming;*
2. *Any expression of the multiset table algebra can be replaced by an equivalent to him expression which uses single-attribute, single-tuple constant tables and only operations of selection, join, projection, union, difference, and renaming.*

3 About Relationship Between Table Algebra of Infinite Tables and Multiset Table Algebra

The 1-multisets are the analogs of ordinary sets. According to this statement table algebra of infinite tables seems to be a subalgebra of multiset table algebra. However according to the proof in the article [4] table algebra of infinite tables does not form the subalgebra of multiset table algebra since it is not closed with respect to the union, projection, and active complement. Let's check it for the union operation.

For this, suppose that a table of table algebra of infinite tables is a pair $\langle t^1, R \rangle$, where t^1 is a 1-multiset (in particular infinite) of tuples of fixed scheme R. In this case, obviously, the set of all tables of table algebra of infinite tables $\boldsymbol{T}$ is a subset of the set of all tables of multiset table algebra Ψ, i.e. $\boldsymbol{T} \subseteq \Psi$.

Find out that the set $\boldsymbol{T}$ is not closed with respect to the union. Bases of 1-multisets t_1^1 and t_2^1 are designated as $\Theta(t_1^1)$ and $\Theta(t_2^1)$ respectively.

The union defined as $\langle t_1^1, R\rangle \cup_{All}^{R} \langle t_2^1, R\rangle = \langle t_1^1 \cup_{All} t_2^1, R\rangle$, where $\langle t_1^1, R\rangle$, $\langle t_2^1, R\rangle \in \boldsymbol{T}$.

Basis of the multiset $t_1^1 \cup_{All} t_2^1$ of the resulting table is equal to union of bases of multisets of input tables: $\Theta(t_1^1 \cup_{All} t_2^1) = \Theta(t_1^1) \cup \Theta(t_2^1)$.

Duplicate tuples, which appear after the implementation of the operation, are not removed from the result. The number of duplicates is given by the following formula:

$$Occ(s, t_1^1 \cup_{All} t_2^1) = \begin{cases} 1 \; if \; s \in \Theta(t_1^1) \setminus \Theta(t_2^1) \; or \; s \in \Theta(t_2^1) \setminus \Theta(t_1^1), \\ 2 \; if \; s \in \Theta(t_1^1) \cap \Theta(t_2^1), \end{cases}$$

where $s \in \Theta(t_1^1) \cup \Theta(t_2^1)$. As a result $t_1^1 \cup_{All} t_2^1$ is a multiset and it is not a 1-multiset, as we expected. Thus, multiset table algebra is not a wider formalism than table algebra of infinite tables.

It is also proved that table algebra is closed with respect to the inner and outer join, semijoin, and aggregate operations of multiset table algebra [5].

4 Equivalence of Table Algebra of Infinite Tables and Corresponding Relational Calculi

Relational calculus is the basis of most relational query languages because, unlike relational (table) algebra, calculus expresses only what must be the result, and does not determine how to get it. Relational calculus is based on first-order predicate calculus. There are two forms of relational calculus: tuple calculus and domain calculus. These forms have been proposed by E. Codd and M. Lacroix with A. Pirotte respectively.

In this research, the classical relational calculi are filled up by functional and predicate signatures on the universal domain (while usually considering only binary predicates and functional signature is generally empty). Syntax of terms, atoms, and formulas of generalised tuple calculus and generalised domain calculus is defined, and the sets of legal formulas are introduced in [3].

Generalised tuple calculus expression has the form

$$E = \{\boldsymbol{x}(R) | \boldsymbol{P}(\boldsymbol{x})\},$$

where

1. formula $\boldsymbol{P}$ is legal,
2. $\boldsymbol{x}$ is the only tuple variable that occurs free in $\boldsymbol{P}$;
3. if $scheme(\boldsymbol{x}, \boldsymbol{P})$ is defined, then $scheme(\boldsymbol{x}, \boldsymbol{P}) = R$, otherwise, $attr(\boldsymbol{x}, \boldsymbol{P}) \subseteq R$.

$scheme(\boldsymbol{x}, \boldsymbol{P})$ and $attr(\boldsymbol{x}, \boldsymbol{P})$ are the scheme (a finite set of attributes) and the set of attributes accordingly. These expressions are expressions of our metalanguage and are not used in the subject language that is being built.

The value of the expression E is the table of the scheme R:

$$\langle \{s | s \in S(R) \wedge \boldsymbol{P}(s/\boldsymbol{x}) - true\}, R \rangle,$$

which contains all rows $s \in S(R)$ such that the formula $\boldsymbol{P}(s/\boldsymbol{x})$ is true.

Generalised domain calculus expression has the form

$$H = \{\boldsymbol{x}_1^{A_1}, ..., \boldsymbol{x}_n^{A_n} | \boldsymbol{P}(\boldsymbol{x}_1^{A_1}, ..., \boldsymbol{x}_n^{A_n})\},$$

where

1. formula $\boldsymbol{P}$ is a legal domain calculus formula with exactly the free variables $\boldsymbol{x}_1^{A_1}, ..., \boldsymbol{x}_n^{A_n}$ (variables over universal domain $\boldsymbol{D}$); each variable $\boldsymbol{x}_i^{A_i}$ is associated with an attribute A_i, where all A_i are pairwise different, $i = \{1, 2, ..., n\}$;
2. $R = \{A_1, ..., A_n\}$ is the scheme of expression and the order of attributes is fixed.

The value of the expression H is the table of the scheme $R = \{A_1, ..., A_n\}$:

$$\langle \{s | s \in S(R) \wedge \boldsymbol{P}(\boldsymbol{d}_1/\boldsymbol{x}_1^{A_1}, ..., \boldsymbol{d}_n/\boldsymbol{x}_n^{A_n}) - true\}, R \rangle,$$

which contains all rows $s \in S(R)$, $s = \{\langle A_1, \boldsymbol{d}_1 \rangle, ..., \langle A_n, \boldsymbol{d}_n \rangle\}$ such that the formula $\boldsymbol{P}(\boldsymbol{d}_1/\boldsymbol{x}_1^{A_1}, ..., \boldsymbol{d}_n/\boldsymbol{x}_n^{A_n})$ is true.

There are two main results. The first concerns the equivalence of table algebra for infinite tables and corresponding relational calculi, and the second – the equivalence of subalgebra of finite tables of the given algebra and corresponding relational calculi in which consideration is limited to only so-called "safe" expressions. The obtained results generalise the classical result about the equivalence of Codd's relational algebra and tuple (domain) calculus. The next theorems are proofed.

Theorem 3. *If F is the expression of table algebra of infinite tables, then it is possible effectively to build an equivalent to it expression E of generalised tuple relational calculus.*

Theorem 4. *If E is an expression of generalised tuple relational calculus, then it is possible effectively to build an equivalent to it expression H of generalised domain relational calculus.*

Theorem 5. *If H is an expression of generalised domain relational calculus, then it is possible effectively to build an equivalent to it expression F of table algebra of infinite tables.*

Theorem 5 proves that table algebra of infinite tables is as expressive as generalised domain relational calculus (in terms of [6]).

Taking into account Theorems 3, 4, and 5 we are getting a basic result.

Theorem 6. *Table algebra of infinite tables, generalised tuple relational calculus, and generalised domain relational calculus are equivalent.*

5 Equivalence of Table Algebra of Finite Tables and Corresponding Relational Calculi

In practice, as usually deals with finite tables. Under the table algebra of finite tables is understood an algebra

$$\langle \boldsymbol{T}', \Omega_{P,\Xi} \rangle,$$

where $\boldsymbol{T}'$ is the set of all finite tables.

A table of the table algebra of finite tables of the scheme R is a pair

$$\langle t, R \rangle,$$

where t is a finite set of tuples of the fixed scheme R.

Generalised relation calculus allows for determining infinite tables. To overcome this demerit, inheriting [6] we considered only, so-called, "safe" expressions of relational calculus. Safe tuple calculus expression E has the following three conditions.

1. If formula $\boldsymbol{P}(s/\boldsymbol{x})$ is true, then $s(A) \in D_{\boldsymbol{P}}$, $A \in R$;
2. For every subformula of $\boldsymbol{P}$ of the form $\exists \boldsymbol{y}(X)\ \ \boldsymbol{G}(\boldsymbol{y}, \boldsymbol{z}_1, ..., \boldsymbol{z}_k)$, $\boldsymbol{G}(s/\boldsymbol{y}, u_1/\boldsymbol{z}_1, ..., u_k/\boldsymbol{z}_k)$ is true implies $s(A) \in D_{\boldsymbol{G}}$, $A \in X$ and $X \subseteq R$, where $\boldsymbol{y}, \boldsymbol{z}_1, ..., \boldsymbol{z}_k$ are all free tuple variables in $\boldsymbol{G}$;
3. For every subformula of $\boldsymbol{P}$ of the form $\forall \boldsymbol{y}(X)\boldsymbol{G}(\boldsymbol{y}, \boldsymbol{z}_1, ..., \boldsymbol{z}_k)$, $s(A) \notin D_{\boldsymbol{G}}$, $A \in X$ implies $\boldsymbol{G}(s/\boldsymbol{y}, u_1/\boldsymbol{z}_1, ..., u_k/\boldsymbol{z}_k)$ is true, where $\boldsymbol{y}, \boldsymbol{z}_1, ..., \boldsymbol{z}_k$ are all the free tuple variables in $\boldsymbol{G}$.

generalised domain calculus expression H is safe if the following three conditions hold.

1. If for constants $d_1, ..., d_n$ formula $\boldsymbol{P}(d_1/\boldsymbol{x}_1^{A_1}, ..., d_n/\boldsymbol{x}_n^{A_n})$ is true, then $d_i \in \boldsymbol{D}_{A_i,\boldsymbol{P}}$, $i = 1, ..., n$, where A_i is attribute from R which is associated with variable $\boldsymbol{x}_i^{A_i}$, $R = \{A_1, ..., A_n\}$;
2. If $\exists \boldsymbol{y}^A(A)\boldsymbol{G}$ is a subformula of $\boldsymbol{P}$, then formula $\boldsymbol{G}(d/\boldsymbol{y}^A)$ is true implies that $d \in \boldsymbol{D}_{A,\boldsymbol{G}}$, d is an arbitrary constant;
3. If $\forall \boldsymbol{y}^A(A)\boldsymbol{G}$ is a subformula of $\boldsymbol{P}$, then $d \notin \boldsymbol{D}_{A,\boldsymbol{G}}$ implies that formula $\boldsymbol{G}(d/\boldsymbol{y}^A)$ is true, d is an arbitrary constant.

Theorem 7. *If F is the expression of table algebra of finite tables, then it is possible effectively to build an equivalent to it safe generalised tuple calculus expression E.*

Theorem 8. *If E is a safe generalised tuple calculus expression then it is possible effectively to build an equivalent to it safe generalised domain calculus expression H.*

Theorem 9. *If H is a safe generalised domain calculus expression then it is possible effectively to build an equivalent to it expression of table algebra of finite tables F.*

Taking into account theorems 7, 8, and 9 we are getting a basic result.

Theorem 10. *Table algebra of finite tables, limited generalised tuple calculus and limited generalised domain calculus are equivalent.*

6 Conclusions

The paper is focused on some theoretical questions of the table databases such as table algebra of infinite tables, multiset table algebra and their relationship, equivalence of table algebra of infinite tables, and corresponding relational calculi. In the future, it plans to investigate the properties of the operations of multiset table algebra and define calculus, which may be used in the formulation of queries on any database with multiset semantics.

References

1. Buy, D., Lysenko, I.: Calculi and extensions of table algebras signature. NDU im. M. Gogol, Nizhyn (2016)
2. Codd, E.F.: A relational model of data for large shared data banks. Comm. ACM (6) (1970)
3. Glushko, I.: Generalization of the classical result of Codd-Lacroix-Pirotte. Int. J. Comput. **1** (2016)
4. Glushko, I.: About relationship between table algebra of infinite tables and multiset table algebra. In: CEUR Workshop Proceedings, vol. 2139, pp. 159–163 (2018)
5. Lysenko, I.: Extended table algebra, extended multiset table algebra, and their relationship. In: Proceedings of the International Conference on Software Engineering SoftEngine 2020, pp. 27–30 (2020)
6. Maier, D.: The Theory of Relational Databases. Computer Science Press, Rockville (1983)
7. Redko, V.V., Borona, Y.J., B.Buy, D., Poliakov, S.S.: Relational Databases: Table Algebras and SQL-Like Language. Vydavnychyi dim "Akademperiodyka", Kyiv (2001)

Digital Technologies and Applications

Problems of Implementation of the Centre for Parallel Simulation Technologies

Serhii Kovalov[1], Oleksandr Miroshkin[2(✉)], Michael Resch[3], and Volodymyr Svjatnyj[1]

[1] Donetsk National Technical University, Shibankova sq. 2, Pokrovsk, Ukraine
{sergiy.kovalov,volodymyr.svyatnyy}@donntu.edu.ua
[2] Ulm University, Helmholtzstraße 16, 89081 Ulm, Germany
miroshkinan@gmail.com
[3] Stuttgart University, Nobelstr. 19, 70550 Stuttgart, Germany
emre@hlrs.de

Abstract. The article considers issues of the establishment of the Ukrainian Centre for Parallel Simulation Technologies in Donetsk National Technical University (Ukraine) and its current functioning, defines the main directions of the Centre's work. As an example of the Centre's research activities, an example of solving the problem of parallel modelling of a technical system is presented to ensure the design of distributed parallel modelling environments for performing parallel modelling of complex dynamic systems, taking into account the capabilities and technical features of the computing MIMD-cluster provided to the Ukrainian university by the High Performance Computing Centre Stuttgart (HLRS).

Keywords: MIMD · simulation model · HPC · Block-oriented simulation language · Parallel Simulation Technologies

1 Introduction

Scientific and technical progress in all subject areas depends on the level of theory and mathematical description of dynamic processes of various physical nature. This includes mining, technological branches of industry, mechanical engineering, electrical engineering, electronics, information and measurement, military equipment, chemical technologies, medicine, and bio-mechanics of living organisms. The implementation of theories in the practical design of automated technical objects and technological installations is crucial. These are defined as cyber-physical systems of Industry-4.0 and complex dynamic systems (CDS) with concentrated and distributed parameters. Modelling methods and tools play a crucial role in ensuring the novelty, quality, reliability, and competitiveness of project solutions. They support all stages of scientific and technical projects, making them essential for success.

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2026
I. Potapov et al. (Eds.): RTC-Digital 2023, CCIS 2647, pp. 155–160, 2026.
https://doi.org/10.1007/978-3-032-04731-1_20

Since 1992, DonNTU has been cooperating with the Institute of Parallel and Distributed Systems (IPVS) and the Computing Centre (HLRS/HPCC) of the University of Stuttgart (SUni) on the problem of system organisation and effective usage of parallel computing resources in CDS simulation. As a generalisation of the acquired experience, a new interdisciplinary direction of joint research has been established - Parallel Simulation Technology (ParSimTech). The central ParSimTech problem is the building of distributed parallel simulation environments (DPSE). The DPSE concept proposed by DonNTU and SUni provides for the fully functional development of parallel methods and algorithms for simulation software. For the further development of ParSimTech capacities in Ukraine, the University of Stuttgart handed over an 80-nodes MIMD cluster. All prerequisites for the development and implementation of the Ukrainian research and training centre for parallel simulation technologies have been created at DonNTU. The centre works in the following main directions:

- **Research and development** of problem-oriented DPSE, parallel equation solvers based on new block computing methods, parallel simulation algorithms, architecturally relevant hierarchical approaches to parallelisation, parallel simulation languages, methods of model support for industrial projects [3].
- In the field of education: **development and implementation of ParSimTech-oriented curricula** for IT specialties for technical universities; training of students and young scientists in technological specialties; training of interested representatives of industry and business, cooperation with international partners.
- **Cooperation with HLRS** in the framework of international projects on the European HPC-strategy [1].

2 Scientific Components of ParSimTech Activities

The main problems, including tasks of development and research, are the following:

- Problem-oriented DPSE, in particular: functionality and structures of environments; input languages of the CDS specification; formal description of topologies and dynamic processes; topological analysers; equation generators; approaches to parallelisation and virtual parallel simulation models; devirtualisation; selection and justification of methods of parallel solution for equations; decomposition of DPSE into subsystems, method of building subsystems; resource load balancing in the DPSE; prospects for building modelling service centres for subject areas. The main components of the DPSE will be developed for the fields of systems biology and network dynamic systems (e.g. mine air networks).
- Equation solvers based on block computational methods (BCM), in particular systems of partial differential equations describing dynamic processes in objects of various topologies; analysis of the effectiveness of BCM solvers in comparison with the results obtained using conventional methods.

- Parallel simulation algorithms for CDS subject areas of Mine ventilation networks critical for safety conditions and Systems biology, in particular, parallel genetic algorithms (PGA) for identification of parameters of biological models based on population structuring.
- Architecturally-relevant hierarchical approaches to the parallelisation of mathematical models of automated CDS, taking into account the type of topologies, the location of automation tools at different levels of the hierarchy, the possibility of mapping to existing and promising MIMD architectures, the inclusion of SIMD components in the parallel resources of the DPSE.
- Parallel modelling languages as the main means of increasing the level of service and friendliness of DPSE resources to developers and users of parallel simulators. The development is based on the analogy between the principles of operation of sequential block-, equation- and object-oriented (BO-, EO-, OO- respectively) simulation languages and MIMD-parallelism.
- Means of simulation support for industrial projects, in particular Industry-4.0: a technique for building parallel simulators of cyber-physical systems; setting re-engineering tasks; model support for the development of an automated system for a safety-critical facility.

3 An Example of an Equation Solver Based on BCM

The research and implementation of block solvers was started with the j-edge model and a small test dynamic network objects with distributed parameters (DNODP), that is described topologically as a graph $G(m, n)$, where $m = 8$ is the number of edges, and $n = 4$ is the number of nodes (Fig. 1). Corresponding system of partial differential equations uses the following designations: Q_j is an airflow and P_j is an air pressure in the j-edge respectively; α_j, β_j, $\beta_{r,j}$, γ_j are the coefficients dependent on aerodynamic parameters of the j-th edge, v is a number of an edges fragment after approximation by spatial coordinates. A detailed description of the model, as well as the transformation of equations describing the processes in its branches, is given in papers [2,4,5]. After the approximation by spatial coordinates, this system is reduced to the form of a spatially discrete simulation model:

$$\begin{cases} \dot{Q}_{j,v} = \alpha_j \left(P_{j,v} - P_{j,v+1}\right) - \beta_j Q_{j,v} \left|Q_{j,v}\right| - \beta_{r,j} Q_{j,v} \left|Q_{j,v}\right|; \\ \dot{P}_{j,v+1} = \gamma_j \left(Q_{j,v} - Q_{j,v+1}\right). \end{cases} \tag{1}$$

Here, the coefficients α_j, $\beta_{r,j}$, γ_j depend on the aerodynamic parameters of the j-edge. We will further consider the equations of air flows (Q-equation) and pressures (P-equation) and introduce the following notations for further formal transformations:

$$\begin{cases} \dot{Q}_{j,v} = F_{j,v}; \\ F_{j,v} = \alpha_j \left(P_{j,v} - P_{j,v+1}\right) - \beta_j Q_{j,v} \left|Q_{j,v}\right| - \beta_{r,j} Q_{j,v} \left|Q_{j,v}\right|; \\ \dot{P}_{j,v+1} = D_{j,v+1}; \\ D_{j,v+1} = \gamma_j \left(Q_{j,v} - Q_{j,v+1}\right). \end{cases} \tag{2}$$

When applying the BCM, we get from (1) and (2) a discrete simulation model (BCM-DSM). For the one-step four-point method, the equations of this model have the following form under the condition $\beta_{r,j} = 0$:

$$\begin{cases} Q_{j,v,n,1} = \frac{\tau}{720}\left(251F_{j,v,n,0} + 646F_{j,v,n,1} - 264F_{j,v,n,2} + 106F_{j,v,n,3} - 19F_{j,v,n,4}\right) + Q_{j,v,n,0}; \\ Q_{j,v,n,2} = \frac{\tau}{90}\left(29F_{j,v,n,0} + 124F_{j,v,n,1} + 24F_{j,v,n,2} + 4F_{j,v,n,3} - F_{j,v,n,4}\right) + Q_{j,v,n,0}; \\ Q_{j,v,n,3} = \frac{3\tau}{80}\left(9F_{j,v,n,0} + 34F_{j,v,n,1} + 24F_{j,v,n,2} + 14F_{j,v,n,3} - F_{j,v,n,4}\right) + Q_{j,v,n,0}; \\ Q_{j,v,n,4} = \frac{2\tau}{45}\left(7F_{j,v,n,0} + 32F_{j,v,n,1} + 12F_{j,v,n,2} + 32F_{j,v,n,3} + 7F_{j,v,n,4}\right) + Q_{j,v,n,0}. \end{cases} \tag{3}$$

$$\begin{cases} F_{j,v,n,0} = \alpha_j\left(P_{j,v,0} - P_{j,v+1,n,0}\right) - \beta_j Q_{j,v,n,0}\left|Q_{j,v,n,0}\right|; \\ F_{j,v,n,1} = \alpha_j\left(P_{j,v,0} - P_{j,v+1,n,1}\right) - \beta_j Q_{j,v,n,1}\left|Q_{j,v,n,1}\right|; \\ F_{j,v,n,2} = \alpha_j\left(P_{j,v,0} - P_{j,v+1,n,2}\right) - \beta_j Q_{j,v,n,2}\left|Q_{j,v,n,2}\right|; \\ F_{j,v,n,3} = \alpha_j\left(P_{j,v,0} - P_{j,v+1,n,3}\right) - \beta_j Q_{j,v,n,3}\left|Q_{j,v,n,3}\right|; \\ F_{j,v,n,4} = \alpha_j\left(P_{j,v,0} - P_{j,v+1,n,4}\right) - \beta_j Q_{j,v,n,4}\left|Q_{j,v,n,4}\right|. \end{cases}$$

$$\begin{cases} P_{j,v+1,n,1} = \frac{\tau}{720}\left(251D_{j,v+1,n,0} + 646D_{j,v+1,n,1} - 264D_{j,v+1,n,2} + 106D_{j,v+1,n,3} - 19D_{j,v+1,n,4}\right) + P_{j,v+1,n,0}; \\ P_{j,v+1,n,2} = \frac{\tau}{90}\left(29D_{j,v+1,n,0} + 124D_{j,v+1,n,1} + 24D_{j,v+1,n,2} + 4D_{j,v+1,n,3} - D_{j,v+1,n,4}\right) + P_{j,v+1,n,0}; \\ P_{j,v+1,n,3} = \frac{3\tau}{80}\left(9D_{j,v+1,n,0} + 34D_{j,v+1,n,1} + 24D_{j,v+1,n,2} + 14D_{j,v+1,n,3} - D_{j,v+1,n,4}\right) + P_{j,v+1,n,0}; \\ P_{j,v+1,n,4} = \frac{2\tau}{45}\left(7D_{j,v+1,n,0} + 32D_{j,v+1,n,1} + 12D_{j,v+1,n,2} + 32D_{j,v+1,n,3} + 7D_{j,v+1,n,4}\right) + P_{j,v+1,n,0}. \end{cases} \tag{4}$$

$$\begin{cases} D_{j,v+1,n,0} = \gamma_j\left(Q_{j,v,n,0} - Q_{j,v+1,n,0}\right); \\ D_{j,v+1,n,1} = \gamma_j\left(Q_{j,v,n,1} - Q_{j,v+1,n,1}\right); \\ D_{j,v+1,n,2} = \gamma_j\left(Q_{j,v,n,2} - Q_{j,v+1,n,2}\right); \\ D_{j,v+1,n,3} = \gamma_j\left(Q_{j,v,n,3} - Q_{j,v+1,n,3}\right); \\ D_{j,v+1,n,4} = \gamma_j\left(Q_{j,v,n,4} - Q_{j,v+1,n,4}\right). \end{cases}$$

The values $Q_{v,n,0}$ and $P_{v+1,n,0}$ at the zero point of block n are defined as variables at the last point of the previous block for $n > 1$ ($t_{n-1,k} = t_{n,0}$). For each edge of the test network object (Fig. 1), systems of equations (3, 4) with the corresponding indices $j = 1, 2, ..., 8$ and $v = 1, 2, ..., M_j$ are generated.

Boundary conditions $PU_1 ... PU_4$ are described by nodal equations According to system (3), each nodal equation of system (4) is written in block form and the sought variables PU_w ($w = 1, 2, 3, 4$) are calculated in the process of solving equations common to the network. The parameters of the branch elements were chosen for the approximation step 40 m. The simulation results shown in Fig. 2 testify to the efficiency of the BCM modelling and coincide with the results obtained using conventional methods [2].

4 Educational Components of ParSimTech Activities

To disseminate knowledge, methods, and developed technologies, ParSimTech scientists carry out the following educational activities:

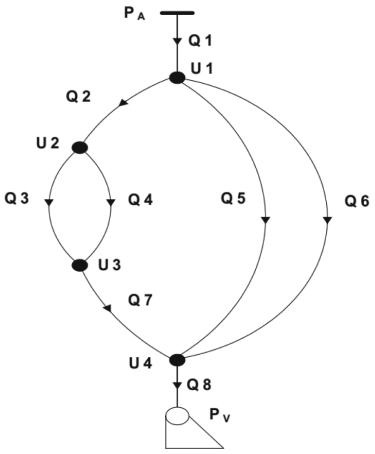


Fig. 1. Graph of the test work object with $m = 8$ edges and $n = 4$ nodes.

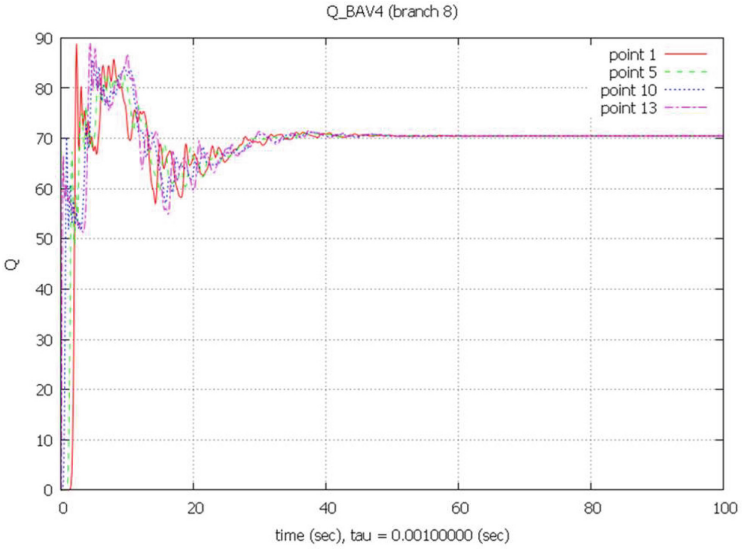


Fig. 2. Dynamic processes in elements 1, 5, 10 and 13 of edge Q_8

– Education of IT students: Developing and implementing of the ParSimTech-oriented curricula for technical university IT programs, including Computer Engineering, Software Engineering, Cybersecurity, etc. The task involves preparing scientific and methodological support for disciplines such as Parallel and Distributed Computing, Parallel Programming, Software Design of Parallel Simulation Environments, Fundamentals of Parallel Modelling of Complex Dynamic Systems, and Parallel Simulation Languages. Additionally, it

includes organising ParSimTech laboratory workshops, engaging students in scientific work related to ParSimTech topics, including bachelor's and master's level theses, and postgraduate studies in ParSimTech topics.
– Training students and young scientists of technological specialties the basics of professionally oriented parallel simulations: preparation of the discipline Fundamentals of parallel simulation of complex dynamic systems; organisation of the Parallel programming workshop at the MIMD cluster; consultations of master's and postgraduate students regarding the use of ParSimTech ideas when completing the topics of qualification works, assistance in conducting research on DPSE-resources; development of means of problem orientation of DPSE in the interests of technological specialties.

5 Conclusions

The main problems of ParSimTech research and training were formulated. Discrete virtual parallel simulation models of dynamic network objects with distributed parameters with 4 possible levels of parallelisation were considered, and the tasks of developing equation solvers based on block computing methods (BCM) were described. We would like to develop cooperation in this scientific area. Specialised software developed by ParSimTech scientists, after appropriate testing and debugging processes, is planned to be open to the scientific community and can be freely used for non-commercial purposes.

References

1. Koller, B., Lewandowski, N.: Stärkung der europäischen hpc-strategie durch die etablierung von nationalen kompetenzzentren und deren vernetzung. Probleme der Simulationstechnik und der rechnergestützten Projektierung **17**(1), 40–45 (2021)
2. Kushnarenko, V., Resch, M., Svjatnyj, V., Wesner, S.: Zur entwicklung des gleichungslösersubsystems der verteilten parallelen simulationsumgebung. In: Tagungsband ASIM 2014–22. Symposium Simulationstechnik, ASIM'2014 in Berlin, Tagungsband 2, vol. 43, pp. 357–363. ARGESIM (2014). https://doi.org/10.11128/arep.43
3. Kushnarenko, V., Miroshkin, O., Svjatnyj, V.: Towards the simulation of dynamic network objects with distributed parameters based on block difference numerical methods. In: 2017 9th IEEE International Conference on Intelligent Data Acquisition and Advanced Computing Systems: Technology and Applications (IDAACS), vol. 2, pp. 1073–1076 (2017). https://doi.org/10.1109/IDAACS.2017.8095250
4. Resch, M., Kuschnarenko, V., Svjatnyj, V.: Zur entwicklung der parallelen simulatoren für dynamische netzobjekte mit verteilten parametern. In: Simulation-2012, pp. 26–34. Kyiv (2012)
5. Svjatnyj, V., Kushnarenko, V.: Ein ansatz zur gleichmäßigen lastverteilung zwischen prozessoren des mimd-simulators der dynamischen netzobjekte mit verteilten parametern. Scientific works of DonNTU, series 'Problems of modeling and design automation' (1(15)), pp. 5–14 (2020)

Application of Digital Twins for High-Tech Enterprises and Infrastructure Objects in the Context of Digital Transformation

Oleksandr Prokhorov(✉), Oleg Fedorovich, and Evgeniy Druzhinin

National Aerospace University Kharkiv Aviation Institute, Chkalova Street, 17, Kharkiv 61070, Ukraine
o.prokhorov@khai.edu

Abstract. This paper contains case studies that help demonstrate the digital twins we have developed and their benefits. Several practical applications of digital twins for high-tech enterprises and infrastructure objects are introduced. Firstly, proposed digital model for analyzing the processes of controlling the flow of semi-finished products and calculating the characteristics of nanoelectronic production. This model is used to make decisions about the management of semi-finished products flows in multicluster technological complexes of any architecture, taking into account resource limitations and risk factors. Secondly, a simulation model of energy consumption at the metallurgical enterprise, which covers the entire metal products manufacture process, has been also developed. As a result of modeling, it was possible to increase the productivity of electric arc furnaces and ladle furnaces and reduce the maximum consumption of active power by the metallurgical enterprise. The third application is related to the creation of an adaptive intelligent system, for monitoring and controlling the technological process for electrochemical corrosion protection of main pipelines. The system provides remote control of electrochemical protection parameters and their optimization, as well as adaptive control of the parameters of cathodic protection stations while taking into account changes in external conditions. Two examples of simulation modeling of infrastructure objects of the street and road network and construction are considered.

Keywords: Digital transformation · digital twin · IoT · simulation modeling

1 Introduction

Digital transformation of business processes and business models is a necessary and relevant stage in the development of high-tech industry. It is based on the development and application of digital platforms, digital twins and implementation of the transition to cyber-physical systems [1]. Digital twin is a key concept for digitization in the Industry 4.0. Industry 5.0 complements the approach of Industry 4.0, using research and innovation as a tool for the transition to a sustainable, human-centered industry. The two key elements of a digital twin are a dynamic simulation model and data that reflects the

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2026
I. Potapov et al. (Eds.): RTC-Digital 2023, CCIS 2647, pp. 161–168, 2026.
https://doi.org/10.1007/978-3-032-04731-1_21

current state of a live system. With the model and the data, it is possible to build powerful digital twin software for experimentation, analysis, and communication [2]. Digital twin uses real-time monitoring data (using IoT) to drive dynamic updating of the model. We would like to present some of our projects related to the creation of digital twins. Development and application of digital twins – description of the behavior in various operating conditions of real materials, products, equipment, technological process and cyber-physical systems. It is based on a family of interconnected mathematical models with a high level of adequacy to real objects and processes. Moreover, the enterprise is represented not by a set of separated digital twins, but by a comprehensive modeling and simulation capacity embracing the full manufacturing process [3]. Therefore, simulation models must take into account the integration of data from the production system in real time.

2 Application of Digital Twins

2.1 Digital Twin of Multicluster Technological Complex in Nanoelectronic

We would like to present the results of our research on modeling of technological process in high-tech nanoelectronic production.

Our project was focused on the simulation model for analyzing the processes of controlling the flow of semi-finished products and calculating the characteristics of nanoelectronic production, which enables search for rational solutions related to managing the flows of semi-finished products in multicluster technological complexes of any architecture, taking into account resource limitations and risk factors [4].

The development and modernization of semiconductor production today is carried out in accordance with the requirements for high productivity, efficiency in processing a wide range of products, which is reflected in the multifunctionality and multivariance of technological equipment. Multicluster technological complexes (MTC) are characterized by a complex structure of objects and processes of multipurpose functioning; multiple structural layout solutions; complex interaction with a changing external environment etc. The effectiveness of the proposed approach is associated with the use of agent-based modeling and original methods of task planning and optimization, which search for rational routes of movement, accumulation and calculation of the performance criteria values depending on the proposed transport service models.

The simulation model (Fig. 1) of automated semiconductor production consists of components, each whereof reflects a separate element of the dynamics of the MTC behavior.

In the developed simulation model, using the class of active objects (agents), an input buffer, an output buffer, a cluster, linear transport are implemented, which enables usage of any number of instances of these classes in the model according to the number of elements involved in the MTC. In addition, simple classes are created in the model: module, wafer, transport, robot, operation.

It should be noted that the model can use a special mechanism to set up search among many alternatives (modules, queues, wafers, etc.) to set various strategies for controlling the flow of semi-finished products that ensure the fulfillment of the tasks set.

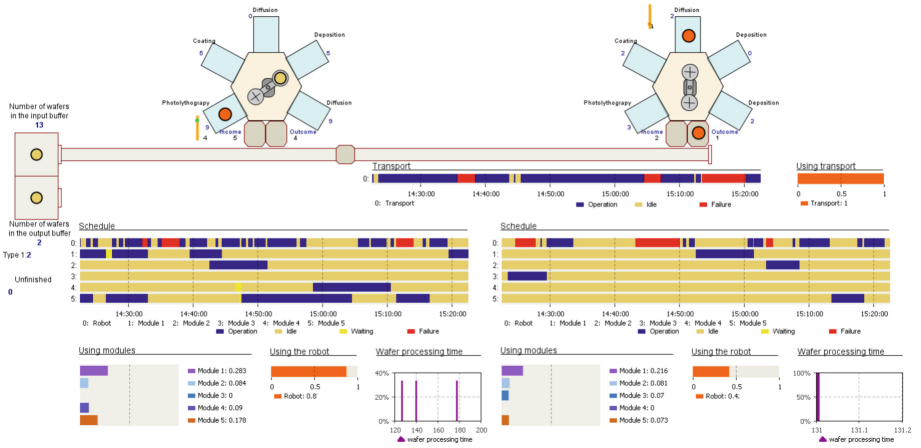


Fig. 1. Simulation with one type of wafers on two clusters.

Also in the model the following transport service models are realized: provide timely, without delays, transport services to MTC, taking into account restrictions on the cost of movement; minimize transportation costs during the planned period of MTC operation, taking into account restrictions on the delivery time of wafers; provide a solution to a compromise problem that minimizes the cost of movement and delivery time of wafers, taking into account restrictions on the cost and time of transport service.

The results obtained mean the achievement of the research goal, i.e. increasing the productivity of automated cluster equipment in semiconductor production based on the control of semi-finished product flows.

The model enables the search for rational solutions related to the control of semi-finished product flows, taking into account resource limitations and failures, and can be used in the design and operation of modern cluster multifunctional, high-performance plants with increased operational reliability.

2.2 Simulation of Energy Consumption Processes at the Metallurgical Enterprise

The task of energy-saving and efficient use of energy is one of the key issues for industrial enterprises within the Industry 5.0 concept framework. The forecasting and planning energy consumption in metallurgical enterprises is a multi-complex problem with probabilistic parameters and components, and the cause-and-effect relationship of energy consumption with each of these parameters is rather complicated and does not always have an unambiguous formal description.

Thereby, modeling is considered the only way to reflect all elements of a metallurgical enterprise's production system in a virtual environment for carrying out numerous experiments. The purpose of these experiments is to carry out predictive calculations, including energy efficiency parameters calculating while decision-making process of certain energy-saving projects and measures implementation [5].

To fully consider all interdependencies, constraints, dynamics and uncertainties, it was decided to create a simulation model of energy consumption at the metallurgical

enterprise. The agent model (Fig. 2) covered whole metal product manufacture processes. The initial data in the construction of the model contained information on the movement of material flows and units of products in steel smelting workshops, preparation of steel ladles, secondary refining, and steel casting.

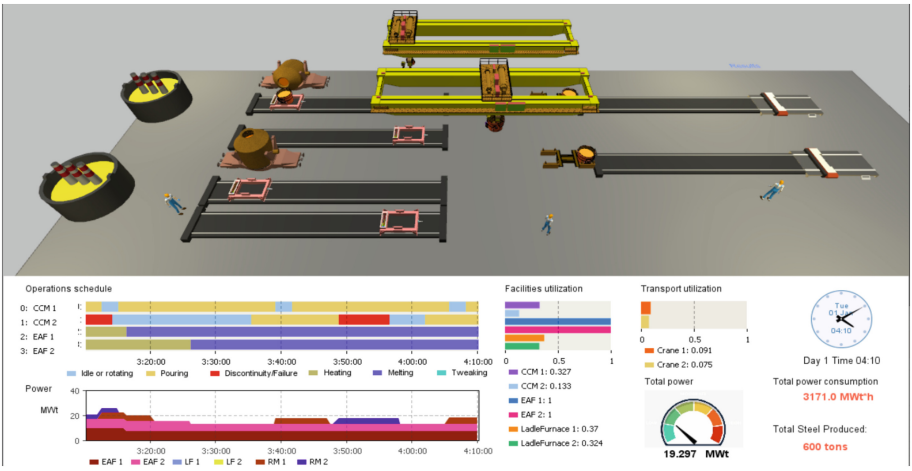


Fig. 2. Predictive modeling of electrical energy consumption at the metallurgical enterprise.

Routes of melting through the units and processing times coincide with the actual production plan at the enterprise. Each electric arc furnace (EAF) and every ladle furnace (LF) act as elements of the regulation model and objects. One of the tasks of modeling energy consumption at the metallurgical enterprise is to increase the productivity of the EAF and the LF by eliminating their idle. The idle can be caused by the discontinuity of steel casting on continuous casting machines (CCM) as a result of emergencies and reducing the maximum consumption of active power by the metallurgical enterprise for a fixed time of day. To do this, we use the existing reserves for regulating of ladle furnace power.

Thanks to multivariate calculations on the simulation model of transport services, the optimal routes of transport are selected depending on the formulations of energy consumption optimization and logistics management problems.

The automated control system for the technological process of steel-making receives signals from sensors and controllers and monitors the duration of technological operations at the production plants of each technological route.

Next, the system calculates the control ranges for EAF and LF that are permissible to ensure the continuity of steel casting during melting a certain steel grade according to the corresponding technological route.

With the help of a software package, it is possible to compare the permissible values and correct the predicted consumption of active power by the metallurgical enterprise for fixed time of day, depending on idle's introduction before starting the EAF and/or the LF. Further, using an automated control system for EAF operation modes, signals are

sent to a special agent-controller of the equipment to start each EAF ready for launching with a time lag that does not exceed the EAF melt duration period preceding the launch.

Observing the modeling process, the following can be obtained: current and total electrical energy consumption; volume of steel production; total load-graph of the metallurgical enterprise; Gant diagram (with an indication of forwarding and late indicators of starting-up and closing dates of technological operations, as well as idle and simulation of emergencies); diagrams and corresponding values of the load factors of the main equipment and vehicles. The model is fully animated in both 2D and 3D. With the help of the model, it is possible to solve a number of tasks, including the assessment of the rationality and efficiency of the existing structure of electrical energy consumption at the enterprise, forecasting the expected levels of electrical energy consumption during changing technology, production conditions, etc.

2.3 Electrochemical Corrosion Protection System of Pipelines Based on IoT and Optimizing Modeling

We have developed an adaptive intelligent system for monitoring and controlling the technological process of electrochemical corrosion protection of main pipelines, which is designed for remote control of electrochemical protection parameters, their optimization and adaptive control of parameters of cathodic protection stations (CPS). In accordance with regulatory and technical documentation, the system ensures that the EChP process is maintained at an optimal level between the destructive zones of "underprotection" and "overprotection", taking into account monitoring data, geological conditions at the pipeline laying site, climatic or seasonal changes and other factors. Our system involves the use of a CPS with inverter-type cathodic converters, a control and measuring point with telemetry equipment supporting modern industrial IoT class communication technologies and special software developed on the basis of intelligent information technologies [6]. In addition to the GSM communication networks often used in the EChP systems, our system also uses an energy-efficient long-range data transmission network LPWAN. As a programming tool for connecting end IoT devices, APIs and cognitive services of the developed platform, Node-RED is used a visual browser-based JavaScript runtime development environment for IoT. The one of the main modes of functioning for the proposed system is the mode of estimating the state of EChP objects in the pipeline system and the formation of control commands for the CPS. For this, an intelligent decision support system is used. The intelligent system allows for real time analysis and diagnostics in the presence of a significant number of significant factors that change dynamically. The intelligent system is also used to support decision making on pipeline corrosion diagnostics and monitoring, monitoring and generating control commands for the EChP facilities, the determination of ranges of protective potential values and other tasks. The solution of the optimization problem is based on the use of the linear programming method. The objective function is to minimize the sum of protective currents of all CPSs in a given optimization area. Figure 3 shows the results of modeling and optimization of the protective parameters of the CPS.

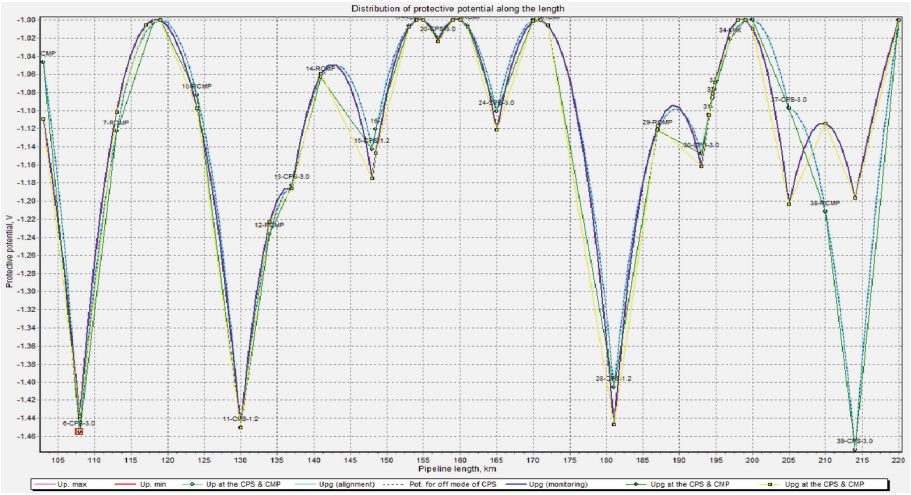


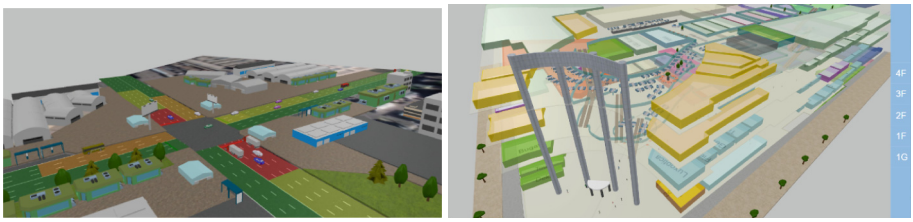
Fig. 3. A new plan for the distribution of the total potential after optimization.

2.4 Simulation Modeling of Infrastructure Objects of the Road Network and Construction

In recent years, there has been an increase in the number of complex infrastructural objects of the road network and buildings: air and railway terminals; city environment, public and residential buildings; shopping malls; parking areas, etc. Indeed, at the stage of designing large infrastructure objects, it is important to assess their capacity and throughput, draw up evacuation plans, business processes. Pedestrian simulation enables the effective development of solutions for planning, managing, and optimizing pedestrian flows in public buildings like airports, railway stations, shopping malls, and stadiums. Road infrastructure is inseparable from the general socio-economic environment in which it is embedded. Territories are developing, built up, the number of registered vehicles is increasing, due to which the volume of road traffic and its composition are constantly changing. Therefore, it is important at the design stage, and then at the operational stage, to be able to assess the limit parameters of the functioning of the road infrastructure in the conditions of the predicted capacity. Simulation modeling is a highly effective tool for evaluating traffic flows and the quality of service functioning in conditions of predicted road and passenger traffic. Making changes to the road network always has socio-economic effects, and these can be extremely severe when experiments are carried out on a real system. That is why it is so important to find optimal solutions before making changes.

A traffic model has been developed for one of Kharkiv's intersections with heavy traffic (Fig. 4, a), which allows carrying out capacity studies and optimizing phases of traffic light regulation. We will receive real traffic data from the traffic control system. Policies obtained in the modeling profession can be loaded into controllers of traffic light regulation equipment. Another model simulates the one of the shopping mall "Nikolsky" in Kharkiv (Fig. 4, b). The model allows you to estimate the filling of the shopping center according to different flows of visitors, especially on weekends and holidays. It provides

a representation of simulated visitor flows, where individual visitors have to visit several shops and, optionally, food zones and restaurants, toilets and a cinema. The model allows you to determine the traffic of visitors to the most popular stores. Also, the model allows you to study the map of the density of visitors in the halls, as well as to study evacuation scenarios.



(a) (b)

Fig. 4. City traffic modeling and optimization of traffic light regulation (a), pedestrian simulation of the shopping mall "Nikolsky" in Kharkiv (b).

3 Conclusions

We presented developed and implemented solutions from various industries. The simulation model for make decisions about the management of semi-finished products flows in multicluster technological complexes are proposed. Using a digital twin for a metallurgical plant, the productivity of electric arc furnaces and ladle furnaces was increased and maximum energy consumption was reduced. A digital twin has been created to monitor and manage the parameters of cathodic protection stations, taking into account changes in external conditions in linear sections of main pipelines. Two examples of simulation modeling of infrastructure objects of the street and road network and construction are considered. We would like to develop cooperation in this and other scientific areas related to the creation of digital twins, simulation modeling, IoT, artificial intelligence and machine learning.

References

1. Pronchakov, Y., Prokhorov, O., Fedorovich, O.: Concept of high-tech enterprise development management in the context of digital transformation. Computation **10**(7), 118 (2022). https://doi.org/10.3390/computation10070118
2. Becue, A., Maia, E., Feeken, L., Borchers, P., Praça, I.: A new concept of digital twin supporting optimization and resilience of factories of the future. Appl. Sci. **10**, 4482 (2020). https://doi.org/10.3390/app10134482
3. Angira, S., Kosasih, E., Zhang, J., Brintrup, A., Calinescu, A.: Digital twins: state of the art theory and practice, challenges, and open research questions. J. Ind. Inf. Integr. **30**, 100383 (2022). https://doi.org/10.1016/j.jii.2022.100383
4. Prokhorov, O., Pronchakov, Y., Fedorovich, O., Kunanets, N.: Modeling of technological process in nanoelectronic production. In: Proceedings of the 2020 IEEE 15th International Conference on Computer Sciences and Information Technologies, CSIT, Zbarazh, Ukraine, pp. 324–327 (2020). https://doi.org/10.1109/CSIT49958.2020.9321926

5. Kiyko, S., Druzhinin, E., Prokhorov, O., Haidabrus, B.: Multi-agent Model of Energy Consumption at the Metallurgical Enterprise. Lecture Notes in Mechanical Engineering, pp. 156–165 (2020). https://doi.org/10.1007/978-3-030-50794-7_16
6. Prokhorov, O., Prokhorov, V., Khussanov, A., Khussanov, Z., Kaldybayeva, B., Turdybekova, D.: Complete integrated automation of the electrochemical corrosion protection system of pipelines based on IoT and big data analytics. Computation **10**, 7:123 (2022). https://doi.org/10.3390/computation10070123

Development and Implementation of Digital Transformation Tools in the Domain of Apparel Design

Oksana Zakharkevich[1(✉)], Tetiana Zhylenko[2(✉)], Julia Koshevko[1(✉)], and Galina Shvets[1(✉)]

[1] Khmelnytskyi National University, 11, Institytska Street, Khmelnytskyi, Ukraine
zakharkevycho@khmnu.edu.ua, juliakoshevko@gmail.com, galyamet@ukr.net
[2] Sumy State University, 2 Rymskogo-Korsakova Street, Sumy, Ukraine
t.zhylenko@phe.sumdu.edu.ua

Abstract. The purpose of the is to develop a range of mobile applications as tools for supporting apparel design and manufacturing processes. Applications for the calculation of different garment types patterns such as "CloStyler", "SHOES Step-by-Step", "RDMK Step-byStep", "NUnderwear" are purposed to reduce the time wasted while constructing patterns and at the same time to increase their accuracy. The applications "MatVed" and "TechLab" primarily aim to support technological processes and increase learning efficiency.

Keywords: Mobile app · Techniques · Sewing

1 Introduction

Many researchers worldwide use mobile apps for learning and teaching. The most prominent of them are Kim Y. and Smith D. [2], Zaranis N., Kalogiannakis M. [12], Mackaya B.J., Anderson J., Hardingb T. [4]. We can see evidence of great variety in the works [1,3,5–9,13]. Some are working on specific issues of using mobile apps as a part of smart technology. Others are studying the possibilities of using mobile applications as 3D scanners.Some of their results might be used in clothing designers' education. However, specific issues of using mobile apps in training future apparel design experts have just now been observed. Thus, the project aims to develop a range of mobile applications as tools for supporting the apparel design and manufacturing processes as well as the educational processes of clothing designers and pattern makers.

2 Methodology

The first step in any application development is market exploration. Thus, we performed a statistical analysis of the existing apps that are or might be used for

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2026
I. Potapov et al. (Eds.): RTC-Digital 2023, CCIS 2647, pp. 169–174, 2026.
https://doi.org/10.1007/978-3-032-04731-1_22

clothing design, pattern drafting, clothing distribution, educational purposes, or any other activity regarding clothing design. Input data for the current research are the search results on Google Play and AppStore by keywords "clothing", "clothing patterns", "clothing design", etc. We discovered that many stages in apparel design are in dire need of specific mobile applications. Besides that, some applications must be replicated in another operating system. It would allow for preparing unified instructions for all the students with no regard to what exactly smartphones they have. On the other hand, some stages of garment design are perfectly equipped with mobile tools in both operating systems. There is a variety of applications to choose from in categories of "Wardrobe organisers", "Online shops apps", "Galleries of clothing styles", and "Galleries of ready-made clothing patterns" Every one of them consists of several apps that have both Android and iOS versions, besides all applications that have not.

3 Results and Discussion

We developed a range of mobile applications that students can use in the educational process of apparel design. Applications for the calculation of different garment types patterns such as "CloStyler", "SHOES Step-by-Step," "RDMK Step-byStep" [10], "NUnderwear" [11] are purposed to reduce the time wasted while constructing patterns and at the same time to increase their accuracy. The mobile applications are concerned with top-wear garments and underwear. Users may choose a pattern drafting method and a garment type in these apps. The applications "MatVed" and "TechLab" primarily aim to support technological processes and increase learning efficiency.

3.1 Applications for the Pattern Design "CloStyler"

The mobile application "CloStyler" (Fig. 1, a) aims to calculate the parameters of the basic blocks of clothing. In the lite version of the app, it is possible to calculate two garment types' constructions: a skirt and a top. The app's pro version allows users to calculate clothing blocks' parameters by three different pattern drafting methods. Eight garment types are available: a top, pants, a suit jacket, a jacket, a skirt, a dress, a blouse, and a shirt. Functional languages are as follows: English and Ukrainian. Users may choose a pattern drafting method as well as a garment type. The input data for calculation are body measurements and the amount of ease. A user might enter the body measurements and amount of ease. The app allows users to complete the text fields by a shorthand method on the basis of what has been typed before. Otherwise, the fields will be autocompleted by zeros.

3.2 "SHOES Step-by-Step"

"SHOES Step-by-Step" is the app for calculating the original design of shoes. Two application languages are used: Ukrainian and Russian. A significant number of construction stages (steps) led to the operator's need to go from the main

page to any of the steps. To work with the "SHOES Step-by-Step application", the user enters the source data and presses the "START CONSTRUCTION" button (Fig. 1, b). The user is provided with an image of the construction drawing, a sequence of formulas, the names of the segments, and their calculated values. Dialog boxes of the developed mobile application are presented in Fig. 1. The input fields will automatically fill with zeros if the user does not enter the source data. The developed application "SHOES Step-by-Step" is available for testing[1].

3.3 "RDMK Step-by-Step"

To work with the application "RDMK Step-by-Step", the user chooses the style of the garment to be designed. The app calculates the garment parameters with set-in sleeves and owlman-style garments. The construction of the owlman style may be performed only after computing the parameters of the basic bodice design (set-in sleeve). After that, the user chooses the "BODICE" or "SLEEVE" button (Fig. 1, c) and enters the source data, and presses the button "START CONSTRUCTION." The user is provided with an image of the construction drawing, a sequence of formulas, the names of the segments, and their calculated values. Many construction steps led to the operator's need to go from the main page to any of the steps in a similar way as shown for the application "SHOES Step-by-Step."

3.4 "NUnderwear"

In the preliminary study of the current research, we developed the mobile application named "NUnderwear" (Fig. 1, d) [11]. The application is the only one on the market of mobile applications in the field of garment design that allows calculating parameters of underwear patterns. The input data for calculation are body measurements and the amount of ease. A user must enter the body measurements and amount of ease. The app allows users to complete the text fields by a shorthand method based on what has been typed before. Otherwise, the fields will be autocompleted by zeros. The order of calculation is performed according to the pattern drafting method. The names of the constructive segments correspond to the points in the given figures of pattern blocks.

3.5 "Hat and Cap"

The developed mobile application "Hat and Cap" is intended for mobile devices on the Android operating system. The primary audience of the application is teachers and students studying fashion design. In the application (Fig. 1, e), the design calculation of two models of balaclavas is available. Entering the "head girth" size feature also allows one to determine the size of their headdress when purchasing ready-made products. The application offers two options for drafting the headwear design according to typical parameters or individual ones.

[1] https://play.google.com/store/apps/details?id=appinventor.aizbirvukladach.Shoescopy&hl=en.

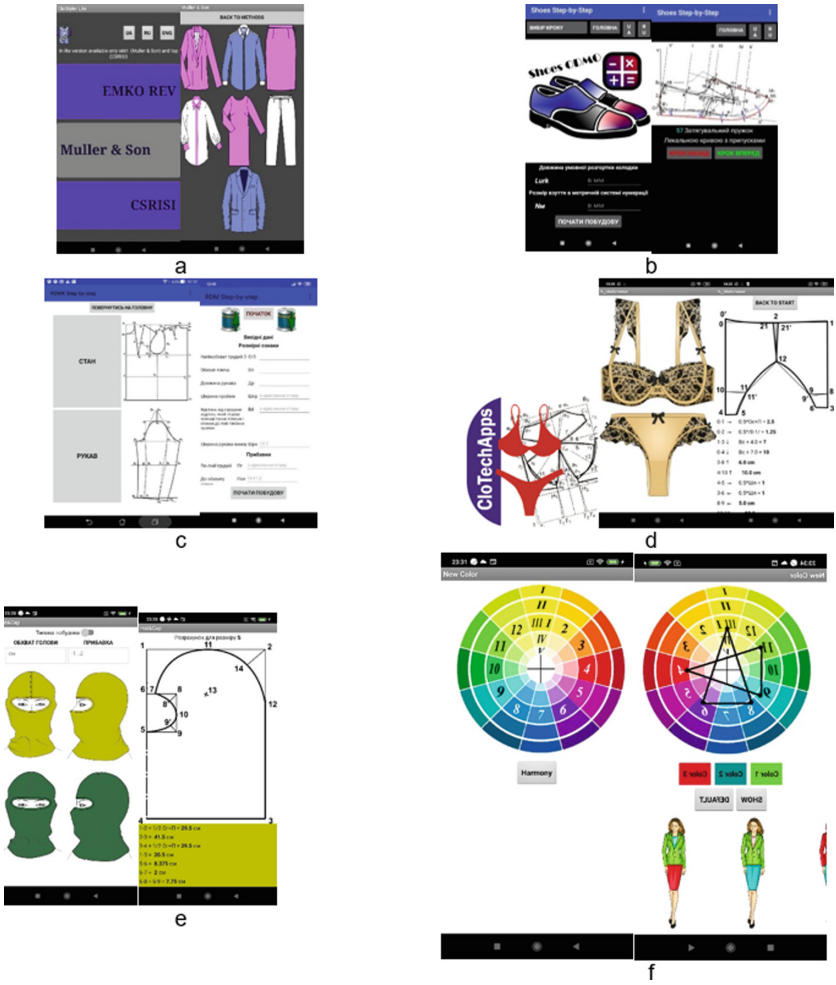


Fig. 1. Developed mobile applications: a) CloStyler; b) "SHOES Step-by-Step"; c) "RDMK Step-by-Step"; d) "NUnderwear"; e) "Hat& Cap"; f) "NewColor".

3.6 "NewColor"

The "NewColor" application provides visualisation of harmonious and balanced color combinations in clothing models, which helps the user create exquisite, noble and elegant combinations. In developing colouristics in a costume ensemble for women, we eliminated the risk of accidental errors due to the human factor.

3.7 Applications for Supporting Technological Processes "TechLab"

The mobile app "TechLab" was developed using MIT App Inventor. We generate a QR code to download the program quickly. The mobile application contains

six laboratory works for studying the features of manufacturing leather and fur garments. Three of them are devoted to studying manufacturing natural and artificial leather clothes. The other three laboratory works are related to manufacturing corresponding elements of natural and artificial fur garments. Thus, the main subjects of laboratory work are:

- the technology for processing pockets;
- technology for processing collar, neck-front facing, and hem;
- technology for processing sleeves.

Every laboratory work includes 3 to 10 variants of manufacturing main product elements' design and sewing techniques. For the beginning to work with the application, the user selects the laboratory work to be completed. The next step is to choose a specific product element (Fig. 1, d). Each product element represents by an appearance and several cross-sections indicated on the external view. By pressing the "TP" button, one can see a technology sequence for manufacturing product elements. The numbers of operations in the sequence refer to all cross-sections within the product element. All assembly schemes are presented in colour variants for easy understanding by the users. Colour conventions are shown in each laboratory work: red is the base material, green is the lining material, pink is the interlining, and blue is the adhesives joining. The user gets acquainted with the technology sequence for manufacturing product elements and puts the operation numbers on the scheme following the sequence. In the end, one can check the correctness of the selected operations by pressing the "green checkmark" under each section. The available language of the mobile app "TechLab" is Ukrainian. After successfully testing the application in higher educational institutions, ensuring its operation in English is advisable. Performing laboratory work, the student strengthens theoretical knowledge, acquires the skills to represent assembly schemes for processing main product elements, and composes a technology sequence for indivisible manufacturing operations, coding seams, and stitches.

3.8 "MatVed"

The mobile app "MatVed" for Android OS is an additional resource for selecting textile materials to manufacture clothes of a particular assortment. "MatVed" is designed to reduce time costs and improve the accuracy of mathematical calculations. The mobile app allows users to calculate the value of the different types of fabrics' properties. The program itself writes all basic calculation algorithms. To work with the application, the user selects the values of fabrics' properties to be determined, enters the input data measured with the special equipment, and presses "CALCULATE". The application provides the ability to compare the calculated values with standard data.

4 Conclusion

The purpose of the project was successfully achieved by developing a range of mobile applications as tools for supporting apparel design and manufacturing

processes. The developed apps allow achieving the same calculation precision as other calculation methods while the speed is much preferable. Besides that, the risk of accidental mistakes due to the human factor is excluded from designing patterns, selecting textile materials, etc.

References

1. Khaddage, F.: iTeach we learn via mobile apps "a casestudy in a business course". In: Ron McBride, M.S. (ed.) Society for Information Technology & Teacher Education International Conference, New Orleans, Louisiana, 25 March 2013, pp. 3225–3233. Association for the Advancement of Computing in Education (AACE), Association for the Advancement of Computing in Education (AACE), Waynesville, NC USA, New Orleans, Louisiana, United States (2013)
2. Kim, Y.: Pedagogical and technological augmentation of mobile learning for young children interactive learning environments. Interact. Learn. Environ. **25**(1), 4–16 (2015). https://doi.org/10.1080/10494820.2015.1087411
3. Lumsden, C.: Using mobile devices for teaching and learning in clinical medicine. Arch. Dis. Child. Educ. Pract. **100**(5), 244–251 (2015). https://doi.org/10.1136/archdischild-2014-306620
4. Mackaya, B.: Mobile technology in clinical teaching. Nurse Educ. Pract. **22**(2), 1–6 (2017). https://doi.org/10.1016/j.nepr.2016.11.001
5. Mouzaa, C.: Bridging the app gap: an examination of a professional development initiative on mobile learning in urban schools. Comput. Educ. **88**, 1–14 (2015). https://doi.org/10.1016/j.compedu.2015.04.009
6. Pereira, C.: A mobile app for teaching formal languages and automata. Comput. Appl. Eng. Educ. **26**(5), 1742–1752 (2018). https://doi.org/10.1002/cae.21944
7. Rosell-Aguilar, F.: A taxonomy and framework for evaluating language learning mobile applications. Calico J. **34**(2), 243–258 (2017). https://doi.org/10.1558/cj.27623
8. Shraim, K.: Perceptions of using smart mobile devices in higher education teaching: a case study from Palestine. Contem. Educ. Technol. **6**(4), 301–318 (2015). https://doi.org/10.30935/cedtech/6156
9. Thomas, R.: Effectiveness of mobile apps in teaching field-based identification skills. J. Biol. Educ. **51**(2), 136–143 (2017). https://doi.org/10.1080/00219266.2016.1177573
10. Zakharkevich, O.: Clostyler – mobile application to calculate the parameters of clothing blocks. In: Ron McBride, M.S. (ed.) IOP Conference Series: Materials Science and Engineering, vol. 1031, p. 012031. Association for the Advancement of Computing in Education (AACE), Waynesville, NC USA, Yambol, Bulgaria (2021). https://doi.org/10.1088/1757-899X/1031/1/012031
11. Zakharkevich, O.: Development of the mobile applications for using in apparel and shoes desing. Vlákna a textil **28**(2), 105–122 (2021)
12. Zaranis, N.: Using mobile devices for teaching realistic mathematics in kindergarten education. Creative Educ. **4**(7A), 1–10 (2013). https://doi.org/10.4236/ce.2013.47A1001
13. Zhylenko, T.: Mobile application to calculate the parameters of top wear basic design. Nauka Innov. **15**(3), 24–34 (2019). https://doi.org/10.15407/scine15.03.024

Mathematical Learning Experiment Using Digital Technologies

Mariia Astafieva, Dmytro Bodnenko, Oleksandra Lokaziuk, Oksana Lytvyn, and Volodymyr Proshkin(✉)

Borys Grinchenko Kyiv Metropolitan University, 18/2 Bulvarno-Kudriavska Street, Kyiv 04053, Ukraine
{m.astafieva,d.bodnenko,o.lokaziuk,o.lytvyn,v.proshkin}@kubg.edu.ua

Abstract. The article examines the benefits of learning experiments using digital technologies in research based mathematics learning. We discuss the didactic purposes and tasks of learning experiments in mathematical education. In order to demonstrate the practical advantage of using digital tools, we describe four specific examples of mathematical learning experiments, based on Geogebra, Desmos and Maple.

Keywords: Mathematical learning experiment · Digital technology · Research-based learning · GeoGebra · Desmos · Maple

1 Introduction

Modern trends in the educational paradigm change are reflected in the Council of Europe document [5], which defines research-based learning as the leading educational methodology of the 21st century. The research study is possible only when students act as active study subjects and can solve cognitive problems. At the same time, mathematical experiments, observations and research are some methods of implementing such an approach to learning. They enable the involvement of students in active search activities and the implementation of research-based learning. The result is not only the solution to educational problems but also the formation of students' motivation to study mathematics, cognitive activity, the ability to work independently and in collaboration, etc. Using digital technologies for conducting experiments contributes to a better understanding of mathematical concepts, the active student's involvement in constructing their knowledge, and developing their research skills.

2 Literature Review

To date, several studies on the problems of experimental mathematics have been implemented. These studies revealed the possibilities of using its various methods, particularly using digital technologies [1,3,4,8,10]. The previous research

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2026
I. Potapov et al. (Eds.): RTC-Digital 2023, CCIS 2647, pp. 175–181, 2026.
https://doi.org/10.1007/978-3-032-04731-1_23

of this study's authors shows the advantages of digital visualisation in forming students' mathematical competence. In addition, it disclosed possibilities of involvement of digital visualisation tools for conducting mathematical experiments using dynamic geometry environments GeoGebra and advanced graphing calculator Desmos [2].

However, despite numerous studies, the possibilities of learning experiments in teaching and learning mathematics and using digital tools for its implementation still need to be presented.

Therefore, the article aimed to demonstrate the possibilities of learning experiments using digital tools for research-based mathematics learning.

3 Research Results

3.1 Aims and Stages of Mathematical Learning Experiment

The mathematical experiment's aim in learning mathematics is to form the student's ability to think mathematically: to experiment, observe, notice patterns, check theoretical results, etc. That is, to give students a full-fledged mathematical experience. A mathematical experiment is used to formulate hypotheses based on observed facts and properties; to check the hypotheses or the correctness of the results established analytically; to visualise mathematical concepts, facts, and principles for a better understanding of them; to predict a possible result to assess whether it is worth a formal proof; to see the result immediately, if its rigorous mathematical proof is not the ultimate goal; to replace (automate) cumbersome manual calculations and complex transformations.

In [8], it has been proven that learning mathematics aims easier to implement if they are supported by an experiment, particularly with the involvement of digital technologies.

We highlight the stages of a mathematical learning experiment using digital technologies are as follows:

1. Determination of the experiment learning aim, formulation of its task as a mathematical research problem, justification of the feasibility of involving digital technologies, and selection of digital tools.
2. Solving the research task: developing a computer model, conducting a mathematical experiment, obtaining its results (formulating a new mathematical hypothesis, setting a problem, developing a certain algorithm, refuting the result obtained analytically).
3. Reflection and conclusions regarding achieving the research and learning aims.

3.2 Examples

Example 1. The mathematical experiment as an element of research-based learning involves changing the key parameters of the built model to observe its behaviour. GeoGebra makes building dynamic models of mathematical structures and geometric objects possible [7]. By creating and researching such

dynamic models, the student can observe the properties of geometric objects that are invariant under various transformations. For example, let's consider the mathematical experiment in GeoGebra when studying "Pascal's theorem in a projective plane".

The student is invited to investigate the collinearity of the three points of intersection of the opposite sides of a hexagon inscribed in a conic (it may be an ellipse, parabola or hyperbola). The simplest image is a convex hexagon inscribed in an ellipse (Fig. 1a). By moving one of the vertices of the hexagon along the conic, the student can observe other configurations: the hexagon with self-intersections (Fig. 1b); another conic (if you move one of the five vertices-points defining the conic, Fig. 1c). But with any transformations, the points of intersection of the opposite sides of the hexagon are on the same straight line (Fig. 1a–c). Thus, the student can summarise the results of the experiment in Pascal's theorem: if six arbitrary points are chosen on the conic and joined by line segments in any order to form a hexagon, then the points of intersection of the pairs of its opposite sides are collinear (lie on a straight line—Pascal line).

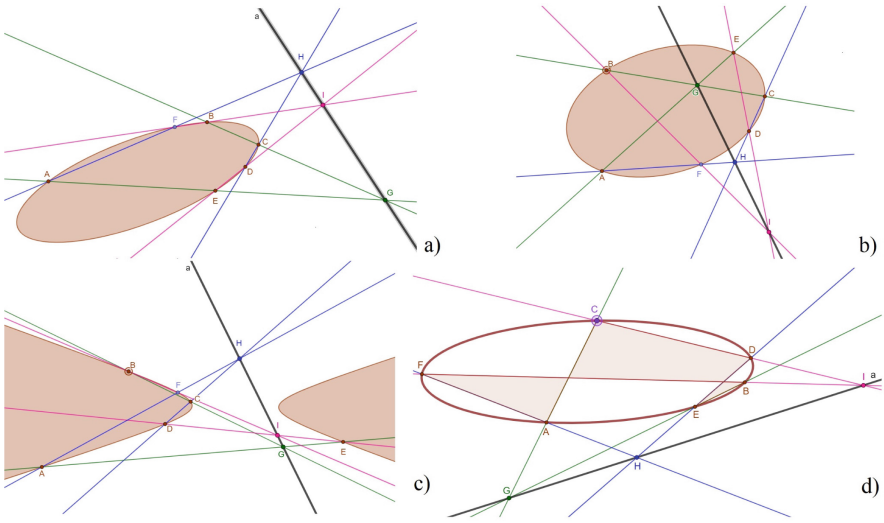


Fig. 1. The dynamic model to Pascal's theorem

If students construct an arbitrary hexagon ABCDEF; find the points of intersection of three pairs of opposite sides of the hexagon; draw the straight line through two of the opposite sides (for example, G and I); by moving the vertices of the hexagon, ensure that the third point of intersection of the opposite sides (point H) falls on the GI line (Fig. 1d). Then, they can found that the sixth vertex of the hexagon (point C in the figure) falls on the conic defined by the other five vertices. That is, we have a statement that is the inverse of Pascal's theorem.

Example 2. One of the concepts of mathematical analysis that causes first-year mathematics students the most difficulty in understanding and mastering is the uniform continuity of the function. The formal definition of the uniform continuity of the function of one variable looks like this:

The function $f(x)$ is called uniformly continuous on the interval X if for any $\varepsilon > 0$ there exists $\delta = \delta(\varepsilon) > 0$ such that from the condition $|x_1 - x_2| < \delta$, the inequality $|f(x_1) - f(x_2)| < \varepsilon$ follows for any x_1 and x_2 from the set X.

A lack of understanding of the essence of the concept leads to the fact that many first-year students cannot correctly reproduce this definition, or, having invented it, they quickly forget it. Therefore, we begin the study of the concept of the uniform continuity of the function with a visual experiment using GeoGebra. Let's describe this experiment.

If the function $f(x)$ is continuous on the interval X, then it is continuous at any point x_0 of this interval. That is, $\forall\varepsilon > 0\ \exists\delta > 0 : |x - x_0| < \delta \Rightarrow |f(x) - f(x_0)| < \varepsilon$.

It is evident that δ depends not only on ε (the smaller ε, the smaller δ) but also on the point x_0. If the value of δ does not rely on the point x_0, that is, the same δ is suitable for all points of the interval X, then the function is uniformly continuous on this interval.

Consider in GeoGebra the graph of the function $f(x) = 1.5\log_{0.5}(x + 0.5)$ on the interval $(-0.5; 2]$. Construct a rectangular frame with height ε and width δ. We move this frame along the graph of the specified function in the direction from right to left. Let's try to make it so that the curve crosses only the vertical sides of the rectangular frame so that the values of the function differ less than by ε. By moving point x along the abscissa axis from right to left, students make sure that at some point, the curve "breaks out" from a rectangular frame on its upper side (Fig. 2a).

This means that it is possible to specify two argument values that differ by less than δ, and the distance between their corresponding function values is greater than ε. To "keep" the points of the graph within the height of the frame, this frame will have to be narrower, i.e. reduce δ (Fig. 2b). You can see that the new frame will not satisfy us for long. Designing increasingly narrow frames will not help because one frame cannot cover the entire infinite curve, no matter how narrow it is. That is, this function is not uniformly continuous on the interval $(-0.5; 2]$.

However, students notice that if we consider the same function on the interval, for instance, $[-0.49; 1]$, i.e. cut it off, it is possible to find a frame with which to "pass" the entire curve. Thus, they reach a hypothesis about the uniform continuity of the function on a closed interval (Cantor's theorem, one of the most important theorems of mathematical analysis).

Finally, we note that preparing the described dynamic model (as well as many others) is difficult. Therefore, we often involve senior students in the development of such models. This is a valuable practice for them.

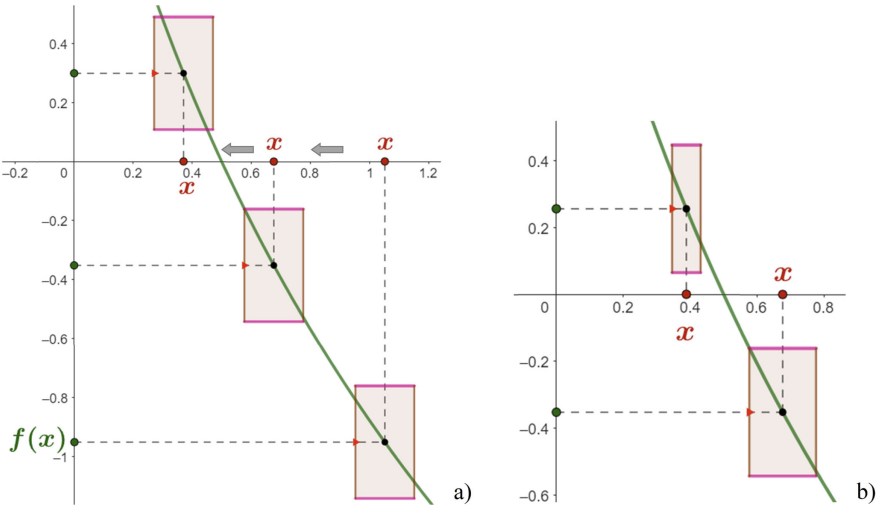


Fig. 2. The dynamic model to definition of the uniform continuity of the function

Example 3. The task of calculating the limit

$$\lim_{x\to\infty}\left(x - x\cos\frac{1}{x}\right)$$

caused difficulties for first-year mathematics students. One of the students suggested using a graphic representation of a function whose limit at infinity should be found (Fig. 3). This made it possible to see the result (0), after which an analytical solution method was found.

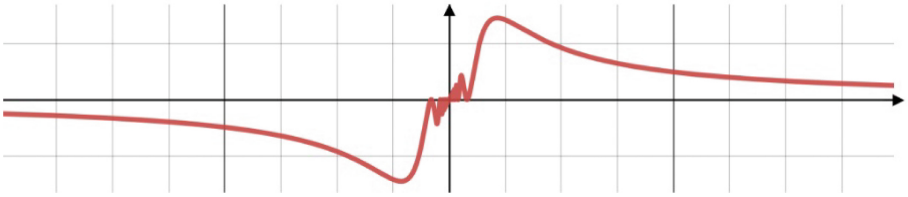


Fig. 3. The graph of the function made in the graphic calculator Desmos [6]

If finding the boundary was not the main task but only an intermediate stage of solving some problem, the result obtained based on a graphic image would be reasonably sufficient.

Example 4. Maple has a wide range of possibilities for conducting educational mathematical experiments in higher mathematics [9]. The example of the Maple application is studying the equilibrium points types of differential equations systems. By changing the system's parameters, the student can visually identify

equilibrium points and analyse the movement of system solutions around these points. By changing the coefficients of the Lorentz system of differential equations, the student analyses the changes in the oscillogram graphs (Fig. 4a) and the appearance of the Lorentz attractor projections on the planes of the three-dimensional phase space (Fig. 4b) based on the solutions in Maple. In this case, these projections are irregular orbits around unstable fixed points.

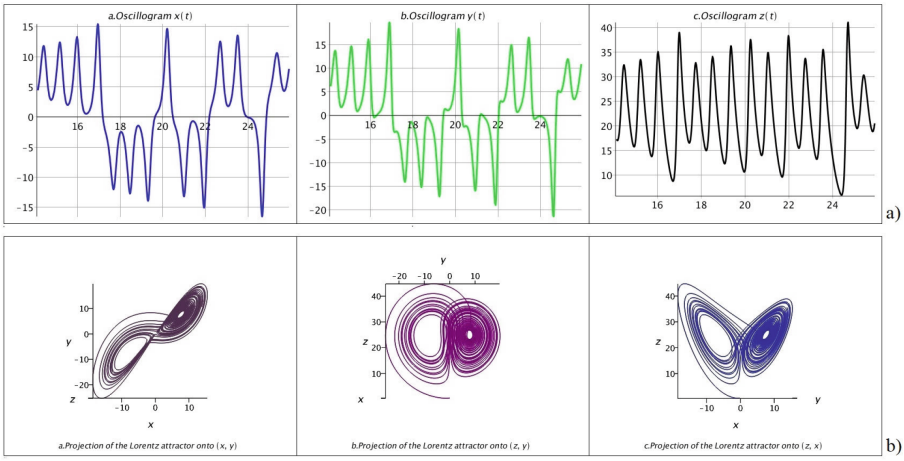


Fig. 4. Investigation of the equilibrium points of differential equations systems

4 Conclusions

It has been demonstrated that learning experiments, as a teaching and learning strategy, have an essential role in the mathematical preparation of students, forming their full-fledged mathematical experience. Digital technologies, in turn, provide a wide range of interactive opportunities for experimentation and exploration.

The possibilities of learning experiments using digital tools for research-based mathematics learning are illustrated by specific examples.

Analysis of the effectiveness of learning experiments using digital technologies in teaching and learning mathematics will be the subject of our further research.

References

1. Arnol'd, V.: Experimental mathematics. MSRI Mathematical Circles Library, vol. 16, American Mathematical Society (2015)
2. Astafieva, M., Bodnenko, D., Proshkin, V., Lytvyn, O.: The use of digital visualization tools to form mathematical competence of students. CEUR Workshop Proc. **2740**, 416–422 (2020)

3. Bailey, D., Borwein, J.: Mathematics by Experiment Plausible Reasoning in the 21st Century (2016)
4. Baker, A.: Experimental mathematics. Erkenn **68**, 331–344 (2008)
5. Competences for democratic culture. Living together as equals in culturally diverse democratic societies. Council of Europe (2016)
6. Desmos Studio, PBC, ATTN: Eli Luberoff, Beaverton, Oregon (2023)
7. Geogebra (Classic), Internationales GeoGebra Institut, Linz, Austria (2023)
8. Kortenkamp, U.: Experimental mathematics and proofs in the classroom. Zentralblatt für Didaktik der Mathematik **36**, 61–66 (2004)
9. Maple 2018: Maplesoft, a division of Waterloo Maple Inc., Waterloo, Ontario (2018)
10. Rogness, J.: Mathematics by experiment plausible reasoning in the 21st century (2011)

Mathpar Project

Gennadi Malaschonok(✉)

National University of Kyiv-Mohyla Academy, 2 Skovorody, Kyiv 04070, Ukraine
malaschonok@ukma.edu.ua
https://fin.ukma.edu.ua

Abstract. We present the Mathpar project. It is based on the MathPartner computer algebra system, which underlies the three areas of research. The first is freely available cloud mathematics MathPartner. The second is a supercomputer runtime that is designed for parallel recursive matrix algorithms for dense and sparse matrices. And the third area is a platform for e-learning in physical and mathematical disciplines, which allows you to abandon the use of paper notebooks and pens.

Keywords: Cloud math · MathPartner · Computer algebra · Supercomputing · GPU computations · e-learning

1 Introduction

The computer era began in Kyiv with the creation of the MESM computer in 1951. It was created in the laboratory of Sergei Lebedev. It was the first computer in continental Europe.

The history of computer algebra systems in Kyiv begins with the creation of the MIR computer at the Institute of Cybernetics of the Ukrainian Academy of Sciences in 1965. Its language ALMIR-65, a later version of which was called ANALYTIC, is considered one of the first computer algebra systems. Around these same years, Martinus J.G. Veltman created Schoonschip (1963) [4], Bell Labs created ALPAK (1964) and ALTRAN (1966), in IBM Jean E. Sammet created FORMAC (1964) [1,2].

One of the members of the team that created the MIR machine was Alexander Adolfovich Letichevsky. Later, under his leadership, a new computer algebra package APS [5] was developed. In the late eighties, I was a member of the working group (RG-24) of the countries of the Council for Mutual Economic Assistance on computer algebra, which he headed.

Computer algebra algorithms, whose history began at Lviv's Institutes of the Academy of Sciences, at Lviv, Kyiv and Tambov universities, formed the basis of the MathPartner Cloud Math, created by the Mathpar team. The project platform is deployed on the server [6].

Today "Mathpar" project has three research directions.

(1) Cloud Math "MathPartner" is a freely available computer algebra system. We use Java, Spring JS, MathJax, LaTeX, Mathpar.

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2026
I. Potapov et al. (Eds.): RTC-Digital 2023, CCIS 2647, pp. 182–189, 2026.
https://doi.org/10.1007/978-3-032-04731-1_24

(2) "MathparDAP" is a supercomputer runtime and library of recursive matrix algorithms. We use OpenMPI, Java, jCUDA.
(3) "MathparEdu" is e-education platform for physics and mathematics. It is used in the educational process at our university, at Kyiv Academic University and the University of Life and Environmental Sciences.

2 What Is Mathpar?

It is a language for writing mathematical texts. On the one hand, it is one of the TeX dialects, and on the other hand, it is the user language of the computer algebra system "MathPartner". The first version of the name of this language was A-TeX (Active TeX). We are doubly grateful to Professor Donald Knuth for the birth of language MathPar, since he created the TeX language and he wrote one of the first computer algebra textbooks, "The Art of Programming", Volume 2, "Seminumerical Algorithms".

Compiling the standard language LaTeX produces only beautiful mathematical text, for example, in PDF format. In addition to this, the compilation of the Mathpar language also gives the result of performing the indicated actions in the computer algebra system MathPartner.

This result also has two representations: the original text format, which can be further used for calculations, and the PDF format for easy viewing of the resulting expressions.

In the same browser window, the user can easily switch between these two displays.

In addition, all mathematical objects get their own names and these names can be used in further calculations. You can use the PRINT operator to see their values. If such an operator is not used, then only the value of the last expression will be displayed.

3 Cloud-Based Computer Algebra System "MathPartner"

MathPartner can be used for scientific and engineering calculations, as well as in high schools and universities. It allows one to carry out both simple calculations (acting as a scientific calculator) and complex calculations with large-scale mathematical objects.

Mathpar is the language of the user. It is a procedural language and it supports a large number of elementary and special functions, as well as matrix and polynomial operators. This service allows you to build function plots and animate them.

Today, the project platform is deployed here [6].

You can get acquainted with the current state of this project using the User's Guide, which can be downloaded on this site, through its Help Pages, as well as with the help of tips that can be found in the side bar.

The basic features of cloud mathematics (see [7]):

- Cloud systems can be accessed from any computer connected to the Internet.
- They provide users with advanced services, including a workbook with a user-friendly interface.
- They provide users with the possibility of solving the problems that cannot be solved on a personal computer. This can be done through the access to parallel software packages installed on a supercomputer.
- An important feature is the support of the REST-Ful protocol for data communication with other Internet-connected devices. Computational results can be obtained by the cloud without human assistance.
- Mathematical calculations in the form of program texts for cloud systems can be stored in generally accessible libraries. Any mathematical problem can be solved by uploading such a text file from a shared library to the cloud workbook.
- Mathematics textbooks and reference books can be complemented with the libraries containing text files with the corresponding algorithms designed for computations in the cloud workbook.
- This can be the basis for creating a very good service for physics and mathematics education. It will include textbooks, problem books, automatic verification of the correctness of problem solving, demonstration of the correct solution of a learning problem, saving the results of tests and exams in the student's record book.

4 Supercomputer Runtime "MathparDAP"

We are developing a new runtime environment for a distributed memory supercomputer [10].

It focuses on recursive matrix algorithms. Since Alexander Karatsuba proposed a new way to multiply numbers and polynomials, and then Volker Strassen [10] proposed a new way to multiply matrices, many new recursive algorithms have appeared. These are QR [9] and LEU factorisation of matrices, Cholesky decomposition [10], calculation of the adjoint and inverse matrix, calculation of the echelon form of a matrix, and others [3].

We also develop matrix algorithms using GPUs. Sergii Sukharskyi's report is devoted to this direction.

We plan to end up with a library of programs for a supercomputer equipped with GPUs.

To explain the features of MathparDAP runtime, we can consider a simple example. It is the inversion algorithm for the non singular triangular matrix.

Let an invertible triangular matrix A of size $n = 2^p$, $p > 0$, be given and it is required to calculate its inverse matrix. Such a matrix can be divided into 4 square blocks. Denote the blocks of the matrix A by $a, 0, c$, and d,; and denote the blocks of the matrix A^{-1} by $x, 0, z$, and k. The product of these matrices must be equal to the identity matrix:

$$A \cdot A^{-1} = \begin{pmatrix} a & 0 \\ c & d \end{pmatrix} \begin{pmatrix} x & 0 \\ z & k \end{pmatrix} = \begin{pmatrix} I & 0 \\ 0 & I \end{pmatrix},$$

Now we can write the equations for blocks

$$x = a^{-1},\ k = d^{-1},\ z = -k \cdot c \cdot x.$$

These are the equations of a block recursive algorithm for calculating the inverse matrix.

Two diagonal blocks are invertible triangular matrices. To calculate the inverse matrices for these blocks, it is enough to recursively call this algorithm. These calculations can be performed independently of each other and simultaneously. The top right block of the inverse matrix is zero. The lower left block can be found using two matrix multiplications, one of which requires changing the sign of the matrix.

It is clear that matrix multiplication is a recursive block procedure that reduces to 8 block matrix multiplications. You can also use the scheme with 7 block multiplications proposed by Strassen.

Thus, the triangular matrix inversion algorithm is reduced to four recursive block procedures. It is shown in Fig. 1

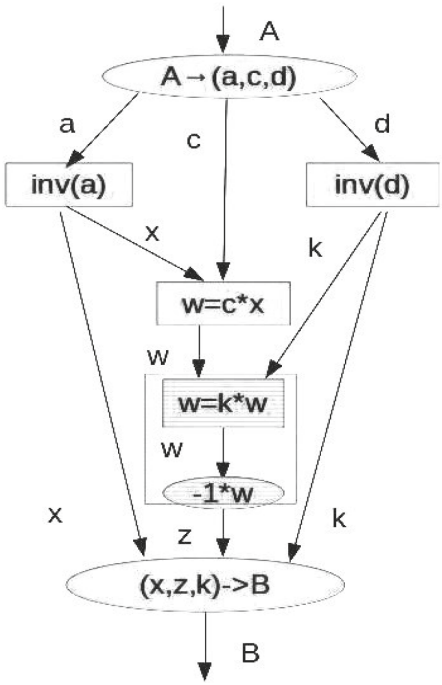


Fig. 1. Graph of the block recursive algorithm for inverting a triangular matrix

MathparDAP runtime is designed to control the computational process of parallel block recursive (PBR) algorithms. It allocates independent recursive

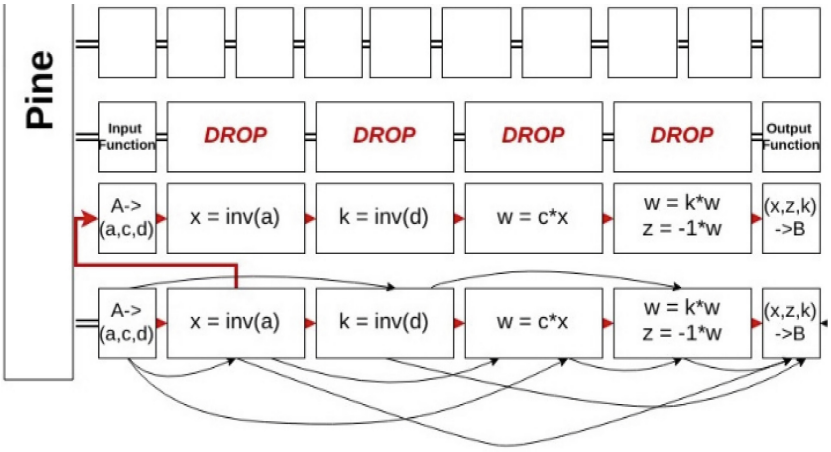


Fig. 2. Mechanism for controlling the computational process

computation units in each processor and sends them to other processors. This division occurs automatically as long as the block sizes remain relatively large. Blocks of small sizes are not profitable to send because of the long time of the transfer. Such small blocks are calculated by a sequential algorithm in one processor.

Let us indicate the class of PBR algorithms for which this runtime is intended in terms of the properties of the algorithm graph.

The PBR computational algorithm can be represented by a directed graph with one input, to the input vertex, and one output, from the output vertex. Each vertex of this graph corresponds to a certain computational block for matrix computations. Each directed edge of the graph indicates the direction of data transfer.

At the same time, the input and output vertices contain computational blocks of low complexity; it does not exceed the complexity of matrix addition. These calculations are performed on the same processor and cannot be transferred to other processors.

As a rule, at the input vertex the matrices that are transmitted to the input are divided into blocks, and at the output vertex matrices are formed from the blocks. These matrices are the result of calculations.

All vertices, except the input and output, contain computational blocks of great complexity. Such blocks are called “drops”. As a rule, their complexity is the complexity of matrix multiplication. Each such block can be transferred to another processor to parallelise calculations. In this case, the computational algorithm of each such block is an independent PBR algorithm.

The management of such a parallel computing process is decentralised. All parallel PBR blocks are sent to free processors. There, PBR blocks of smaller sizes are again allocated and sent to free processors. This continues until either

the free processors are exhausted, or the block size is so small that sending it to another processor will not speed up the calculations.

The computational graph for one level of recursion, like Fig. 1, is called an "amine". The list of all amines that are calculated in a certain processor is called a "pine" (see at Fig. 2). The name DAP is derived from the names of three main objects: drop, amine and pine. The Fig. 2 shows how the graph of the triangular matrix inversion algorithm is expanded in depth and the blocks are saved at the next step of the recursion.

The first experiments with MathparDAP runtime were carried out with a block-recursive version of the Cholesky decomposition. They showed good scaling [8].

5 Platform for Physical and Mathematical Education "MathparEdu"

Today you have many different ways of writing mathematical texts. Most mathematicians and physicists prefer the LaTeX language and its dialects. But humanity still does not have a generally accepted computer language for storing mathematical knowledge. It is still not possible to make a copy of the text in LaTeX and perform the corresponding calculations automatically.

Such a possibility today is provided by the freely available cloud mathematics MathPartner. Its user language is a dialect of LaTeX and it allows you to do the corresponding calculations. You can create electronic textbooks that contain "active math expressions". You can read such a textbook and calculate mathematical expressions at the same time. Moreover, these expressions can be copied into your "cloud notebook" and used for your own calculations.

The main thing that is achieved is the deepening of the student's knowledge as a result of a multiple increase in the number of problems that the student can solve during study time.

Firstly, he can now perform many actions not with his hands, but with the help of operators available to him.

Secondly, he will have access to checking the correctness of the solution of the educational problem, so he can check his solution and correct mistakes.

Thirdly, for training, he can ask to show him a variant of the correct solution in order to further independently solve similar problems.

In addition, you can save the teacher from checking homework and checking exam papers, since this is very easy to automate.

Such an educational process will make it possible to obtain an analysis of statistics on the quality of education in one group of students, in one educational institution and in the whole state. It will allow you to compare the quality of education in different countries.

6 Conclusion

We described the Mathpar project and the opportunities it provides to a wide range of users, from professional mathematicians to schoolchildren. This project continues the direction of research that has been formed in Kyiv since the mid-twentieth century.

The development of computer mathematics gives impetus to the development of all scientific and technical areas, including those necessary for the defence of the country.

Therefore, this year we have opened a new master's degree in Computer Mathematics.

The following academic disciplines are studied here: "Introduction to computer mathematics", "The architecture of cloud mathematics", "Parallel programming", "Supercomputing with dynamic control", "Matrix calculations", "Matrix algorithms for supercomputing", "Polynomial calculations".

The teaching material, laboratory assignments, and student research activities are largely based on the Mathpar project.

We plan further development of this project. The multilingual platform of the project can be used for the development of science and education in other countries. We invite other universities and scientific organisations to cooperate in all three directions of Mathpar project presented above.

Acknowledgments. Thanks to the Institute of Cybernetics of the National Academy of Sciences of Ukraine for the opportunity to conduct computational experiments on supercomputer SKIT.

Thanks to the administration of the National University of Kyiv Mohyla Academy for the provided server on which Cloud Mathematics MathPartner is hosted.

Thanks to the Simons Foundation for the partial support of my research in 2022–2023 years.

References

1. Buchberger, B., Collins, G.E., Loos, R., (eds.): Computer Algebra: Symbolic and Algebraic Computations, 2nd edn. Springer, New York (1983)
2. von zur Gathen, J., Gerhard, J.: Modern Computer Algebra. Cambridge University Press (2013)
3. Higham, N.J.: Accuracy and Stability of Numerical Algorithms, 2nd edn. Society for Industrial and Applied Mathematics, Philadelphia, PA, USA (2002)
4. Hooft, G., Veltman, M.J.G.: Elucidation of the quantum structure of electroweak interactions. Usp. Fiz. Nauk **170**(11), 118–1224 (2000)
5. Kapitonova, Y.V., Letichevsky, A.A.: Mathematical theory of design of computer systems, Moscow. Nauka, Phys. Math. Lit. (1988). ISBN 5-02-013777-4
6. "MathPartner" Cloud Mathematics. http://mathpar.ukma.edu.ua
7. Malaschonok, G.I.: MathPartner computer algebra. Program. Comput. Softw. **43**(2), 112–118 (2017). https://doi.org/10.1134/S0361768817020086

8. Malaschonok, G.I., Sidko, A.A.: Supercomputer environment for recursive matrix algorithms. Program Comput. Soft. **48**, 90–101 (2022). https://doi.org/10.1134/S0361768822020086, arxiv:2303.11017
9. Schönhage, A.: Unitäre Transformationen großer Matrizen. Numer. Math. **20**, 409–417 (1973)
10. Strassen, V.: Gaussian elimination is not optimal. Numer. Math. **13**(4), 354–356 (1969)

Two-Step Singular Value Decomposition on a Single GPU with CUDA

Serhii Sukharskyi(✉)

National University of Kyiv-Mohyla Academy, 2 Skovorody Street, Kyiv 04070, Ukraine
s.sukharskyi@ukma.edu.ua
https://github.com/sserhio

Abstract. A new way of performing singular value decomposition on a graphic processing unit based on a series of Householder transformations and a Givens rotation matrix principle is proposed and implemented. NVIDIA CUDA programming model and architecture combined with jCUDA library is used for the implementation part. First results have confirmed GPU efficiency claims by previous researchers and demonstrated a great performance in general.

Keywords: SVD · Householder transformation · Givens rotation matrix · CUDA · GPU

1 Introduction

Singular value decomposition (SVD) is a prominent algorithm that allows us to process large amounts of data and extract the main features from it. Its main applications are recommender systems in services like Netflix, image processing, principal component analysis and natural language processing. It also heavily relies on high performance computations, therefore, requires a lot of expensive computing resources. Although much research has been conducted on optimisation of the existing processes, little has been done on proposing new algorithms designed specifically for more efficient graphic processing units (GPUs). To prove the importance and potential of GPUs we may remember new TOP-500 leader—supercomputer Hewlett Packard Enterprise Frontier, or OLCF-5, the world's first and fastest exascale supercomputer which combines 4 GPUs and one CPU into a node and has more than 9 thousands of them.

This research aims to propose a fast and reliable SVD algorithm and alongside with implementation designed to be computed on a CUDA-enabled graphic processor. Regarding recent publications on this topic, we should definitely note the importance of Lahabar and Narayanan paper [1] who were among the first to start working in this field. They proposed a great algorithm that performs bidiagonalisation completely on a GPU and uses a hybrid implementation for the diagonalisation part that splits the computations between the CPU and the GPU. We propose a different approach where in the bidiagonalisation part we use a CPU to control the flow of the algorithm and GPU to perform intense operations relying on optimised CUDA libraries, while the diagonalisation is

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2026
I. Potapov et al. (Eds.): RTC-Digital 2023, CCIS 2647, pp. 190–196, 2026.
https://doi.org/10.1007/978-3-032-04731-1_25

performed solely on a GPU in our case. Another important research on this topic was conducted by Malashonok and Semylitko [2]. They proposed a GPU-based diagonalisation algorithm, however, during the experiments phase discovered that it poorly uses available graphic processor resources and most of the cores just wait while small amount of cores perform intense operations. In this research we propose a way of dealing with this issue.

2 The Algorithm

2.1 Definition

There is a matrix A of size $m \times n$ filled with complex or double values. Factorization of the form

$$A = U\Sigma V^T,$$

where U is an $m \times m$ orthogonal matrix, Σ is a diagonal $m \times n$ matirx with non-negative diagonal elements in decreasing order, and V is $n \times n$ orthogonal matrix, is called Singular Value Decomposition.

2.2 Bidiagonalisation

In order to get a two-diagonal matrix, we use Householder transformation, which is a linear vector space transformation used in linear algebra to calculate orthogonal factorisations:

$$A = QR, A = UDV^T,$$

where Q, V, U - orthogonal (unitary for the field of complex numbers), R - is a right triangular, and D is a two-diagonal matrices. We may easily compose orthogonal Householder matrix, multiplication of which by a defined vector x transforms it to $y = Px$, in a way to affect only one component alongside with setting other components of this vector to 0. For example, we need to set to 0 all the components except the first one:

$$y_1 = (||x||, 0, ..., 0) \text{ or } y_2 = (-||x||, 0, ..., 0).$$

in order to decrease computation error, this option is used:

$$y = (-sign(x_1)||x||, 0, ..., 0),$$

$$u = x - y = (x_1 + sign(x_1)||x||, x_2, ..., x_n)$$

These formulas are used in the Algorithm 1 described below. As it was mentioned before, we use jCUDA technology and list use CPU instructions to control the flow of the algorithm. All the informations is stored only on a GPU and each next command uses previously calculated data without reading and writing it back and forth from and to a graphic card gaining an important speed-up.

Algorithm 1: Bidiagonalization algorithm

Input: A
Output: $U,\ A,\ V$

begin
  N=$size(A)$-1
  i=0
  while $i < N$ **do**
    x = the low part of i-th column of matrix A, starting from i-th row
    x_1 = the low part of x: the column without one upper component (of i-th row)
    if $norm_{x1} > 10^{-10}$ **then**
      apply $norm_x$ to i element of x
      if U *is not initialized* **then**
        $U = I - (2/norm_x^2)xx^T$
      else
        $U = U - (2/norm_x^2)(Ux^T)x$
      $A_{i+1} = A_i - (2/norm_x^2)x(x^T A_i)$
    $x = i$ row of matrix A starting from $(i+1)$-th column
    x_1 = the right part of x: the row without one left component
    if $norm_{x1} > 10^{-10}$ **then**
      apply $norm_x$ to $i+1$ element of x
      if V *is not initialized* **then**
        $V = I - (2/norm_x^2)x^T x$
      else
        $V = V - (2/norm_x^2)x^T(xV)$
      $A_{i+1} = A_i - (2/norm_x^2)(A_i x^T)x$
    $i = i + 1$
  return $[U, A, V]$

2.3 Diagonalization

An iterative matrix processing is conducted in this step. We iterate over the matrix and set to zero either row above diagonal or a column below it. Imagine we have a matrix A.

$$A = \begin{bmatrix} d_1 & ud_1 & 0 & 0 \\ 0 & d_2 & ud_2 & 0 \\ 0 & 0 & d_3 & ud_3 \\ 0 & 0 & 0 & d_4 \end{bmatrix}$$

First two iterations will affect first and second squares of matrix A. We talk about two iterations, since they have a common element d_2, therefore can not be calculated in parallel. After composing a Givens rotation matrix and applying it to matrix A, we will set ud_1 value to 0, however, new value under d_1 will appear.

$$A_1 = \begin{bmatrix} d'_1 & 0 & 0 & 0 \\ x & d'_2 & ud_2 & 0 \\ 0 & 0 & d_3 & ud_3 \\ 0 & 0 & 0 & d_4 \end{bmatrix}$$

Then, we repeat this process until all the elements under and below diagonal are zeros. In the process of transforming the main matrix, we also compute left and right multiplier matrices, which are identity matrices at the beginning, and are affected by Givens matrix multiplication during the algorithm execution, forming an important part of the solution by the end of the process. These matrices are then multiplied by appropriate left and right multiplier matrices from the bidiagonalisation process and result in two feature matrices which are an essential parts of a singular value decomposition. As we can see from the formulas below, only two rows or columns are affected. Therefore, we do not perform whole matrix multiplication operation, which is quite expensive in terms of calculation time and resources, but simply perform appropriate elements multiplications, gaining an important performance boost.

$$L1 = G \begin{pmatrix} c & s & 0 & 0 \\ -s & c & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \cdot L \begin{pmatrix} 1 & 2 & 3 & 1 \\ 2 & 3 & 1 & 2 \\ 3 & 1 & 2 & 3 \\ 1 & 2 & 3 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} (2\cdot\ s+c) & (3\cdot\ s+2\cdot\ c) & (s+3\cdot\ c) & (2\cdot\ s+c) \\ (2\cdot\ c-s) & (3\cdot\ c-2\cdot\ s) & (c-3\cdot\ s) & (2\cdot\ c-s) \\ 3 & 1 & 2 & 3 \\ 1 & 2 & 3 & 1 \end{pmatrix};$$

$$R1 = R \begin{pmatrix} 1 & 2 & 3 & 1 \\ 2 & 3 & 1 & 2 \\ 3 & 1 & 2 & 3 \\ 1 & 2 & 3 & 1 \end{pmatrix} \cdot G \begin{pmatrix} c & -s & 0 & 0 \\ s & c & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} (2\cdot\ s+c) & (2\cdot\ c-s) & 3 & 1 \\ (3\cdot\ s+2\cdot\ c) & (3\cdot\ c-2\cdot\ s) & 1 & 2 \\ (s+3\cdot\ c) & (c-3\cdot\ s) & 2 & 3 \\ (2\cdot\ s+c) & (2\cdot\ c-s) & 3 & 1 \end{pmatrix};$$

The entire process is parallelised by processing $N/2$ blocks simultaneously. The first thread that performs calculations starts the process by calculating first two blocks, since they have a common element. As soon as it has finished the first two blocks, it moves to the next one and another thread starts working on the first two blocks. This process repeats until there are free threads. Each thread starts to work and does that until no calculations need to be made in all the blocks. These threads are called main threads. As it was noted by Malashonok and Semylitko [2], despite working with only part of a large matrix, this approach tends to be slow, since each thread apart from working with main matrix has

to work with two columns of large left and right matrices. In order to deal with this issue, we added a group of additional threads to each main thread. The process is started not on N/2 threads, but on much more, depending on a matrix size and it allows to use them in such a way that each additional thread works only with a part of left and right multiplier matrices. For example, if we are working with a 128×128 matrix and use 64 main threads, and the maximum number of threads within a block is 1024, then we use not just 64 threads, but 64 groups of 16 threads each. Checks are added to determine which actions should be performed only by the main thread, and which can be done by additional. For instance, applying the rotation matrix to main matrix is done only by the main thread, while the others do not spend time on this. This approach has demonstrated significant acceleration. However, a large number of threads also adds certain delays with the program completion time, due to the need to wait until all threads finish their work.

And last but not least is memory usage. Threads in one block use block shared memory, which is quite fast. Each thread also uses local memory. Since matrices can be really large they may not fit one block, thus multiple blocks should be used. Due to a common element between blocks a synchronisation between blocks using a global memory is required.

It is important to note, that current version of the algorithm works with square matrices only, however, the algorithm itself supports rectangular matrices and it is just a programming part that should be extended.

3 Experiments

Experiments were held for each part separately and for the whole program. It is important to note, that the first part leaves the results on a GPU and a second part continues the process using those results. We have measured computation time and error depending on a matrix size and precision. The computational experiments were carried out on a graphic processor GA104-300-A1 (Ampere architecture) with 5888 CUDA cores, maximum of 1024 threads, 8 GB of GDDR6 (256 bit; 448 GB/s), and 48 KB of shared block memory. As for the central processor, it was Intel Core i5-12600K 3.7 GHz combined with 16 GB of DDR4 3600 MHz RAM. Input matrices were filled in with generated Double numbers from 0 to 100. For each matrix size there were multiple runs, which allowed to see the deviation both in time and error measurements.

3.1 Computation Time

First of all calculation time was investigated.

As we can see from Table 1, bidiagonalization performance is quite impressive—matrix of size 1024 is calculated for less than a second. As for the diagonalisation, results are not as impressive. As we can see, on small matrices $128 \cdot 128$ and on $256 \cdot 256$, calculation time is low. However, on bigger matrices

Table 1. Computation time for the SVD algorithm

Size	128	256	512	1024	2048
Bidiagonalization	0.060	0.103	0.216	0.873	6.557
Deviation ($\pm t$)	0.005	0.009	0.022	0.009	0.008
Diagonalization	0.131	0.338	1.280	11.886	64.122
Deviation ($\pm t$)	0.016	0.022	0.076	0.802	2.667
Algorithm	0.191	0.441	1.496	12.759	70.679

calculations take longer than expected. We tend to relate it with global memory usage in order to synchronise common matrix parts between two blocks. Another possible reason is that the amount of blocks and block size was not optimal, however our experiments with these parameters have not result in dramatic speed-up.

3.2 Accumulation of Computational Error

Table 2. Computational error for the SVD algorithm

Size	128	256	512	1024	2048
Bidiagonalization	$4.47 \cdot 10^{-13}$	$3.69 \cdot 10^{-13}$	$5.33 \cdot 10^{-13}$	$1.28 \cdot 10^{-12}$	$1.78 \cdot 10^{-12}$
Deviation ($\pm t$)	$1.065 \cdot 10^{-13}$	$4.25 \cdot 10^{-14}$	$4.28 \cdot 10^{-14}$	$1.17 \cdot 10^{-14}$	$1.03 \cdot 10^{-13}$
Diagonalization	$5.28 \cdot 10^{-11}$	$4.51 \cdot 10^{-11}$	$1.73 \cdot 10^{-5}$	$5.44 \cdot 10^{-5}$	$5.45 \cdot 10^{-6}$
Deviation ($\pm t$)	$7.22 \cdot 10^{-13}$	$6.16 \cdot 10^{-12}$	$8.35 \cdot 10^{-6}$	$7.56 \cdot 10^{-6}$	$3.2 \cdot 10^{-6}$
Algorithm	$7.007 \cdot 10^{-12}$	$8.64 \cdot 10^{-12}$	$2.63 \cdot 10^{-6}$	$2.18 \cdot 10^{-4}$	$1.06 \cdot 10^{-6}$

Table 2 shows that in general, there is quite a good calculations precision, especially on a relatively small square matrices of size 128 and 256. Bidiagonalisation once again demonstrates great results on all matrix sizes. Diagonalisation is precise on small matrices as well, but there is a precision drop in diagonalisation part for larger matrices, which affects calculation accuracy of the algorithm, though, it is still a solid accuracy. Precision drop is seen on the same matrix sizes as calculation time increase. Therefore, it seems to be a related issue to the one mentioned in the previous section.

4 Conclusions

We have described, implemented, and investigated the SVD algorithm, designed specifically for a CUDA-enabled GPU. This is the case where the matrix can be factorised on a single GPU. Householder transformations proven to be fast and reliable and allow working with large matrices keeping the high accuracy. An

iterative diagonalisation process, solely on a graphic processor using Givens rotation matrices, also shows good performance on relatively small matrices, however drops the performance on a larger ones and therefore needs to be improved.

Our further research will focus resolving the synchronisation issue as well as on creating such an algorithm that allows itself to be scaled to a cluster in which each processor is equipped with one or more graphics cards. This will make it possible to factorise larger matrices.

References

1. Lahabar, S., Narayanan, P.J.: Singular value decomposition on GPU using CUDA. In: IEEE International Symposium on Parallel and Distributed Processing, Rome, Italy, 2009, pp. 1–10 (2009). https://doi.org/10.1109/IPDPS.2009.5161058
2. Malaschonok, G., Semylitko, M.: Parallel SVD algorithm for a three-diagonal matrix on a video card using the Nvidia CUDA architecture. NRPCOMP **4**, 16–22 (2021). https://doi.org/10.18523/2617-3808.2021.4.16-22
3. Malaschonok, G., Sukharskyi, S.: A GPU-based orthogonal matrix factorization algorithm that produces a two-diagonal shape. NRPCOMP **4**, 10–15 (2021). https://doi.org/10.18523/2617-3808.2021.4.10-15
4. Numerical linear algebra and matrix factorizations. Cornell University. http://pi.math.cornell.edu/~web6140/TopTenAlgorithms/Householder.html
5. Persson, O.: Householder reflectors and givens rotations. In: MIT 18.335J/6.337J Introduction to Numerical Methods (2006). https://math.dartmouth.edu/~m116w17/Householder.pdf

New Evolutionary Methods for Solving Optimization Problems in the Digital Transformation Processes

Vitaliy Snytyuk and Nataliia Tmienova(✉)

Taras Shevchenko National University of Kyiv, Volodymyrs'ka Str. 64/13, Kyiv 01601, Ukraine
tmyenovox@gmail.com

Abstract. The life of each person, as well as society, is determined by the decisions made. Each such decision is aimed at optimizing social, economic, technological or technical processes. The use of computer science and computer technology to solve optimization problems is the basis of modern digital transformations. A lot of practical problems are optimization problems with or without constraints. The objective functions in such problems are in most cases complex, non-differentiable and polyextreme. Application of classical methods of continuous and discrete optimization in this case is impossible. In such cases, evolutionary algorithms, which do not require restrictions on the objective function and guarantee only probabilistic convergence to the global optimum, are used. Usually, such algorithms converge slowly, because a significant number of steps in them are performed in the wrong direction. A feature of the known evolutionary algorithms is that one or two parent solutions are used to generate potential offspring solutions. We propose methods in which a larger number of offsprings are used for the generation of offsprings, which allows considering more information about the features of the optimum search area and speeds up the search process. The advantages of the proposed methods are a deeper study of the area in the neighborhood of promising solutions and a reduction in the number of algorithm steps in unpromising directions. There are the Method of Deformed Stars and the Method of Fractal Structuring. These methods are parametric and allow for improvement. They can be used to solve placement and packaging problems, traveling salesman problem, and any objective function optimization problems.

Keywords: Digital transformation · decision making · optimization · computer science · method of deformed stars · method of fractal structuring

1 Introduction

In modern world, digital transformations accompany development processes in many areas, in particular in those that determine a safe human habitat. Ensuring food or environmental security, maintaining the state of critical infrastructure facilities in proper condition, producing a sufficient amount of energy of various types - these are only a small part of the problems, the solution of which is based on solving the problems of identifying unknown dependencies and their optimization, because solutions of these tasks are the basis of decision-making processes.

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2026
I. Potapov et al. (Eds.): RTC-Digital 2023, CCIS 2647, pp. 197–204, 2026.
https://doi.org/10.1007/978-3-032-04731-1_26

A large number of modern practical problems belong to the class of constraint satisfaction problems. The target functions in such tasks are, as a rule, non-differentiable and (or) polyextreme dependencies. The use of classical methods of continuous optimization and in many cases discrete optimization is impossible. Combinatorial optimization methods, evolutionary algorithms, etc. are used to solve such tasks. The functional dependencies can be set tabularly or algorithmically. In this case evolutionary algorithms are most often preferred.

Exactly, the use of this algorithms does not require strict target functions constraints, but does not guarantee the finding of a global optimum, although according to certain conditions there is a probability convergence. The obtained solutions are considered suboptimal.

Historically, the first methods of evolutionary optimization were genetic algorithms and evolutionary strategies. These methods allowed to consider optimization problems differently and expanded the subject base of optimization technologies. Also, these methods were based on the ideas of natural evolution. In particular, genetic algorithms traditionally use the principle that the best parents tend to have better children. Two parental potential solutions are involved in generating potential solutions-offspring. In evolutionary strategies, potential offspring solutions are generated around a single parent solution. This is the main difference between genetic algorithms and evolutionary strategies in the generation of offspring solutions.

We propose methods [1–7] in which a larger number of offsprings are used for the generation of offsprings, which allows considering more information about the features of the optimum search area and speeds up the search process. The advantages of the proposed methods are a deeper study of the area in the neighborhood of promising solutions and a reduction in the number of algorithm steps in unpromising directions.

2 Formulation of the Problem

Let us consider the problem of finding the minimum value of the multivariable function:

$$\begin{aligned} &f(x_1, x_2, \ldots, x_n) \to \min, \\ &\overline{x} = (x_1, x_2, \ldots, x_n), \\ &\overline{x} \in D \subset R^n, \end{aligned} \tag{1}$$

where n is the dimension of the search space, D is some area.

The problem (1) is equivalent to the search problem:

$$\begin{aligned} &Arg \min f(\overline{x}), \\ &x \in D. \end{aligned} \tag{2}$$

Let some unknown point $x^* = (x_1^*, x_2^*, \ldots, x_n^*) \in D$ be its solution.

Note that the function f can be undifferentiated, polyextreme, as well as given tabularly or algorithmically. In such cases, the methods of classical optimization are ineffective.

We propose to use the idea that underlies all evolutionary algorithms, namely the population approach, which consists of solving problems (1), (2) by evolution the population of potential solutions.

In contrast to the evolutionary strategy, where one parental solution generates a set of offsprings solutions from which the best ones are chosen, and the genetic algorithm, where a pair of parental solutions generates offsprings solutions, we propose not to be limited to one or two solutions. The reason for this is the fact that simultaneously taken into account and analyzed the number of potential solutions contains more information about the study area and will allow the generation of offsprings solutions more efficiently.

Another hypothesis is that there is a motion direction of the potential solution of the problem (1), (2), which will allow with high probability and quickly finding the optimal solution of the problem, and there is an area in neighborhood of potential solution to be investigated in more detail because it has solutions that are better than the parent's potential solution.

The main constructive elements of the solution search for the problem (1), (2) are compression, rotation and some other transformations of figures formed by the connection of points, which are the potential solutions and which we will call deformed stars.

So, let's consider the main idea of MODS for n-dimensional case [3, 7].

3 Algorithm of the Method of Deformed Stars

Let's present the Algorithm of the Method of Deformed Stars in the form of the following steps.

Step 1. Perform the initialization of the algorithm parameters. $t = 0$.

Step 2. Generate m potential solutions in the area D (population P_t).

Step 3. Form m triangles (stars) T_i, $i = \overline{1, m}$.

Step 4. For each T_i find the vertex in which the function f acquires the minimum value and consider it as the first vertex.

Step 5. Find the centroid of each triangle.

Step 6. For each triangle T_i we find a modified triangle R_i, $i = \overline{1, m}$.

Step 7. For each triangle T_i we find the triangle Q_i obtained by rotating T_i around the first vertex.

Step 8. For each triangle T_i we find the triangle U_i obtained as a result of the operation of compressing the vertices to the first vertex.

Step 9. $T = t+1$. We form a population P_t that will include all potential solutions from P_{t-1}, as well as the vector of solutions $\left(\overline{y}^{1i}, \overline{y}^{2i}, \overline{y}^{3i}, \overline{t}^{2i}, \overline{t}^{3i}, \overline{z}^{2i}, \overline{z}^{3i}\right) \forall\, i \in \{1, \ldots, m\}$. So $|P_t| = 8m$.

Step 10. Check whether all elements P_t belong to D. If this is not the case, then perform the necessary transformations.

Step 11. For all potential solutions from P_t, find the values of the function f and sort the potential solutions in ascending order of these values.

Step 12. We leave in P_t only the best m solutions and check the fulfillment of the stop criteria.

Step 13. If the stop condition is not executed, then go to step 3, else we have the end of algorithm.

Note that the stop criteria can be:

I. achieving a given number of iterations;
II. the maximum distance between solutions in one population is less than the specified distance;
III. the absolute value of the deviation of minimum values of the function in neighboring populations is less than the specified value;
IV. others.

When implementing the method of deformed stars, a much smaller number of steps are performed in the wrong direction, in contrast to the genetic algorithm and evolutionary strategy as classical representatives of the evolutionary paradigm.

The accuracy of the obtained solutions is, on average, higher than the accuracy of competing algorithms due to a deeper study of the area D.

Next, we will consider another method of global optimization - the method of fractal structuring [6].

4 Algorithm of the Method of Fractal Structuring in n-dimensional Case

Step 1. Perform the initialization of the algorithm parameters. Iteration is $t = 1$.

Step 2. Generate a sample of uniform distributed in the hypercube points$P_t = \{(a_1^1, a_2^1, ..., a_m^1), ..., (a_1^n, a_2^n, ..., a_m^n)\}, a_i^j \in (0, 1),\ \forall i = \overline{1, m}, \forall j = \overline{1, n}.$

Step 3. Find the value of the function f at the points in the sample P_t and get $f_1, f_2, ..., f_n$.

Step 4. For each j-th hypersphere we generate 7 offsprings solutions (points) that will lie on its surface and are the centers of the hyperspheres with radius $r = \frac{r}{t+1}$. Next we generate a random vector $(x_1, x_2, ..., x_{k-1}, x_{k+1}, ..., x_m)$, such that

$$(x_i \in (a_i^j - r, a_i^j + r),\ i \neq k)\ \ (\sum_{\substack{i=1 \\ i \neq k}}^{m} (x_i - a_i^j)^2 < r^2),$$

and calculate

$$x_k = a_k^j + random\{-1; 1\} \cdot (r^2 - \sum_{\substack{i=1 \\ i \neq k}}^{m} (x_i - a_i^j)^2)$$

Step 5. Suppose, that the best offspring solutions may lie in the middle of the rectangle sides for which the segment connecting the points a i b is a diagonal. To generate them, we generate a random number $r \in \{-1,\ 1\}$, where (–1) corresponds to the solution a, 1 corresponds the solution b and the random number $q \in \{1, 2, ..., m\}$. Generate the descendant vector as follows:

$$c = (\chi(r = -1)a_1 + \chi(r = 1)b_1, ..., \frac{a_q + b_q}{2}, ..., \chi(r = -1)a_m + \chi(r = 1)b_m).$$

Step 6. Just as there may be an even better solution between the best potential solutions, so, given the relief of many polyextreme functions, the best solution may be among the worst potential solutions. So, let's choose $2n$ the worst solutions from the set P_v. We find the average value of the objective function f_{ave} for the n best solutions.

Step 7. At this step, probabilistic generation and recording in a new population of solution, in which the corresponding values of the objective function are not promising? Are provided. If the set of the worst solutions have been exhausted and the stop condition is not met, then reduce the temperature $T = \frac{T}{2}$, $t = t + 1$ and go to step 4.

If the stop condition is satisfied, then we have the end of the algorithm.

5 Computational Experiment

To determine the efficiency of the Method of Deformed Stars (MODS), a comparative analysis of its results was performed with the results of classical evolutionary methods: genetic algorithm, evolutionary strategy, differential evolution (see Table 1).

The experiments were performed for different stop conditions of the algorithm: by the number of iterations (100 and 300), by the closeness of the differences between the mean and maximum values of fitness functions for the neighboring populations. The average execution time of each of the algorithms at 100 iterations (averaged over 100 program launches) was investigated, and the number of resulting values that reached the specified accuracy and those that did not reach was calculated.

All algorithms were tested for ten functions: Ackley, Beale, Goldstein-Price, Booth, Levi N.13, Threehump camel, Matyas, McCormick, Schaffer N. 2 and Bukin N.6.

According to the data, at 100 iterations results MODS-3, MODS-4, MODS-5 is more accurate, and other algorithms in many cases have not found a solution with acceptable accuracy [1–3]. Using the second condition, namely the limited absolute difference between the mean values of the fitness function of two neighboring populations, there is a similar trend. MODS-3 showed the best average run time (in seconds) per 100 runs. Experiments show that, in general, MODS-4, MODS-5 methods have about 50 iterations enough to find a sufficiently accurate solution, which cannot be said about genetic algorithms, evolutionary strategies or differential evolution.

The Method of Fractal Structuring also demonstrated effective results.

Analyzing the obtained results and taking into account the main ideas and principles underlying the developed algorithms, we conclude that the application of such algorithms to solving practical problems is not only a rational choice of the researchers, but also a way to solve the problem relatively accurately and quickly.

Table 1. Experiments results.

Test function	Global minimum	100 iterations, accuracy $= 10^{-6}$											
		Genetic algorithm	Accuracy achieved	Evolution strategy	Accuracy achieved	Differential evolution	Accuracy achieved	MODS-3	Accuracy achieved	MODS-4	Accuracy achieved	MODS-5	Accuracy achieved
	$f(x_1, x_2)$	$f(x_1, x_2)$	N	$f(x_1, x_2)$	N	$f(x_1, x_2)$	N	$f(x_1, x_2)$	N	$f(x_1, x_2)$	N	$f(x_1, x_2)$	N
Ackley	0	0.0002	–	0.0	72	0.0	81	0.0	46	0.0	19	0.0	26
Beale	0	0.0128	–	0.0	16	0.0015	–	0.0	56	0.0	15	0.0	22
Goldstein–Price	3	3.0018	–	3.0002	–	3.1193	–	3.0	27	3.0	14	3.0	19
Booth	0	0.0	14	0.0	15	0.0101	–	0.0	32	0.0	12	0.0	20
Levi N.13	0	0.0	15	0.0	20	0,0000	–	0.0	65	0.0	15	0.0	25
Three-hump camel	0	0,000	–	0.0	9	0.0	80	0.0	22	0.0	10	0.0	13
Matyas	0	0.0009	–	0.0	12	0.0011	–	0.0	34	0.0	9	0.0	19
McCormick	–1.913	–1.913	–	0.0	–	–4.897	–	–1.913	–	–1.913	–	–1.913	–
Schaffer N. 2	0	0.0	79	0.0	22	0,0001	–	0.0	46	0.0	16	0.0	17
Bukin N.6	0	0.0014	–	0.0069	–	0.8597	–	0.0017	–	0.0025	–	0.0042	–
F_{avg}, accuracy $= 10^{-6}$													
Beale	0	0.0	14	0.3976	–	0.0018	–	0.0	53	0.0	16	0.0	21
Goldstein–Price	3	3.0	–	5.1224	–	3.0258	–	3.0	39	3.0	12	3.0	17
Booth	0	0.0	16	0.8913	–	0.0079	–	0.0	38	0.0	10	0.0	17
Levi N.13	0	0,0	–	0.5064	–	0,0000	–	0.0	66	0.0	17	0.0	31
Three-hump camel	0	0.0	15	0.7692	–	0.0019	–	0.0	28	0.0	10	0.0	12
Matyas	0	0.0	11	0.1628	–	0.0134	–	0.0	39	0.0	9	0.0	16
McCormick	–1.9133	–1.913	–	–1.8901	–	–1.911	–	–1.9132	–	–1.9132	–	–1.9132	–
Schaffer N. 2	0	0.0	18	0.4018	–	0.0081	–	0.0	53	0.0	17	0.0	21
Bukin N.6	0	0.2952	–	5.27	–	3.0309	–	0.0078	–	0.0003	–	0.0093	–

Another feature of such algorithms is their parametric structure, which allows to carry out parametric optimization of algorithms depending on the problem to be solved, the type and amount of input data, and the desired result.

Another feature is the relatively simple implementation of algorithms and the possibility of their effective application to solving multidimensional problems. We remind once again that such effective results of the application of the developed methods can be explained by fewer steps in the wrong directions and a deeper study of promising areas, in contrast to the well-known classical evolutionary methods.

It should be taken into account that using an ellipse, ellipsoid, and hyperellipsoid as the basic figure can significantly improve the accuracy of problem solving, while the solution time can be significantly increased. We believe that this direction of research and development is extremely promising.

6 Conclusions

In this paper we propose the Method of Deformed Stars and the Method of Fractal Structuring.

These methods can be attributed to the class of Evolutionary Computation technologies, which in turn are representatives of Computer Science. A feature of such methods is the possibility of their application to solving problems with minimal restrictions on the initial data and the objective function. Such properties are characteristic of problems related to Digital Transformation. Output data, which are given in a table; unknown objective function with unknown properties; the presence of quantitative and qualitative variables - such circumstances testify in favor of the use of Evolutionary Computation technologies.

The first proposed method is called the Method of Deformed Stars. According to this method, a population of potential solutions is generated to solve the problem of optimization the objective function in some rectangular hyperparallelepiped (perhaps after normalization). We will randomly form a set of stars (triangles, quadrilaterals, pentagons or n-sided polygons), where the vertices are the generated potential solutions. At each vertex of the star, we find the value of the objective function. In the next step, we perform a number of operations for each star. The best vertex generates an offspring vertex in the direction from the star center, and the other vertices generate offsprings in the direction of the best vertex. This operation allows to explore the search area in the perspective direction and analyze the area around the best point in more detail. To generate new points – potential solutions, we also perform rotation operations around the best vertex and rotation around the centroid of the star. The new points form a population of a new generation and form stars again. The iterative process ends when a certain stop condition is met.

The second method is called the Method of Fractal Structuring. In this method, as in the previous one, we generate a population of n random points - potential solutions. Further, each such point produces several offspring-solutions (3–10 or more), which are placed randomly at a distance r from the parent solution. Among the parent potential solutions and offspring-solutions, we determine the 2n best ones. For these potential solutions, we will randomly form pairs, the average values of which will already be

elements of the new era and will allow us to explore the search area more deeply. In order to minimize the chances of hitting a local optimum, let's give small random increments to the worst 2n solutions and compare the value of the objective function at the obtained points with the average value of the objective function for the best solutions. If the generated points turn out to be promising, then we will write them into the population of the new era, if not, then we will discard them with a certain probability, moreover, with each iteration, this probability will increase. Thus, we will obtain the population of the new era. In the next step, its elements will generate offsprings, which will be already at a distance of r/2 from the parent solutions. This iterative process will end when a certain stopping criterion is met.

Summing up, we note that algorithmization and experimental studies with repository functions of the proposed methods proved their effectiveness. The methods are parametric and allow for improvement. They can be used to solve placement and packaging problems, traveling salesman problem, and any objective function optimization problems.

References

1. Snytyuk, V.: Method of deformed stars for multi-extremal optimization. one- and two-dimensional cases. In: International Conference Mathematical Modeling and Simulation of Systems. MODS 2019 Advances in Intelligent Systems and Computing, vol. 1019, Springer, Cham, pp. 122–130 (2020)
2. Antonevych, M., Didyk, A., Snytyuk, V.: Optimization of functions of two variables by deformed stars method. In: 2019 IEEE International Conference on Advanced Trends in Information Theory (ATIT), pp. 475–480 (2019)
3. Tmienova, N., Snytyuk, V.: Method of deformed stars for global optimization. In: 2020 IEEE 2nd International Conference on System Analysis & Intelligent Computing (SAIC), pp.1–4 (2020)
4. Snytyuk, V., Antonevych, M., Didyk, A., Tmienova, N.: Choosing the best parameters for method of deformed stars in n-dimensional space. Adv. Inf. Technol. **1**(1), 24–28 (2021)
5. Antonevych, M., Tmienova, N., Snytyuk, V.: Models and evolutionary methods for objects and systems clustering. In: Int Sol, CEUR Workshop Proceedings, vol. 3018, pp. 37–47 (2021)
6. Antonevych, M., Snytyuk, V., Tmienova, N.: The method of fractal structuring as on evolutionary method of global optimization. In: CEUR Workshop Proceedings, vol. 3179, pp. 69–78 (2021)
7. Snytyuk, V., Antonevych, M., Didyk, A., Tmienova, N.: The method of deformed stars as a population algorithm for global optimization. Stud. Comput. Intell. **1022**, 229–247 (2022)

Algebraic Approach to Molecular Modelling and Its Application in Studies of the Effect of Radiation Therapy on Cancer Treatment

Yuliia Tarasich(✉)

V.M.Glushkov Institute of Cybernetics of the NAS of Ukraine,
Akademika Glushkova Avenue 40, Kyiv 03187, Ukraine
yutarasich@gmail.com

Abstract. The article proposes a new approach to molecular modelling. This approach is based on the formalism of behavioural algebra and insertion modelling. Biological systems and processes of chemical interaction between substances are considered as complex hybrid systems. We see one of the directions of application of this approach in the possibilities of modelling the radiation therapy process - starting from modelling the accelerators' work and ending with modelling the interaction of the particles' beam with the matter at the level of quantum interactions. Finally, the article presents an example of applying the algebraic approach to the radiation therapy formalisation process at the upper level.

Keywords: Algebraic modelling · Proton therapy · Molecular modelling · Formal methods · Insertion modelling

1 Introduction

Despite the rapid development of the chemical industry and science, discoveries in the field of health care, the emergence of drugs and therapeutics based on nanotechnology and the development of radiation therapy technologies, the safety of biomedical applications of the latest products, and the search for new methods and approaches to the diagnosis and treatment of cancer are an open issue. We believe that one of the safe and fast methods of researching the behaviour of new materials and tools and selecting the best candidates is the modelling of relevant processes, in particular, computer molecular modelling based on mathematical models.

Hybrid models and methods for systems biology and medicine (including working with formal hybrid specifications such as temporal and hybrid automata), and combining models by integrating combinatorial and continuous constraints and using machine learning to design models and define their parameters [3,8] are essential steps in solving open problems in the field of modelling and research of relevant processes and systems. Mathematical models are successfully used to study oxygen transport, tumour angiogenesis, and various cancer treatment methods [2,5]. However, despite the increasing availability of

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2026
I. Potapov et al. (Eds.): RTC-Digital 2023, CCIS 2647, pp. 205–211, 2026.
https://doi.org/10.1007/978-3-032-04731-1_27

existing protein and nucleic acid data, modelling methods, and tools [9], developing and using a wide range of combined methods and tools for modelling and computing large molecular systems remains an open issue.

As one of the possible solutions, we propose using an algebraic approach and the corresponding formal methods, which today have proven their effectiveness in many other areas. The main idea of our research is to apply the technology of algebraic modelling and quantum chemical apparatus for modelling and verification of problems of organic chemistry, in particular, modelling and verification of different approaches to the question of researching the effect of radiation therapy on cancer treatment.

2 Algebraic Approach and Algebraic Modelling System

The proposed algebraic approach is implemented in the system of insertion modelling, which is based on the theory of agents and environments, launched by O.A. Letichevsky and D. Gilbert [6]. The basic idea of the proposed approach is the interaction of agents in certain environment. The environment can also be an agent that interacts with similar agents in the higher-level environment. Each agent has its own type, determined by the agent's attributes. Each attribute is typed and belongs to a certain theory that defines predicates and operations. We can identify arithmetic, symbolic, bit, and byte attributes.

To solve the problem of the formula's executability in the chosen theory, are using the solver systems and systems for automatic proof of theorems, which are the basis of the algebraic approach in modelling.

The interaction of agents in the environment is determined by behaviour. Behaviour represents a tree of possible actions of the agent in the environment, respectively, behavioural expression is determined by a formula that is built from actions and behaviours using behavioural algebra operations: 1) prefixing operation "." - determines that some action x precedes the behaviour B. It is written as $x.B$; 2) nondeterministic choice "+" - defines alternative behaviours. It is written as $A + B$. Algebra is also extended by the sequential "$A; B$" and by the parallel "$A||B$" composition of behaviours [6].

The semantics of the action of each agent is defined as a triple $B = \langle P, A, Q \rangle$, where P is a precondition, presented in the form of a formula in certain theory, Q is a postcondition, and A is a process that visualises the transition of the agent between states. In other words, if the precondition is performed, the agent's state will change according to the postcondition. The precondition is determined by the predicate, which is a Boolean expression over the formulas of the corresponding theories, for example, - equality or inequality in linear arithmetic. Postcondition changes the environment and also uses predicates, assignment operators, and operations in the chosen theory.

A behavioural equation is an equality in which on the right side is the name of the behaviour, and on the left is the expression in the behaviour algebra over actions and other behaviours. With a behavioural equation, we can describe some process of interaction of agents in some environment. This process can describe

the processes and phenomena at the level of physical or chemical terms. Finding a scenario in the form of a sequence of actions that leads to the desired property is possible by solving the behavioural equation. The solution is with the help of algebraic modelling. To find the solution, the technique of unfolding equations is used, which is used in the technique of unfolding rewriting rules [7].

For modelling hybrid systems, the system of insertion modelling was expanded with the possibility of analytical solving of differential equations, the operators of which are the executing algebraic specifications.

Note that multi-level modelling is a valuable achievement in the theory of agents and environments for research in the natural sciences. It means the possibility to model the agents that are an environment for other agents and so on. So, we can move from the study of behaviour in quantum interactions to chemical reactions and further consider the essence of a higher level of abstraction, such as cellular interactions.

3 Algebraic Modelling of the Particle Acceleration Process and Passage of the Particle Beam in the Substance

The modelling of this experiment involves the interaction of such agents as Particles and Substance. In the first stage, we consider the change in the state of the agents' Particles inserted in a certain environment, which is an accelerator (cyclotron, synchrotron). By changing the state of agents, we mean changing the values of the attributes of agents under the influence of the environment and interactions between the agents themselves. In the second stage, we consider the passage processes of a beam of accelerated particles through substance and possible interactions between particles and substance.

Modelling the interaction of accelerated particles with individual atoms, molecules, and organic and inorganic substances at the level of quantum-chemical interaction mechanisms requires determining the properties and structure of the particles themselves and the atoms that make up the substances. Since the structure and energy states of atoms depend on the quantitative and qualitative characteristics of protons and electrons as integral components of each atom/ion, protons and electrons in this experiment are considered both as components of the atom - its main attributes and as separate agents. Accordingly, we define such types of agents as PROTON, ATOM, MOLECULE and SUBSTANCE.

For this experiment, a proton was chosen as the accelerated particle. The following attributes are defined for the agent of the PROTON type – mass (*mass*), charge (*charge*), momentum (*momentum*), velocity (*velocity*), the radius of motion (*motionRadius*), energy (*energy*), particle frequency (*particleFrequency*). Accordingly, the type of agent PROTON will be formalised with the following list of attributes:

```
PROTON:obj(mass:real, charge:real, momentum:real, velocity:real,
motionRadius:real, energy:real, particleFrequency:real)
```

The structure and "characteristics" of molecules and substances are considered using both methods as the method of valence bonds (OTs) and, in terms of the method of molecular orbitals (MO). The structure of the substance is set by a set of agents of molecule type, which, in turn, contain a function, reflecting the set of atoms of which the given molecules are composed. The properties of atoms are determined by the structure of their nuclei and the number and allocations of orbital electrons. Accordingly, the main attributes that characterise substances, in turn, are the structure of their molecular orbitals and nuclear models of atoms that are part of a substance that allows considering all processes at the level of quantum interactions. To determine the value of the physically absorbed dose to the main attributes of the agent, we add the radius (to determine the area of the beam), density, mass braking capacity, and the length of the segment that the protons must pass.

The task of the first stage of the experiment is to model the process of the growth for the particles' energy and the constancy of the radius of the orbit by a corresponding increase in the magnetic field and the frequency of the accelerating fields, as well as the particles' achievement of the energy required for irradiation in the accelerator. So, at first, we consider inserting PROTON-type agents into an environment that models the operation of an accelerator. Thus, the main attributes of the environment are the induction of the magnetic field, the frequency of the accelerating field, the operating time of the accelerator, the radius, and the energy the particle must acquire before leaving the synchrotron. The induction of the magnetic field, the accelerating field's frequency, and the time can be specified not only by specific values but also by possible interval values.

To calculate ionisation losses and protons' scatter on atoms and nuclei, we need to know the proton velocity v or its impulse p, which can be calculated by knowing its kinetic energy E. Let's describe the process of calculating these attributes in the form of a behavioural equation:

```
PROTONS_DATA = ((GetParticles_v);(GetParticles_p)),
```

where `GetParticles_v` and `GetParticles_p` are actions of the PROTON type agent. First, we find the proton velocity v – action `GetParticles_v`, and then the impulse p – action `GetParticles_p`.

```
GetParticles_v = (Forall (i:int)( 1<=i<=N)->
"PROTON \#part1: action 'Get_Velocity';"
(particles(i).velocity = particles(i).charge *
magneticFieldInduction * particles(i).mass)),
```

In this case, the velocity of the particle at the time of "departure" of the particles from the accelerator is determined by the formula: $v = qBm$. The proton impulse is determined by the ratio $v/c = pc/(E + mc^2)$.

```
GetParticles_p = (Forall (i:int)( 1<=i<=N)->
{"PROTON \#part1: action ' Get_momentum ';" }
(particles(i).momentum = (particles(i).velocity *
```

```
(particles(i).energy +particles(i). mass *c*c))/c*c)),
```

The second step is to model the interaction of a proton beam with the matter. At this stage, we start working with agents such as SUBSTANCE. In this case, the agent of the SUBSTANCE type acts as an environment for the agent type PROTON. The main task of the stage is to determine the characteristics of the proton beam and dose delivery systems to the patient.

Sequences of possible interactions of protons with substance (inelastic interaction with electrons of atoms (inhibition, braking ability), elastic interaction with atomic nuclei (scattering), nuclear reactions) are considered in the form of behavioural equations. For each interaction, we determine the transition using a hybrid scheme, i.e. the addition of a differential equation that determines the change in the number of particles carried through a unit surface, depending on time; change in the energy of radiation absorbed by the substance per unit mass; differential angular distribution of particles after passing a layer of matter with a certain thickness, etc.

For example, the linear brake capacity of the particle at depth x is determined by the formula: $S = (-dE/dx)$, $E(x) = (R - x)/\alpha)^{1/p}$, where $R = \alpha * E^p$, $\alpha = 1.9 * 10^{-3}$ for the proton and $1.73 * 10^{-3}$ for the α-particle, $p = 1.8$ for the proton and 1.5 for α-particle.

Then the action of calculating the linear brake capacity of the particle will be formalised as follows:

```
GetParticles = (Forall (i:int, j:int)
((1<=i<= partNum) && (1<=j<= subNum) &&
(x<=substances(j).l)) -> (DIFF(1)),
```

where $DIFF(1)$ is the differential equation that is presented in the form:

```
1: -d(particles(i).E)= substances(j).massBrakingCapacity*d(x)
```

Solving differential equations in the insertion modelling system is considered as a part of the software package of the algebraic modelling system (APS) in the Aplan language [1,4]. Special converter and Solver were developed for formal models that contain differential equations. So, the files with actions and differential equations will be transferred to the converter of differential equations (we have two files for this case: *.ACT and *.diff), which will convey the prepared equations to the Solver by completing the necessary transformations. The resulting equations will be substituted in the *.ACT file instead of DIFF (¡Number¿). Other mathematics systems, such as Maple, can also be used for solving equations. Solving or using an approximation of the solutions of these equations will determine the step of algebraic modelling. Given that the state of the agents and the environment will be defined as symbolic, we will determine the presence of the script, provided it exists, in symbolic form. Using algebraic methods, we can further determine an example of a scenario of reachability of the desired property (for example, the required characteristics of the proton beam, etc.) with specific attributes of the environment. Thus, we can consider the behaviour of the model

for different initial values (irradiation time, the angle of the particles' "injection" to the substance, the length of the particles' way in the substance, etc.), which will allow us to analyse and study the necessary indicators to improve the characteristics of the particle's beam and systems dose delivery to the patient.

4 Conclusions

The implementation of innovative technologies and research tools in cancer treatment, particularly in radiation therapy, has brought it to a qualitatively new level in recent years. However, despite significant successes, this issue requires active experimental research, proof of safety, and successful application results. One of the effective, safe, and low-budget methods that allow solving this problem is computer molecular modelling. Multilevel modelling based on a simulation or probabilistic approach is a daunting task, as it requires processing a huge amount of information. Using an algebraic approach allows us to abstract from information of varying complexity and solve complex problems by deducing relevant knowledge using formal methods. The main advantage of using algebraic modelling is that it will provide an opportunity to derive consequences from the laws and, therefore - can offer new facts and theories that will allow us to solve complex problems.

At this research stage, a method of applying algebraic modelling to formalise the structure of organic and inorganic substances has been developed. The work of particle accelerators (cyclotron, synchrotron) and the physics of the interaction of a proton beam with matter was formalised at the upper level. The next stage is to model the interaction of beams of accelerated particles (protons, ions) with amino acids/proteins/cells, and in particular, the modelling of possible interactions of protons with cellular structures at the quantum interaction level as a prerequisite for activating the process of programmed cell death.

References

1. APS and IMS (2023). https://apsystems.org.ua/
2. Dahle, T., et al.: Monte Carlo simulations of a low energy proton beamline for radiobiological experiments. Acta Oncol. **56**(6), 779–786 (2017). https://doi.org/10.1080/0284186X.2017.1289239
3. Dobbe, R., Tomlin, C.J.: Hybrid systems modeling for (cancer) systems biology. bioRxiv **035022** (2015). https://doi.org/10.1101/035022
4. Insertion model creator (2023). https://rd.litsoft.com.ua/
5. Jones, B.: Clinical radiobiology of proton therapy: modeling of RBE. Acta Oncol. **56**(11), 1374–1378 (2017). https://doi.org/10.1080/0284186X.2017.1343496
6. Letichevsky, A., Gilbert, D.: A model for interaction of agents and environments. In: Bert, D., Choppy, C., Mosses, P.D. (eds.) WADT 1999. LNCS, vol. 1827, pp. 311–328. Springer, Heidelberg (2000). https://doi.org/10.1007/978-3-540-44616-3_18
7. Letichevsky, A., Letychevskyi, O., Peschanenko, V.: The non-deterministic strategy of rewriting. Control Syst. Comput. **6**, 53–58 (2013)

8. Lincoln, P., Tiwari, A.: Symbolic systems biology: hybrid modeling and analysis of biological networks. In: Alur, R., Pappas, G.J. (eds.) HSCC 2004. LNCS, vol. 2993, pp. 660–672. Springer, Heidelberg (2004). https://doi.org/10.1007/978-3-540-24743-2_44
9. Pirhadi, S., Sunseri, J., Koes, D.R.: Open source molecular modeling. J. Mol. Graph. Model. **69**, 127–143 (2016). https://doi.org/10.1016/j.jmgm.2016.07.008

Computer Design and 3D Printing of New-Generation Multifractals

Ivan Grabar[1(✉)] and Yuri Kubrak[2]

[1] Polissia National University, Staryi Blvd 7, Zhytomyr 10008, Ukraine
ivan-grabar@ukr.net
[2] Zhytomyr Polytechnic State University, Cherniachovskiy 103, Zhytomyr 10005, Ukraine

Abstract. Among the large number of algorithms and technologies for synthesizing multifractals, a special place is occupied by algorithms based on the Brownian dynamics of a point in a field of N attraction forces. It is shown that the boundary points of the paths of such motion form fractal sets. This allows us to significantly enrich the existing synthesis capabilities, to obtain the dependence of fractal parameters on the parameters of the field of gravity, in an isotropic or anisotropic formulation, with the ability to synthesize almost any isotropic or anisotropic mono- or multifractals with predefined service-useful properties. A list of parameters controlling the properties of synthesis is given: the number and relative position of the attractive forces, their field characteristics, coordinates of the points of application, characteristics of the field component tensors, transparency component of the field barriers, perturbation of the coordinates of the points of application of the forces, etc. The obtained dependences of the influence of control parameters on the characteristics of synthesized fractal structures allow us to come close to the correct formulation and solution of the inverse problem of fractal analysis for new generation mano- and multifractals.

Keywords: Synthesis of multifractals · isotropic and anisotropic multifractals · Brownian dynamics · inverse problem of fractal analysis · computer modeling of multifractals

1 Actuality

The development of modern engineering and technology stimulates the design of materials and structures with predetermined properties: self-similarity, high porosity, sufficient residual strength in compression, bending, and shearing. These materials should also have a low specific gravity and a large free surface area. It is fractal structures that best meet these requirements. Traditionally, monofractals are considered to be more studied - geometric objects with an infinite number of generations, each subset of which is self-similar to a subset of the previous generation and the set of the first generation, i.e., the elements of each subsequent generation are a reduced image of the previous generations. When moving to the next generations, the initial figure is repeatedly reduced (snowflakes, tree crowns, respiratory organs of living beings, etc.).

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2026
I. Potapov et al. (Eds.): RTC-Digital 2023, CCIS 2647, pp. 212–220, 2026.
https://doi.org/10.1007/978-3-032-04731-1_28

In the case of anisotropic fields, we obtain an anisotropic multifractal, where the multilevel self-similarity is preserved, but has its own peculiarities. It is to reveal these features that it is necessary to develop algorithms for synthesizing multifractals in wide ranges of control parameters. Our algorithms allow such conditions to be realized, as will be shown below. The visualization of anisotropic multifractals with a gradual change in the parameters of the field of attractive forces and the location of their application points allows us to quantify these features.

One of the most common areas of application of mono- and multifractals is in the following: fuel mixture spraying technologies, turbulence control of liquids and gases, construction of new generation fractal filters, new technologies for oil and natural gas production [1, 2], etc.

In spite of the rapid development of algorithms and technologies for synthesizing fractal structures [3–5], they allow us to only partially imitate Nature, the great creator of fractals. In fact, there are several methods for constructing fractals [3–6]:

1. Graphical - define an infinite sequence of graphical transformations of some initial sets (sequential cutting of parts of a segment generates a Cantor set, parts of a square - a Serpinsky set, replacing a segment with a triangular or rectangular pulse generates a Koch curve, etc.).
2. Selection of special functional mappings, most often in complex functions (for example, Julio sets).
3. By computer graphics methods, infinite formations of self-similar subsets according to predefined algorithms by points 1–2.
4. Numerical studies of nonlinear systems far from equilibrium have revealed strange attractors with a fractal nature (Lorenz, Feigenbaum, Haken, and others).

We have developed algorithms for the synthesis of multifractal sets that differ significantly from those given in paragraphs 1–3, with maximum approximation to natural processes. They are based on the Brownian dynamics of a point in a field of N attraction forces. In such nonlinear dynamics, the boundary points of the trajectories of such motion form fractal sets [7–9]. This makes it possible to quantitatively study the dependence of fractal characteristics on the parameters of the field of attraction, and also allows us to synthesize almost any isotropic or anisotropic mono- or multifractals with predefined service-useful properties. We have found that the controlling parameters of synthesis are: the number and relative position of the attractive forces, their field characteristics, coordinates of the points of application, characteristics of the field component tensors, transparency component of the field barriers, perturbations of the coordinates of the points of application, etc. In our algorithms, the force fields of each of the N gravitational forces can be individualized. This allows us to synthesize a multifractal with a vector of fractal dimensions D { D_1, D_2, D_3,…, D_N }. As a rule, for such multifractals, it is important to achieve the closure conditions of the first generation subsets to further obtain integral samples in their D3 printing.

2 Main Material

2.1 Isotropic Fields of Attraction

Algorithms allow us to synthesize fractal sets in homogeneous isotropic fields. Conventionally, such fractals will be called isotropic. Figures 1, 2, 3, 4, 5–6 show examples of the synthesis of multifractals in isotropic fields of attraction. The captions under the figures indicate the number of gravitational forces N.

Fig. 1. One-dimensional multifractal, N = 2. It makes it possible to synthesize any macro-discrete spectra in synergistic systems far from equilibrium.

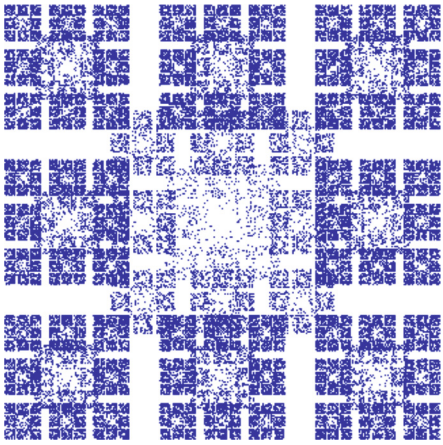


Fig. 2. A two-dimensional multifractal in an isotropic field of central forces, N = 9 (Sierpinski multifractal carpet with an additional force of attraction in the center of the matrix).

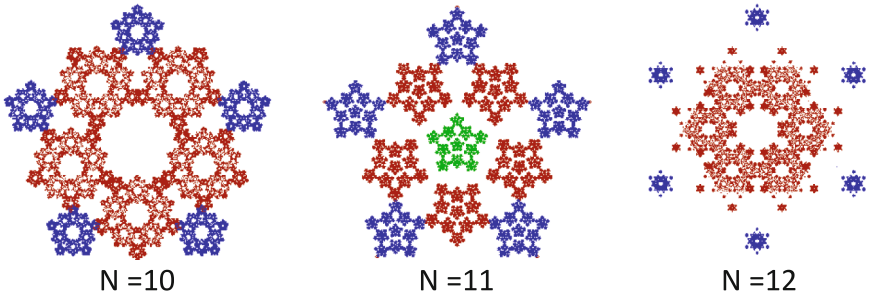


Fig. 3. Two-dimensional multifractals in an isotropic field of central forces placed on concentric rings. The number of central forces is 10, 11 and 12, respectively.

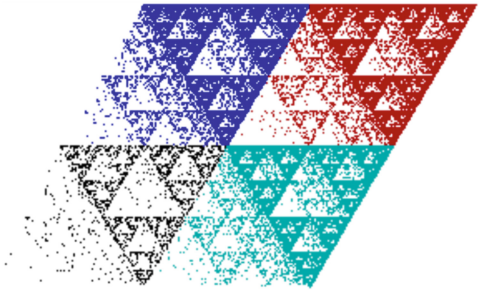


Fig. 4. A two-dimensional multifractal in an isotropic field of central forces with control of the transparency of the barrier of one force, N = 4.

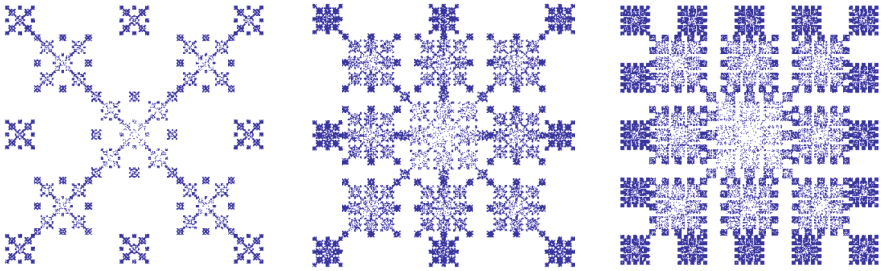


Fig. 5. Multifractal symmetric isotropic 5 × 5 lattices with the size ratios of the first generation sets 1:2:4.

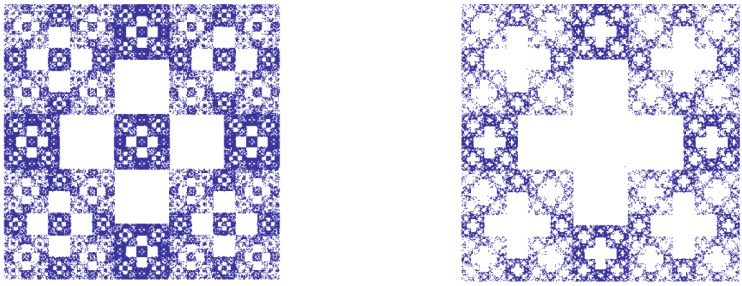


Fig. 6. Contour isotropic closed 3 × 3 multifractal lattices for N = 9 and N = 8.

2.2 Isotropic Fields of Attraction

In the more complex case of synthesis, the algorithms allow us to create anisotropic force fields for each of the central forces. Conventionally, we call such multifractals anisotropic. Figures 7–8 show examples of anisotropic multifractals synthesized under conditions of orthotropic anisotropy of some of the central forces. It is easy to identify anisotropic multifractals by the characteristic difference in the linear sizes of subsets of

the first (and subsequent) generations of multifractals in mutually perpendicular directions. In addition to orthotropic anisotropy, which is the easiest to implement, we have used oblique: rhombic and hexagonal anisotropic fields.

The algorithms allow setting the following synthesis control parameters: N_i – number of central forces; m_i - parameters of the force field of each central force; $(X_iY_iZ_i)$ - coordinates of the points of application of the central forces; ε_i - transparency coefficients of the barriers of each central force, as well as control of the anisotropy of the field of each force. This provides virtually unlimited possibilities for synthesizing mono- and multifractals with a set of properties necessary for practice. For some classes of multifractals synthesized by us, we obtained functional dependencies between the parameters of the field of central forces, the coordinates of the points of application of these forces, the transparency coefficients of the barriers of these forces, and the required service characteristics - components of the fractal dimension vector of the multifractal D { D_1, D_2, D_3,…, D_N}, total porosity, strength characteristics, aerodynamic and hydraulic resistance, etc. Some of these dependencies were obtained in quadratic form, and some were obtained as correlation dependencies [9]. These results allow us to be cautiously optimistic about the possibility of correctly solving the inverse problem of multifractal analysis. From our point of view, the complete solution of the inverse problem will allow the synthesis of any multifractal for a given set of the specified service characteristics.

Algorithms based on modeling the attractor of Brownian dynamics of a point in a field of N central forces have been brought to the technology of synthesis and 3D printing: mono- and multifractal lattices, multifractal antennas, rings, linear, planar and volumetric mono- and multifractals with predetermined characteristics of porosity, fractal dimensions, residual strength, aerodynamic and hydrodynamic drag, etc. For a minimum value of $N = 2$ (Fig. 1), the algorithms allow us to synthesize linear multifractals such as the Cantor set. For $N \geq 2$, the algorithms allow us to synthesize linear mono- or

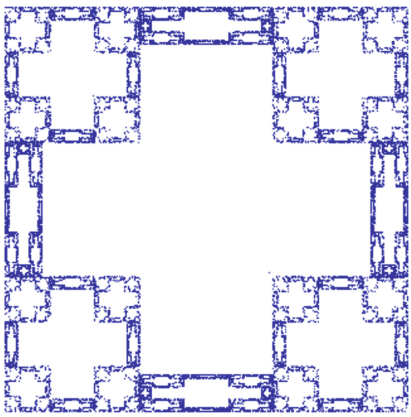


Fig. 7. A two-dimensional multifractal synthesized in an anisotropic field of central forces on a 3x3 lattice, $N = 8$ (anisotropic Sierpinski carpet, four corner central forces have isotropic fields, four lateral forces have anisotropic fields).

multifractals of any complexity and characteristics, for $N \geq 3$ - respectively, flat, and for $N \geq 4$ - three-dimensional.

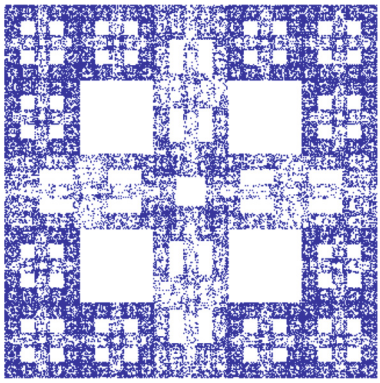


Fig. 8. A two-dimensional multifractal in an anisotropic field of central forces on a 5×5 lattice. The anisotropic fields of the elements a_{13}, a_{31}, a_{35}, a_{53}. $N = 16$.

2.3 Control of Coordinates of Points of Application of Attraction and Anisotropy of Their Field

The control of the coordinates of the points of application of attractive forces and the field anisotropy significantly enriches the possibilities of synthesizing multifractals. Figures 9–10 show examples of multifractal synthesis for cases where the coordinates of the points of application of attractive forces (corner points of synthesized fractals) and the anisotropy of their field are used as control parameters.

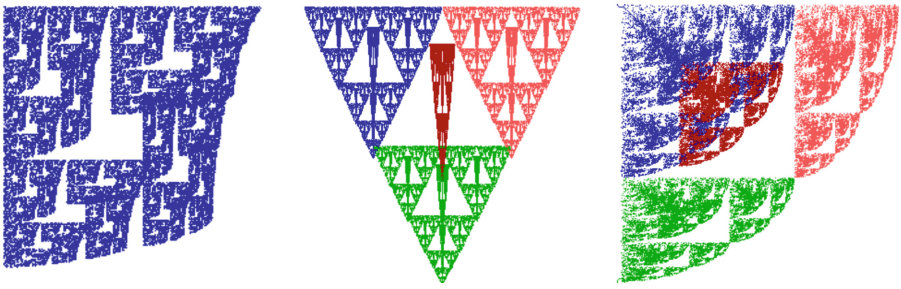


Fig. 9. Synthesis of multifractal gratings of curved profile by varying the anisotropy of the field of attractive forces and shifting their application centers.

2.4 3D Printing of Multifractals

Most of these multifractals were reproduced by 3D printing. We have fabricated solid-state multifractal lattices with dimensions $a0 \times a0 = 100 \times 100$ mm and a thickness

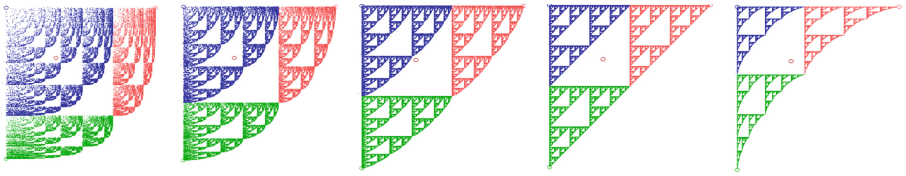


Fig. 10. Synthesis of multifractal lattices for N = 3 with windows of curved profile obtained by changing the anisotropy coefficient of the attractive field F2 and F3, respectively: 2.57; 1.55; 1.18; 1; 0.66 and fulfillment of the conditions for closure of the first generation subsets.

of 3 mm. A necessary condition for their 3D printing is the closure of subsets of the first generation of the multifractal. The fractal sets shown in Fig. 2 and Fig. 6 - Fig. 10 correspond to such conditions.

As 3D printing has shown, out of an infinite number of fractal generations, this technology has limitations related to the finite pixel size of the 3D printer. We obtained a condition for reproducing the maximum generation number of a fractal. This approximation gives a good correspondence with the experimental results for both isotropic and anisotropic multifractals. Thus, for monofractal lattices on an *n* x *n* matrix, the maximum generation number J_{max}, where cells can still be reproduced during their 3D printing, is determined by the following dependence:

$$J_{max} = \frac{\ln[\frac{a_{min}}{a_0}]}{\ln[\frac{1}{n}]}$$

a_0 is the size of the fractal lattice, a_{min} is the minimum pixel size of the 3D printer, and n is the size of the lattice.

Example: for a 3×3 multifractal lattice with dimensions $a_0 * a_0 = 100 \times 100$ mm in the form of a closed Sierpinski carpet for a 3D printer resolution of $a_{min} = 0.1$ mm.

$$J_{max} = \frac{\ln[\frac{0,1}{100}]}{\ln[\frac{1}{3}]} = 6,2878$$

Thus, under the given conditions, it is possible to reproduce a multifractal with a resolution of up to the sixth generation. At a_{min}= 1 mm, = J_{max} 4.191, i.e., the resolution will be only four generations of the multifractal.

The table shows the limit values of J_{max} generations that can be reproduced in 3D printing depending on the size of the lattice and the size of the minimum pixel amin of the 3D printer for a square lattice with a square side a_0= 100 mm (Table 1).

To obtain a high-quality image of a multifractal, it is necessary to repeat the iterative process of Brownian point dynamics for at least 5 million iterations.

The demonstrated results have been brought to technology:

- Design of multifractals of any complexity with predefined service and operational characteristics (structure, structure, fractal dimension tensor, porosity, residual strength, aerodynamic and hydraulic resistance, etc.;)
- Refinement and adjustment of parameters to customer requirements;

Table 1. The maximum possible number of generations of windows reproduced by 3D printing fractal lattices.

Lattice	a_{min}, mm	0,1	0,2	0,5	1
2 × 2	J_{max}	9,97	8,91	7,64	6,64
3 × 3	J_{max}	6,29	5,65	4,82	4,19
4 × 4	J_{max}	4,98	4,48	3,82	3,32

– 3D printing of multifractals from polymers and composites.

The results of computer modelling using our algorithms made it possible to construct the tensor mechanics of mono- and multifractals with the ability to determine the service-useful parameters of multifractals for predefined parameters, i.e., for the first time in the world, to come as close as possible to the correct solution of the inverse problem of mono- and multifractal analysis [9].

We evaluated the synthesis capabilities of our algorithms and estimated the power of the set of synthesized multifractals as Q - 10^{40}.

3 Conclusions

1. It is shown that the Brownian dynamics of a point in the field of N attractive forces allows us to synthesize fractal structures with an almost unlimited range of properties. In this case, the number of subsets of the J-th generation of the multifractal formed by the number of N attractive forces is NJ.
2. The condition for the maximum reproduction of the number of multifractal generations by 3D printing technologies is obtained and its dependence on the number of attraction forces, fractal dimensions, and pixel size of the 3D printer is shown.
3. The proposed synthesis algorithms allow the processes of design, synthesis, and experimental study and application of multifractals in modern technologies to be brought to a new stage - the conscious synthesis of multifractals with predetermined properties.

It remains an open question why the Brownian dynamics of a point in a field of N attractive forces with isotropic or anisotropic fields gives rise to superordered fractal sets? It is the dynamics of a point in the field of N attractive forces when they are chaotically switched on that gives rise to a multifractal, while deterministic sequential switches on the attractive forces give rise to limit cycles [9].

References

1. Balankin, A., Patiño Ortiz, J., Patiño, O.M.: Inherent features of fractal sets and key attributes of fractal models. Fractals **30**(4), 2250082 (2022). https://doi.org/10.1142/S0218348X22500827
2. Balankin, A., Ramírez-Joachin, J., González-López, G., Gutierrez, H.S.: Formation factors for a class of deterministic models of pre-fractal pore-fracture networks. Chaos Solitons Fractals **162**(9), 112452 (2022). https://doi.org/10.1016/j.chaos.2022.112452

3. Mandelbrot, B.: The Fractal Geometry of Nature. W.H. Freeman, San Francisco (1982)
4. Haken H.: Synergetics. An introduction. Nonequilibrium phase transitions and self-organization in physics, chemistry. Springer-Verlag. Berlin Heidelberg New York Tokyo (1983)
5. Devaney, R.: Cantor and Sierpinski, Julia and fatou: complex topology meets complex dynamics. Not. Am. Math. Soc. **51**(1), 9–15 (2004)
6. Taylor, T.D., Hudson, C., Anderson, A.: Examples of using binary cantor sets to study the connectivity of sierpiński relatives. Fractals **20**(1), 61–75 (2012)
7. Grabar, I., Grabar, O.: Fractal Ornaments. Polissia National University, Zhytomyr (2021)
8. Grabar, I., Kubrak, Y.: Synthesis of multifractals by brownian dynamics of a point in a field of N central forces (virtual). In: 16th Chaotic Modeling and Simulation International Conference (2023)
9. Grabar, I.: Synthesis of Multifractals. Polissya National University, Zhytomyr (2023)

Application of an Algebraic Approach to the Formalization of Complex Biochemical Processes at the Level of Quantum Interactions

Hanna Soloshenko(✉)

Kherson State University, 27 University Street, Kherson 73000, Ukraine
hannasoloshenko@gmail.com

Abstract. We apply algebraic modelling techniques and the quantum-chemical apparatus for modelling and checking the problems of organic chemistry, particularly the modelling of chemical reactions. The article presents the first steps of the research, where we consider the approach to modelling complex biochemical processes as hybrid systems. The formalization and analysis of properties are considered using an insertion modelling platform.

Keywords: Computational chemistry · algebraic modeling · molecular models · chemical reaction model · formal methods · insertion modeling

1 Introduction

Modelling of complex biochemical processes today occupies one of the main roles in research related to the manufacture of drugs, virus-cell interaction, cancer therapy, etc. To date, a significant number of methods, algorithms, and tools have been developed for building molecular models of biological systems [1–3]. Still, the presence of open problems in the fields of organic and inorganic chemistry, physics, and biology indicate the need to improve old ones and find new approaches and methods for studying properties of organic and inorganic substances and mechanisms of their interaction.

The presence of both discrete and continuous dynamics in biochemical processes makes it attractive to use hybrid systems for their simulation. The popular use of ordinary differential equation models has its limitations because they only capture continuous dynamics, whereas we observe many discrete dynamical phenomena in biology, such as gene switching or mutations. Hybrid systems modelling provides a framework in which both continuous and discrete dynamics can be modelled and analyzed. In addition, this approach provides methods for developing approximations and abstractions of complex dynamics that are amenable to analysis.

Research in the modelling of complex substances at the level of quantum interaction of atoms and molecules is relevant since the construction of such models allows studying the existing properties of the model and predicting those that have not yet been studied. Repeated confirmation of this are Nobel prize awards in chemistry, physics, and medicine for research at the interface of biochemistry and molecular biology, as those that solve

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2026
I. Potapov et al. (Eds.): RTC-Digital 2023, CCIS 2647, pp. 221–227, 2026.
https://doi.org/10.1007/978-3-032-04731-1_29

the main chemical problems of humanity. In 2002, the Nobel Prize in Chemistry was awarded for the discovery of powerful analytical methods for the study of biological macromolecules [4]. In 2021, Benjamin List and David Macmillan won the Chemistry Prize for developing a new precise tool for molecular design: organocatalysis [5–8].

The rapid development of machine learning methods, systems for automatic theorem proving and solving mathematical problems, combining the latest achievements with the theory of agents and environments [9], which was created by O.A. Letychevskyi and D. Gilbert [10], allowed to develop and use the algebraic modelling method and the corresponding software tools based on it for modelling various systems [11, 12]. The multi-level algebraic method allows you to consider the essences of different levels of complexity, and therefore to consider the behaviour of biological objects in real-time, modelling and considering not individual scenarios, but a set of possible behaviours. Thus, we get the opportunity to analyze the properties of substances, and model and study complex biochemical processes and their results, which can be especially valuable in the fields of pharmacology, medicine, and chemistry. In the following sections of the article, we give examples of the application of the algebraic approach to the formalization of substances, chemical reactions and processes, as well as provide brief information about the systems and tools used.

2 Modeling System and Algebraic Approach

The Insertion Model Creator modelling system, which uses symbolic modelling methods, including algebraic and deductive-formal methods for solving complex problems, was used to model and simulate models. Insertion Model Creator includes several subsystems and libraries for implementing algebraic formal methods and integration with other software systems.

The main features of the system are modelling technology; model-based development; verification of the reachability of model properties [13]. Insertion Model Creator design is based on the insertion modelling system IMS [12], which in its turn is based on the algebraic programming system APS [9]. Insertion modelling focuses on building models and studying the interaction of agents and environments in complex multi-agent systems [14]. The Basic Protocol Specification Language (BPSL) is used to represent requirements specifications [15–17]. The central concept of this language is the concept of a basic protocol, which is a sequence diagram with pre- and post-conditions as logical formulas, which are interpreted in the scope of different theories and different data types. The semantics of BPSL allow the creation of concrete and symbolic models at different levels of abstraction. For mathematical refinements, we use a transition system for an agent, which is the most abstract mathematical concept modelling system that evolves over time. As part of the insertion modelling method, we use the specifications of the behaviour algebra for formalization [14]. As part of the modelling system, problems with a continuous component also can be solved.

3 Formalization of Substances and Chemical Reaction Processes Using an Algebraic Approach

We consider the substance model as a multi-agent environment in which agents interact at different levels of abstraction. Agents of higher levels are environments for other agents, which in turn can also be environments. We will consider the formalization of the substance model in the insertion modelling system, in order to study and analyze the specified properties, using the algebraic modelling of this multi-level system.

Agents that interact in matter, as an environment of the highest level, are characterized by a set of attributes that we will define in the process of formalization. Each agent performs a certain set of actions under certain conditions, which forms its behaviour. We will compile behavioural equations for interacting agents based on arbitrary initial attribute values.

Thus, the structure of SUBSTANCE-type agents is given by a set of MOLECULE-type agents X, which, in turn, contain a functional representing the set of atoms that make up the given molecules. Atoms, in turn, are represented by the type of agents ELEMENT, whose attributes fully characterize the chemical elements of periodical table. The main attributes of the ELEMENT agent are quantum numbers principal quantum number (n), orbital angular momentum quantum number (l), magnetic quantum number (m_l), spin quantum number (m_s), electronegativity, valence (valence), nuclear charge (Z), relative atomic mass (A_r), electronic configuration of an atom.

The MOLECULE agent contains the following attributes: the set of atoms it consists of, the electronic configuration of the molecule, bond length (d_bond), bond energy (BondEnergy), dipole moment (DipoleMoment), molar mass (M_r), bond order (bondMO – by the method of molecular orbitals, bondV – by the method of valence bonds).

Accordingly, we can consider substances according to the structure of their molecular orbitals, nuclear models of the atoms that make up their composition. The amount of substance in moles, concentration per volume unit is also specified (these parameters can be set as initial values of the experiment or determined during simulation). As an example, consider the formalization of the substance represented by sodium hydroxide NaOH:

substance1.molecules == molecule1 &&
substance1.nu == 35 &&
molecule1.atomsNum == 3 &&
molecule1.atoms(1).name == H &&
molecule1.atoms(2).name == O &&
molecule1.atoms(3).name == Na &&
molecule1.bondV(molecule1.atoms(1), molecule1.atoms(2)) == 1 &&
molecule1.bondV(molecule1.atoms(2), molecule1.atoms(3)) == 1 &&
Elements(1).name == H &&
Elements(1).n == 1 &&
Elements(1).A_r == 1.00794 &&
Elements(1).electronegativity == 2.2 &&
Elements(1).Orbital(1,0,1) == 1 &&

Elements(1).m_s(1,0,1,1) == 0.5 &&
Elements(1).Z == 1 &&
…

The initial values needed for the simulation, such as the formula/structure of the molecule/substance, we can get from the MOL file using a special parser.

Having received information about the structure of a molecule/substance, we have data on the values of its main attributes (molecular weight, saturation, density, activation energy, etc.), including the values of the attributes of atoms, and therefore we can consider and model reactions at the electronic level, guided by the laws of quantum chemistry and physics.

Interactions of agents (atoms, molecules, substances) are formalized using formal actions. Their possible sequences are considered as behavioural equations. In particular, for each interaction, we determine the transition using a hybrid system, that is, adding differential equations that determine the change in the number of molecules that have reacted, depending on time, the reaction rate depending on the concentration of substances, temperature, pressure, activity of catalysts, etc.

Attributes can be specified not by specific values, but by possible intervals of values. The medium of interaction for agents of the substance/molecule type can be a certain reaction or another substance. Thus, we observe the change in the environment and the value of the attributes of the agents when the parameters of temperature, pressure, etc. change.

As an example, consider the dissociation reaction of substances with the formula EOH.

This reaction will be represented by the following behaviour:

```
DISSOCIATION_EOH = (
(dissociationEOH1 + !dissociationEOH1.dissociationEOH2 +
        !dissociationEOH2);
(dissociationResultEOH1.dissociationResultEOH2.reactionRateCoeff.
getConcentrationToTime))
```

where the actions of dissociationEOH1, dissociationEOH2 determine which bond will be broken depending on what kind of substance it is (base or acid); dissociationResultEOH1, dissociationResultEOH2 - determination of the first and second products of the reaction, respectively; reactionRateCoef - calculation of the reaction rate constant; getConcentrationToTime - determining the change in concentration of a substance over time).

As an example, consider an action that describes the determination of one of the reaction products:

```
dissociationResultEOH1 = (Forall (i:int, j:int, rNum:int, k:int).
(1 <= rNum <= reactNum &&
1 <= i <= reactions(rNum).reagents(1).molecules.atomsNum &&
1 <= j <= reactions(rNum).reagents(1).molecules.atomsNum &&
i ! = j &&
reactions(rNum).type == dissociation &&
reactions(rNum).reagents(1).molecules.formula == EOH &&
```

reactions(rNum).reagents(1).molecules.
bondV(reactions(rNum).reagents(1).molecules.atoms(i),
reactions(rNum).reagents(1).molecules.atoms(j)) == 0 &&
IFTHE(reactions(rNum).reagents(1).molecules.atoms(i).name ! = O,
k = = i, k = = j)) >
(Reactions(rNum).products(1).molecules.atoms(1)) =
reactions(rNum).reagents(1).molecules.atoms(k)).

That is, we "review" all reactants of given reactions that correspond to the given formula. If it's a dissociation reaction (reactions(rNum).type == dissociation), the substance with the given formula EOH dissociates (reactions(rNum).reagents(1).molecules.formula == EOH) and a bond between some two atoms reagent is broken (reactions(rNum).reagents(1).molecules.bondV(i,j) == 0), we find the atom that will be "detached" from the compound and write it down as one of the results of the reaction.

When modelling reactions, we take into account the continuous component, that is, the gradual change of the medium depending on the change in time, temperature, and pressure, which is determined by the differential equation.

Thus, the change in the concentration of a substance over time due to the course of a chemical reaction will be formalized as follows:

```
getConcentrationToTime = Operator(
(Forall (i:int)(1 <= I <= reactNum) ->
        "MOLECULE # M1: action ' getConcentration';"
 (DIFF(2)),

2: d(reactions(i).reagents(i).Concentration) / d(reactions(i).reactTime)
  = (-1) * reactions(i).rateCoeff * reactions(i).reagents(i).Concentration
```

where the change in the concentration of a substance over time due to the course of a chemical reaction is determined by the kinetic equation of the reaction: $\frac{dC}{dt} = k \cdot C_A^{n_A} \cdot C_B^{n_B} \cdot \cdots \cdot C_i^{n_i}$.

Since the reaction under consideration is a first-order reaction, the reaction rate can be calculated as follows: $\frac{dC}{dt} = -k \cdot C$.

Solving or using an approximation of the solution of this equation will determine the algebraic modelling step. At the same time, the state of agents and the environment will be defined as non-specific, but symbolic, since there are intervals of values of the attributes of the environment.

Having this formalization, we run the model in the Insertion Modeling Creator system to determine the reachability of some properties.

Such a property can be the destruction of diseased tumour cells and the resistance of the cell to the invasion of the virus. This property is determined by some target formula F.

At each step, we determine the satisfiability of the formula F & S, where S is the current state of the simulation. If the condition holds, then there is a scenario that leads to the target formula. We receive this scenario in the form of a track - a sequence of actions of agents in a formalized environment.

Using inverse modelling, we can find the initial conditions of the environment for which the target formula is achievable. This gives the exact concentration of substances to achieve formula F.

In practical terms, it can be the exact distribution of ingredients of future drugs or treatment conditions. In the search for a substance with certain properties, these are the initial values and the scenario of obtaining such a substance.

4 Conclusions

The results of the research allow us to claim the possibility of applying the algebraic approach and the tools used to research and solve the considered and similar problems, in particular, the modelling of complex biological and biochemical systems.

The main advantage of the proposed approach is that the knowledge is represented in algebraic form, in particular, as behavioural equations. Thus, in contrast to traditional simulation and probabilistic modeling, it is necessary to solve the corresponding equations, what in many cases allows for avoiding the problem of exponential explosion.

Hybrid systems modelling provides a framework in which both continuous and discrete dynamics can be modelled and analyzed. In addition, this approach provides methods for developing approximations and abstractions of complex dynamics that are amenable to analysis.

At this stage of the research, we have developed a formalization technique that allows modelling simple and complex organic and inorganic substances, analyzing and researching the main characteristics of substances based on their atomic structure and quantum interactions between their structural elements, simulating the flow of various types of chemical reactions between substances. It is planned to continue research in the formalization of complex chemical reactions, interactions of substances in organic chemistry, biochemistry, etc. These processes will be considered from the point of view of modelling complex hybrid systems using an algebraic approach.

References

1. Beentjes, C.H.L., Baker, R.E.: Quasi-Monte Carlo methods applied to tau-leaping in stochastic biological systems. Bull Math Biol. (2019)
2. Hughes-Oliver, J.M., et al.: ChemModLab: a web-based cheminformatics modelling laboratory. In: Silico Biol. pp. 61–81 (2012)
3. Morency, L.P., Gaudreault, F., Najmanovich, R.: Applications of the NRGsuite and the molecular docking software FlexAID in computational drug discovery and design. Methods Mol. Biol. **1762**, 367–388 (2018). https://doi.org/10.1007/978-1-4939-7756-7_18. PMID: 29594781
4. The Nobel Prize in Chemistry. https://www.nobelprize.org/prizes/lists/all-nobel-prizes-in-chemistry/
5. Levitt, M.: The birth of computational structural biology. Nat. Struct. Biol. **8**, 392–393 (2001)
6. Karplus, M.: Spinach on the ceiling: a theoretical chemist's return to biology. Annu. Rev. Biophys. Biomol. Struct. **35**, 1–47 (2006)
7. Johnson, P.: Warshel Fêted by Royal Society of Chemistry (2012). https://dornsife.usc.edu/news/stories/warshel-feted-by-royal-society-of-chemistry/

8. Mathpal, D., et al.: Pharmacophore modeling, docking and the integrated use of a ligand- and structure-based virtual screening approach for novel DNA gyrase inhibitors: synthetic and biological evaluation studies. RSC Adv. **25**;11(55), 34462–34478 (2021). https://doi.org/10.1039/d1ra05630a. October 2021
9. APS & IMS. https://apsystems.org.ua/. Accessed 28 Feb 2023
10. Letichevsky, A., Gilbert, D.: A model for interaction of agents and environments. Recent Trends Algebr. Dev. Tech. 311–328 (2000)
11. Letychevskyi, O., Peschanenko, V., Volkov, V.: Algebraic virtual machine and its applications. Technol. Educ. Res. Ind. 23–41 (2002)
12. Letichevsky, A., Letychevskyi, O., Peschanenko, V.: Insertion modeling and its applications. Comput. Sci. J. Moldova **24**(3), 357–370 (2016)
13. Modeling system Model Creator. https://rd.litsoft.com.ua/
14. Letichevsky, A., Gilbert, D.: A model for interaction of agents and environments. In: Bert D., Choppy C., Mosses P.D. (eds) Recent Trends in Algebraic Development Techniques. WADT 1999, LNCS, vol. 1827, pp. 311–328 (1999)
15. Baranov, S., Jervis, C., Kotlyarov, V., Letichevsky, A., Weigert, T.: Leveraging UML to deliver correct telecom applications. In: Lavagno, L., Martin, G., Selic, B., (eds.) UML for Real: Design of Embedded Real-Time Systems. Kluwer Academic Publishers, Amsterdam (2003)
16. Letichevsky, A.A., et al.: System specification with basic protocols. Cybern. Syst. Analyses (2005)
17. Letichevsky, A., Kapitonova, J., Letichevsky, A., Jr., Volkov, V., Baranov, S., Weigert, T.: Basic protocols, message sequence charts, and the verification of requirements specifications. Comput. Netw. **49**(5), 661–675 (2005)

Author Index

A
Adamson, Duncan 90
AlZoubi, Alaa 47
Areshkov, Pavlo 54
Astafieva, Mariia 175

B
Babudzhan, Ruslan 26
Babych, Tetiana 3
Bodnenko, Dmytro 175

C
Cannings, Nigel 60
Clapham, Peter 54
Constance, Tracy Gooodluck 60
Cope, Wei 54

D
de Angelis, Marco 113
Dev, Kapil 20
Dmytriyeva, Olga 97
Druzhinin, Evgeniy 161

F
Fedorovich, Oleg 161
Fedosova, Irina 12
Ferson, Scott 113
Frolova, Alina 54

G
Glackin, Cornelius 60
Grabar, Ivan 212

H
Havryliuk, Maksym 3
Horyslavets, Dmytro 54
Hrytsiuk, Petro 3
Huskova, Vira 97

I
Ibrahim, Nasir 47

J
Joslin, Emily 54

K
Kardash, Oksana 3
Khoroshylov, Serhii 39
Koshevko, Julia 169
Kovalov, Serhii 155
Kubrak, Yuri 212

L
Larin, Oleksiy 76
Levytska, Tetiana 12
Lokaziuk, Oleksandra 175
Lugo, Hector Estrada 113
Lysenko, Iryna 146
Lytvyn, Oksana 175

M
Makarenko, Aleksander 127
Malaschonok, Gennadi 182
Maltby, Harry 60
Mantelos, Harry Sapranidis 104
Martincorena, Inigo 54
McCafferty, James 54
Miroshkin, Oleksandr 155
Moniri, Mansour 60

O
O'Dea, Emily 54
Oliver, Charlotte 54

P
Patelli, Edoardo 113
Paterson, Anna 54
Piatykop, Olena 12
Potapov, Igor 135

© The Editor(s) (if applicable) and The Author(s), under exclusive license to Springer Nature Switzerland AG 2026
I. Potapov et al. (Eds.): RTC-Digital 2023, CCIS 2647, pp. 229–230, 2026.
https://doi.org/10.1007/978-3-032-04731-1

Prokhorov, Oleksandr 161
Pronina, Olha 12
Proshkin, Volodymyr 175
Przybilla, Moritz 54

R
Rajwadi, Marvin 60
Ralph, Jason F. 135
Redka, Mykhailo 39
Resch, Michael 155
Romantsova, Olena 54
Ryndyk, Yaroslav 54

S
Sandrakov, Gennadiy V. 119
Shapovalova, Mariia 26
Shvets, Galina 169
Skorobohatov, Oleksandr 54
Skryabina, Anna 84
Snytyuk, Vitaliy 197
Soloshenko, Hanna 221
Stegantseva, Polina 84
Sukharskyi, Serhii 190
Svjatnyj, Volodymyr 155

T
Tarasich, Yuliia 205
Teague, Jon 54
Tmienova, Nataliia 197
Triommatis, Theofilos 135
Tukalo, Mykhailo 54

V
Viazovychenko, Yuliia 76
Vodka, Oleksii 26

W
Wall, Julie 60

Z
Zakharkevich, Oksana 169
Zhogoleva, Nadia 69
Zhylenko, Tetiana 169